

Verification Continuum™

Library Developer's Guide

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Preface

The Verdi Library Developer's Guide discusses how to create symbol libraries to display gate-level schematics in Novas' *nSchema*.

Customer Support

For any online access to the self-help resources, you can refer to the documentation and searchable knowledge base available in SolvNetPlus.

To obtain support for your Verdi product, choose one of the following:

- Open a case through SolvNetPlus.

Go to <https://solvnetplus.synopsys.com/s/contactsupport> and provide the requested information, including:

- Product L1 as Verdi
- Case Type

Fill in the remaining fields according to your environment and issue.

- Send an e-mail message to verdi_support@synopsys.com.

Include product name (L1), sub-product name/technology (L2), and product version in your e-mail, so it can be routed correctly.

Your e-mail will be acknowledged by automatic reply and assigned a Case number along with Case reference ID in the subject (ref:....ref).

For any further communication on this Case via e-mail, please send e-mail to verdi_support@synopsys.com and ensure to have the same Case ref ID in the subject header or else it will open duplicate cases.

Note: In general, we need to be able to reproduce the problem in order to fix it, so a simple model demonstrating the error is the most effective way for us to identify the bug. If that is not possible, then please provide a detailed explanation of the problem along with complete error and corresponding code, if any/permissible.

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

Library Developer's Guide

This document describes how to create symbol libraries to display gate-level schematics in Novas' *nSchema*.

This document is organized as follows:

- *Novas Symbol Library Introduction*
- *Creating and Verifying a Symbol Library*
- *Use Symbol Libraries*
- *Novas Map File Format*
- *Symbol Library Warning Messages*

Novas Symbol Library Introduction

Novas Symbol Library Overview

To generate gate-level schematics in *nSchema*, you must have the **Novas Symbol Library** for library cells. With this symbol library, Novas displays gate-level cells using meaningful symbols and provides the functional and pin type information needed to operate correctly with trace-related commands in *nSchema*.

Since the Novas Symbol Library only requires functional information (rather than timing information), you can use a single symbol library for different libraries with the same cell name and pin name. A map file is used to map different libraries that have different cell/pin name styles to the **Novas Symbol Library**.

Refer to the *Symbol Library Installation* chapter of the *Installation and System Administration Guide* for details on installing symbol libraries and the list of provided libraries.

Novas places the symbol library by default in the <NOVAS_INSTALL_DIR>/share/symlib directory. Each library consists of one directory.

<ASIC>.lib++ Symbol cells are located in this directory.

Why are Symbol Libraries Important?

If the gate cells in the design are without the symbol mappings, executing **Active Fan-in Cone**, **Fan-in Cone**, and **Fan-out Cone** in *nSchema* does not function properly for the design. Indeed, the schematics only displays squares/blocks instead of logic cells and only one level is traced at a time.

If you have symbol mappings for gate cells, logic gates are differentiated from sequential cells.

Creating and Verifying a Symbol Library

Choose one of the Novas utilities and use the following steps to create a symbol library for Novas' *nSchema*.

- `syn2SymDB`
 - Create the symbol library from Synopsys library (`*.lib`).
- `map2SymDB`
 - Create a map file for the library.
 - Create the symbol library from the map file.

`syn2SymDB`

Creating the Symbol Library From Synopsys Library (`*.lib`)

1. Locate Synopsys `.lib` files for the appropriate standard cells for the design.
2. Create a directory to place the symbol libraries. The mapped symbol libraries can be placed anywhere. However, it is recommended to use a common area similar to where you place other tool support files. Try to keep all tool requirements local to a standard cell library.
3. `cd` to the directory created in step 2.
4. Run `syn2SymDB` (located in Novas install bin directory) on the `.lib` files to create the Novas equivalent symbol libraries. Refer to *Usage and Examples* described in the next section for the details of this utility.
5. Repeat for all standard cell libraries being used.

Usage and Examples

Use the `syn2SymDB` program to translate your Synopsys library files into Novas' symbol library.

NOTE: Novas only supports `.lib` files, not `.slib` files.

Usage:

syn2SymDB [option] synFile [synFile2 ...]

Options	Explanation
-h -help	Prints out the help message.
-L	Converts all cell names to lowercase.
-U	Converts all cell names to uppercase.
-l	Converts all pin names to lowercase.
-u	Converts all pin names to uppercase.
-allcase	Generates libraries with different combinations of case conversion for cell names and pin names. They are libraries with no case conversion. Also, the libraries with cell/pin names converted to upper cases, upper/lower cases, lower/ upper cases, and lower/lower cases respectively.
-m	Generates the mapping file.
-vb	<i>Verbose</i> mode.
-o libName	Specifies the output library name.
-skst	Skips cell generated by state table.
-skfl	Skips cell with logic expression in ff()/latch().
-nowarn <ALL POLARITY STATETABLE>	ALL: Suppresses all warning messages. POLARITY: Suppresses warning message for polarity. STATETABLE: Suppresses warning message for state table.
-stop	Stop program when warning for information lost (for <i>nLint</i> only).
synFile	Specify the input SYNOPSYS library file.

Examples

```
syn2SymDB -o my_symbol tsmc.lib
```

Create a symbol library `my_symbol.lib`++ from `tsmc.lib`.

```
syn2SymDB -o my_symbol /home/symbol_lib/syn1.lib  
/home/symbol_lib/syn2.lib
```

Create/merge a symbol library `my_symbol.lib`++ from multiple Synopsys libraries `syn1.lib` and `syn2.lib`.

map2SymDB

Creating a Map File for the Library

The map file maps all your library cells to the corresponding Novas symbols. Novas provides a generic file (`generic.sym`) that contains the most popular ASIC cells for your quick reference. When you create the map file, you can map a library cell directly to the corresponding cell in Novas generic symbol file (`generic.sym`). If the Novas generic symbol file contains no corresponding cell, you need to define the cell explicitly. Use the `map2SymDB` program to translate the map file into Novas Symbol Library.

The following example is part of a map file that includes cells that directly map to the Novas generic library along with one explicitly defined cell.

```
CELLMAP E_AND2 AN2;
PINMAP I0 A
PINMAP I1 B
PINMAP O Z
; where E_AND2, I0, I1, O is Novas generic library cell/pins
; and AN2, A, B, Z is user's cell/pins. The above statement
; tells Novas to map AN2 to E_AND2 and pin A, B, Z to I0, I1,
; O respectively.
;
CELLMAP E_AND2 AN2P;
PINMAP I0 A
PINMAP I1 B
PINMAP O Z
; Since Novas only uses cell's function, the difference
; between user cell AN2 and AN2P is just the driving
; capability. You can map both of them to E_AND2
;
```

NOTE: Symbols for generic-mapped cells contain a fixed pin sequence set by default in the Novas generic library. You cannot change this pin sequence by changing the pin order in the map file.

The pin order for *Verilog* module declarations follows the pin order in the map file.

```
Cell E_DBHRBN DFF;
D INPUT L D
CKB INPUT L C L
RB INPUT B AR L
Q OUTPUT R L
QB OUTPUT R L L
; Explicitly defined cell which cannot find an equivalent
; one in the generic library
```

NOTE: The symbol of the explicitly defined cell contains a pin sequence following the pin order in the map file. The pin locations read from top to bottom (for pin positions on the left/right) and left to right (for pin positions on the top/bottom).

Creating Symbol Library From the Map File

After preparing the map file of the library, use map2SymDB executable to generate the Novas Symbol Library.

Usage:

map2SymDB: [options] mapFile

Options	Explanation
-h -help	Prints out the help message.
-L	Converts all cell names to lowercase.
-U	Converts all cell names to uppercase.
-l	Converts all pin names to lowercase.
-u	Converts all pin names to uppercase.
-v	Generates symbol library in Verilog HDL format with the default file <i><Library Name>.v</i> .
-vb	<i>Verbose</i> mode.
-b generic_sym_file	Specifies the generic symbol file name instead of the default file <i>generic.sym</i> .
-kp	Keeps the pin order as the pin definition in the map file.
-pu	Partial update: Updates existing symbols with the symbol shape, cell type, and port attributes only. The option gets the change information from the <i>.map</i> file and updates the existing symbol library specified with -o .
-o LibraryName	Creates the name of the library.
mapFile	The mapping file or symbol file.

The map2SymDB program creates the following directory:

<Library Name>.lib++/

Perform the following steps to display all the cells in the Verdi Symbol Library in the *nSchema* window:

1. Invoke Verdi with any design and select **Tools -> New Schematic** command to invoke the *nSchema* window.
2. In the *nSchema* window, type *L* to select the **Schematic -> Load Symbol** command.
3. The *Load Symbol* form appears. Enter the following data:
 - **Library Path:** Enter the name of the directory where the library resides.
 - **Library Name:** Enter the library name.

The *nSchema* window displays all the cells in this library.

NOTE: The cell names and pin names in the symbol library are case-sensitive. If some symbols change to a rectangle or if some symbols are missing, verify the cell names and pin names.

Use Symbol Libraries

The following environment variables specify the symbol library you want to use and where it is located.

- *NOVAS_LIBS*
- *NOVAS_LIBPATHS*

NOVAS_LIBS

`NOVAS_LIBS` specifies Novas symbol libraries you want to use for the design. You can specify multiple libraries by adding them after a colon, a semicolon, or a space.

The following example prompts Novas to use the **TOSHIBA** and **TSMC** libraries:

```
>setenv NOVAS_LIBS "TOSHIBA TSMC"
```

If `NOVAS_LIBS` is not specified, Novas uses the default library, which contains both **TOSHIBA**-like and **VLSI**-like cells.

NOVAS_LIBPATHS

`NOVAS_LIBPATHS` specifies the directories where Novas can search for symbol libraries. You can specify multiple directories by adding them after a space. For example:

```
>setenv NOVAS_LIBPATHS "<directory_A> <directory_B>"
```

If `NOVAS_LIBPATHS` is not specified, then the default directory of either `<NOVAS_INSTALL_DIR>/share/symlib/32` or `<NOVAS_INSTALL_DIR>/share/symlib/64` is used, depending on which version is invoked.

In addition to environment variables, you might also specify symbol library name and symbol library path in **Preferences -> Schematics -> Symbol Lib**. After you exit Verdi, the settings are kept in `novas.rc` file.

Novas searches symbol library according to the following precedence.

1. Environment variables (`NOVAS_LIBPATHS` and `NOVAS_LIBS`)

2. The resource `.rc` file in the path specified for your design file.

NOTE: You might specify design path in *nTrace*, **File -> Import Design -> From Library -> Working Directory** or **File -> Import Design -> From File -> Default Directory**.

3. The resource `.rc` file you currently used.

NOTE: Novas searches its resource file according to the following sequence:

- a. The resource file specified by **-rcFile** option.
- b. The resource file specified by `NOVAS_RC` (it was named `DEBUSSYRC`) environment variable.
- c. `./novas.rc`
- d. `<Home Dir>/novas.rc`
- e. `<NOVAS_INSTALL_DIR>/etc/novas.rc`

If symbol library name and symbol library path are not found in the above environment variable and resource file, no symbol library is used.

Novas Map File Format

Novas uses the map file to map the cells to the corresponding Novas symbols. Novas has two types of map file formats: generic cell mapping and explicit cell map definitions. You can use either or both in the map file.

Generic Cell Mapping

Generic cell mapping maps the cells directly to the corresponding generic cells. The recommendation is to you use this method to map the cells if you can find the corresponding ones in Novas generic library.

Syntax:

```
CELLMAP <Generic_cell_name> <Your_cell_name>
PINMAP <Generic_cell_pin_name> <Your_cell_pin_name>
or
PINMAP <Generic_cell_pin_name> <Your_cell_pin_name> [position
[pin type [polarity]]]
POLARITY <input_pin> <output_pin> <input_output_polarity>
EQUATION <Function expression>
```

where

<Generic_cell_name>	The generic cell to which you want to map
<User-defined_cell_name>	User-defined cell name
<Generic_cell_pin_name>	Generic cell pin name. Note that if the character '+' is specified, this pin exists but is a dummy for the symbol, that is not shown on the symbol display.
<User-defined_cell_pin_name>	User-defined cell pin name. Note that if the character '-' is specified, the generated symbol does not have this pin. For example, all latches and flip-flops in Novas generic cells have both Q and QB pins. If you have any flip-flop or latch that only has the Q pin, you can use this method to get rid of the QB pin.

The following choices are optional. They are used if your cell is similar to the generic cell with the exception of some pins.

position: (L | R | B | T)

L	Left of the symbol
R	Right of the symbol
B	Bottom of the symbol
T	Top of the symbol

pin_type: (AS | AR | SS | SR | C | D | E | TE | TO | COM | L | R | IL)

AS	Asynchronous Set pin
AR	Asynchronous Reset pin
SS	Synchronous Set pin
SR	Synchronous Reset pin
C	Clock pin
D	Data pin
E	Control pin
TE	Tri-state Enable pin
TO	Tri-state Output pin
COM	Combinational Output pin
L	Latched Output pin
R	Registered Output pin
IL	Inferred Latched Output pin

Polarity: (L | H | R | F)

L	Low active (or inverted)
H	High active
R	Rising-edge active
F	Falling-edge active

Input_output_polarity: (+/-/X)

+	For output the same as Input
-	For input invert output
X	For output cannot be decided

Function Expression:

Combination equation that consists of output and inputs.

Example 1

This example shows mapping to a 2-1 multiplexer.

```
CELLMAP E_MUX2 mx21d1;  
PINMAP A i0  
PINMAP B i1  
PINMAP S s  
PINMAP O z  
POLARITY i0 z +  
POLARITY i1 z +  
EQUATION z=s&i0+!s&i1
```

Example 2

This cell is a high-active D-Latch with only a Q output.

```
CELLMAP E_LDP llthqa;  
PINMAP D d  
PINMAP CK ck  
PINMAP Q q  
PINMAP QB -  
POLARITY D Q +
```

Example 3

This cell is a 2-input AND gate with inverted inputs.

```
CELLMAP E_AND2 AN2I2;  
PINMAP O Y  
PINMAP I0 NA L D L  
PINMAP I1 NB L D L  
POLARITY NA Y -  
POLARITY NB Y -
```

Example4

VDD and VSS pins exist in the Verilog module but are dummy for the symbol.

Verilog module:

```
module my_and2(O, A, B, VDD, VSS)  
input A, B, VDD, VSS;  
output O;  
and2 (O, A, B);  
endmodule
```

The map file:

```
CELLMAP E_AND2 my_and2;  
PINMAP O O  
PINMAP I0 A  
PINMAP I1 B
```

```

PINMAP + VDD
PINMAP + VSS
POLARITY A 0 +
POLARITY B 0 +

```

Explicit Cell Map Definitions

If you cannot find an equivalent cell to your cell in Novas generic library, you can use the explicit cell definition format to map your cell. The cell pin I/O type and pin position, which are optional in generic cell mapping, are required for specification in the explicit cell definition format.

Syntax

```

Cell <Your_cell_name> cell_type;
<Your_pin_name> pin_iotype pin_position [pin_type[pin_state]];
POLARITY <input_pin> <output_pin> <input_output_polarity>

```

Or

```

Cell <Your_cell_name> <Boolean_function>
POLARITY <input_pin> <output_pin> <input_output_polarity>
EQUATION <Function expression>

```

where cell type:

Simple Gate	AND NAND OR NOR XNR XOR BUF INV BUF D
Tri-state	BUFIF0 BUFIF1 NOTIF0 NOTIF1
AOI Gate	AOI21 AOI22 AOI211 AOI221 AOI2111 AOI31 AOI32 AOI33 AOI311 AOI321 AOI322 AOI331 AOI332 AOI41 AOI44
OAI Gate	OAI21 OAI22 OAI211 OAI221 OAI2111 OAI31 OAI32 OAI33 OAI311 OAI321 OAI322 OAI331 OAI332 OAI41 OAI44
Function Block	AXO21 ((a&b) ^ c) OXO21 ((a b) ^ c) XOA21 ((a ^ b) & c) XOO21 ((a ^ b) c) MAJ23 ~((a&b) (a&c) (b&c)) AOAI211 ~(((a&b) c) & d) OAOI211 ~(((a b) & c) d) AOIOI ~((a&b) ~(c d)) OAI AI ~((a b) & ~(c&d)) AAIOI ~((a&b) ~(c&d)) A3OI2OI ~((a&b&c) ~(d e)) A2OI3OI ~((a&b) ~(c d e)) OOIAI ~((a b) & ~(c d)) O3AI2AI ~((a b c) & ~(d&e)) O2AI3AI ~((a b) & ~(c&d&e))

Clock Gating Cell	GATECLK
GENERATOR(2 or 3 stage)	GENERATOR (((((a&b) c)&d) e) or ((((((a&b) c)&d) e)&f) g)
Flip-Flop	DFF - For any type of flip-flop, latch
Latch	LATCH
Multiplexer	MUX
Multiplier	MUL Example: CELL MUL44 MUL A[3:0] INPUT L D B[3:0] INPUT L D C[7:0] OUTPUT R COM
Bus keeper/holder	BUS_KEEPER (one inout pin with/without one control pin)
Bi-directional gate	BD (one input, one output, one inout and one control pin)
Others	MACRO - Other cells that do not belong to the above type

NOTE: The following cases for type of *BUS_KEEPER* and *BD* are supported:

```

; (BUS_KEEPER --- bus keeper/holder)
CELL E_BK1 BUS_KEEPER
IO INOUT L D
CELL E_BK2
IO INOUT
E INPUT
BUS_KEEPER
L D H
L D H
; (BD --- Bi-directional gate)
CELL E_BDE BD
OE INPUT B TE
A INPUT R D
Z OUTPUT R COM
PAD INOUT L TO

```

Others definition/mapping with different pin number/pin_iotype/
pin_position for *BUS_KEEPER/BD* type is drawn as MACRO
symbol.

pin_iotype: (INPUT | OUTPUT | INOUT)

INPUT	INPUT	pin
OUTPUT	OUTPUT	pin
INOUT	BIDIRECTION	pin

The following choices are optional. They are used if your cell is similar to the generic cell with the exception of some pins.

position: (L | R | B | T)

L	Left of the symbol
R	Right of the symbol
B	Bottom of the symbol
T	Top of the symbol

pin_type: (AS | AR | SS | SR | C | D | E | TE | TO | COM | L | R | IL)

AS	Asynchronous Set pin
AR	Asynchronous Reset pin
SS	Synchronous Set pin
SR	Synchronous Reset pin
C	Clock pin
D	Data pin
E	Control pin
TE	Tri-state Enable pin
TO	Tri-state Output pin
COM	Combinational Output pin
L	Latched Output pin
R	Registered Output pin
IL	Inferred Latched Output pin

Polarity: (L | H | R | F)

L	Low active
H	High active
R	Rising-edge active
F	Falling-edge active

Input_output_polarity: (+/-/X)

+	For output the same as Input
-	For input invert output
X	For output cannot be decided

Function Expression:

Combination equation that consists of output and inputs.

Example

This cell is a D-Latch with active-low reset and active-low clock.

```
Cell E_DBHRBN DFF;
D INPUT L D
CKB INPUT L C L
RB INPUT B AR L
Q OUTPUT R L
QB OUTPUT R L L
;
;This cell is a 2-input AND gate
CELL i_and AND
A INPUT L D
B INPUT L D
Z OUTPUT R COM
POLARITY A Z +
POLARITY B Z +
EQUATION Z=A&B
; another 2-input AND gate;
CELL i_and2 Z=A&B
POLARITY A Z +
POLARITY B Z +
; 2-input NAND gate
CELL i_nand NAND
A INPUT L D
B INPUT L D
Z OUTPUT R COM L
POLARITY A Z -
POLARITY B Z -
EQUATION Z=!(A&B)
;
; Another 2-input NAND gate
CELL i_nand2 Z=!(A&B)
;
; 2-input XOR
CELL i_xor XOR
A INPUT L D
B INPUT L D
Z OUTPUT R COM
POLARITY A Z x
POLARITY B Z x
EQUATION Z=A^B
```

NOTE: Refer to generic .sym for more examples.

Symbol Library Warning Messages

Variables

- \$LibName: Library Name
- \$CellName: Cell Name
- \$PortName: Port Name
- \$DirName: Directory Name
- \$FileName: File Name
- \$Polarity: Polarity
- \$Reason: Failure Reason
- \$Expression: Functional Expression
- \$Keyword: SYNOPSIS Keyword
- \$FL_Attribute: Attribute of ff() or latch(). \$FL_Attribute represents clear, preset, clocked_on, enable, and so on

Symbol Library Warning Messages

1. *Polarity (\$Polarity) for (\$PortName) and (\$PortName) exists in cell (\$CellName) already, combine it with specified polarity (\$Polarity) to be (x).*

Reported when the polarity information conflicts. For example:

```
POLARITY A Z +  
POLARITY A Z -
```

2. *Port(\$PortName) is not created in cell view(\$CellName).*

The referred port is not defined in the cell.

3. *Pin name \$PortName is duplicated in cell \$CellName!*

Reported when:

- The port name is duplicated in either *map2SymDB* or *syn2SymDB*.
- Pin information already exists before creating symbol in RTL symbol.

syn2SymDB Warning Messages

1. *Can't process [\$FL_Attribute : \$Expression] well in cell(\$CellName).
syn2SymDB cannot process the attributes of ff() or latch().*

2. *Not support multiple ff()/latch(), some information may be lost in cell(\$CellName).*
3. *Program stop for information lost.*
When the **-stop** option is active, *syn2SymDB* stops when there is any information lost. It is used for *nLint*.
4. *Skip cell(\$CellName) for flag(-skst).*
Skip cell with statetable when the **-skst** option is active.
5. *Skip cell(\$CellName) for flag(-skfl).*
Skip cell when *syn2SymDB* cannot process the attributes of *ff()* or *latch()*.
6. *Not support multiple statetables, some information may be lost in cell(\$CellName).*
7. *Statetable has multiple clocks in cell(CellName).*
8. *Not support keyword (\$Keyword) well, some information may be lost in cell(\$CellName).*
9. *file(\$FileName) is invalid, the .slib file is not supported, please use the SYNOPSIS .lib file.*
10. *Unknown cell type-- \$CellName: \$Expression, convert to macro cell.*
syn2SymDB recognizes if a cell is an equation cell but the function expression is wrong. The cell is converted to a macro cell.
11. *Cells do not exist in the specified input library.*
There is no cell in the specified input library.
12. *Failed to find the \$CellName cell in the library.*
After specifying the **-pu** option for partial update, the cell cannot be found in the original library.
13. *The port name or port number do not match the existing \$CellName cell.*
After specifying the **-pu** option for partial update, the cell port name or port number does not match the definition of the original library.
14. *The right-side port number should be 2 in the TRD \$CellName cell.*
15. *Failed to find the \$PortName port in the existing \$CellName cell.*
The *\$PortName* port cannot be found in the *\$CellName* cell definition.
16. *The \$PinName pin is ignored in the \$CellName cell.*
The *\$PinName* pin was specified in the **-ignorepin** option. This pin is ignored in the *\$CellName* cell.

17. *Types are not specified for the \$BusName bus. A bus will be created according to the pin list of the \$BusName bus.*
18. *Failed to find the matched type \$BusType for the \$BusName bus. A bus will be created according to the pin list of the \$BusName bus.*

syn2SymDB Error Messages

1. *The \$LibName library does not exist.*
The specified library file does not exist.
2. *Failed to open the \$LibName library.*
The specified library file cannot be opened.
3. *Failed to open the \$LibName *.gz file to read.*
4. *Failed to open the \$LibName file to read.*

map2SymDB Warning Messages

1. *Failed to open the \$FileName file because of \$Reason.*
2. *Directory(\$DirName) does not exist.*
3. *(\$DirName) is not a directory.*

map2SymDB Error Messages

1. *Failed to open the \$FileName file because of \$ErrorType.*

Database Error

1. *Failed to create library \$LibName(\$Reason)!*
2. *Cannot save (cell) to symbol library(\$LibName)!*
Save cell failed.
3. *Cannot save (info) to symbol library(\$LibName)!*
Save library failed.
4. *Cannot close symbol library(\$LibName)!*

Library Cell Listing/Definition

Refer to the *Symbol_Library.pdf* file in the <NOVAS_INSTALL>/doc directory for the complete Library Cell Listing and Definitions.

Library Cell Listing

Debussy provides the following generic symbol library cells for schematic generation.

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Combinational Functions

[AND Gates](#)

[NAND Gates](#)

[OR Gates](#)

[NOR Gates](#)

[XOR Gates](#)

[XNOR Gates](#)

[AOI Gates](#)

[OAI Gates](#)

[COMPLEX Gates](#)

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AND Gates

E_AND2	2-input AND gate
E_AND3	3-input AND gate
E_AND4	4-input AND gate
E_AND5	5-input AND gate
E_AND6	6-input AND gate
E_AND7	7-input AND gate
E_AND8	8-input AND gate
E_AND9	9-input AND gate
E_AND10	10-input AND gate
E_AND11	11-input AND gate
E_AND12	12-input AND gate
E_AND13	13-input AND gate
E_AND14	14-input AND gate
E_AND15	15-input AND gate
E_AND16	16-input AND gate

NAND Gates

E_NAN2	2-input NAND gate
E_NAN3	3-input NAND gate

E_NAN4	4-input NAND gate
E_NAN5	5-input NAND gate
E_NAN6	6-input NAND gate
E_NAN7	7-input NAND gate
E_NAN8	8-input NAND gate
E_NAN9	9-input NAND gate
E_NAN10	10-input NAND gate
E_NAN11	11n-input NAND gate
E_NAN12	12-input NAND gate
E_NAN13	13-input NAND gate
E_NAN14	14-input NAND gate
E_NAN15	15-input NAND gate
E_NAN16	16-input NAND gate

OR Gates

E_OR2	2-input OR gate
E_OR3	3-input OR gate
E_OR4	4-input OR gate
E_OR5	5-input OR gate
E_OR6	6-input OR gate
E_OR7	7-input OR gate
E_OR8	8-input OR gate

E_OR9	9-input OR gate
E_OR10	10-input OR gate
E_OR11	11-input OR gate
E_OR12	12-input OR gate
E_OR13	13-input OR gate
E_OR14	14-input OR gate
E_OR15	15-input OR gate
E_OR16	16-input OR gate

NOR Gates

E_NOR2	2-input NOR gate
E_NOR2	3-input NOR gate
E_NOR2	4r-input NOR gate
E_NOR2	5-input NOR gate
E_NOR2	6-input NOR gate
E_NOR2	7-input NOR gate
E_NOR2	8-input NOR gate
E_NOR9	9-input NOR gate
E_NOR10	10-input NOR gate
E_NOR11	11-input NOR gate
E_NOR12	12-input NOR gate
E_NOR13	13-input NOR gate

E_NOR14	14-input NOR gate
E_NOR15	15-input NOR gate
E_NOR16	16-input NOR gate

XOR Gates

E_XOR2	2-input XOR gate
E_XOR3	3-input XOR gate
E_XOR4	4-input XOR gate
E_XOR5	5-input XOR gate

XNOR Gates

E_XNR2	2-input XNOR gate
E_XNR3	3-input XNOR gate
E_XNR4	4-input XNOR gate
E_XNR5	5-input XNOR gate

AOI Gates

E_AOI21	2,1 AND-OR-INVERTER gate
E_AOI211	2,1,1 AND-OR-INVERTER gate
E_AOI221	2,2,1 AND-OR-INVERTER gate

E_AOI321	3,2,1 AND-OR-INVERTER gate
E_AOI31	3,1 AND-OR-INVERTER gate
E_AOI32	3,2 AND-OR-INVERTER gate
E_AOI22	2,2 AND-OR-INVERTER gate
E_AOI2X3	2,2,2 AND-OR-INVERTER gate
E_AOI2X4	2,2,2 AND-OR-INVERTER gate
E_AOI2X6	2,2,2,2,2,2 AND-OR-INVERTER gate
E_AOI2X8	2,2,2,2,2,2,2,2 AND-OR-INVERTER gate
E_AOI33	3,3 AND-OR-INVERTER gate
E_AOI3X3	3,3,3 AND-OR-INVERTER gate
E_AOI3X4	3,3,3,3 AND-OR-INVERTER gate
E_AOI44	4,4 AND-OR-INVERTER gate
E_AOI4X3	4,4,4 AND-OR-INVERTER gate
E_AOI4X4	4,4,4,4 AND-OR-INVERTER gate
E_AOAI211	AND2 into OR2 into NAND2 gate

OAI Gates:

E_OAI21	2,1 OR-AND-INVERTER Gate
E_OAI211	2,1,1 OR-AND-INVERTER Gate
E_OAI221	2,2,1 OR-AND-INVERTER Gate
E_OAI321	3,2,1 OR-AND-INVERTER Gate
E_OAI31	3,1 OR-AND-INVERTER Gate

E_OAI32	3,2 OR-AND-INVERTER Gate
E_OAI22	2,2 OR-AND-INVERTER Gate
E_OAI2X3	2,2,2 OR-AND-INVERTER Gate
E_OAI2X4	2,2,2,2 OR-AND-INVERTER Gate
E_OAI2X6	2,2,2,2,2 OR-AND-INVERTER Gate
E_OAI2X8	2,2,2,2,2,2,2 OR-AND-INVERTER Gate
E_OAI33	3,3 OR-AND-INVERTER Gate
E_OAI3X3	3,3,3 OR-AND-INVERTER Gate
E_OAI3X4	3,3,3,3 OR-AND-INVERTER Gate
E_OAI44	4,4 OR-AND-INVERTER Gate
E_OAI4X3	4,4,4 OR-AND-INVERTER Gate
E_OAI4X4	4,4,4,4 OR-AND-INVERTER Gate
E_OAOI211	OR2 into AND2 into NOR2 Gate

COMPLEX Gates:

E_FN01	NAND with inverted input
E_FN02	Inverting 2 of 3 Majority
E_FN03	AND2, NOR2 into NOR2
E_FN04	NAND2, OR2 into NAND2
E_FN05	NOR with inverted input
E_FN06	AND2, NAND2, into NOR2
E_FN07	OR2, NOR2, into NAND2

E_FN08	AND3, NOR2, into NOR2
E_FN09	AND2, NOR3, into NOR2
E_FN10	OR3, NAND2, into NAND2
E_FN11	OR2, NAND3, into NAND2

GENERATOR Gates:

E_PAR8E	8-bit even parity generator
E_PAR8O	8-bit odd parity generator

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BUFFER Gates

E_BUF1	Buffer
E_BUFT1	Tri-State buffer, active-HIGH
E_BUF1B	Tri-State buffer, active-LOW
E_TRD2C	Tri-State internal drive
E_BUF8D	Buffer with 8 Outputs
E_BUF16D	Buffer with 16 Outputs
E_BUF2X	2 individual Buffers
E_BUF4X	4 individual buffers
E_BUF8X	8 individual buffers
E_BUF16X	16 individual buffers

E_CKBUF	CLOCK Dual BUFFER
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INVERTER Gates

E_INV	Inverter
E_INVT	Active-HIGH tri-state inverter
E_INVB	Aactive-LOW tri-state inverter
E_INV2	2-output inverter
E_INV4	4-output inverter
E_INV6	6-output inverter
E_INV8	8-output inverter
E_INV16	16-output inverter
E_INVX2	2 individual inverters
E_INVX4	4 individual inverters
E_INVX6	6 individual inverters
E_INVX8	8 individual inverters
E_INVX16	16 individual inverters

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DECODER Gates

E_DEC2	2 to 4 Decoder
E_DEC2I	2 to 4 Decoder, inverted outputs

E_DEC2E	2 to 4 Decoder, active-HIGH enable
E_DEC2EI	2 to 4 Decoder, active-HIGH enable, inverted outputs
E_DEC2ED	2 to 4 Decoder, active-HIGH enable, with complementary outputs
E_DEC2EB	2 to 4 Decoder, active-LOW enable
E_DEC2EBI	2 to 4 Decoder, active-LOW enable, inverted outputs
E_DEC2EBD	2 to 4 Decoder, active-LOW enable, with complementary outputs
E_DEC3	3 to 8 Decoder
E_DEC3I	3 to 8 Decoder, inverted outputs
E_DEC3E	3 to 8 Decoder, active-HIGH enable
E_DEC3EI	3 to 8 Decoder, active-HIGH enable, inverted outputs
E_DEC3ED	3 to 8 Decoder, active-HIGH enable, with complementary outputs
E_DEC3EB	3 to 8 Decoder, active-LOW enable
E_DEC3EBI	3 to 8 Decoder, active-LOW enable, inverted outputs
E_DEC3EBD	3 to 8 Decoder, active-LOW enable, with complementary outputs

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Adders

ADDER/SUBSTRATOR Gates:

E_FA1	1-Bit full adder
E_FA2	2-Bit full adder
E_FA4	4-Bit full adder
E_HA1	1-Bit half adder
E_ADDSUB	1-bit Adder/Subtractor
E_ADRSUB2	2-bit Adder/Subtractor

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Multiplexers

MULTIPLEXER Gates:

E_MUX2	2-1 multiplexer
E_MUX2I	2-1 multiplexer, inverted output
E_MUX2E	2-1 multiplexer, active-HIGH enable
E_MUX2EI	2-1 multiplexer, active-HIGH enable, inverted output
E_MUX2ED	2-1 multiplexer, active-HIGH enable, complementary outputs
E_MUX2X4ED	Four 2-1 multiplexers, active-HIGH enable, complementary outputs
E_MUX2EB	2-1 multiplexer, active-LOW enable
E_MUX2EBI	2-1 multiplexer, active-LOW enable, inverted output
E_MUX2EBD	2-1 multiplexer, active-LOW enable, complementary outputs

E_MUX2X4EBD	Four 2-1 multiplexers, active-LOW enable, complementary outputs
E_MUX3	3-1 multiplexer
E_MUX3I	3-1 multiplexer, inverted output
E_MUX3E	3-1 multiplexer, active-HIGH enable
E_MUX3EI	3-1 multiplexer, active-HIGH enable, inverted output
E_MUX3ED	3-1 multiplexer, active-HIGH enable, complementary outputs
E_MUX3EB	3-1 multiplexer, active-LOW enable
E_MUX3EBI	3-1 multiplexer, active-LOW enable, inverted output
E_MUX3EBD	3-1 multiplexer, active-LOW enable, complementary outputs
E_MUX4	4-1 multiplexer
E_MUX4I	4-1 multiplexer, inverted output
E_MUX4E	4-1 multiplexer, active-HIGH enable
E_MUX4EI	4-1 multiplexer, active-HIGH enable, inverted output
E_MUX4ED	4-1 multiplexer, active-HIGH enable, complementary outputs
E_MUX4EB	4-1 multiplexer, active-LOW enable
E_MUX4EBI	4-1 multiplexer, active-LOW enable, inverted output
E_MUX4EBD	4-1 multiplexer, active-LOW enable, complementary outputs
E_MUX5	5-1 multiplexer
E_MUX5I	5-1 multiplexer, inverted output

E_MUX5E	5-1 multiplexer, active-HIGH enable
E_MUX5EI	5-1 multiplexer, active-HIGH enable, inverted output
E_MUX5ED	5-1 multiplexer, active-HIGH enable, complementary outputs
E_MUX5EB	5-1 multiplexer, active-LOW enable
E_MUX5EBI	5-1 multiplexer, active-LOW enable, inverted output
E_MUX5EBD	5-1 multiplexer, active-LOW enable, complementary outputs
E_MUX8	8-1 multiplexer
E_MUX8I	8-1 multiplexer, inverted output
E_MUX8E	8-1 multiplexer, active-HIGH enable
E_MUX8EI	8-1 multiplexer, active-HIGH enable, inverted output
E_MUX8ED	8-1 multiplexer, active-HIGH enable, complementary outputs
E_MUX8EB	8-1 multiplexer, active-LOW enable
E_MUX8EBI	8-1 multiplexer, active-LOW enable, inverted output
E_MUX8EBD	8-1 multiplexer, active-LOW enable, complementary outputs

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Comparator Gates

E_CMP4	4-bit Comparator (equal only)
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E_CMP4ED	4-bit Comparator (equal & below), active-HIGH enable, complementary output
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RAM/ROM Gates

E_RAM1	1-bit RAM Cell with inverting and tri-state inverting output
E_RAM16	16-bit RAM Cell
E_RAM32	32-bit RAM Cell
E_ROM16	16-bit ROM Cell
E_ROM32	32-bit ROM Cell

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LATCH Function

D-Latches Cells:

normal

E_LDN	D-Latch, active-LOW enable
E_LDNRB	D-Latch, active-LOW enable, active-LOW reset
E_LDNSB	D-Latch, active-LOW enable, active-LOW set

E_LDNRSB	D-Latch, active-LOW enable, active-LOW set & reset
E_LDNR	D-Latch, active-LOW enable, active-HIGH reset
E_LDNS	D-Latch, active-LOW enable, active-HIGH set
E_LDNRS	D-Latch, active-LOW enable, active-HIGH set & reset
E_LDP	D-Latch, active-HIGH enable
E_LDPRB	D-Latch, active-HIGH enable, active-LOW reset
E_LDPSB	D-Latch, active-HIGH enable, active-LOW set
E_LDPRSB	D-Latch, active-HIGH enable, active-LOW set & reset
E_LDPR	D-Latch, active-HIGH enable, active-HIGH reset
E_LDPS	D-Latch, active-HIGH enable, active-HIGH set
E_LDPRS	D-Latch, active-HIGH enable, active-HIGH set & reset

with multi-bit

E_LDN4	4-bit D-Latch, active-LOW enable
E_LDN4RB	4-bit D-Latch, active-LOW enable, active-LOW reset
E_LDN4SB	4-bit D-Latch, active-LOW enable, active-LOW set

E_LDN4RSB	4-bit D-Latch, active-LOW enable, active-LOW set & reset
E_LDN4R	4-bit D-Latch, active-LOW enable, active-HIGH reset
E_LDN4S	4-bit D-Latch, active-LOW enable, active-HIGH set
E_LDN4RS	4-bit D-Latch, active-LOW enable, active-HIGH set & reset
E_LDN8	8-bit D-Latch, active-LOW enable
E_LDN8RB	8-bit D-Latch, active-LOW enable, active-LOW reset
E_LDN8SB	8-bit D-Latch, active-LOW enable, active-LOW set
E_LDN8RSB	8-bit D-Latch, active-LOW enable, active-LOW set & reset
E_LDN8R	8-bit D-Latch, active-LOW enable, active-HIGH reset
E_LDN8S	8-bit D-Latch, active-LOW enable, active-HIGH set
E_LDN8RS	8-bit D-Latch, active-LOW enable, active-HIGH set & reset
E_LDP4	4-bit D-Latch, active-HIGH enable
E_LDP4RB	4-bit D-Latch, active-HIGH enable, active-LOW reset
E_LDP4SB	4-bit D-Latch, active-HIGH enable, active-LOW set
E_LDP4RSB	4-bit D-Latch, active-HIGH enable, active-LOW set & reset
E_LDP4R	4-bit D-Latch, active-HIGH enable, active-HIGH reset

E_LDP4S	4-bit D-Latch, active-HIGH enable, active-HIGH set
E_LDP4RS	4-bit D-Latch, active-HIGH enable, active-HIGH set & reset
E_LDP8	8-bit D-Latch, active-HIGH enable
E_LDP8RB	8-bit D-Latch, active-HIGH enable, active-LOW reset
E_LDP8SB	8-bit D-Latch, active-HIGH enable, active-LOW set
E_LDP8RSB	8-bit D-Latch, active-HIGH enable, active-LOW set & reset
E_LDP8R	8-bit D-Latch, active-HIGH enable, active-HIGH reset
E_LDP8S	8-bit D-Latch, active-HIGH enable, active-HIGH set
E_LDP8RS	8-bit D-Latch, active-HIGH enable, active-HIGH set & reset

with tri-state

E_LDNT	D-Latch, active-LOW enable, active-HIGH enable tri-state output
E_LDNTRB	D-Latch, active-LOW enable, active-LOW reset, active-HIGH enable tri-state output
E_LDNTSB	D-Latch, active-LOW enable, active-LOW set, active-HIGH enable tri-state output
E_LDNTRSB	D-Latch, active-LOW enable, active-LOW set & reset, active-HIGH enable tri-state output
E_LDNTR	D-Latch, active-LOW enable, active-HIGH reset, active-HIGH enable tri-state output

E_LDNTS	D-Latch, active-LOW enable, active-HIGH set, active-HIGH enable tri-state output
E_LDNTRS	D-Latch, active-LOW enable, active-HIGH set & reset, active-HIGH enable tri-state output
E_LDPT	D-Latch, active-HIGH enable, active-HIGH enable tri-state output
E_LDPTRB	D-Latch, active-HIGH enable, active-LOW reset, active-HIGH enable tri-state output
E_LDPTSB	D-Latch, active-HIGH enable, active-LOW set, active-HIGH enable tri-state output
E_LDPTRSB	D-Latch, active-HIGH enable, active-LOW set & reset, active-HIGH enable tri-state output
E_LDPTR	D-Latch, active-HIGH enable, active-HIGH reset, active-HIGH enable tri-state output
E_LDPTS	D-Latch, active-HIGH enable, active-HIGH set, active-HIGH enable tri-state output
E_LDPTRS	D-Latch, active-HIGH enable, active-HIGH set & reset, active-HIGH enable tri-state output

with scan test

E_LDNM	D-Latch, active-LOW enable, multiplexed with positive-edge-trigger
E_LDNM RB	D-Latch, active-LOW enable, active-LOW reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDNM SB	D-Latch, active-LOW enable, active-LOW set, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDNM RSB	D-Latch, active-LOW enable, active-LOW set & reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDNM R	D-Latch, active-LOW enable, active-HIGH reset, multiplexed with D-type flip-flop, positive-edge-trigger

E_LDNMS	D-Latch, active-LOW enable, active-HIGH set, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDNMRS	D-Latch, active-LOW enable, active-HIGH set & reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDNW2	D-Latch, active-LOW enable, 2 write ports
E_LDNST	D-Latch with Scan Test Input, active-LOW enable
E_LDNSTRB	D-Latch with Scan Test Input, active-LOW enable, active-LOW reset
E_LDNSTSB	D-Latch with Scan Test Input, active-LOW enable, active-LOW set & reset
E_LDNSTRSB	D-Latch with Scan Test Input, active-LOW enable, active-LOW set
E_LDNSTR	D-Latch with Scan Test Input, active-LOW enable, active-HIGH reset
E_LDNSTS	D-Latch with Scan Test Input, active-LOW enable, active-HIGH set & reset
E_LDNSTRS	D-Latch with Scan Test Input, active-LOW enable, active-HIGH set
E_LDPM	D-Latch, active-HIGH enable, multiplexed with positive-edge-trigger
E_LDPMRB	D-Latch, active-HIGH enable, active-LOW reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDPMsb	D-Latch, active-HIGH enable, active-LOW set, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDPMRSB	D-Latch, active-HIGH enable, active-LOW set & reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDPMR	D-Latch, active-HIGH enable, active-HIGH reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDPMs	D-Latch, active-HIGH enable, active-HIGH set, multiplexed with D-type flip-flop, positive-edge-trigger

E_LDPMRS	D-Latch, active-HIGH enable, active-HIGH set & reset, multiplexed with D-type flip-flop, positive-edge-trigger
E_LDPW2	D-Latch, active-HIGH enable, 2 write ports
E_LDPST	D-Latch with Scan Test Input, active-HIGH enable
E_LDPSTRB	D-Latch with Scan Test Input, active-HIGH enable, active-LOW reset
E_LDPSTSB	D-Latch with Scan Test Input, active-HIGH enable, active-LOW set & reset
E_LDPSTRSB	D-Latch with Scan Test Input, active-HIGH enable, active-LOW set
E_LDPSTR	D-Latch with Scan Test Input, active-HIGH enable, active-HIGH reset
E_LDPSTS	D-Latch with Scan Test Input, active-HIGH enable, active-HIGH set & reset
E_LDPSTRS	D-Latch with Scan Test Input, active-HIGH enable, active-HIGH set

other

E_DBHD	D-Latch, complementary enables
E_DLHMP	Master-slave latch, with additional D-input for slave stage
E_DLHMPW2	Master-slave latch, 2 write ports

SR Latches Cells:

E_LSRG	SR with COMMON GATED, active-HIGH set & reset
E_LSRGB	SR with COMMON GATED, active-LOW set & reset

E_NRL	SR latch, NOR type
E_NDL	SR latch, NAND type

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FLIP/FLOP

DFF Cells:

normal

E_DFNDFF, negative-edge- trigger	
E_DFNRB	DFF, negative-edge-trigger, active-LOW reset
E_DFNSB	DFF, negative-edge-trigger, active-LOW set
E_DFNRSB	DFF, negative-edge-trigger, active-LOW set & reset
E_DFNR	DFF, negative-edge-trigger, active-HIGH reset
E_DFNS	DFF, negative-edge-trigger, active-HIGH set
E_DFNRS	DFF, negative-edge-trigger, active-HIGH set & reset
E_DFP	DFF, positive-edge-trigger
E_DFPRB	DFF, positive-edge-trigger, active-LOW reset
E_DFPSB	DFF, positive-edge-trigger, active-LOW set
E_DFPRSB	DFF, positive-edge-trigger, active-LOW set & reset
E_DFPR	DFF, positive-edge-trigger, active-HIGH reset

E_DFPS	DFF, positive-edge-trigger, active-HIGH set
E_DFPRS	DFF, positive-edge-trigger, active-HIGH set & reset

with load enable

E_DFN	DFF, negative-edge-trigger, Load
E_DFNCL	DFF, negative-edge-trigger, active-LOW reset, Load
E_DFNCL	DFF, negative-edge-trigger, active-LOW set, Load
E_DFNCLR	DFF, negative-edge-trigger, active-LOW set & reset, Load
E_DFNC	DFF, negative-edge-trigger, active-HIGH reset, Load
E_DFNC	DFF, negative-edge-trigger, active-HIGH set, Load
E_DFNCL	DFF, negative-edge-trigger, active-HIGH set & reset, Load
E_DFP	DFF, positive-edge-trigger, Load
E_DFPCL	DFF, positive-edge-trigger, active-LOW reset, Load
E_DFPCL	DFF, positive-edge-trigger, active-LOW set, Load
E_DFPCLR	DFF, positive-edge-trigger, active-LOW set & reset, Load
E_DFPC	DFF, positive-edge-trigger, active-HIGH reset, Load
E_DFPC	DFF, positive-edge-trigger, active-HIGH set, Load
E_DFPCL	DFF, positive-edge-trigger, active-HIGH set & reset, Load

with tri-state output

E_DFN	DFF, negative-edge-trigger with Tri-State
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E_DFNR	DFF, negative-edge-trigger with Tri-State, active-LOW reset
E_DFNTS	DFF, negative-edge-trigger with Tri-State, active-LOW set
E_DFNR	DFF, negative-edge-trigger with Tri-State, active-LOW set & reset
E_DFNT	DFF, negative-edge-trigger with Tri-State, active-HIGH reset
E_DFNT	DFF, negative-edge-trigger with Tri-State, active-HIGH set
E_DFNR	DFF, negative-edge-trigger with Tri-State, active-HIGH set & reset
E_DFP	DFF, positive-edge-trigger with Tri-State
E_DFPTR	DFF, positive-edge-trigger with Tri-State, active-LOW reset
E_DFPTS	DFF, positive-edge-trigger with Tri-State, active-LOW set
E_DFPTRS	DFF, positive-edge-trigger with Tri-State, active-LOW set & reset
E_DFPT	DFF, positive-edge-trigger with Tri-State, active-HIGH reset
E_DFPT	DFF, positive-edge-trigger with Tri-State, active-HIGH set
E_DFPTR	DFF, positive-edge-trigger with Tri-State, active-HIGH set & reset

with multi-bit

E_DFN4	4Bit DFF, negative-edge-trigger
E_DFN4RB	4Bit DFF, negative-edge-trigger, active-LOW reset
E_DFN4SB	4Bit DFF, negative-edge-trigger, active-LOW set
E_DFN4RSB	4Bit DFF, negative-edge-trigger, active-LOW set & reset
E_DFN4R	4Bit DFF, negative-edge-trigger, active-HIGH reset

E_DFN4S	4Bit DFF, negative-edge-trigger, active-HIGH set
E_DFN4RS	4Bit DFF, negative-edge-trigger, active-HIGH set & reset
E_DFN8	8Bit DFF, negative-edge-trigger
E_DFN8RB	8Bit DFF, negative-edge-trigger, active-LOW reset
E_DFN8SB	8Bit DFF, negative-edge-trigger, active-LOW set
E_DFN8RSB	8Bit DFF, negative-edge-trigger, active-LOW set & reset
E_DFN8R	8Bit DFF, negative-edge-trigger, active-HIGH reset
E_DFN8S	8Bit DFF, negative-edge-trigger, active-HIGH set
E_DFN8RS	8Bit DFF, negative-edge-trigger, active-HIGH set & reset
E_DFP4	4Bit DFF, positive-edge-trigger
E_DFPRB4	4Bit DFF, positive-edge-trigger, active-LOW reset
E_DFPSB4	4Bit DFF, positive-edge-trigger, active-LOW set
E_DFPRSB4	4Bit DFF, positive-edge-trigger, active-LOW set & reset
E_DFPR4	4Bit DFF, positive-edge-trigger, active-HIGH reset
E_DFPS4	4Bit DFF, positive-edge-trigger, active-HIGH set
E_DFPRS4	4Bit DFF, positive-edge-trigger, active-HIGH set & reset
E_DFP8	8Bit DFF, positive-edge-trigger
E_DFP8RB	8Bit DFF, positive-edge-trigger, active-LOW reset
E_DFP8SB	8Bit DFF, positive-edge-trigger, active-LOW set
E_DFP8RSB	8Bit DFF, positive-edge-trigger, active-LOW set & reset
E_DFP8R	8Bit DFF, positive-edge-trigger, active-HIGH reset
E_DFP8S	8Bit DFF, positive-edge-trigger, active-HIGH set
E_DFP8RS	8Bit DFF, positive-edge-trigger, active-HIGH set & reset

with scan test

E_DFN	DFF with SCAN, negative-edge-trigger
E_DFNZR	DFF with SCAN, negative-edge-trigger, active-LOW reset
E_DFNZS	DFF with SCAN, negative-edge-trigger, active-LOW set
E_DFNZRS	DFF with SCAN, negative-edge-trigger, active-LOW set & reset
E_DFNZ	DFF with SCAN, negative-edge-trigger, and Output enable
E_DFNZ	DFF with SCAN, negative-edge-trigger, active-LOW reset, and Output enable
E_DFNZT	DFF with SCAN, negative-edge-trigger, active-LOW set, and Output enable
E_DFNZTRSB	DFF with SCAN, negative-edge-trigger, active-LOW set & reset, and Output enable
E_DFP	DFF with SCAN, positive-edge-trigger
E_DFPZR	DFF with SCAN, positive-edge-trigger, active-LOW reset
E_DFPZS	DFF with SCAN, positive-edge-trigger, active-LOW set
E_DFPZRS	DFF with SCAN, positive-edge-trigger, active-LOW set & reset
E_DFPZ	DFF with SCAN, positive-edge-trigger, and Output enable
E_DFPZ	DFF with SCAN, positive-edge-trigger, active-LOW reset, and Output enable
E_DFPZT	DFF with SCAN, positive-edge-trigger, active-LOW set, and Output enable
E_DFPZTRSB	DFF with SCAN, positive-edge-trigger, active-LOW set & reset, and Output enable

other

E_DDBTNB	DFF with double clock, active-LOW set & reset
E_DFFZRSB	DFF active-HIGH scan, positive-edge-trigger, active-low set & reset, active-HIGH output enable
E_DFFZRSBLD	DFF active HIGH scan, positive-edge-trigger, active HIGH D load, active-low set & reset
E_DFFNLRB	DFF, active-HIGH scan, positive-edge-trigger, active-LOW set & reset

JK Flip Flop:

normal

E_JKF	JK Flip-Flop, negative-edge-trigger
E_JKFNR	JK Flip-Flop, negative-edge-trigger, active-LOW reset
E_JKFNS	JK Flip-Flop, negative-edge-trigger, active-LOW set
E_JKFNRS	JK Flip-Flop, negative-edge-trigger, active-LOW set & reset
E_JKFN	JK Flip-Flop, negative-edge-trigger, active-HIGH reset
E_JKFN	JK Flip-Flop, negative-edge-trigger, active-HIGH set
E_JKFNR	JK Flip-Flop, negative-edge-trigger, active-HIGH set & reset
E_JKF	JK Flip-Flop, positive-edge-trigger
E_JKFPR	JK Flip-Flop, positive-edge-trigger, active-LOW reset
E_JKFPS	JK Flip-Flop, positive-edge-trigger, active-LOW set
E_JKFPRS	JK Flip-Flop, positive-edge-trigger, active-LOW set & reset

E_JKFP	JK Flip-Flop, positive-edge-trigger, active-HIGH reset
E_JKFP	JK Flip-Flop, positive-edge-trigger, active-HIGH set
E_JKFPR	JK Flip-Flop, positive-edge-trigger, active-HIGH set & reset

tri-state output

E_JKFNZ	JK Flip-Flop, negative-edge-trigger, active-HIGH output enable
E_JKFNZRB	JK Flip-Flop, negative-edge-trigger, active-LOW reset, active-HIGH output enable
E_JKFNZSB	JK Flip-Flop, negative-edge-trigger, active-LOW set, active-HIGH output enable
E_JKFNZRSB	JK Flip-Flop, negative-edge-trigger, active-LOW set & reset, active-HIGH output enable
E_JKFNZR	JK Flip-Flop, negative-edge-trigger, active-HIGH reset, active-HIGH output enable
E_JKFNZS	JK Flip-Flop, negative-edge-trigger, active-HIGH set, active-HIGH output enable
E_JKFNZRS	JK Flip-Flop, negative-edge-trigger, active-HIGH set & reset, active-HIGH output enable
E_JKFPZ	JK Flip-Flop, positive-edge-trigger, active-HIGH output enable
E_JKFPZRB	JK Flip-Flop, positive-edge-trigger, active-LOW reset, active-HIGH output enable
E_JKFPZSB	JK Flip-Flop, positive-edge-trigger, active-LOW set, active-HIGH output enable
E_JKFPZRSB	JK Flip-Flop, positive-edge-trigger, active-LOW set & reset, active-HIGH output enable

E_JKFPZR	JK Flip-Flop, positive-edge-trigger, active-HIGH reset, active-HIGH output enable
E_JKFPZS	JK Flip-Flop, positive-edge-trigger, active-HIGH set, active-HIGH output enable
E_JKFPZRS	JK Flip-Flop, positive-edge-trigger, active-HIGH set & reset, active-HIGH output enable

with scan test

E_JKFNST	JK Flip-Flop, active-HIGH scan, negative-edge-triggered
E_JKFNSTRB	JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-LOW reset
E_JKFNSTSB	JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-LOW set
E_JKFNSTRSB	JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-LOW set & reset
E_JKFNSTR	JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-HIGH reset
E_JKFNSTS	JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-HIGH set
E_JKFNSTRS	JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-HIGH set & reset
E_JKFNSCRSB	JK Flip-Flop, active-LOW scan, negative-edge-trigger, active-LOW set & reset, active-LOW clock enable
E_JKFPST	JK Flip-Flop, active-HIGH scan, positive-edge-triggered
E_JKFPSTRB	JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-LOW reset
E_JKFPSTSB	JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-LOW set

E_JKFPSTRSB	JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-LOW set & reset
E_JKFPSTR	JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-HIGH reset
E_JKFPSTS	JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-HIGH set
E_JKFPSTRS	JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-HIGH set & reset
E_JKFPSCRSB	JK Flip-Flop, active-LOW scan, positive-edge-trigger, active-LOW set & reset, active-LOW clock enable

T Flip/Flop Cells:

normal

E_TFF	T Flip-Flop, negative-edge-triggered
E_TFFNR	T Flip-Flop, negative-edge-triggered, active-LOW reset
E_TFFNS	T Flip-Flop, negative-edge-triggered, active-LOW set
E_TFFNRS	T Flip-Flop, negative-edge-triggered, active-LOW set & reset
E_TFFN	T Flip-Flop, negative-edge-triggered, active-HIGH reset
E_TFFN	T Flip-Flop, negative-edge-triggered, active-HIGH set
E_TFFNR	T Flip-Flop, negative-edge-triggered, active-HIGH set & reset
E_TFF	T Flip-Flop, positive-edge-triggered
E_TFFPR	T Flip-Flop, positive-edge-triggered, active-LOW reset
E_TFFPS	T Flip-Flop, positive-edge-triggered, active-LOW set

E_TFFPRS	T Flip-Flop, positive-edge-triggered, active-LOW set & reset
E_TFFP	T Flip-Flop, positive-edge-triggered, active-HIGH reset
E_TFFP	T Flip-Flop, positive-edge-triggered, active-HIGH set
E_TFFPR	T Flip-Flop, positive-edge-triggered, active-HIGH set & reset

MULTIPLEXED FLIP/FLOP Cells:

E_MFFRSB	Multiplexed-FF, active-LOW set & reset
E_MBNFNQ	Multiplexed-FF, individual selects and double outputs

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I/O Functions

INPUT PAD Cells:

E_IPAD	Input PAD
E_CIPAD	Clock Input PAD (Inverted)
E_IPADS	Input PAD with scan path
E_CDDR	CLOCK INPUT CLOCK DRIVER

BIDIRECT PAD Cells:

E_BDEBTB	Bi-directional IO buffer, active-LOW test enable
E_BDE	Bi-directional IO buffer, active-HIGH output enable

E_BDEB	Bi-directional IO buffer, active-LOW output enable
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OUTPUT PAD Cells:

E_OPADTEB	Tri-State, output buffer, active-LOW output enable, active-LOW test enable
E_OPAD	Output PAD
E_OPADE	Output PAD, active-HIGH output enable
E_OPADEB	Output PAD, active-LOW output enable
E_VDDPAD	Power pad
E_VSSPAD	Ground pad

OSCILLATOR PADS Cells:

E_OSCX	with Feedback Register
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Miscellaneous Functions

E_CLOAD1	clock load cell
E_MAXCAP	maximum capacitance symbol
E_NETWTH	net width symbol

E_RP01D1	repeater cell
E_WEIGHT	weight icon
E_VLOAD	voltage load cell
E_PASS	pass cell
E_PD	pull down cell
E_PU	pull up cell
E_TIE	tie up/down cell
E_DELAY	Delay cell

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Combinational Functions

AND

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NAND

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OR

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NOR

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XNR

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XOR

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AOI

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[E_AOI211](#)
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[E_AOI2X4](#)
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[E A0I3X4](#)
[E A0I44](#)
[E A0I4X3](#)
[E A0I4X4](#)
[E A0AI211](#)

OAI

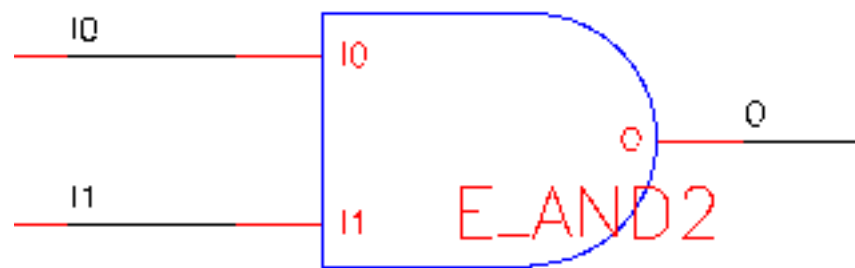
[E OAI21](#)
[E OAI211](#)
[E OAI221](#)
[E OAI321](#)
[E OAI31](#)
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[E OAI22](#)
[E OAI2X3](#)
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[E OAI3X3](#)
[E OAI3X4](#)
[E OAI44](#)
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COMPLEX

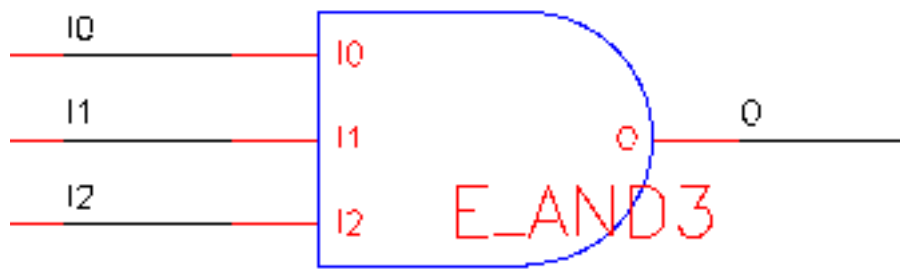
[E FN01](#)
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GENERATOR

[E PAR8E](#)
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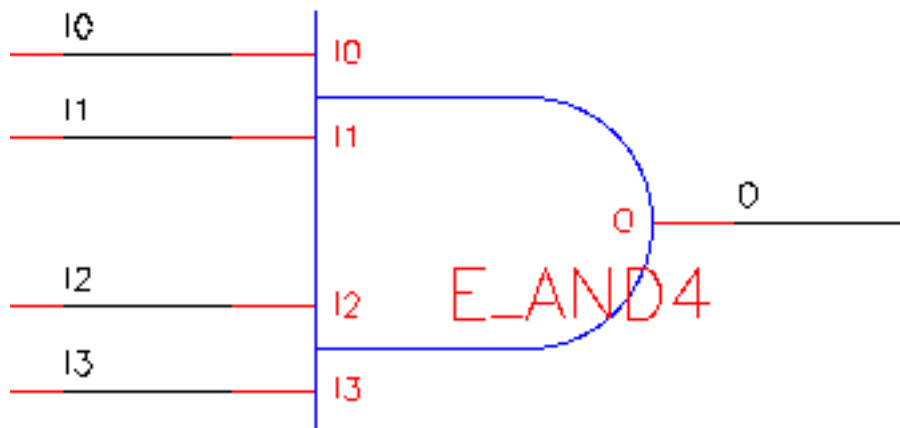
E_AND2 : 2-input AND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

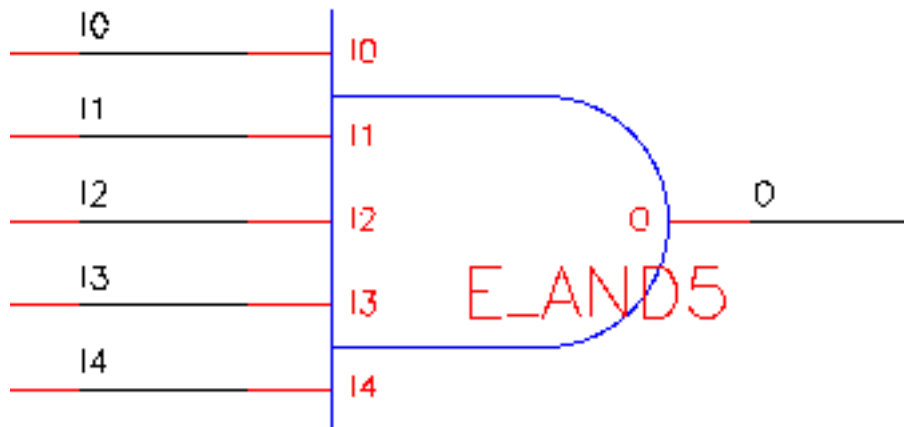
E_AND3 : 3-input AND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
O	OUTPUT	data

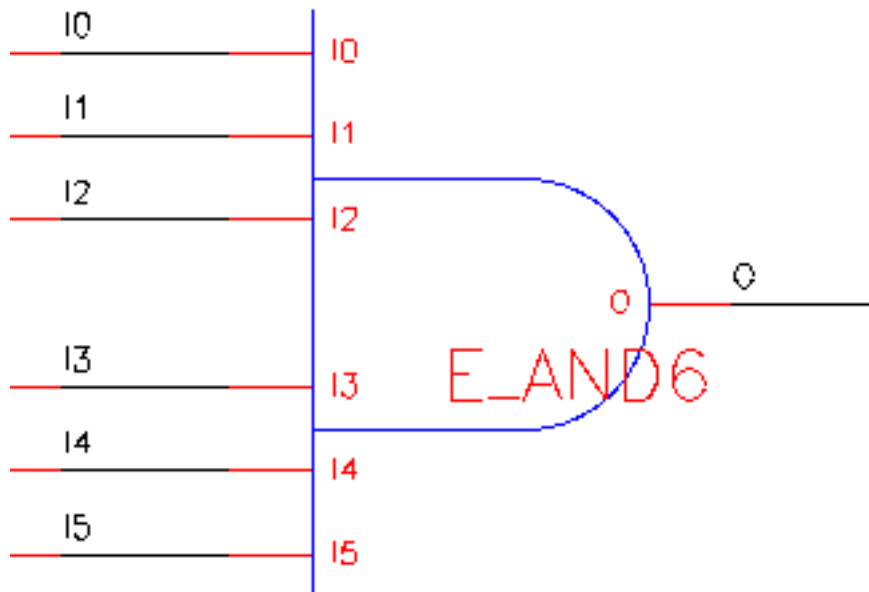
E_AND4 : 4-input AND



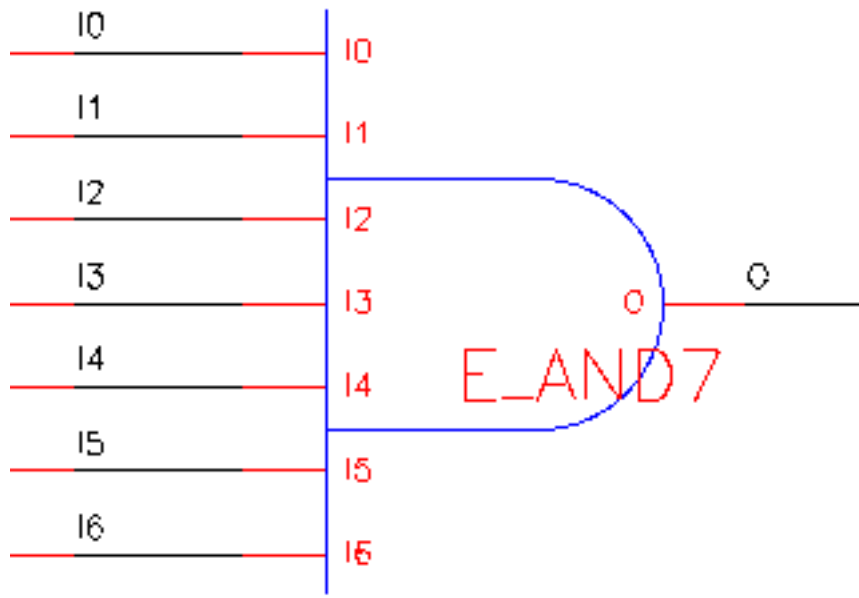
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O	OUTPUT	data

E_AND5 : 5-input AND

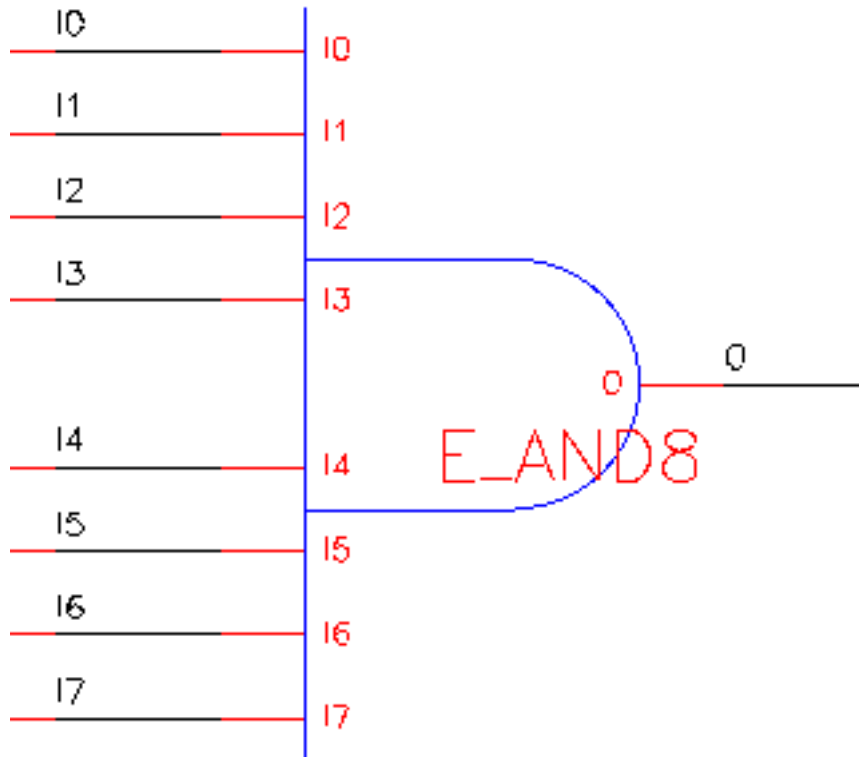
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
O	OUTPUT	data

E_AND6 : 6-input AND

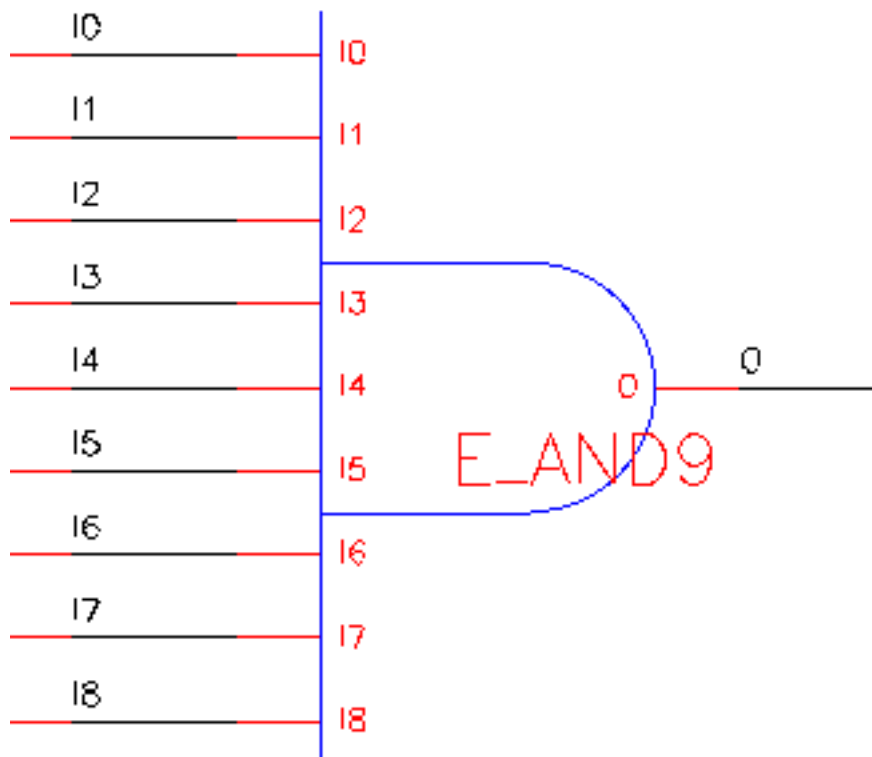
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
O	OUTPUT	data

E_AND7 : 7-input AND

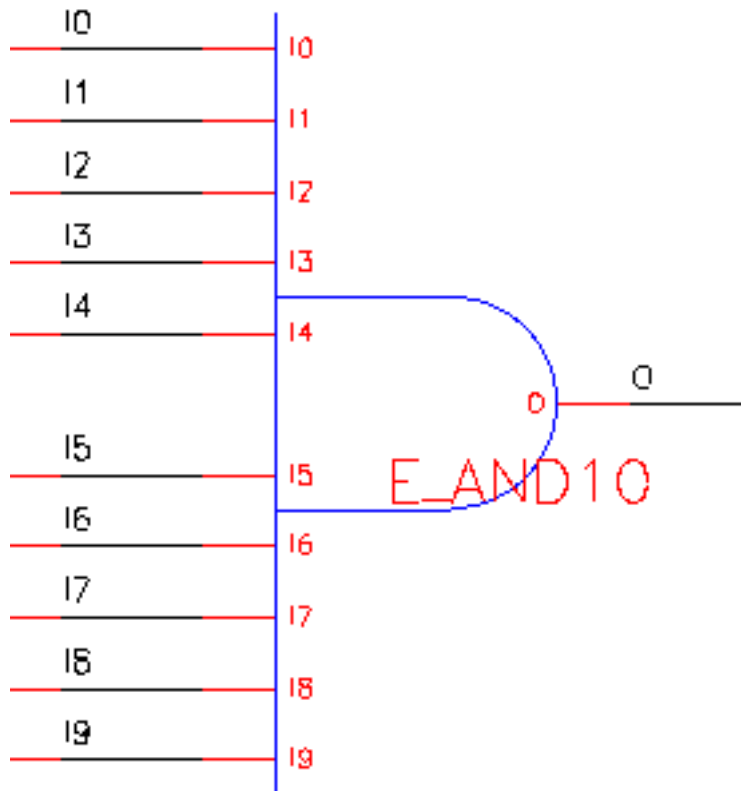
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
O	OUTPUT	data

E_AND8 : 8-input AND

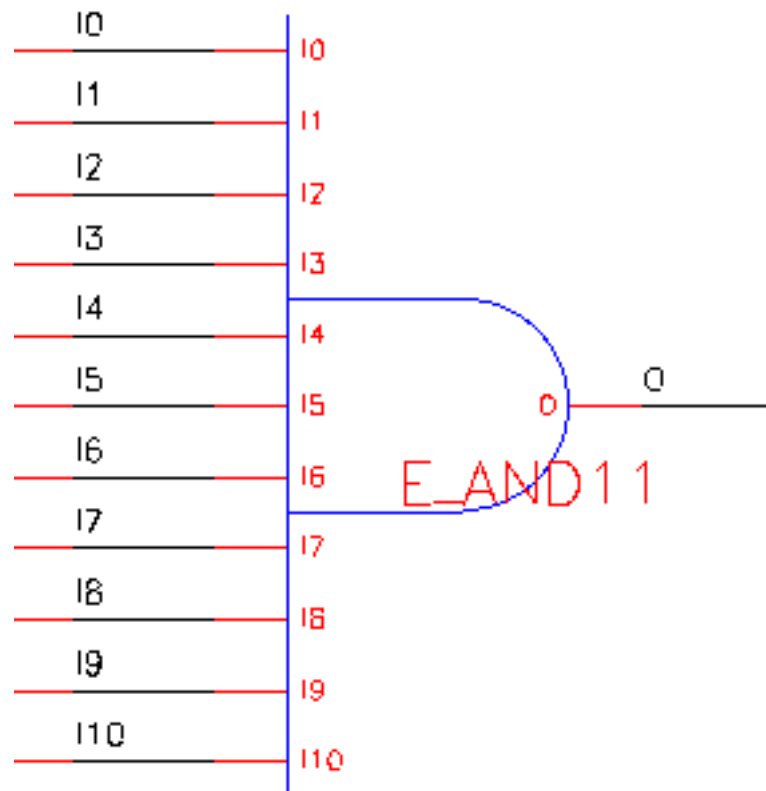
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O	OUTPUT	data

E_AND9 : 9-input AND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data

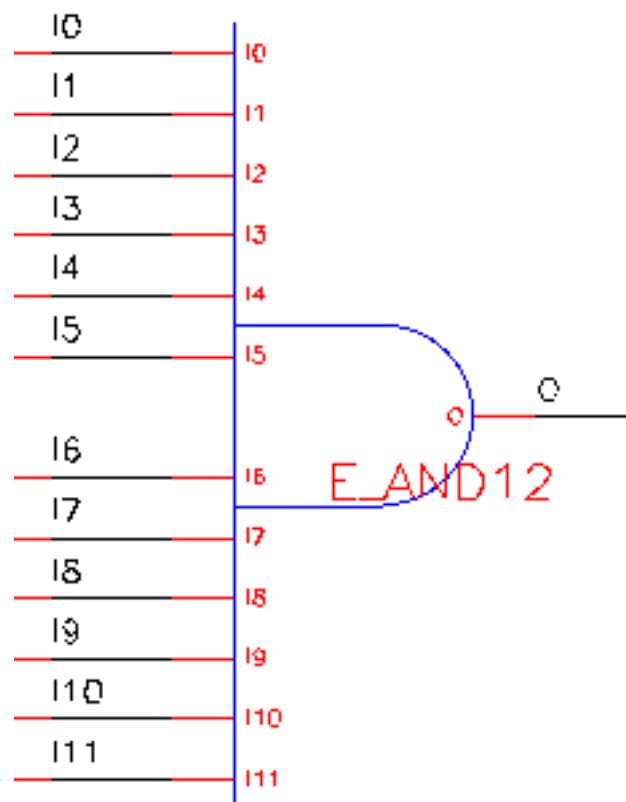
E_AND10 : 10-input AND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
O	OUTPUT	data

E_AND11 : 11-input AND

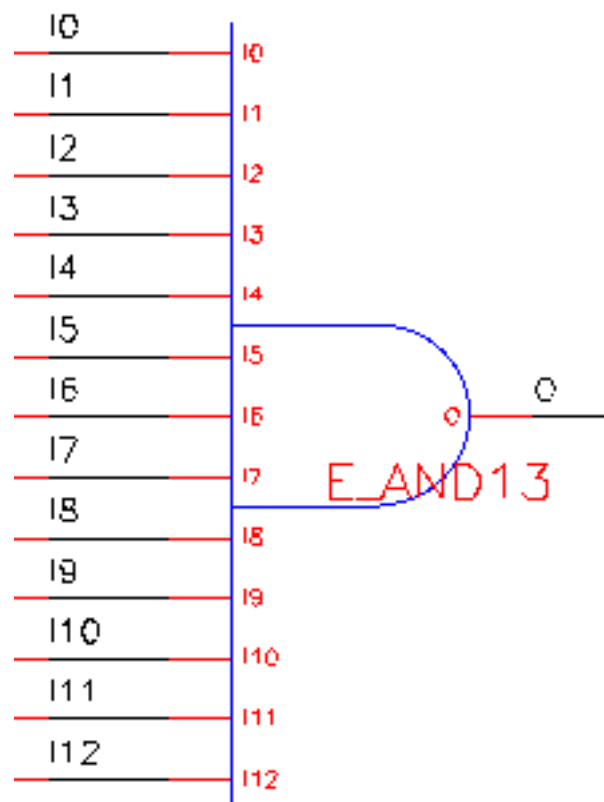
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

O	OUTPUT	data
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E_AND12 : 12-input AND

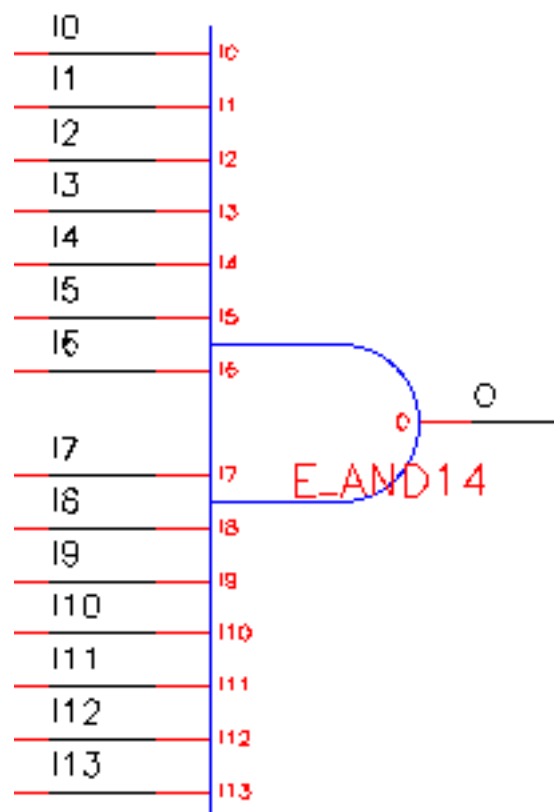
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data
I11	INPUT	data

I11	INPUT	data
O	OUTPUT	data

E_AND13 : 13-input AND

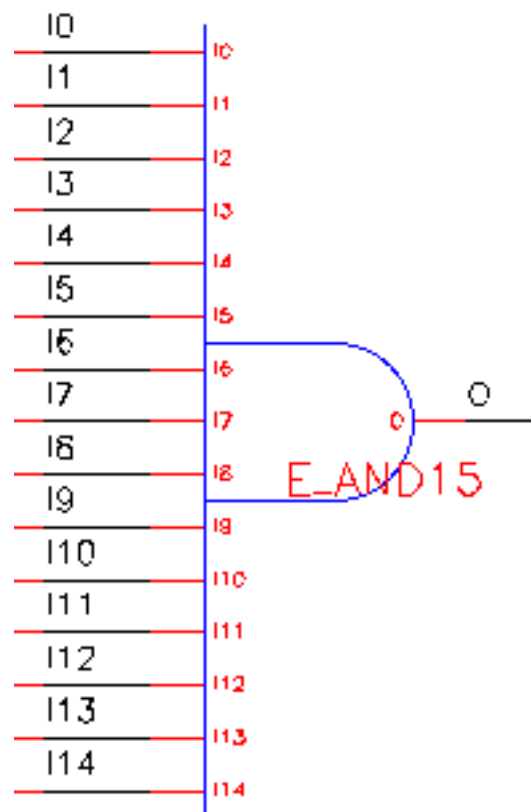
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
O	OUTPUT	data

E_AND14 : 14-input AND

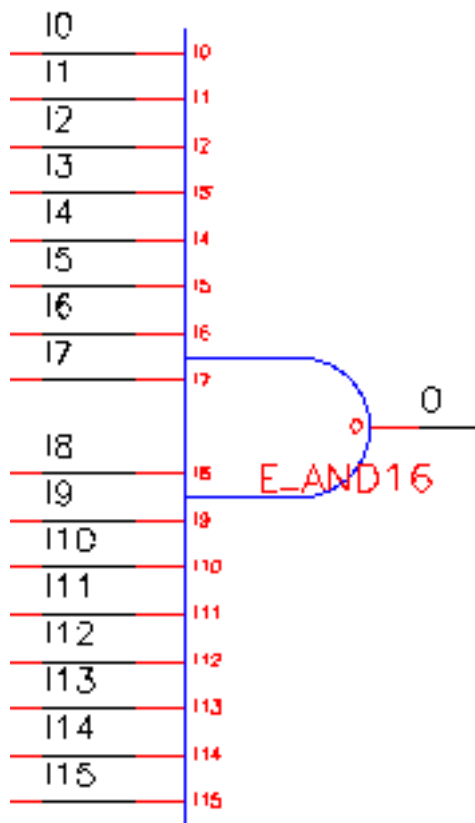
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
O	OUTPUT	data

E_AND15 : 15-input AND

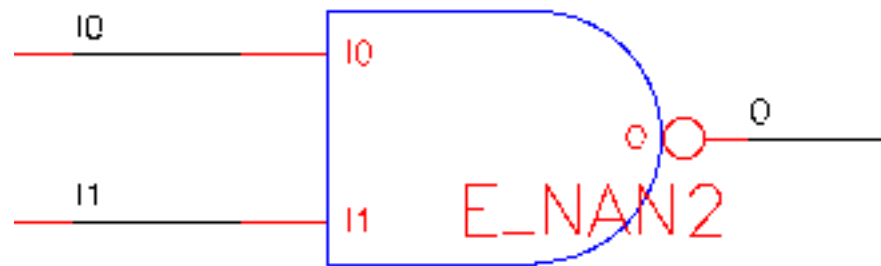
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
O	OUTPUT	data

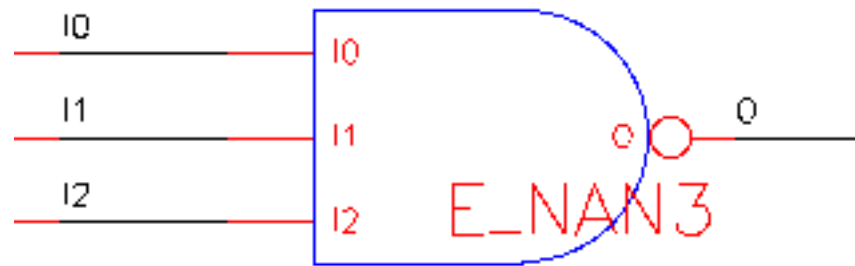
E_AND16 : 16-input AND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

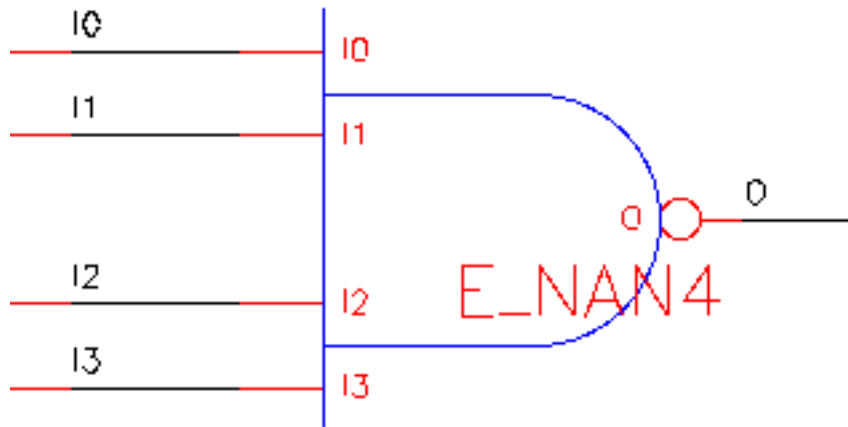
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
I15	INPUT	data
O	OUTPUT	data

E_NAN2 : 2-input NAND

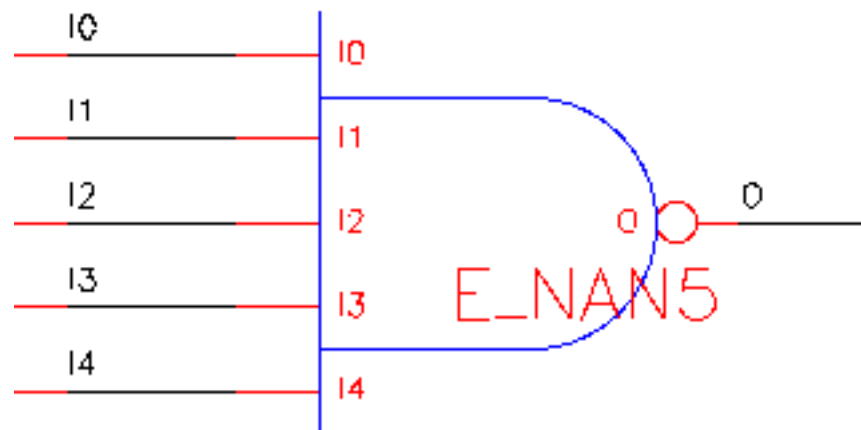
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_NAN3 : 3-input NAND

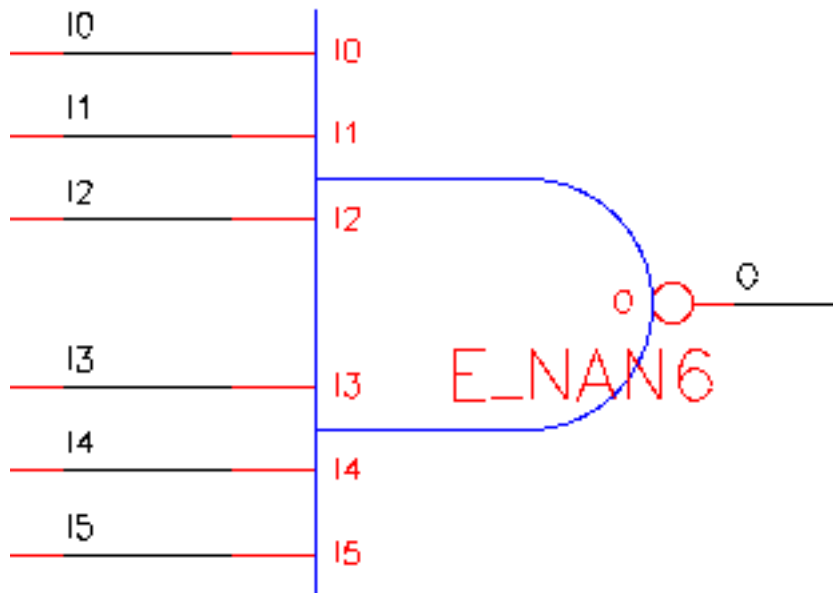
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
O	OUTPUT	data

E_NAN4 : 4-input NAND

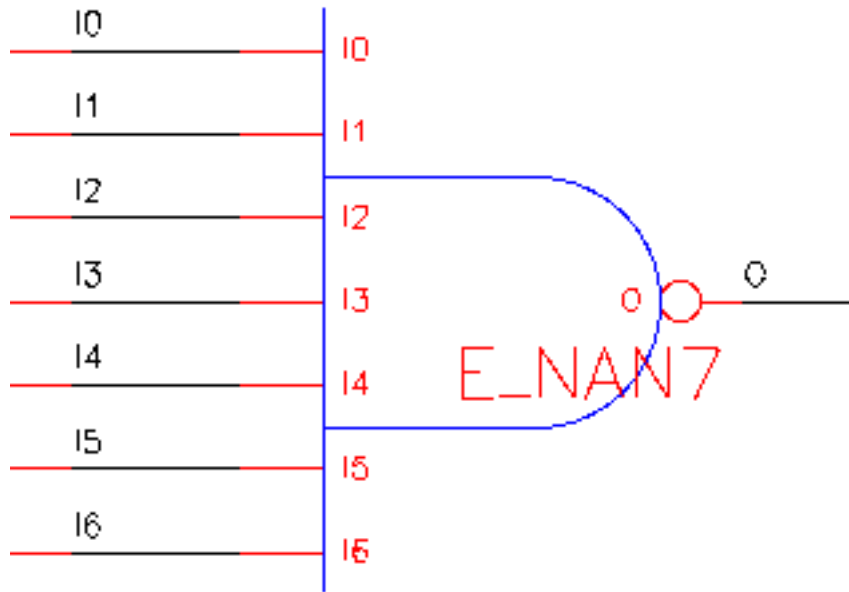
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O	OUTPUT	data

E_NAN5 : 5-input NAND

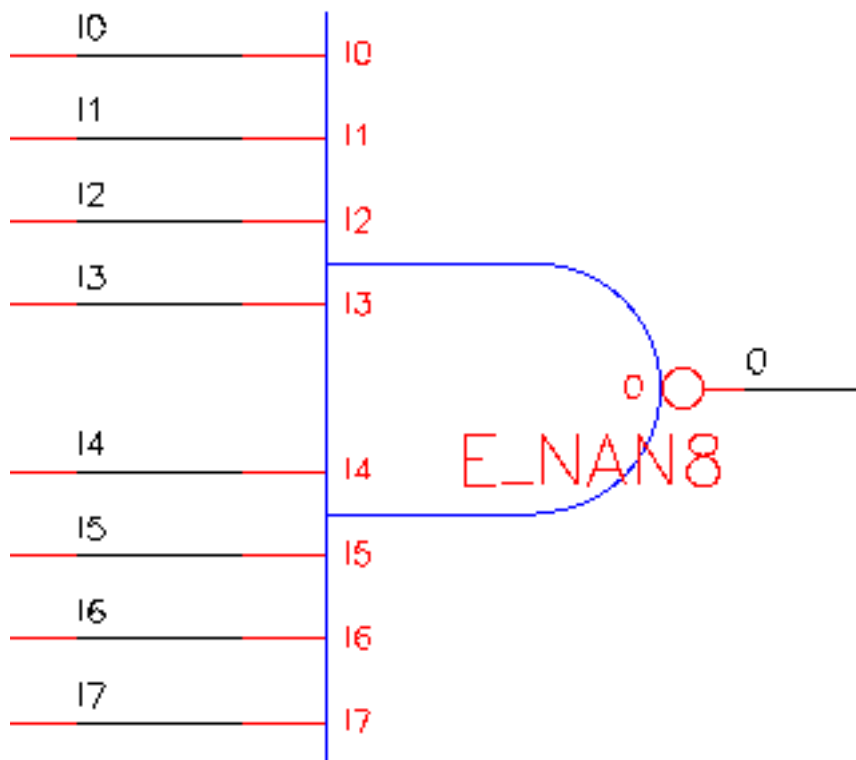
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
O	OUTPUT	data

E_NAN6 : 6-input NAND

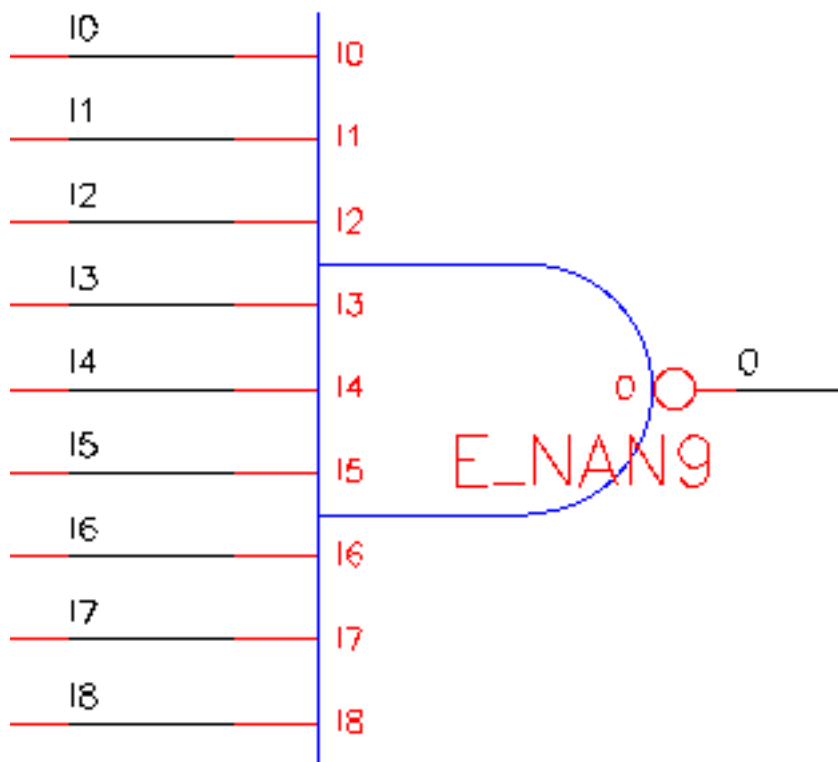
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
O	OUTPUT	data

E_NAN7 : 7-input NAND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
O	OUTPUT	data

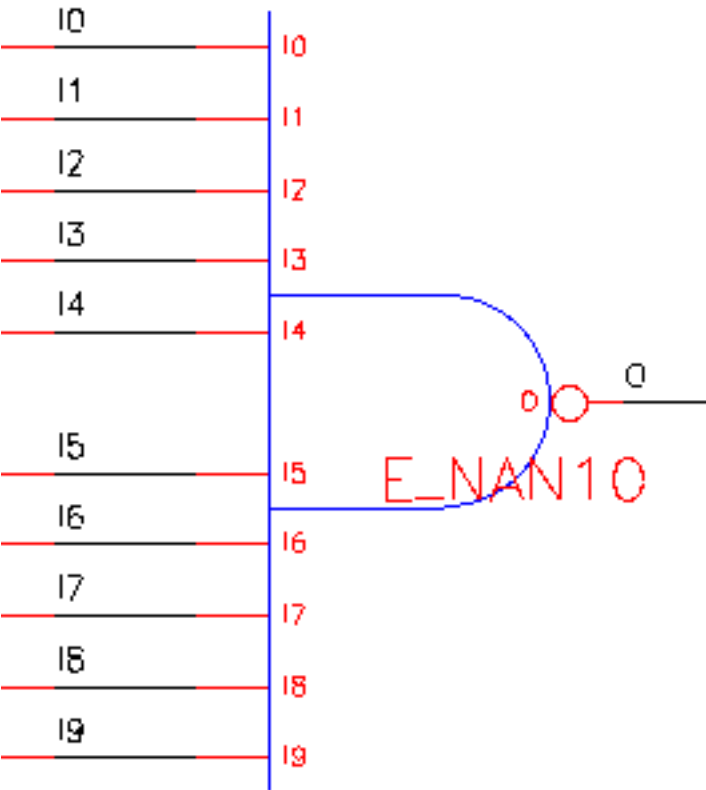
E_NAN8 : 8-input NAND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O	OUTPUT	data

E_NAN9 : 9-input NAND

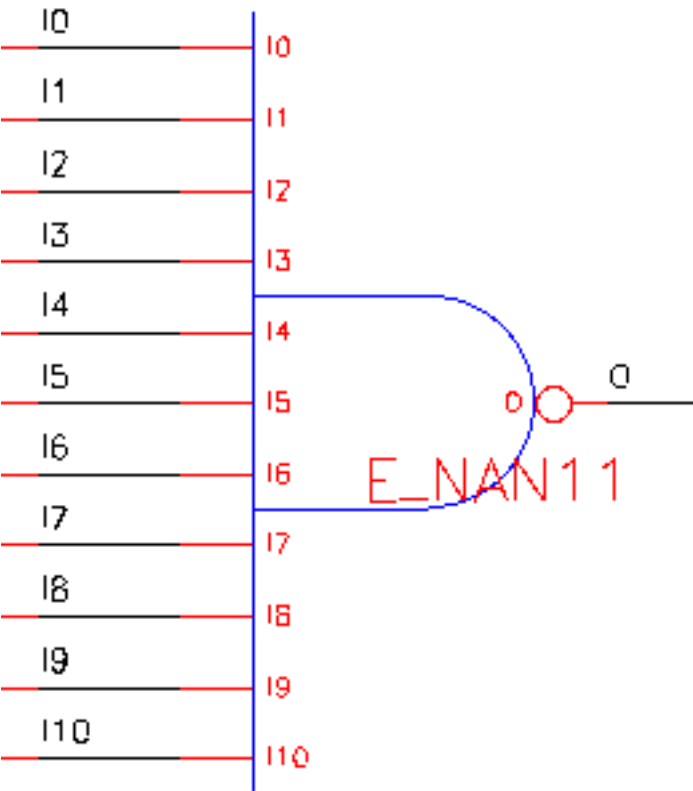
Pin	Direction	Description
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I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
O	OUTPUT	data

E_NAN10 : 10-input NAND



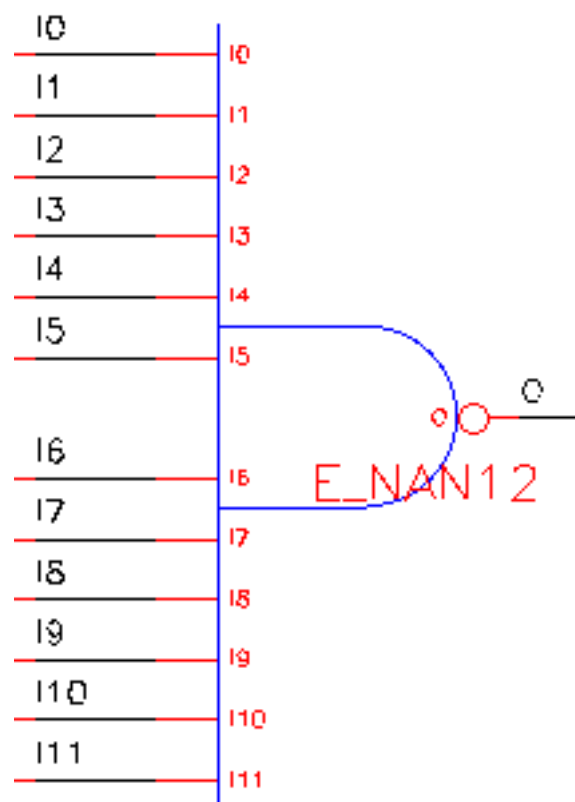
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
O	OUTPUT	data

E_NAN11 : 11-input NAND



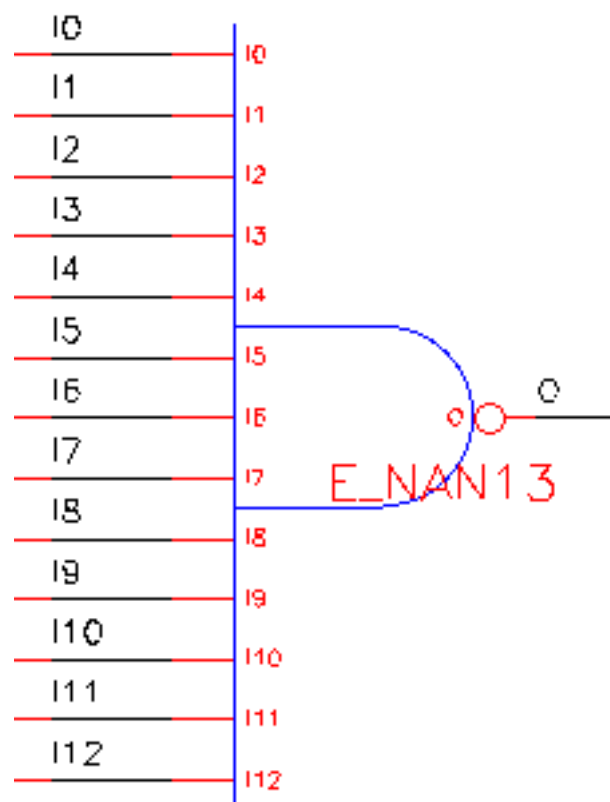
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

O	OUTPUT	data
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E_NAN12 : 12-input NAND

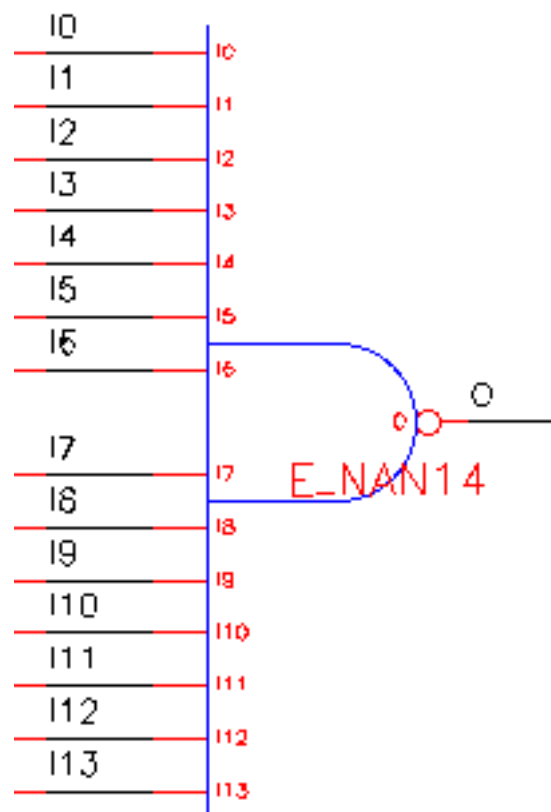
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
O	OUTPUT	data

E_NAN13 : 13-input NAND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

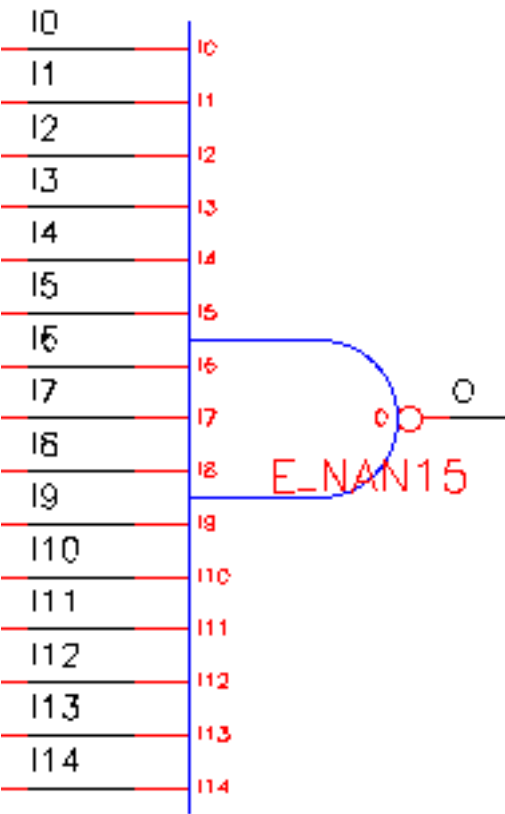
I11	INPUT	data
I12	INPUT	data
O	OUTPUT	data

E_NAN14 : 14-input NAND

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
O	OUTPUT	data

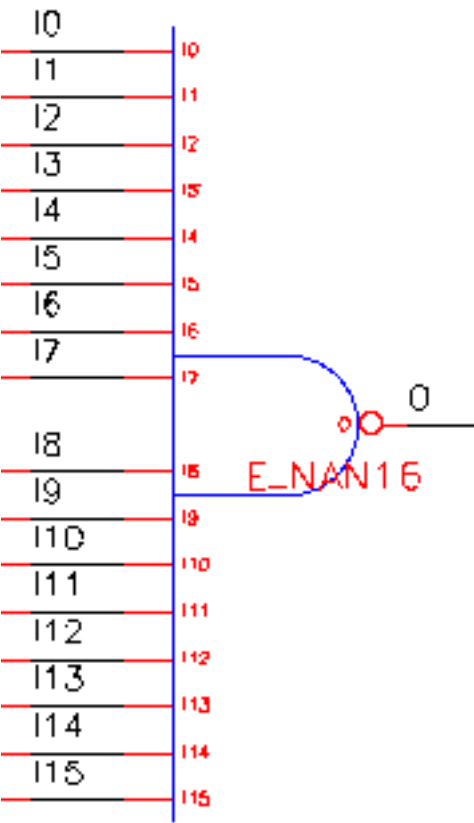
E_NAN15 : 15-input NAND



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

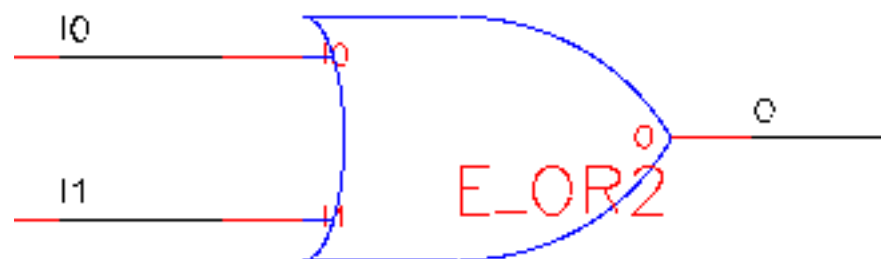
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
O	OUTPUT	data

E_NAN16 : 16-input NAND

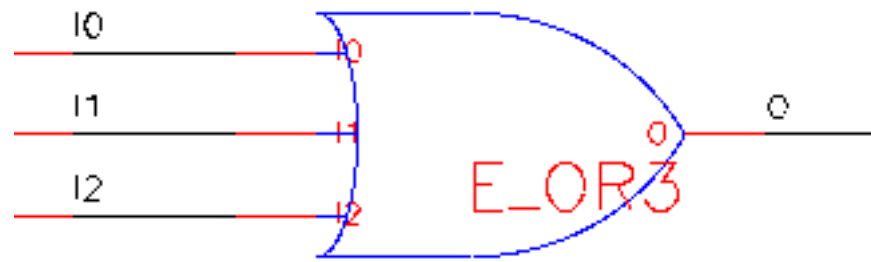


Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

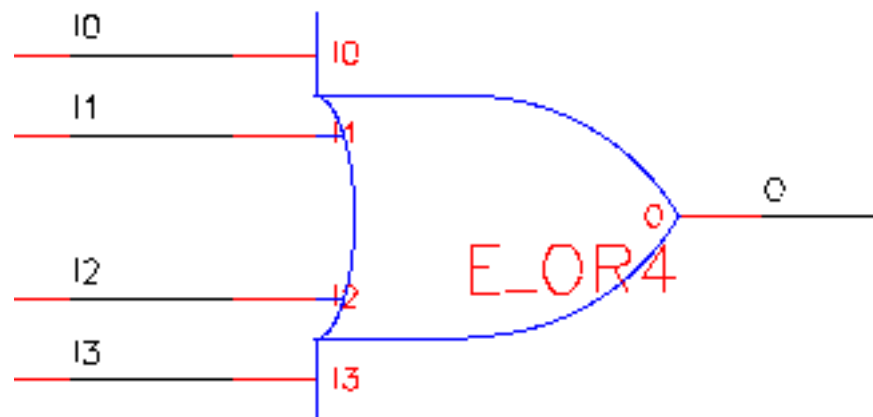
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
I15	INPUT	data
O	OUTPUT	data

E_OR2 : 2-input OR

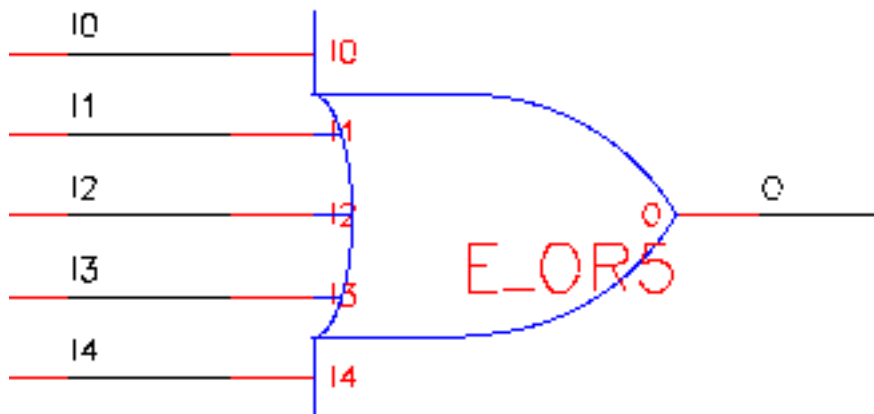
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_OR3 : 3-input OR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
O	OUTPUT	data

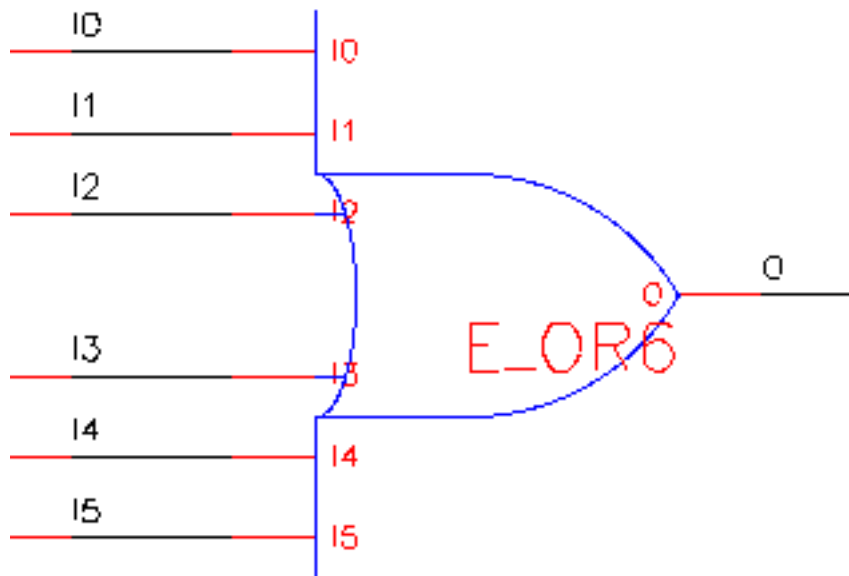
E_OR4 : 4-input OR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O	OUTPUT	data

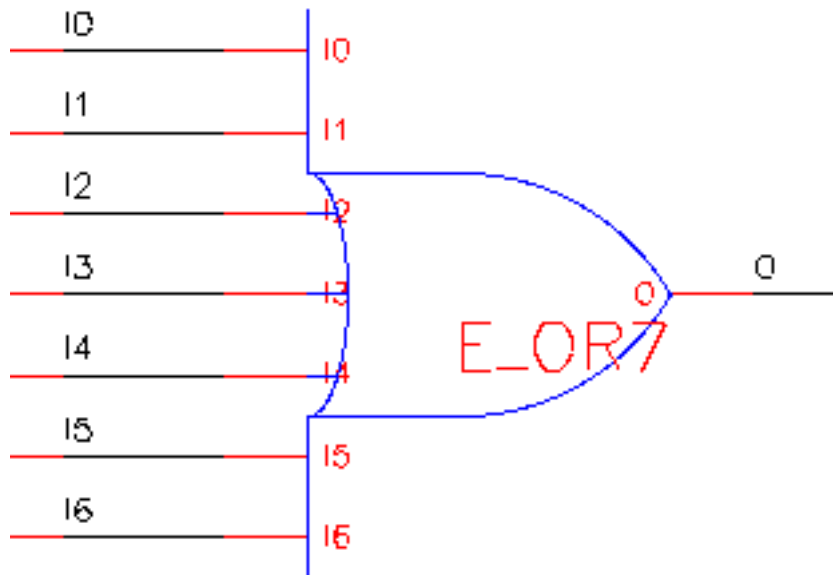


E_OR5 : 5-input OR

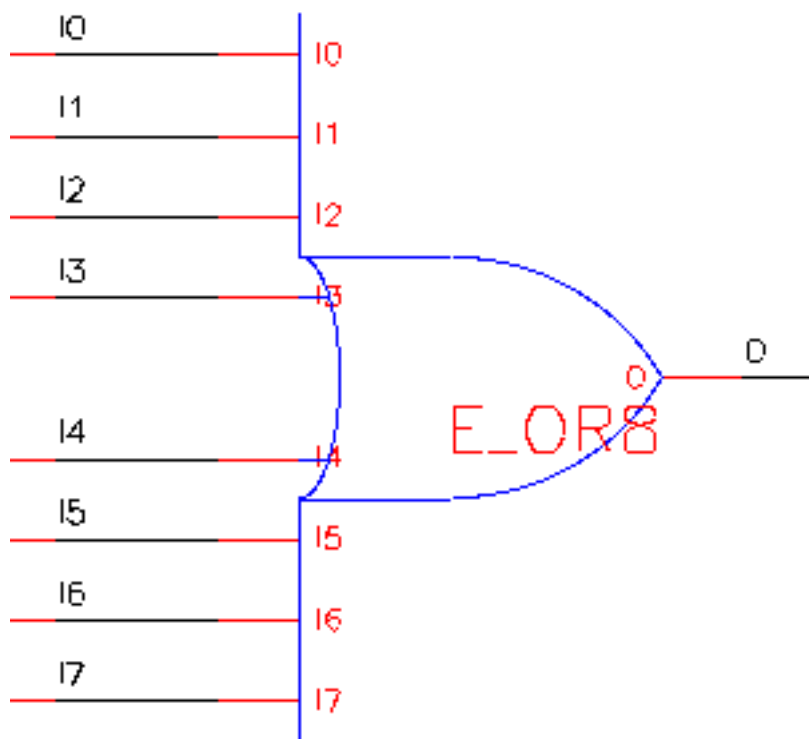
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
O	OUTPUT	data

E_OR6 : 6-input OR

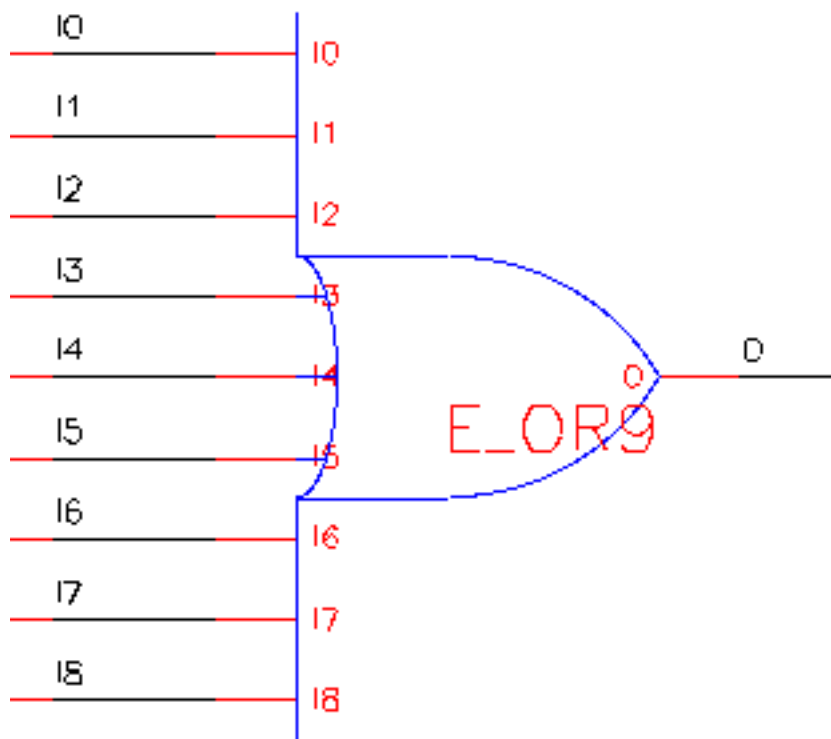
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
O	OUTPUT	data

E_OR7 : 7-input OR

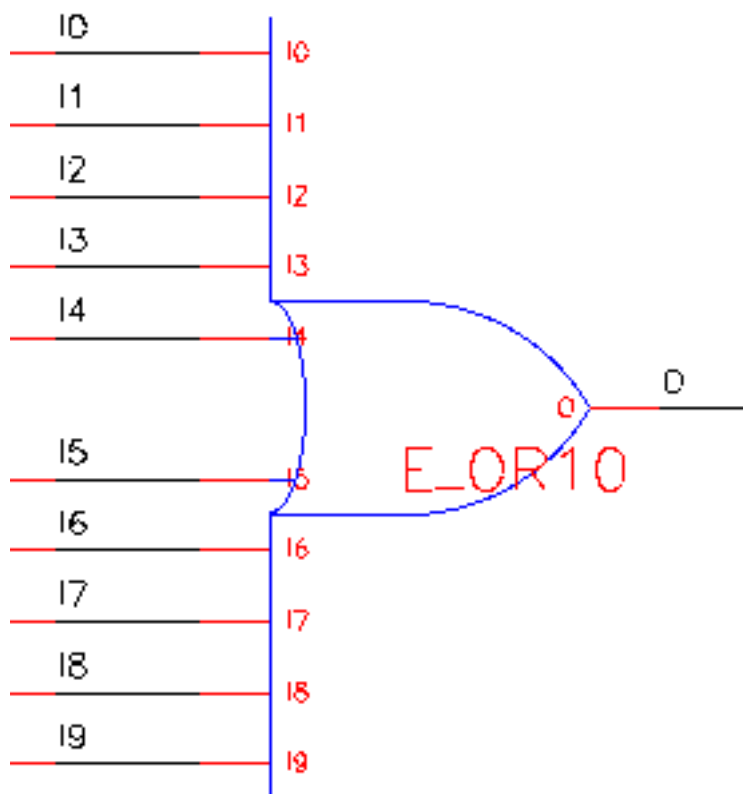
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
O	OUTPUT	data

E_OR8 : 8-input OR

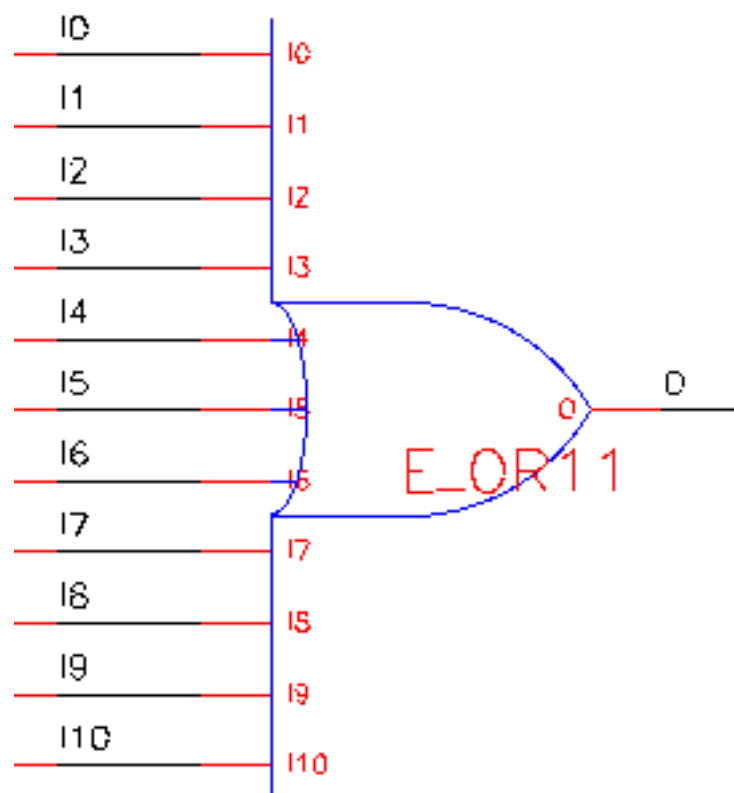
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O	OUTPUT	data

E_OR9 :

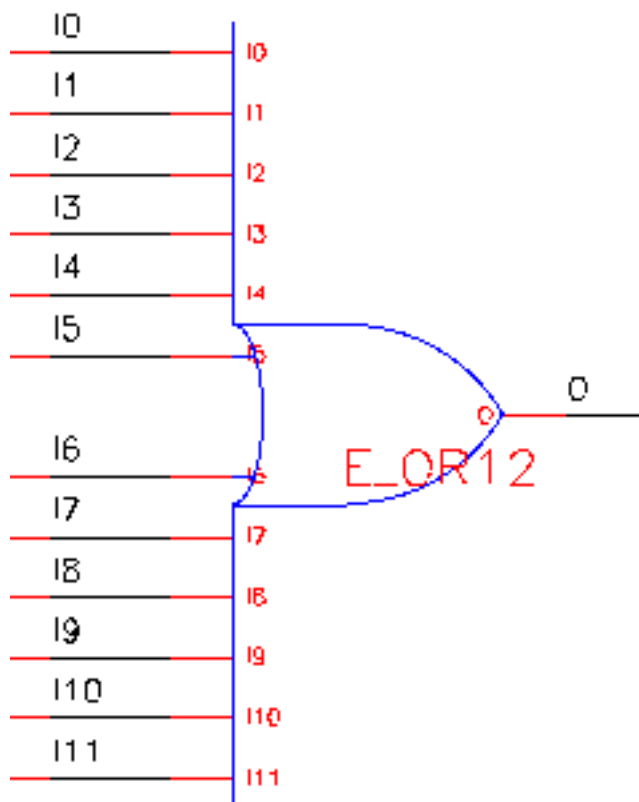
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
O	OUTPUT	data

E_OR10 : 10-input OR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
O	OUTPUT	data

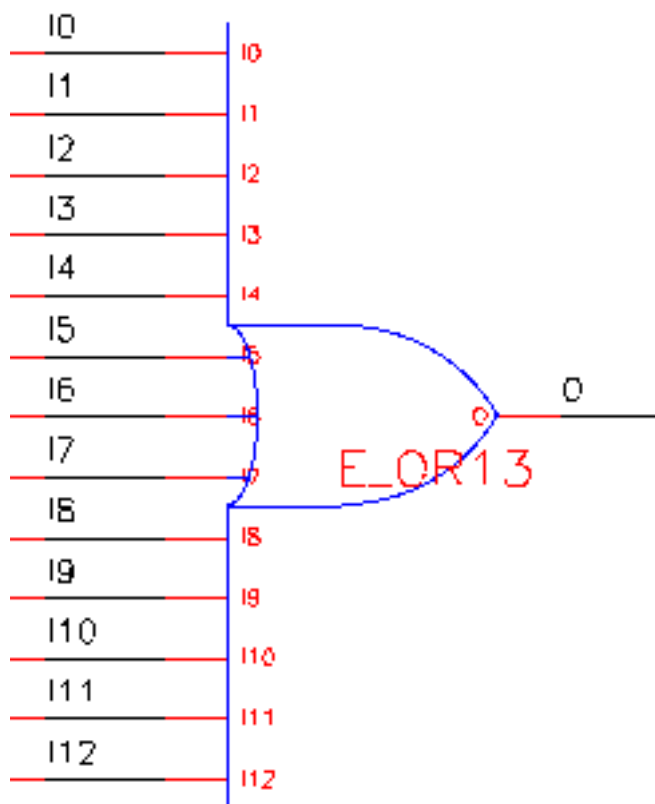
E_OR11 : 11-input OR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data
O	OUTPUT	data

E_OR12 : 12-input OR

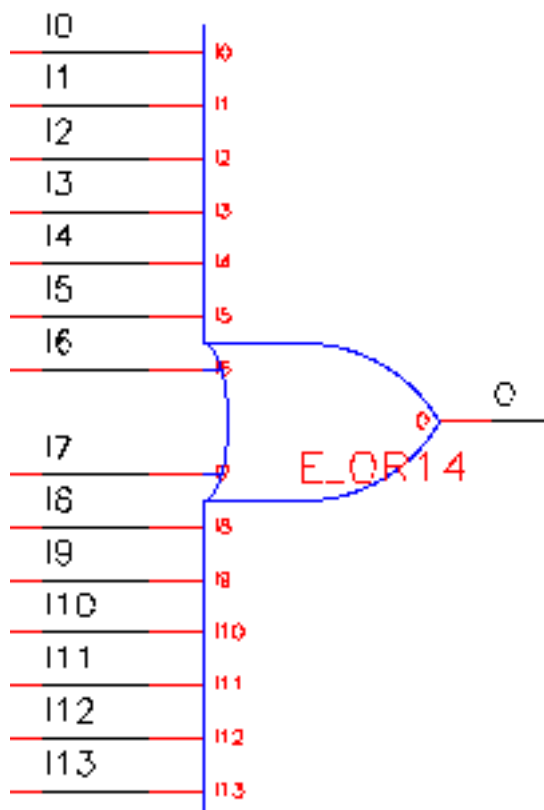
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data
I11	INPUT	data

I11	INPUT	data
O	OUTPUT	data

E_OR13 : 13-input OR

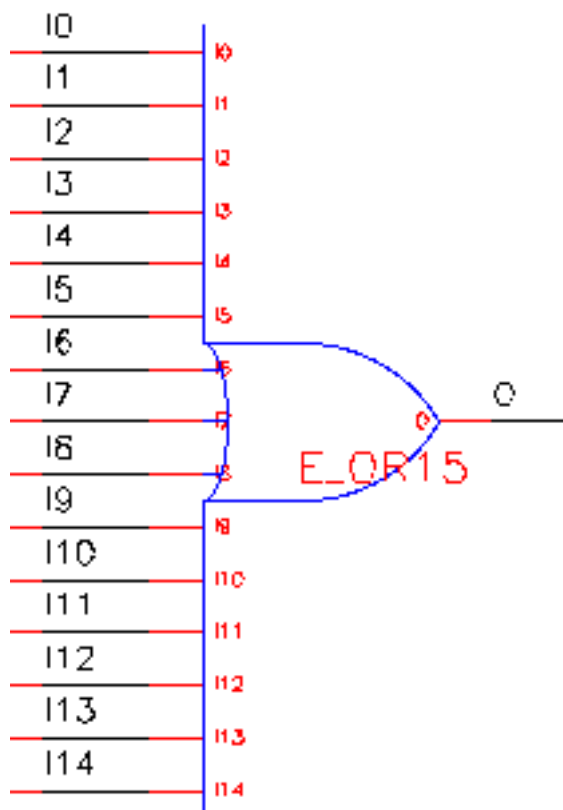
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data
I11	INPUT	data

I12	INPUT	data
O	OUTPUT	data

E_OR14 : 14-input OR

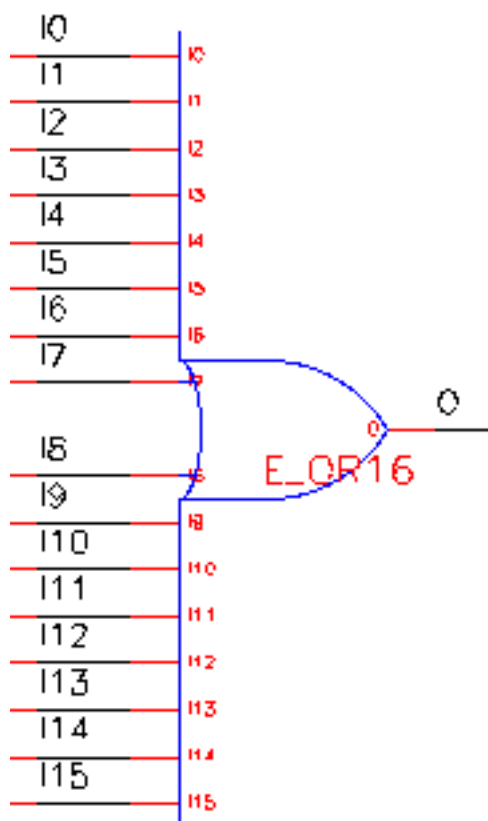
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
O	OUTPUT	data

E_OR15 : 15-input OR

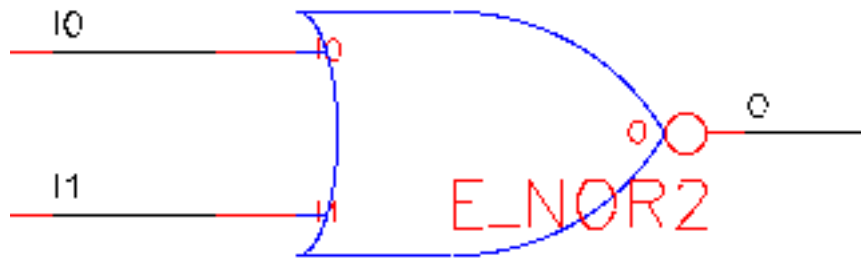
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
O	OUTPUT	data

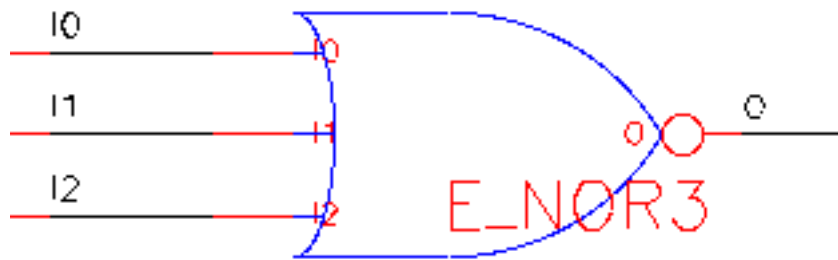
E_OR16 : 16-input OR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

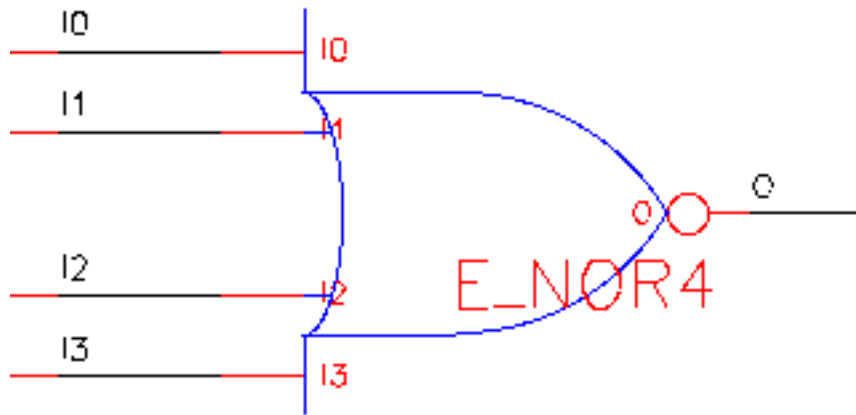
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
I15	INPUT	data
O	OUTPUT	data

E_NOR2 : 2-input NOR

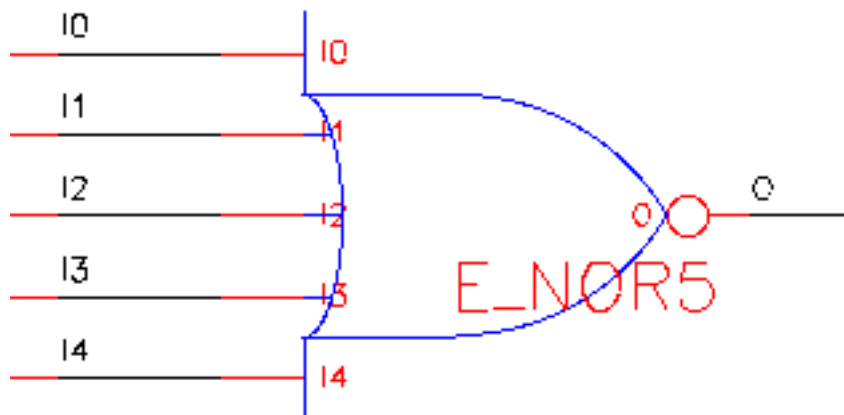
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_NOR3 : 3-input NOR

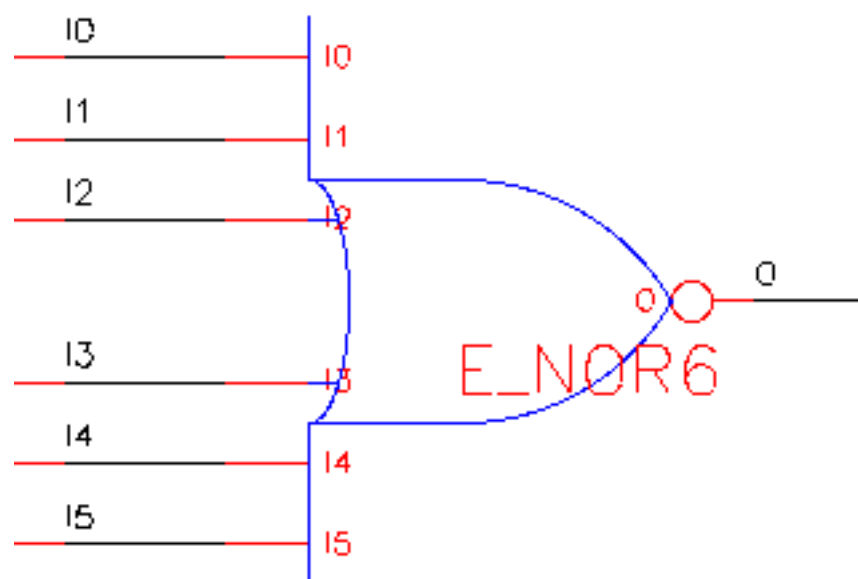
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
O	OUTPUT	data

E_NOR4 : 4-input NOR

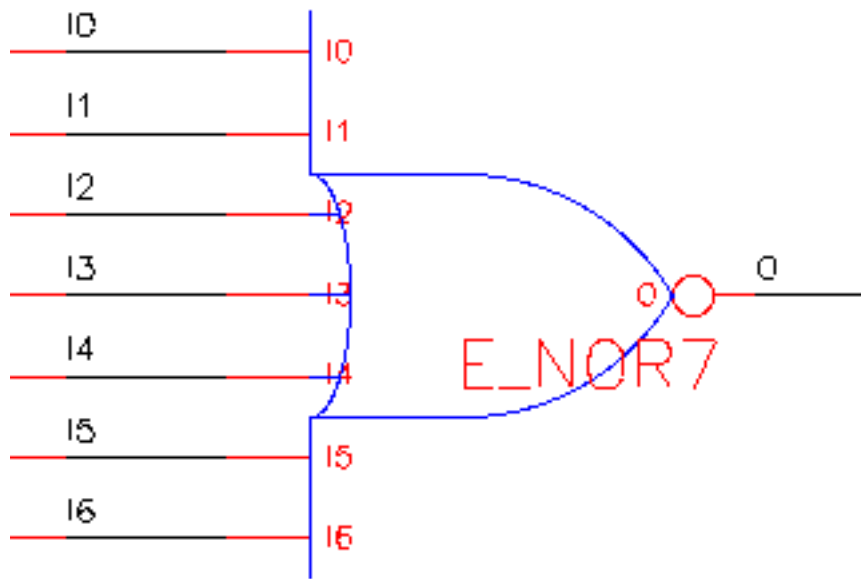
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O	OUTPUT	data

E_NOR5 : 5-input NOR

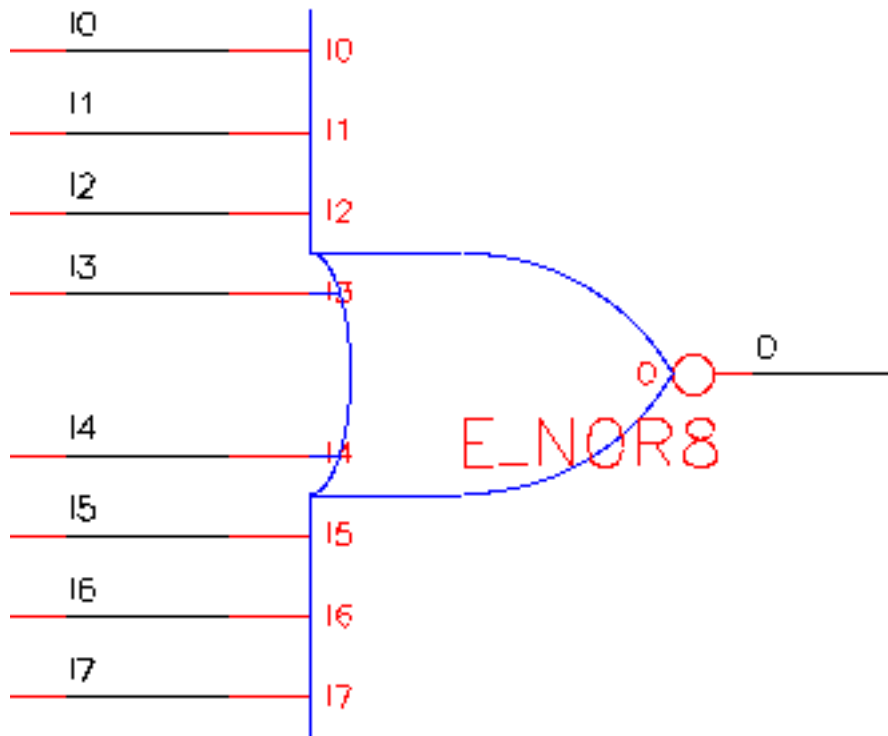
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
O	OUTPUT	data

E_NOR6 : 6-input NOR

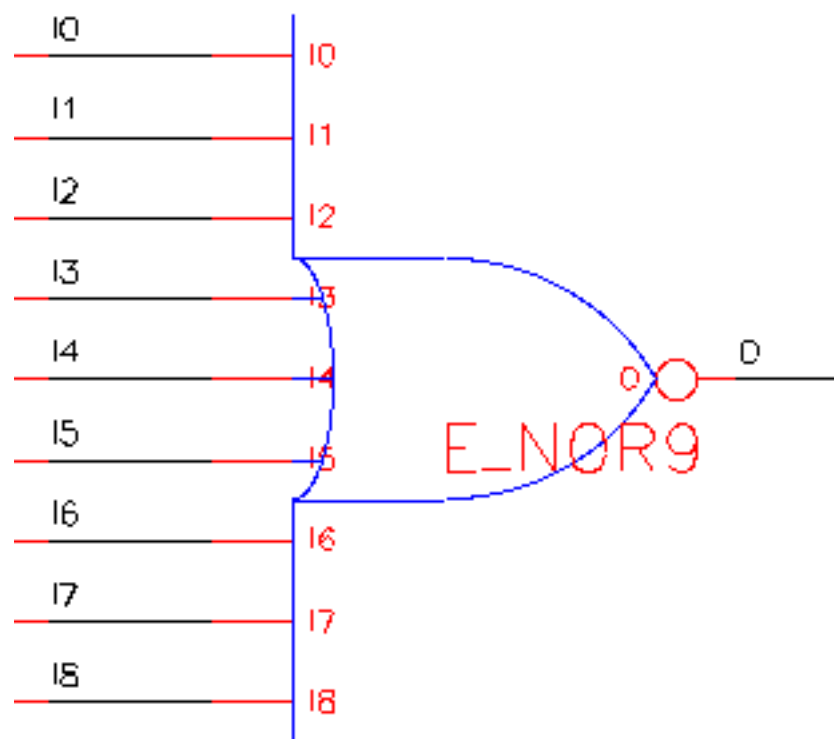
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
O	OUTPUT	data

E_NOR7 : 7-input NOR

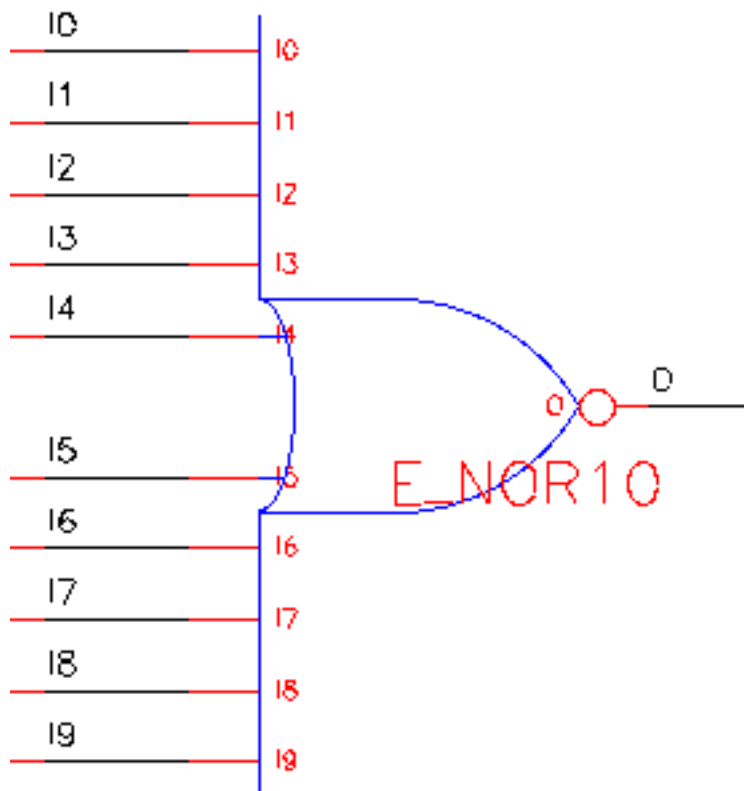
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
O	OUTPUT	data

E_NOR8 : 8-input NOR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O	OUTPUT	data

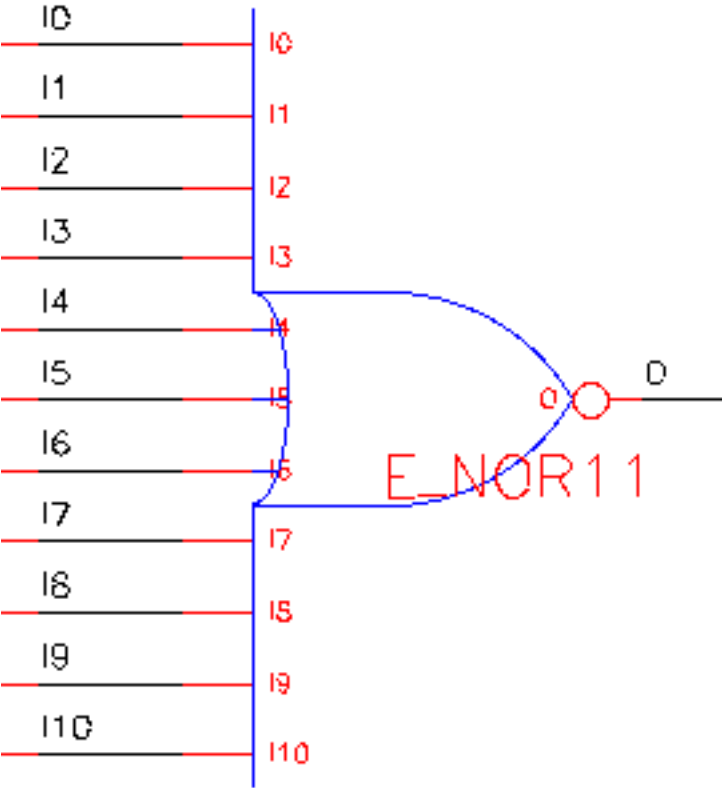
E_NOR9 : 9-input NOR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
O	OUTPUT	data

E_NOR10 : 10-input NOR

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
O	OUTPUT	data

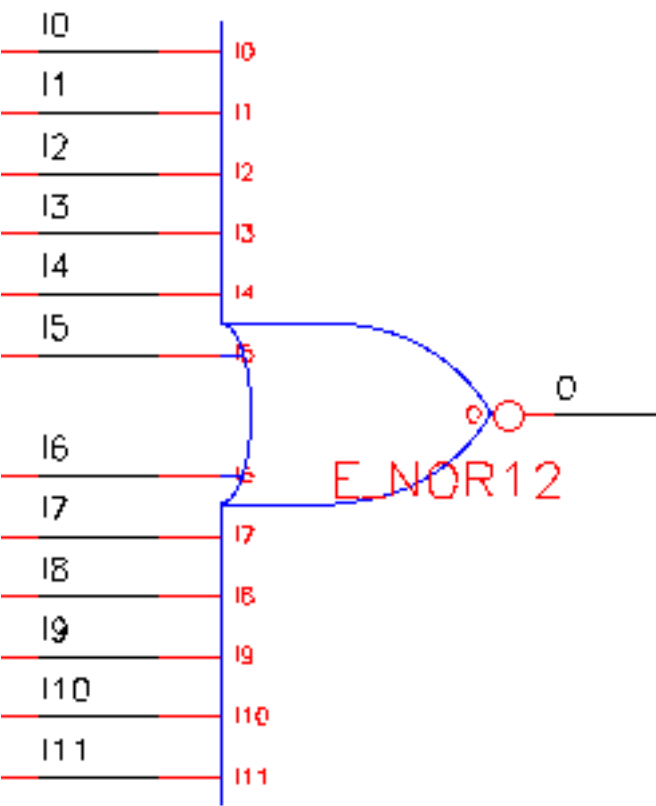
E_NOR11 : 11-input NOR



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

O	OUTPUT	data
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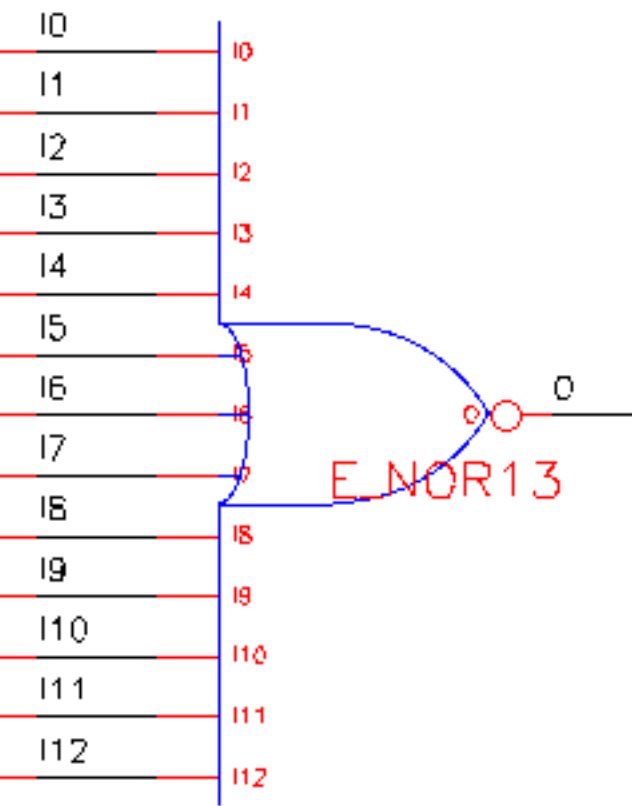
E_NOR12 : 12-input NOR



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
O	OUTPUT	data

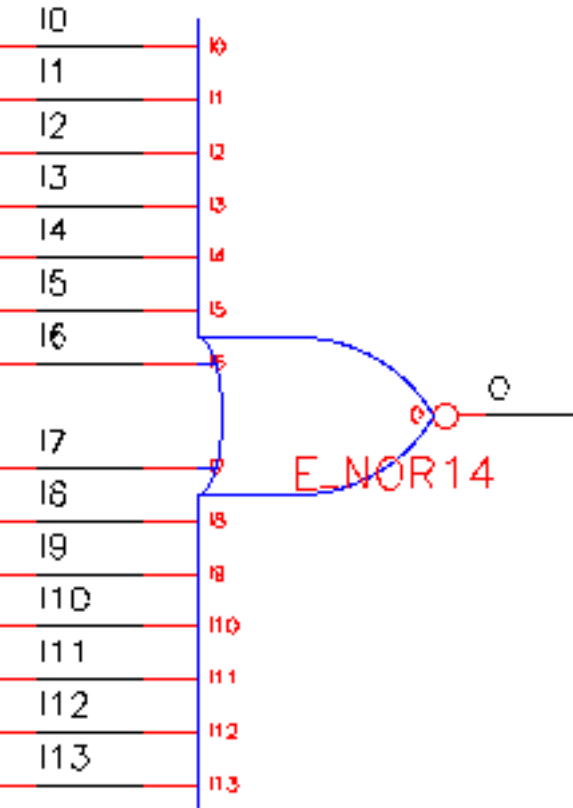
E_NOR13 : 13-input NOR



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
O	OUTPUT	data

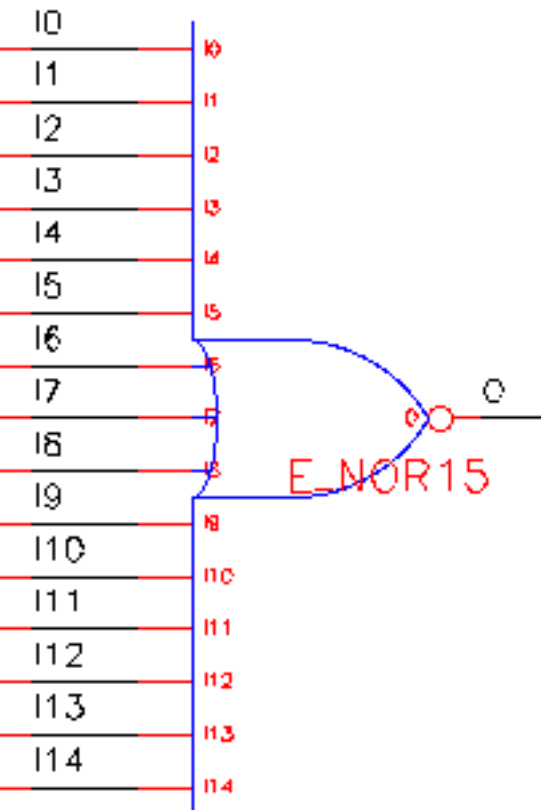
E_NOR14 : 14-input NOR



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
O	OUTPUT	data

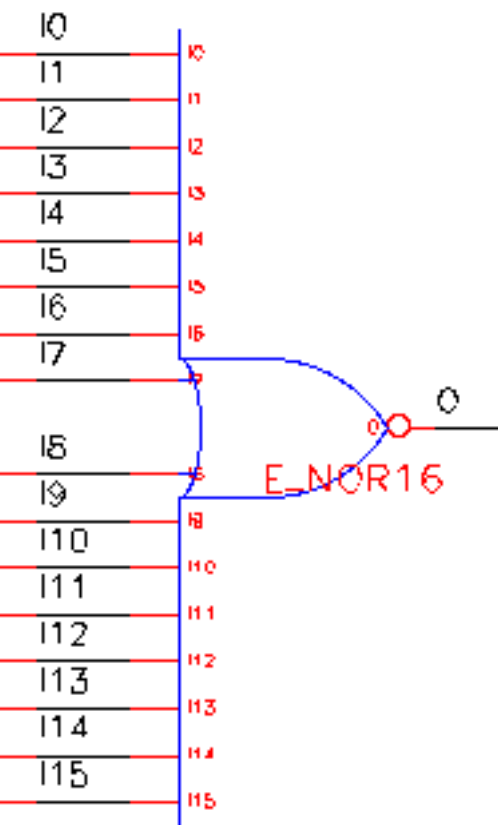
E_NOR15 : 15-input NOR



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

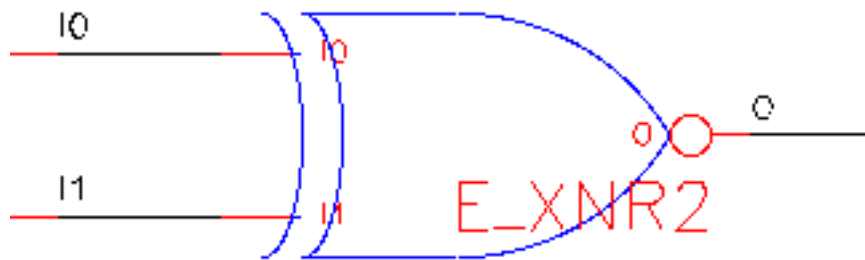
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
O	OUTPUT	data

E_NOR16 : 16-input NOR

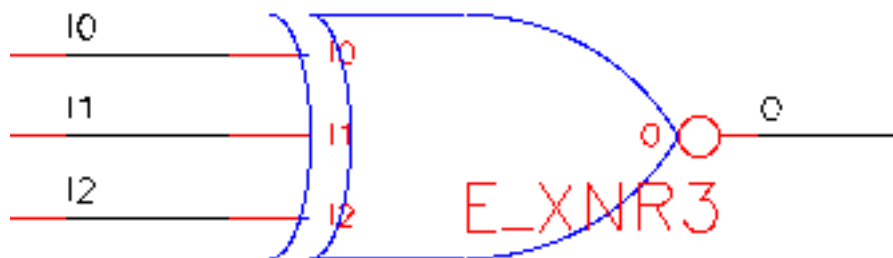


Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data
I10	INPUT	data

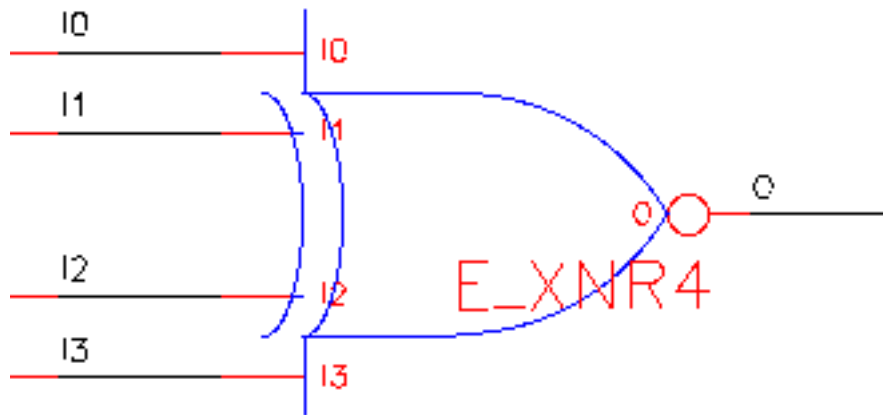
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
I15	INPUT	data
O	OUTPUT	data

E_XNR2 : 2-input XNR

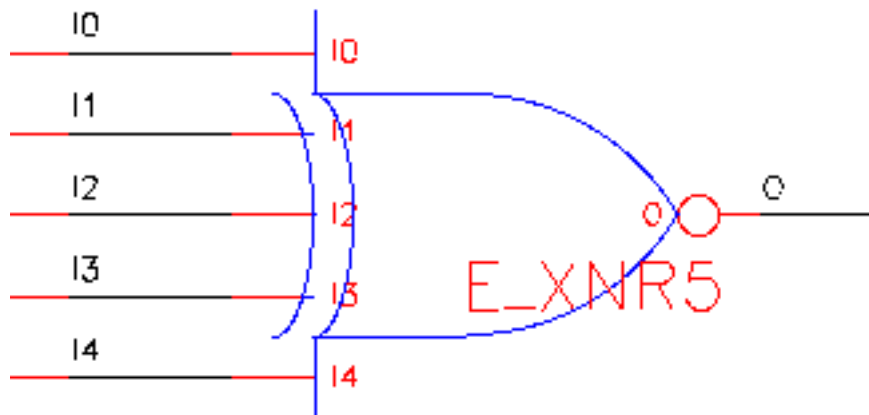
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_XNR3 : 3-input XNR

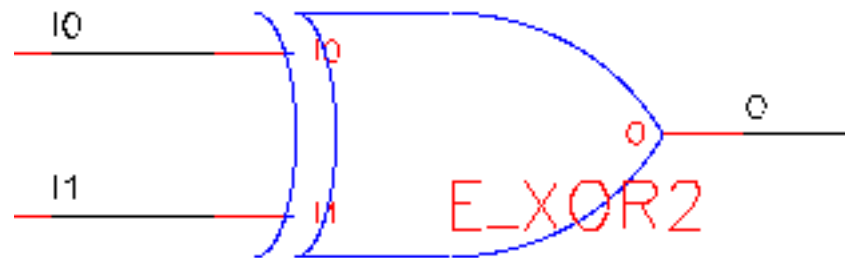
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
O	OUTPUT	data

E_XNR4 : 4-input XNR

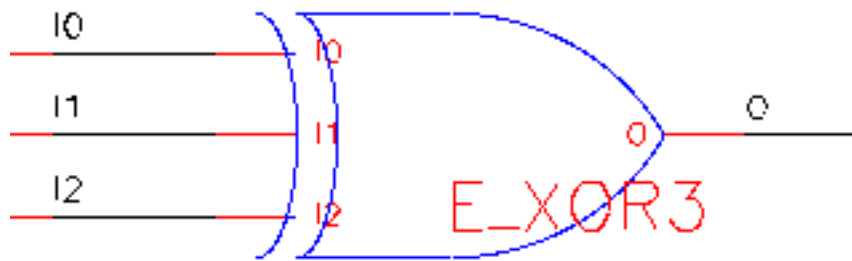
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O	OUTPUT	data

E_XNR5 : 5-input XNR

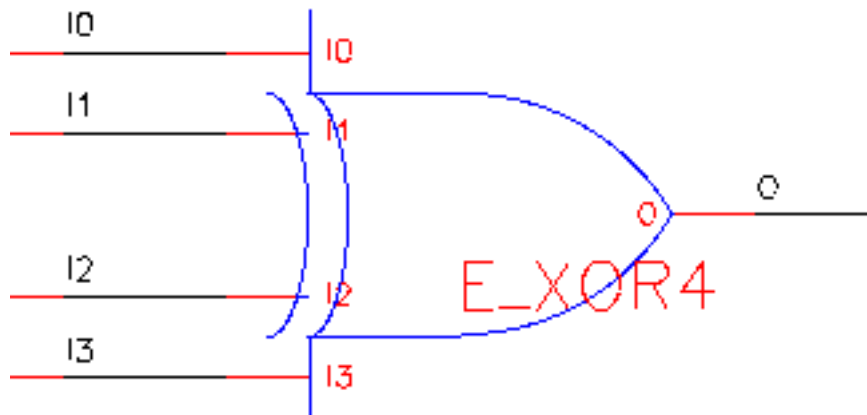
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
O	OUTPUT	data

E_XOR2 : 2-input XOR

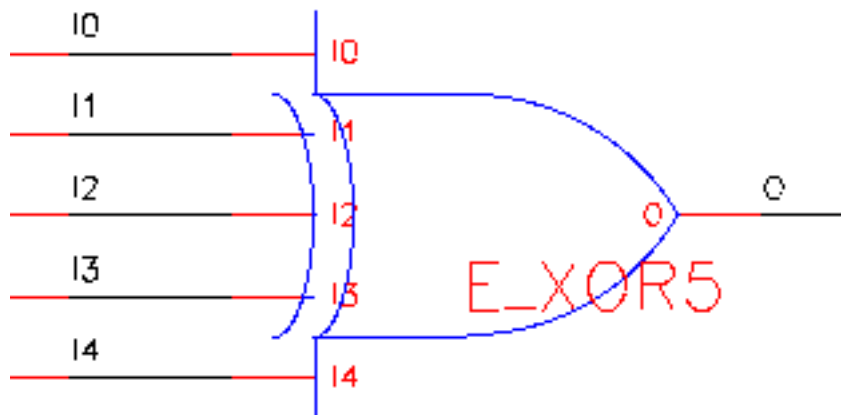
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_XOR3 : 3-input XOR

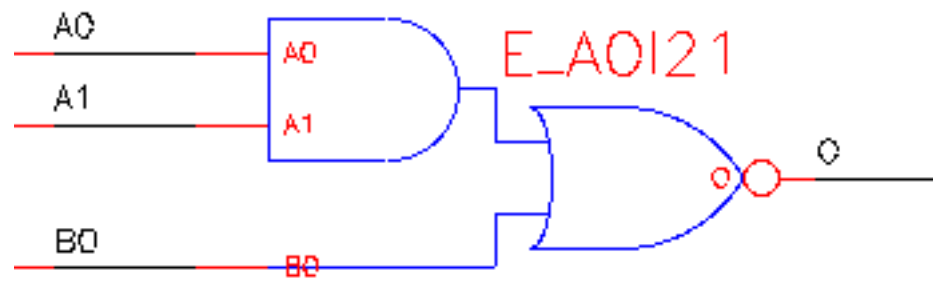
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
O	OUTPUT	data

E_XOR4 : 4-input XOR

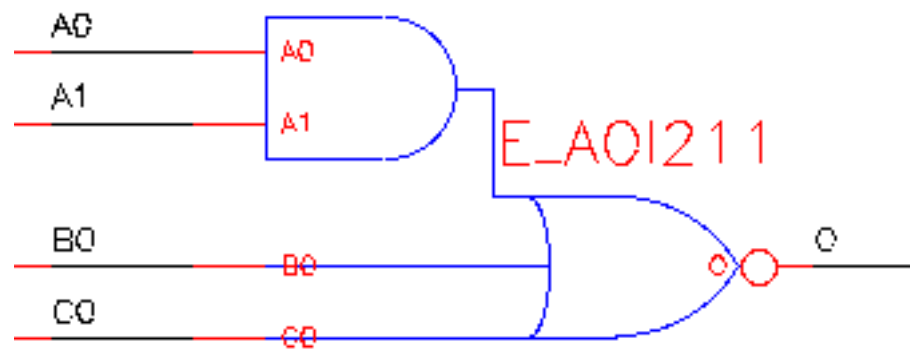
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O	OUTPUT	data

E_XOR5 : 5-input XOR

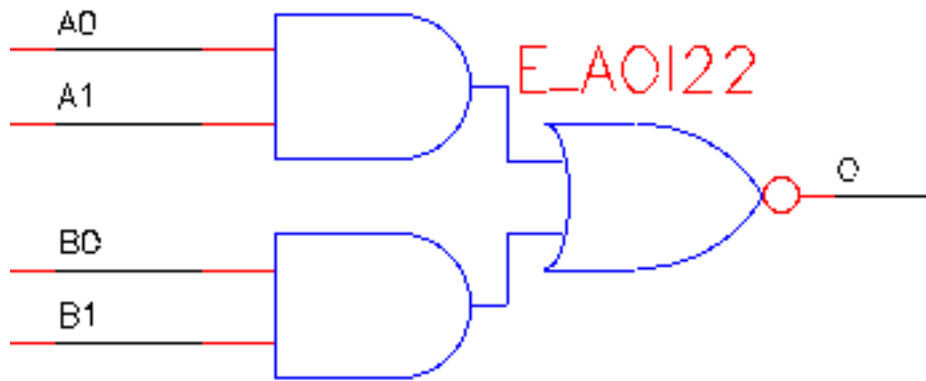
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
O	OUTPUT	data

E_AOI21 : 2,1 AND-OR-INVERTOR

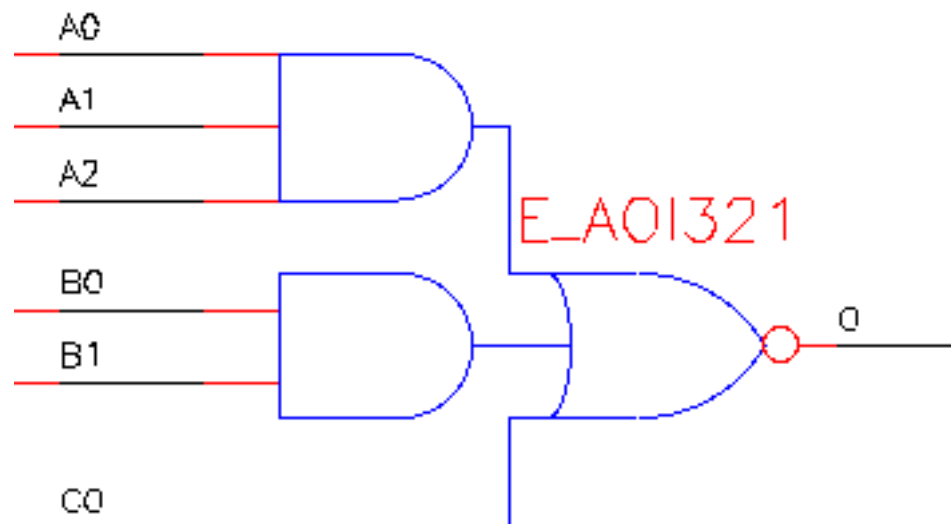
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
O	OUTPUT	data

E_AOI211 : 2,1,1 AND-OR-INVERTOR

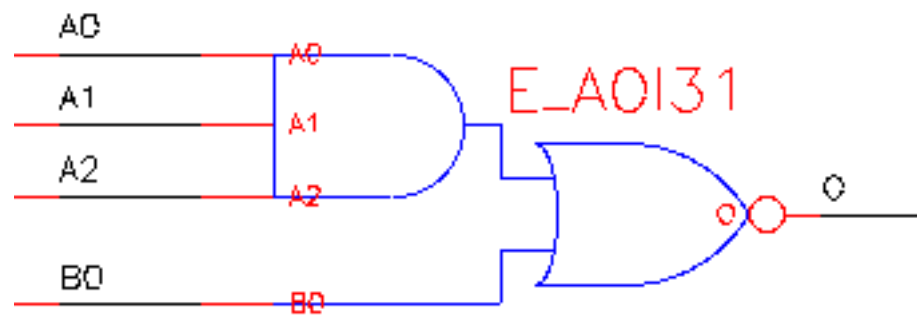
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
C0	INPUT	data
O	OUTPUT	data

E_AOI22 : 2,2 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

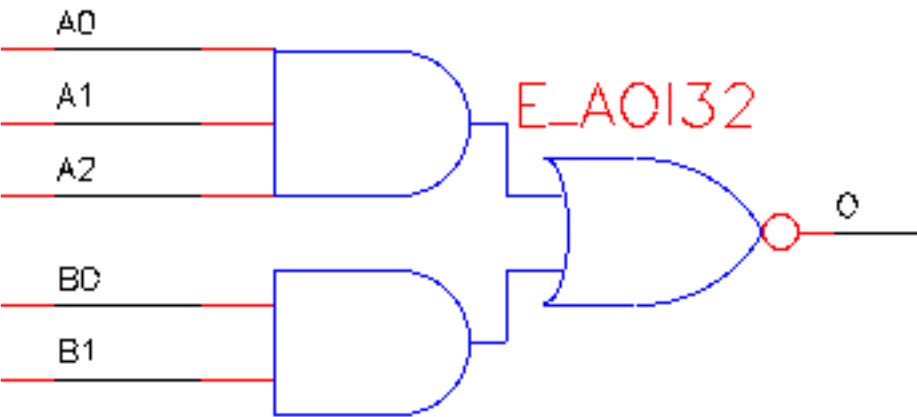
E_AOI321 : 3,2,1 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
O	OUTPUT	data

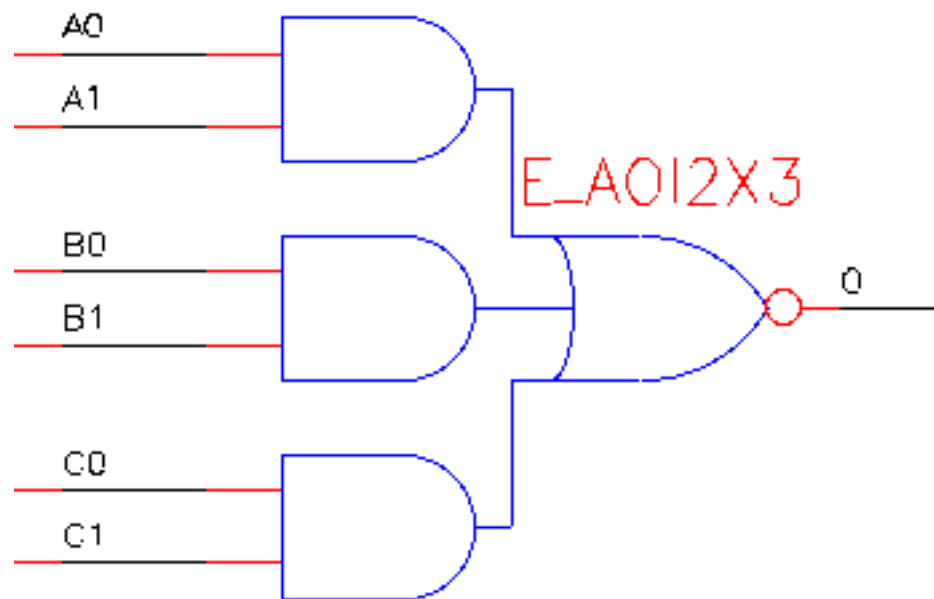
E_AOI31 : 3,1 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
O	OUTPUT	data

E_AOI32 : 3,2 AND-OR-INVERTOR

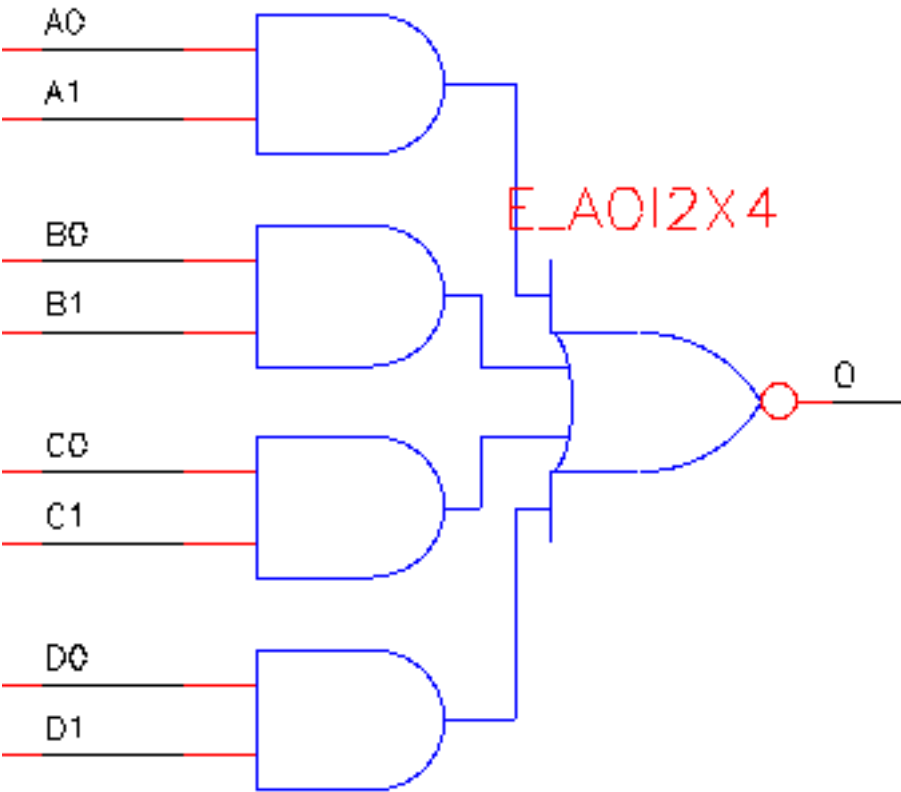


Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

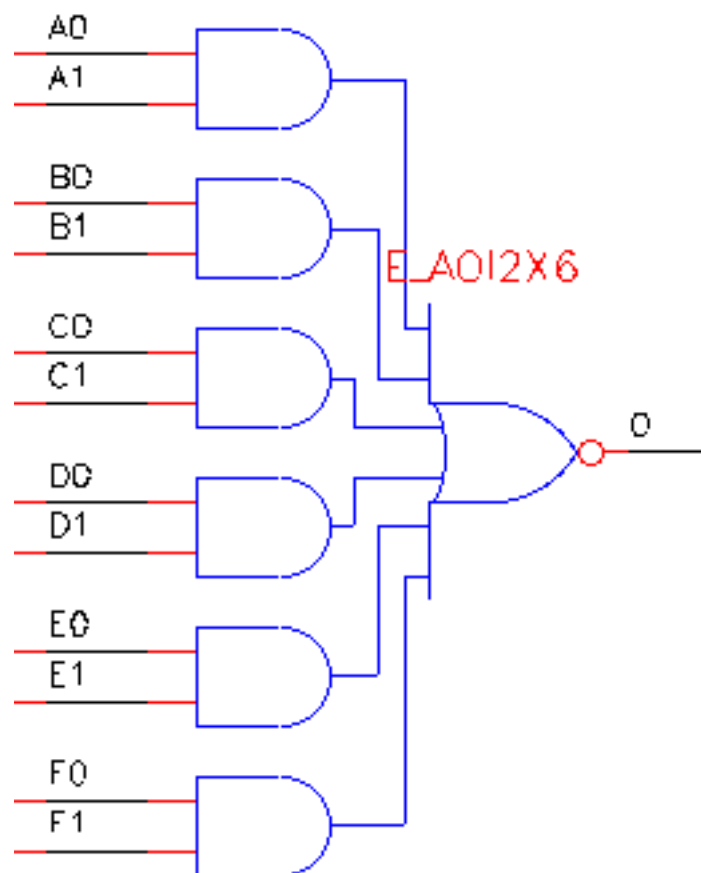
E_AOI2X3 : 2,2,2 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
O	OUTPUT	data

E_AOI2X4 : 2,2,2,2 AND-OR-INVERTOR



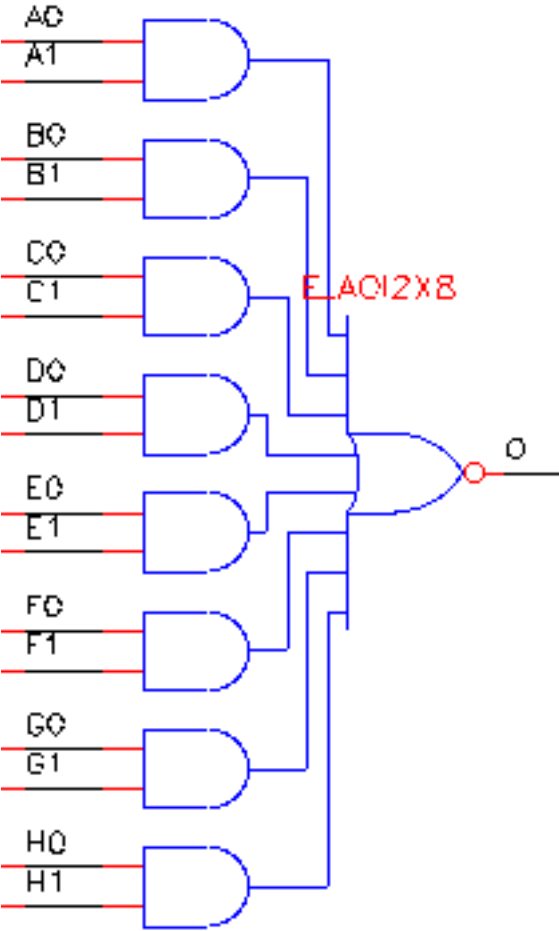
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
D0	INPUT	data
D1	INPUT	data
O	OUTPUT	data

E_AOI2X6 : 2,2,2,2,2 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
D0	INPUT	data
D1	INPUT	data
E0	INPUT	data
E1	INPUT	data

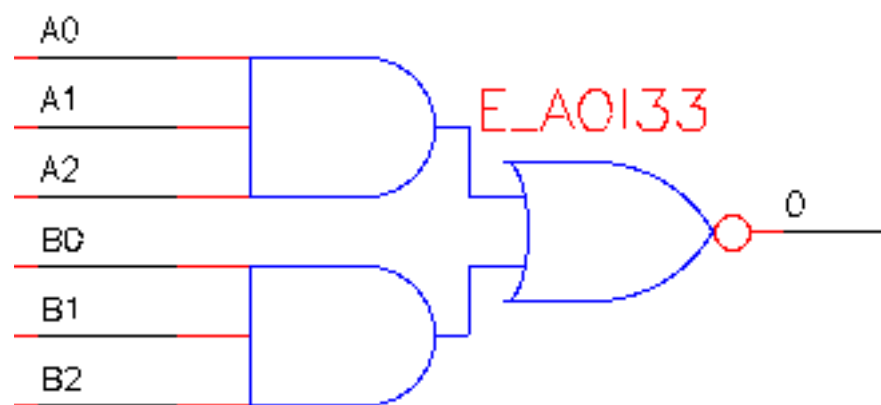
F0	INPUT	data
F1	INPUT	data
O	OUTPUT	data

E_AOI2X8 : 2,2,2,2,2,2,2,2 AND-OR-INVERTOR

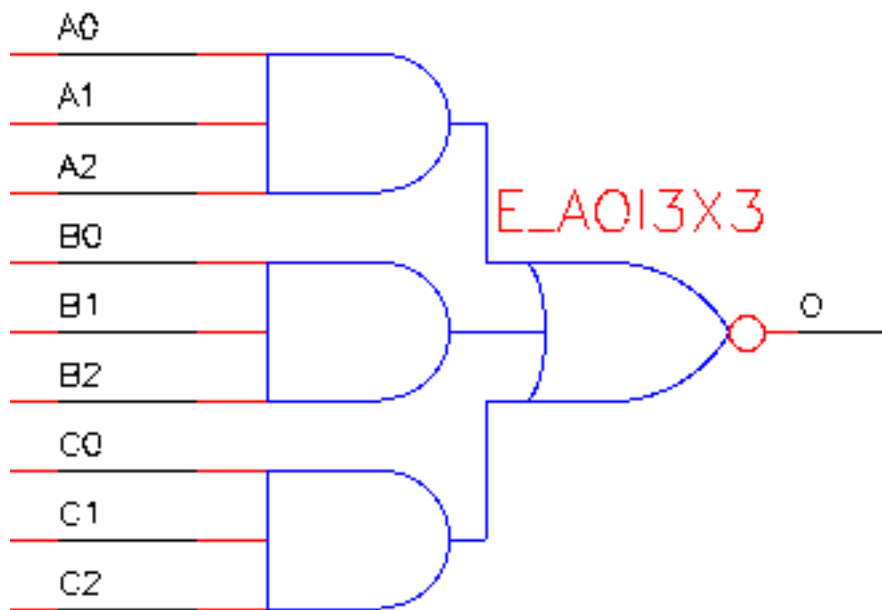


Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
D0	INPUT	data
D1	INPUT	data
E0	INPUT	data

E1	INPUT	data
F0	INPUT	data
F1	INPUT	data
G0	INPUT	data
G1	INPUT	data
H0	INPUT	data
H1	INPUT	data
O	OUTPUT	data

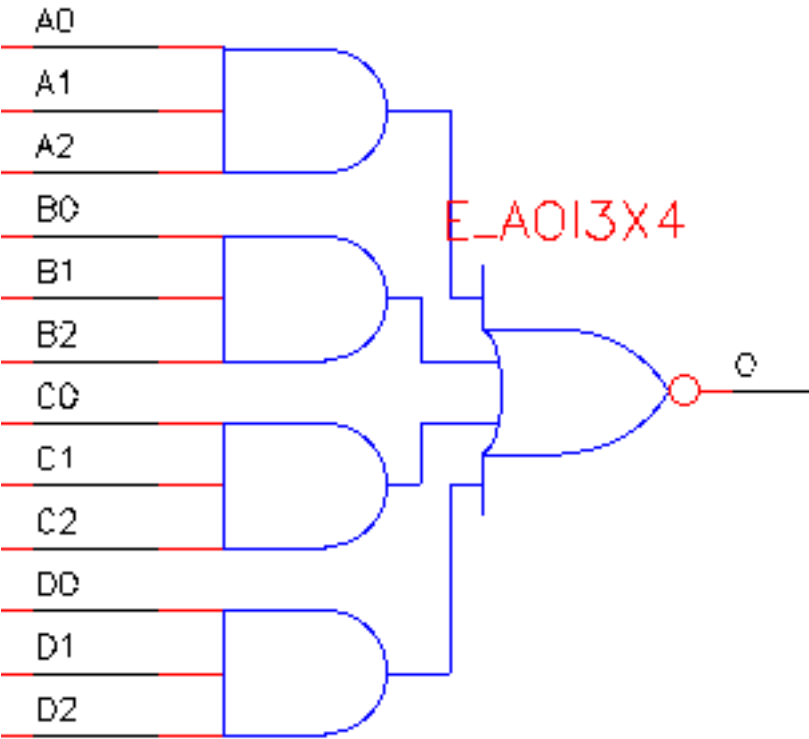
E_AOI33 : 3,3 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
O	OUTPUT	data

E_AOI3X3 : 3,3,3 AND-OR-INVERTOR

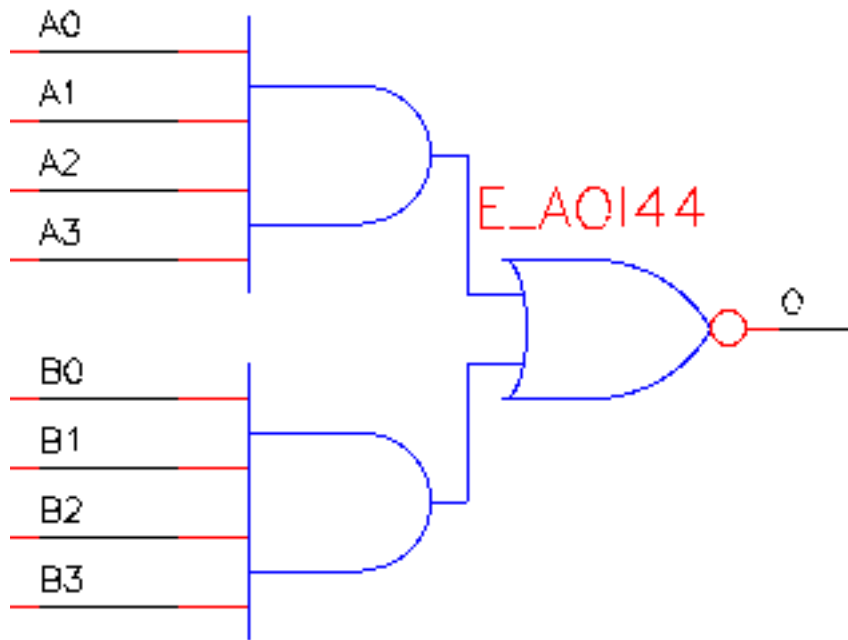
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
C0	INPUT	data
C1	INPUT	data
C2	INPUT	data
O	OUTPUT	data

E_AOI3X4 : 3,3,3,3 AND-OR-INVERTOR



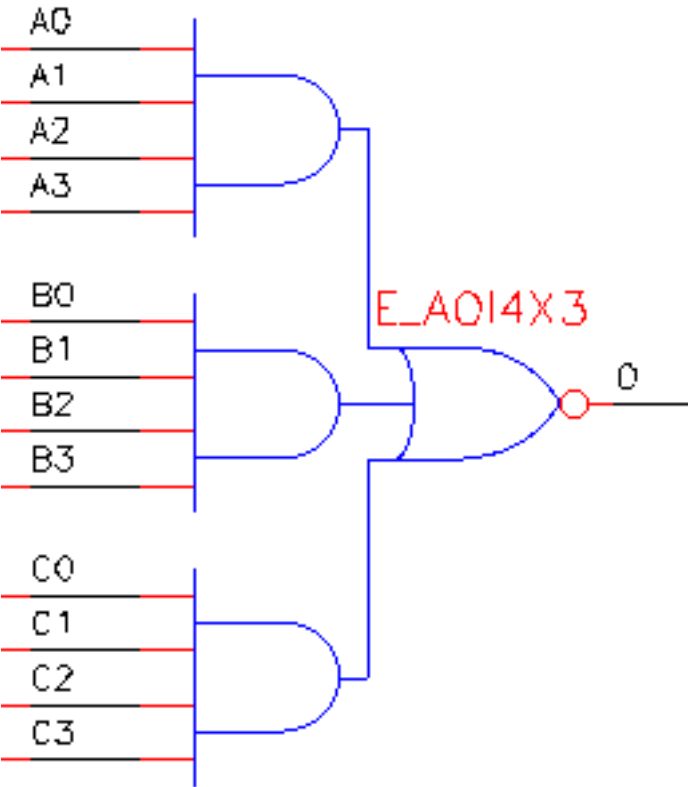
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
C0	INPUT	data
C1	INPUT	data
C2	INPUT	data
D0	INPUT	data
D1	INPUT	data

D2	INPUT	data
O	OUTPUT	data

E_AOI44 : 4,4 AND-OR-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
A3	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
B3	INPUT	data
O	OUTPUT	data

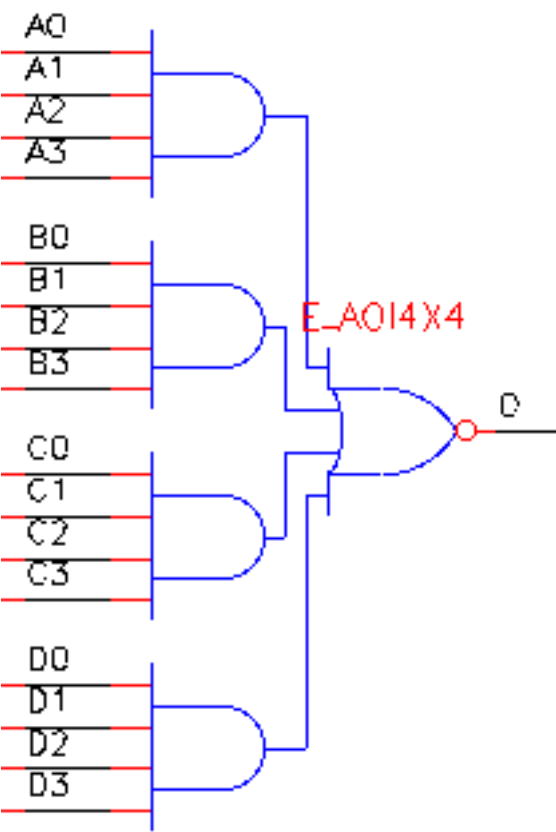
E_AOI4X3 : 4,4,4 AND-OR-INVERTOR



Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
A3	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
B3	INPUT	data
C0	INPUT	data
C1	INPUT	data
C2	INPUT	data

C3	INPUT	data
O	OUTPUT	data

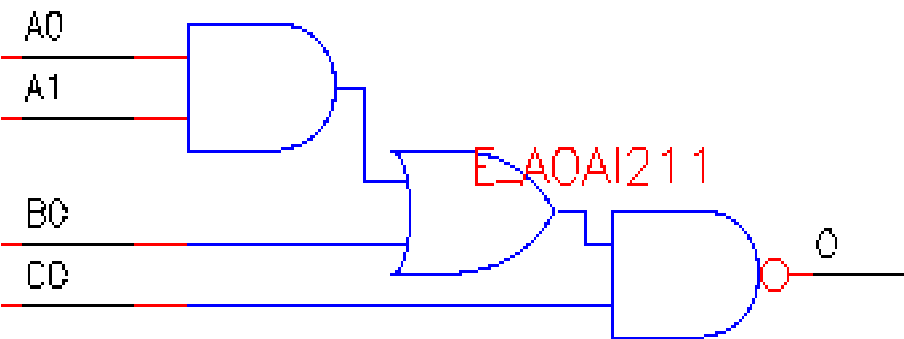
E_AOI4X4 : 4,4,4,4 AND-OR-INVERTOR



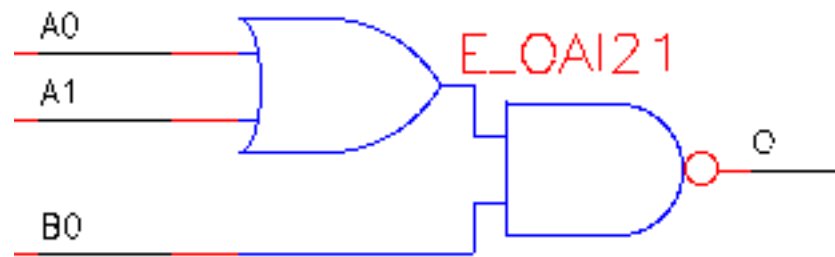
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
A3	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
B3	INPUT	data
C0	INPUT	data
C1	INPUT	data

C2	INPUT	data
C3	INPUT	data
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
O	OUTPUT	data

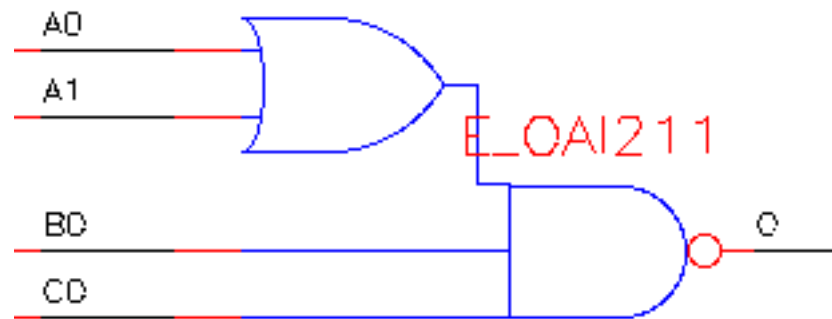
E_AOI211 : AND2 into OR2 into NAND2



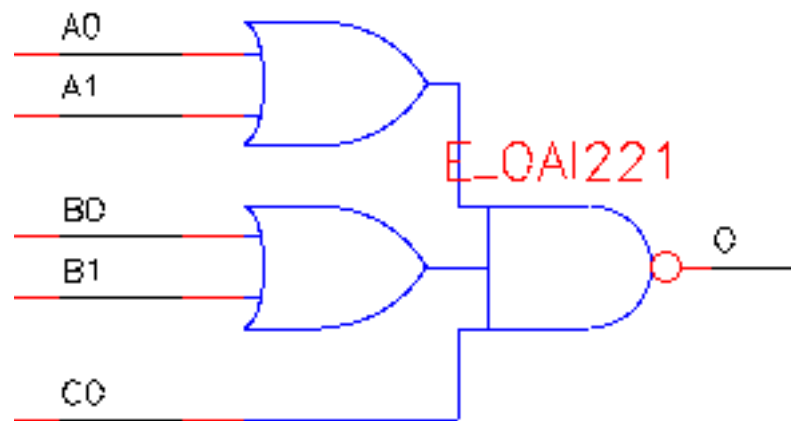
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
C0	INPUT	data
O	OUTPUT	data

E_OAI21 : 2,1 OR-AND-INVERTOR

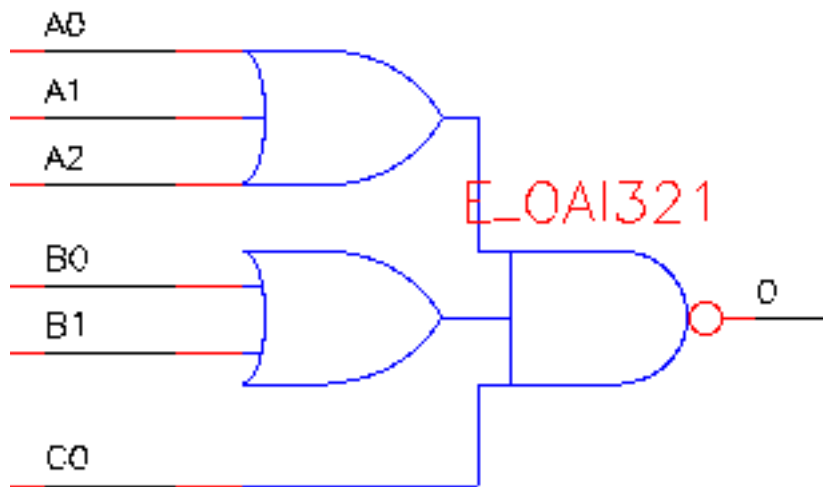
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
O	OUTPUT	data

E_OAI211 : 2,1,1 OR-AND-INVERTOR

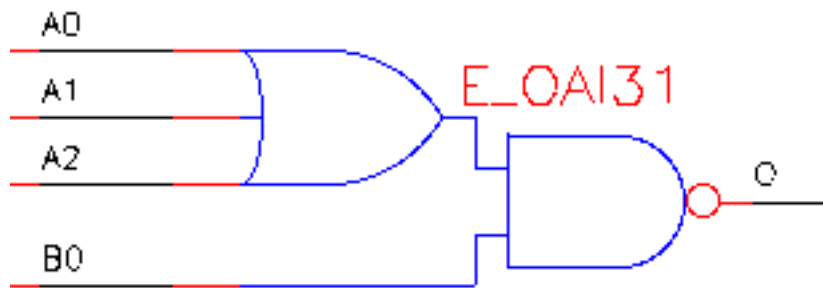
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
C0	INPUT	data
O	OUTPUT	data

E_OAI221 : 2,2,1 OR-AND-INVERTOR

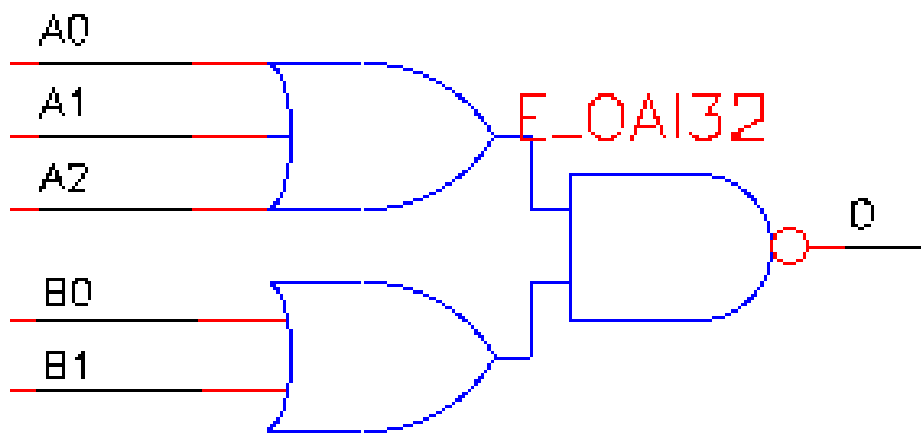
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
O	OUTPUT	data

E_OAI321 : 3,2,1 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
O	OUTPUT	data

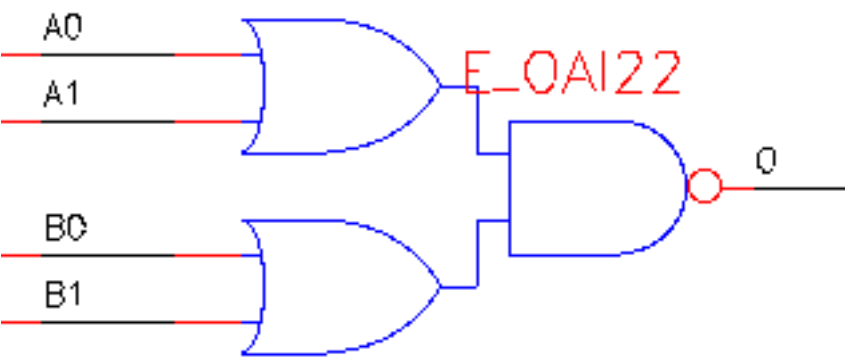
E_OAI31 : 3,1 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
O	OUTPUT	data

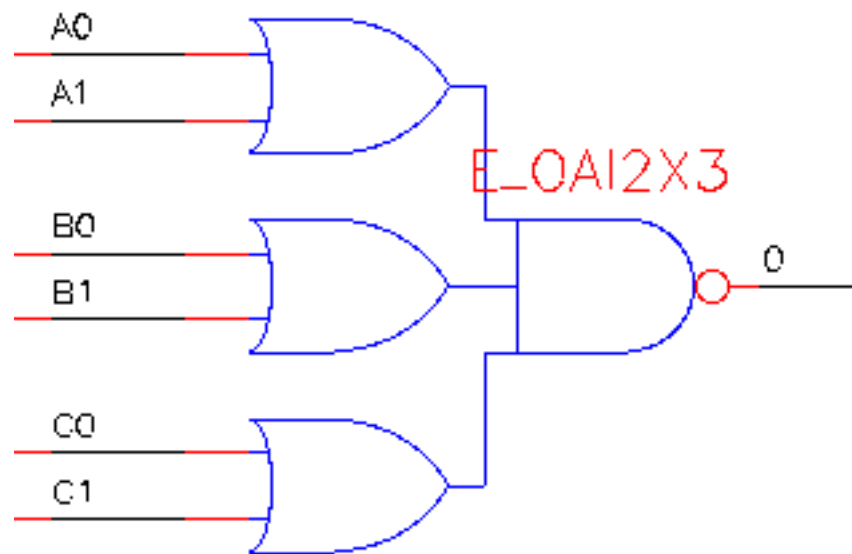
E_OAI32 : 3,2 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

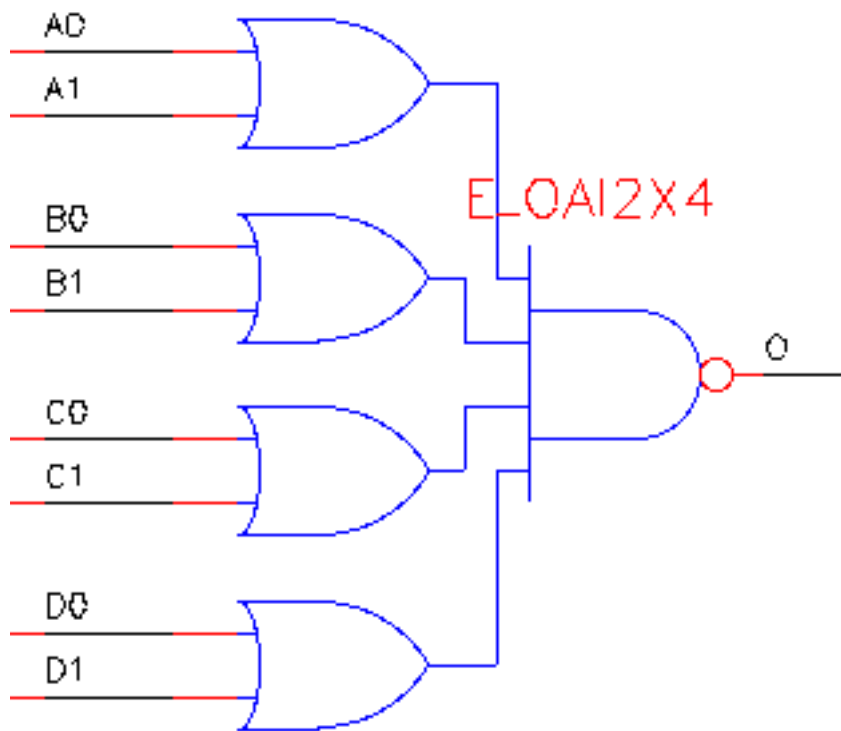
E_OAI22 : 2,2 OR-AND-IN



Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

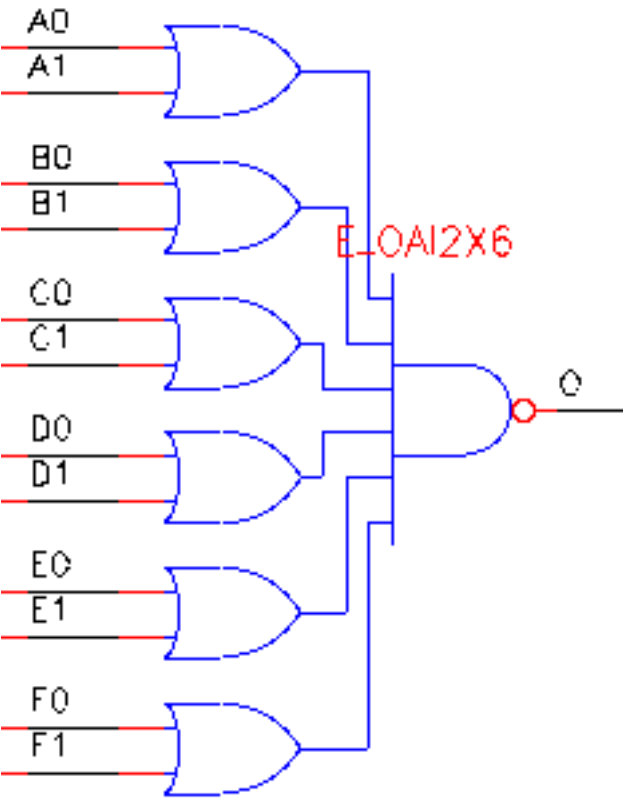
E_OAI2X3 : 2,2,2 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
O	OUTPUT	data

E_OAI2X4 : 2,2,2,2 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
D0	INPUT	data
D1	INPUT	data
O	OUTPUT	data

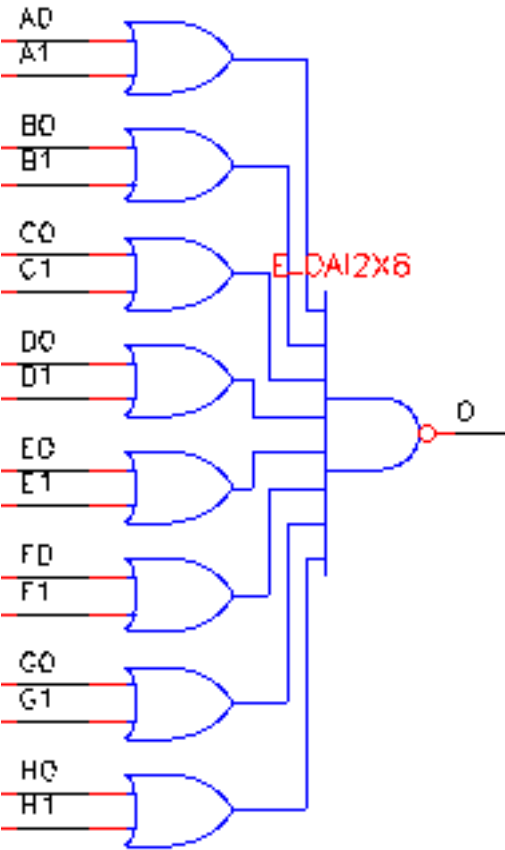
E_OAI2X6 : 2,2,2,2,2 OR-AND-INVERTOR



Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
D0	INPUT	data
D1	INPUT	data
E0	INPUT	data
E1	INPUT	data
F0	INPUT	data

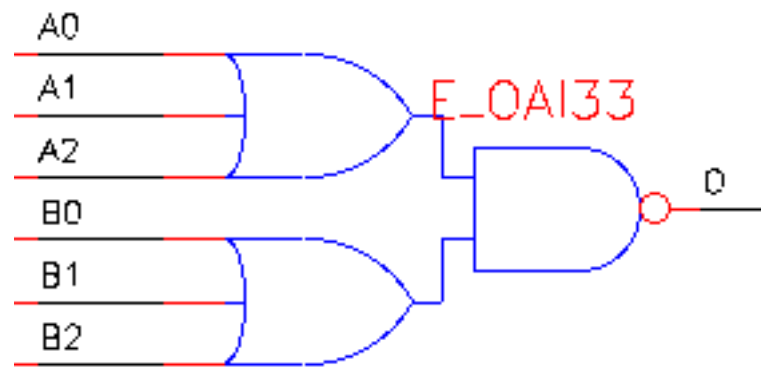
F1	INPUT	data
O	OUTPUT	data

E_OAI2X8 : 2,2,2,2,2,2,2,2 OR-AND-INVERTOR

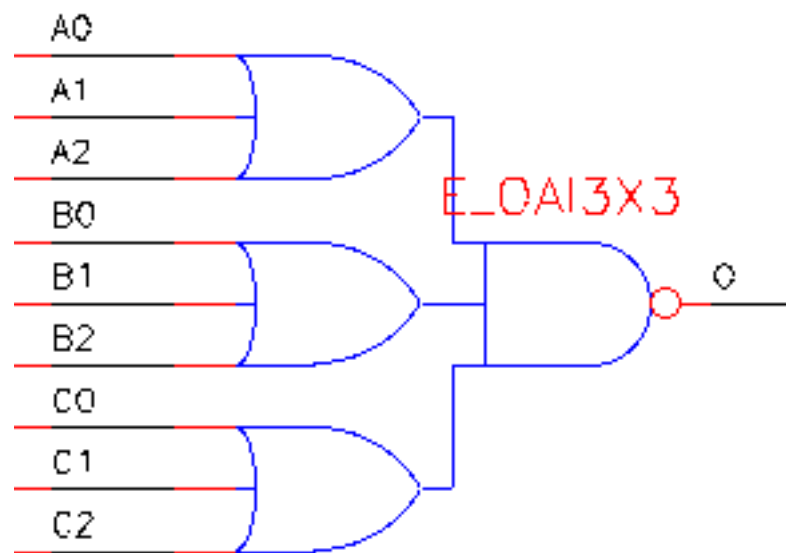


Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
C0	INPUT	data
C1	INPUT	data
D0	INPUT	data
D1	INPUT	data
E0	INPUT	data
E1	INPUT	data

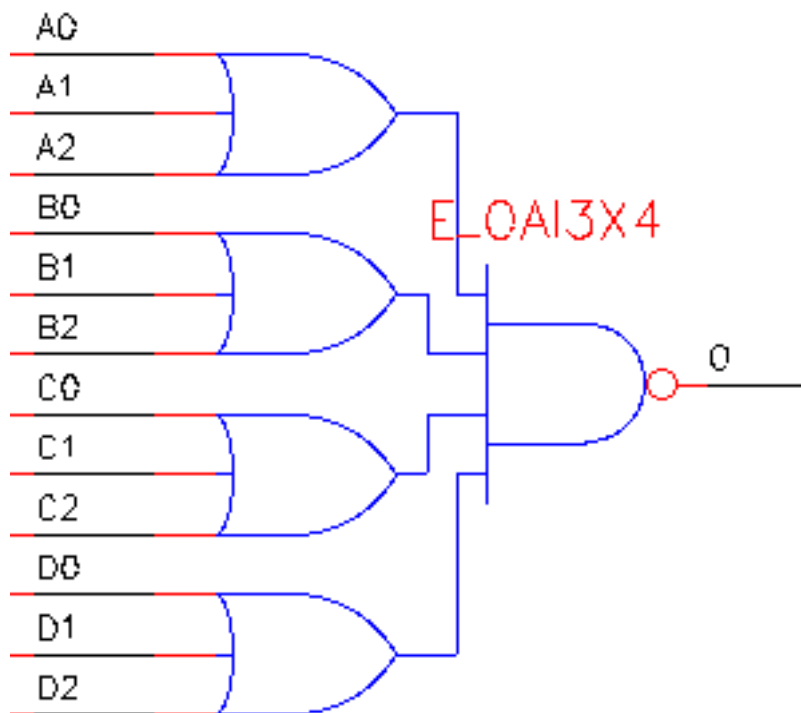
F0	INPUT	data
F1	INPUT	data
G0	INPUT	data
G1	INPUT	data
H0	INPUT	data
H1	INPUT	data
O	OUTPUT	data

E_OAI33 : 3,3 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
O	OUTPUT	data

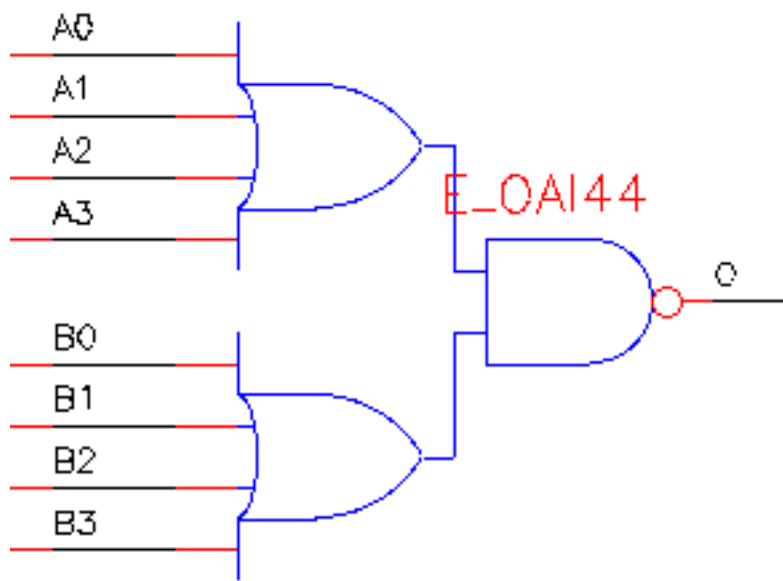
E_OAI3X3 : 3,3,3 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
C0	INPUT	data
C1	INPUT	data
C2	INPUT	data
O	OUTPUT	data

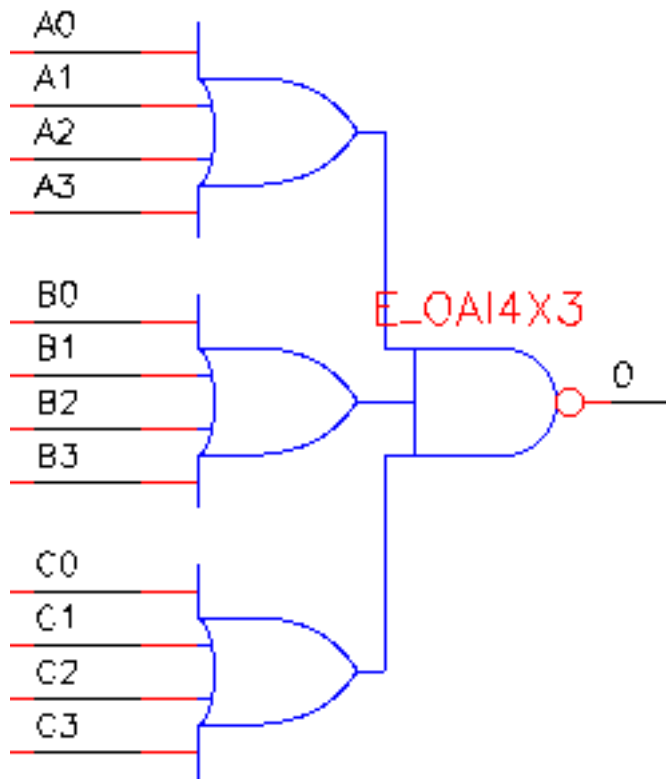
E_OAI3X4 : 3,3,3,3 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
C0	INPUT	data
C1	INPUT	data
C2	INPUT	data
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data

O	OUTPUT	data
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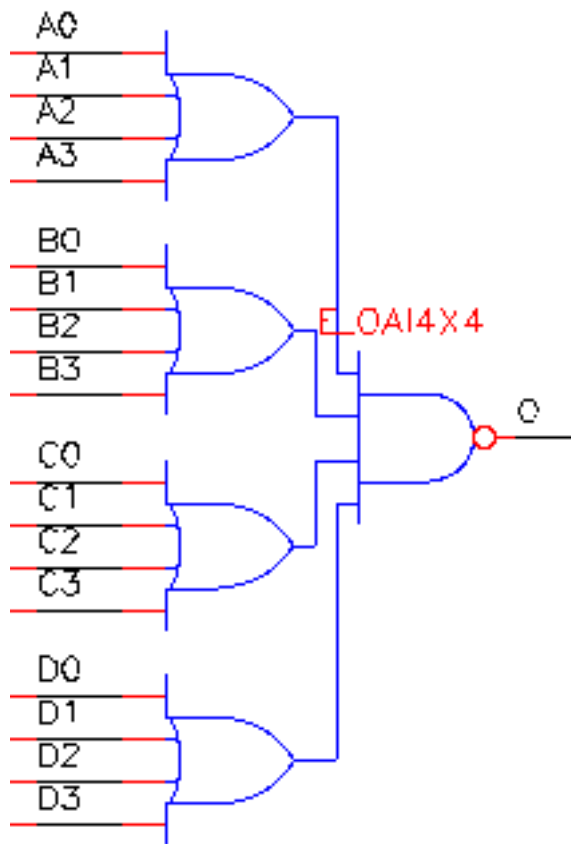
E_OAI44 : 4,4 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
A3	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
B3	INPUT	data
O	OUTPUT	data

E_OAI4X3 : 4,4,4 OR-AND-INVERTOR

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
A3	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
B3	INPUT	data
C0	INPUT	data
C1	INPUT	data
C2	INPUT	data

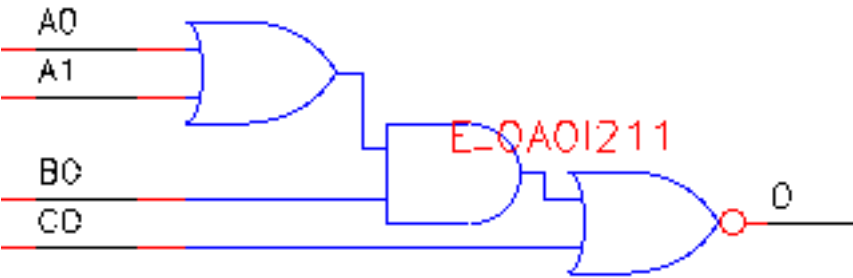
C3	INPUT	data
O	OUTPUT	data

E_OAI4X4 : 4,4,4,4 OR-AND-INVERTOR

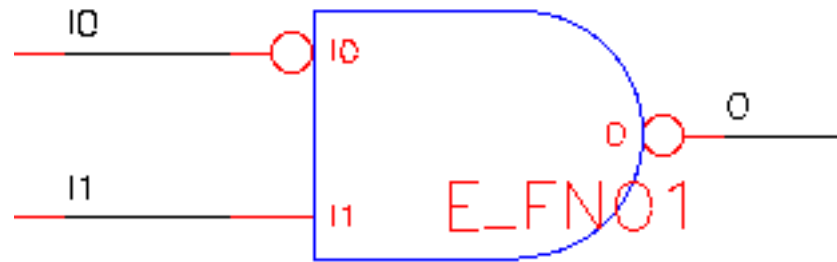
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
A3	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
B3	INPUT	data
C0	INPUT	data
C1	INPUT	data

C2	INPUT	data
C3	INPUT	data
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
O	OUTPUT	data

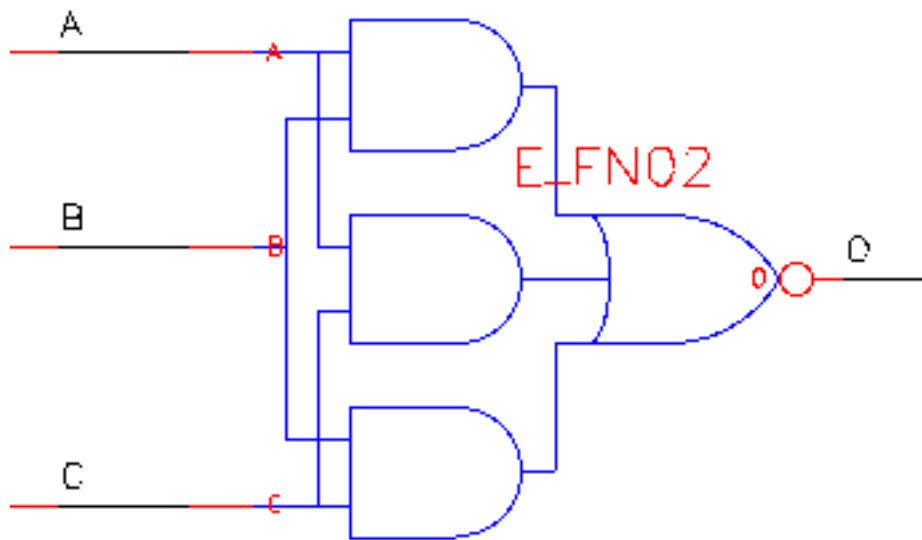
E_OAOI211 : OR2 into AND2 into NOR2



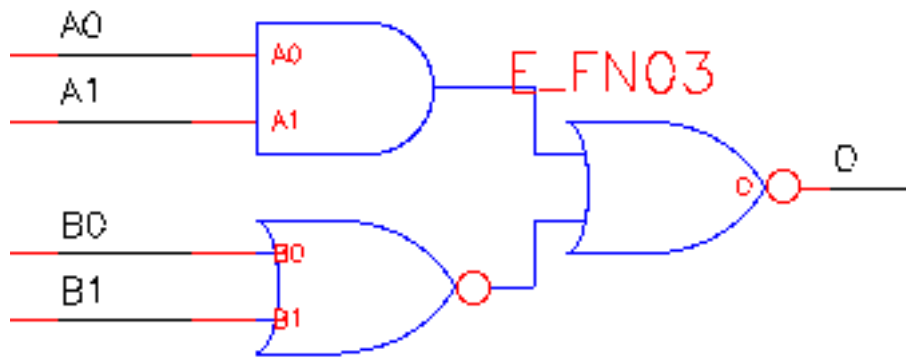
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
C0	INPUT	data
O	OUTPUT	data

E_FN01 : NAND with inverted input

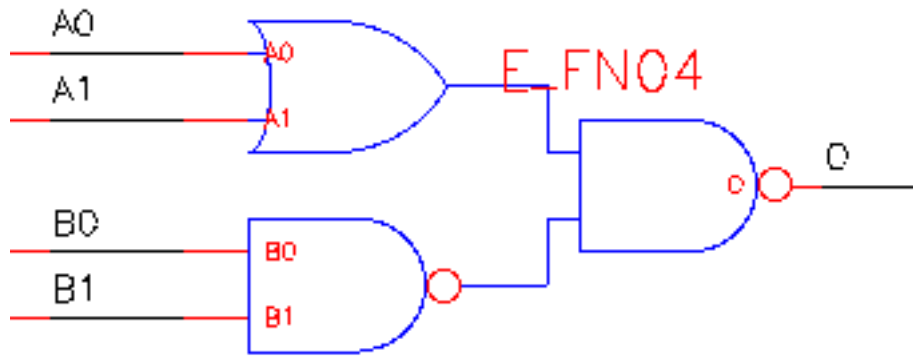
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_FN02 : Inverting 2 of 3 Majority

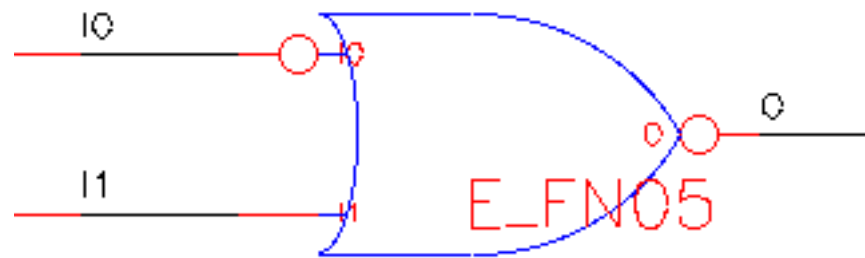
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
O	OUTPUT	data

E_FN03 : AND2, NOR2 into NOR2

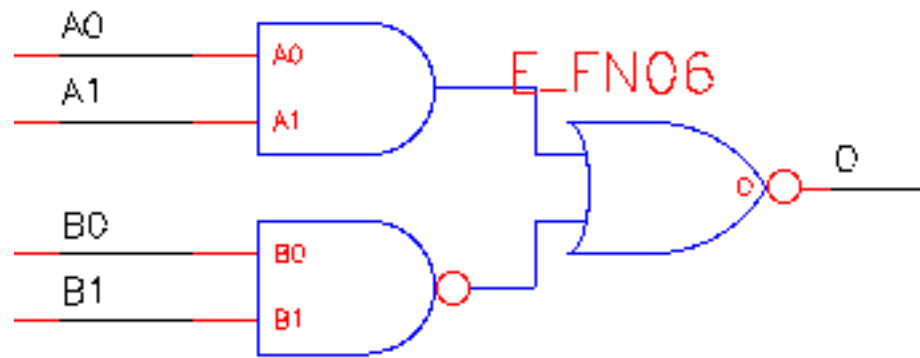
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

E_FN04 : NAND2, OR2 into NAND2

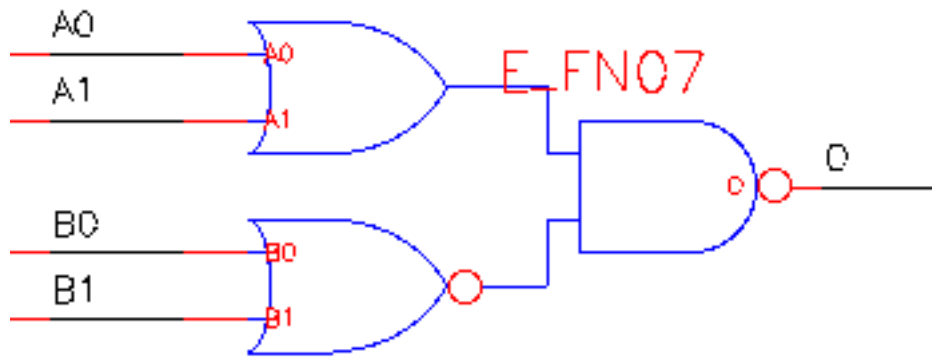
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

E_FN05 : NOR with inverted input

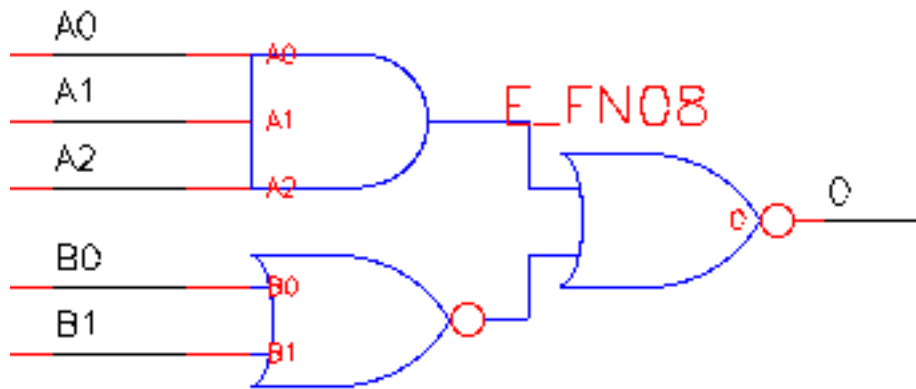
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O	OUTPUT	data

E_FN06 : OR2, NOR2 into NAND2

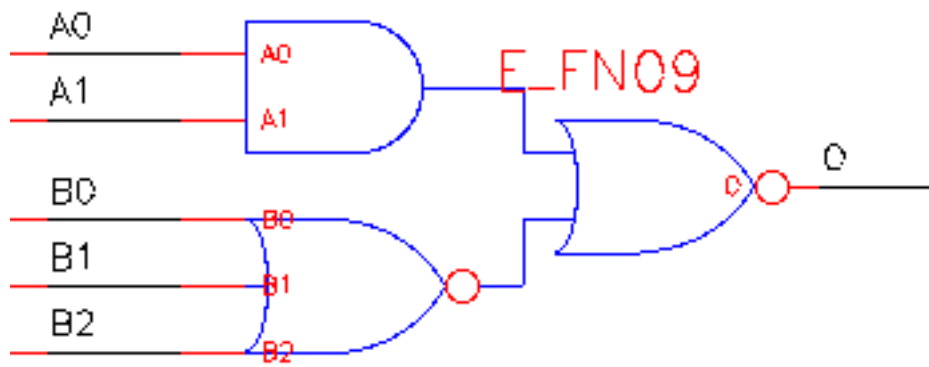
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

E_FN07 : AND3, NOR2 into NOR2

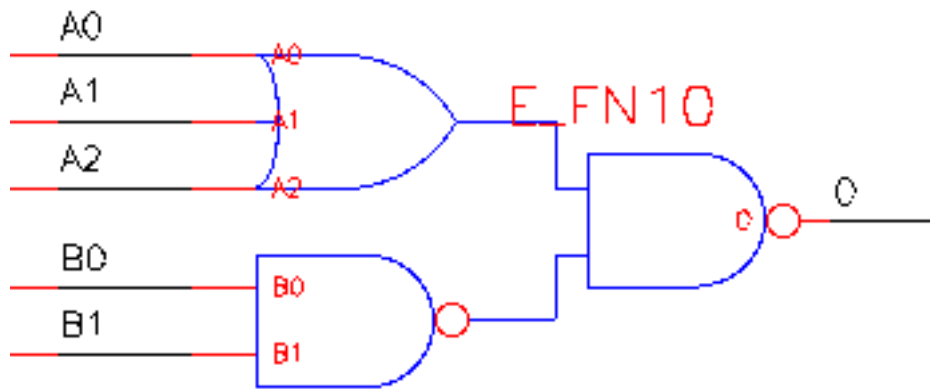
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

E_FN08 : AND3, NOR2 into NOR2

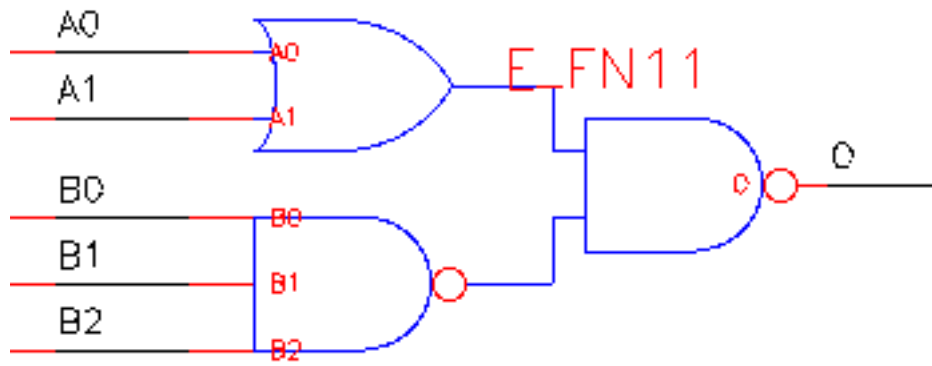
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

E_FN09 : AND2, NOR3 into NOR2

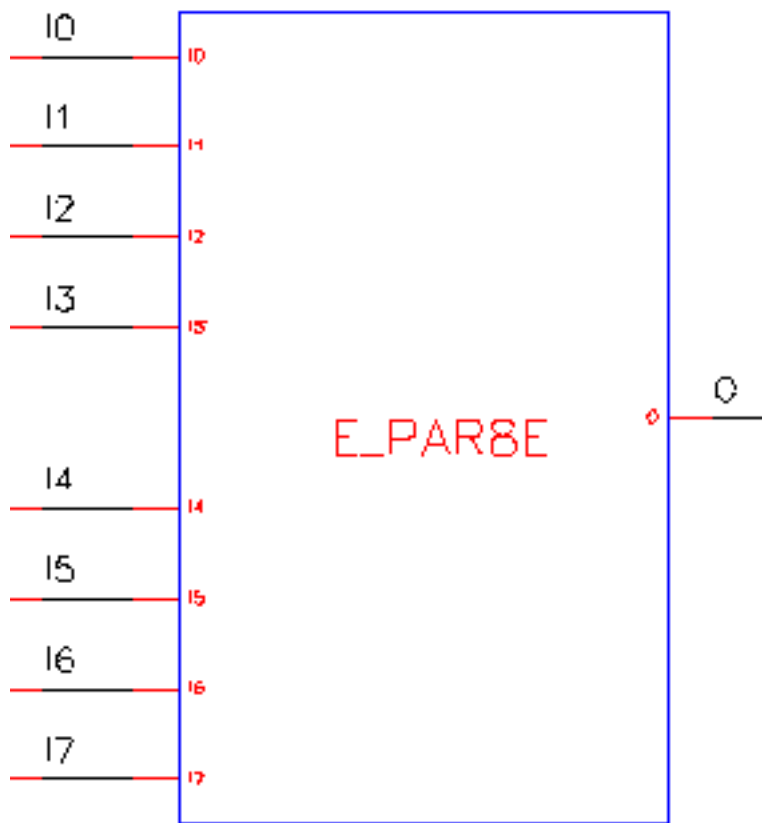
Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
O	OUTPUT	data

E_FN10 : OR3, NAND2 into NAND2

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
A2	INPUT	data
B0	INPUT	data
B1	INPUT	data
O	OUTPUT	data

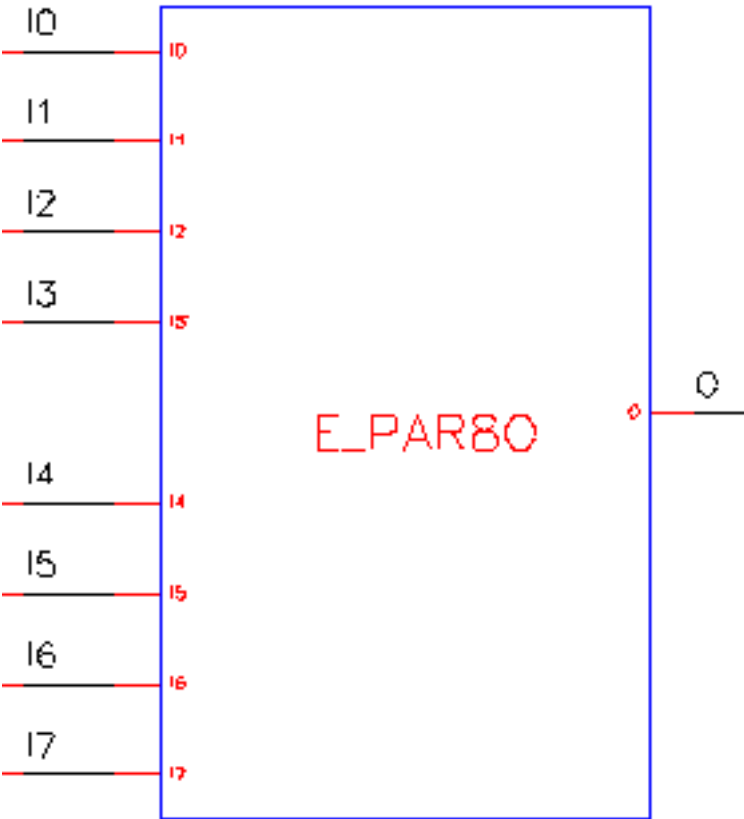
E_FN11 : OR2, NAND3 into NAND2

Pin	Direction	Description
A0	INPUT	data
A1	INPUT	data
B0	INPUT	data
B1	INPUT	data
B2	INPUT	data
O	OUTPUT	data

E_PAR8E : 8-bit even parity generator

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O	OUTPUT	data

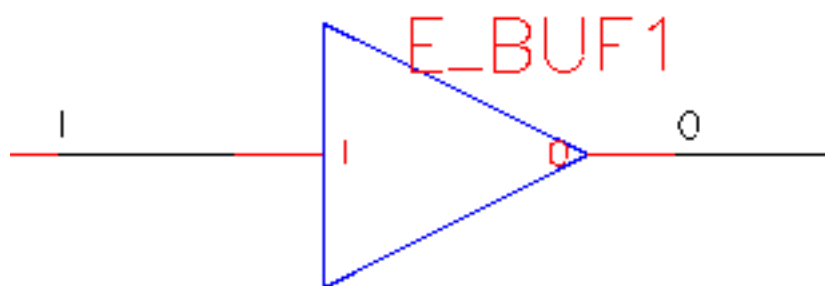
E_PAR80 : 8-bit odd parity generator



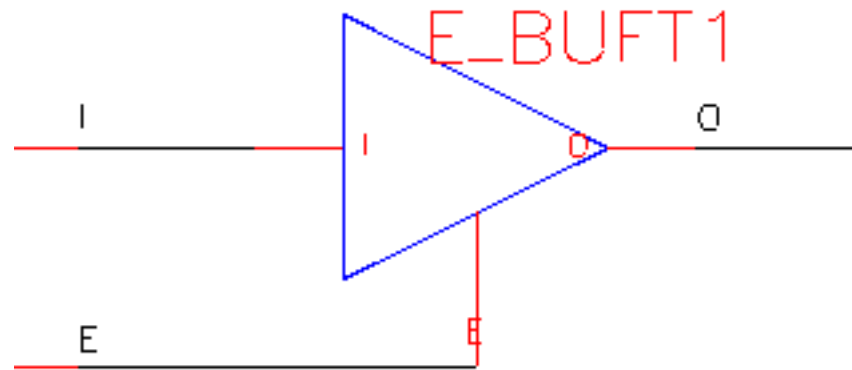
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O	OUTPUT	data

Buffers

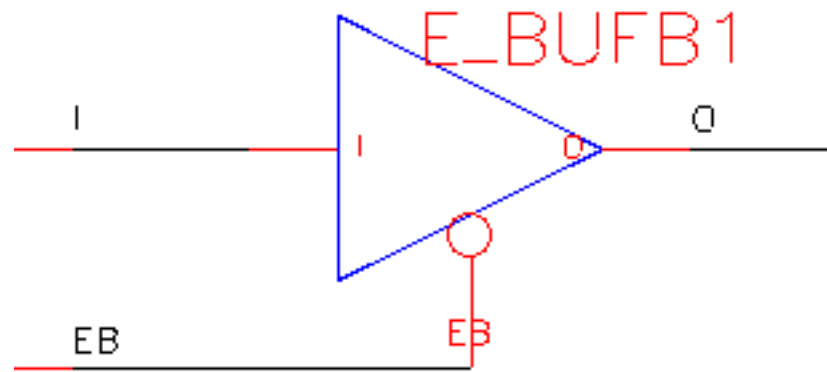
[E_BUF1](#)
[E_BUFT1](#)
[E_BUFBI](#)
[E_TRD2C](#)
[E_BUFD1](#)
[E_BUFD2](#)
[E_BUFD4](#)
[E_BUFD8](#)
[E_BUFD16](#)
[E_BUFX2](#)
[E_BUFX4](#)
[E_BUFX8](#)
[E_BUFX16](#)
[E_CKBUF](#)

E_BUF1 : Buffer

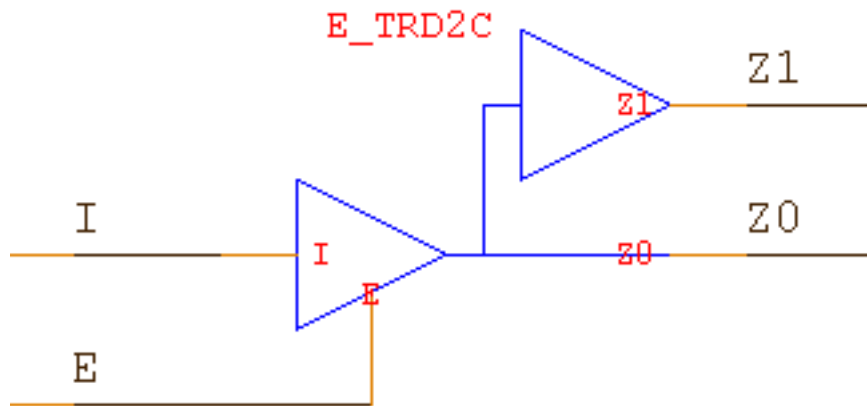
Pin	Direction	Description
I	INPUT	data
O	OUTPUT	data

E_BUFT1 : Tri-state buffer

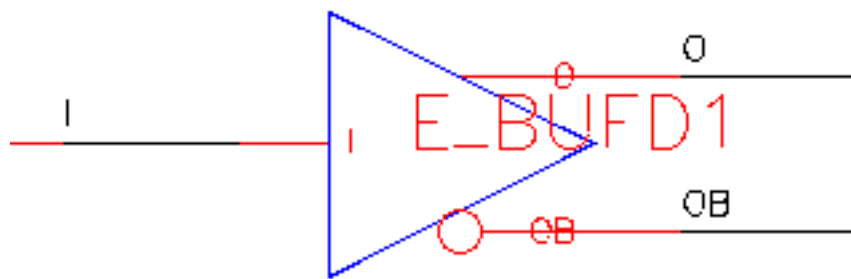
Pin	Direction	Description
I	INPUT	data
E	INPUT	output enable, active-HIGH
O	OUTPUT	data

E_BUF1 : Tri-state buffer

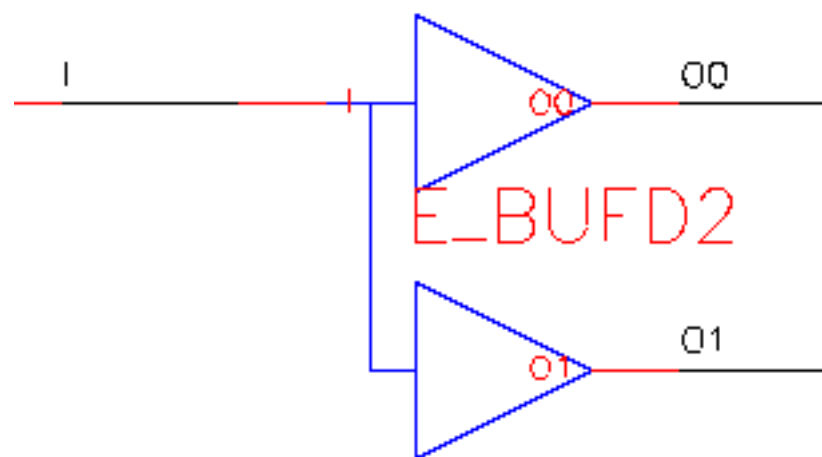
Pin	Direction	Description
I	INPUT	data
EB	INPUT	output enable, active-LOW
O	OUTPUT	data

E_TRD2C : Tri-State internal drive

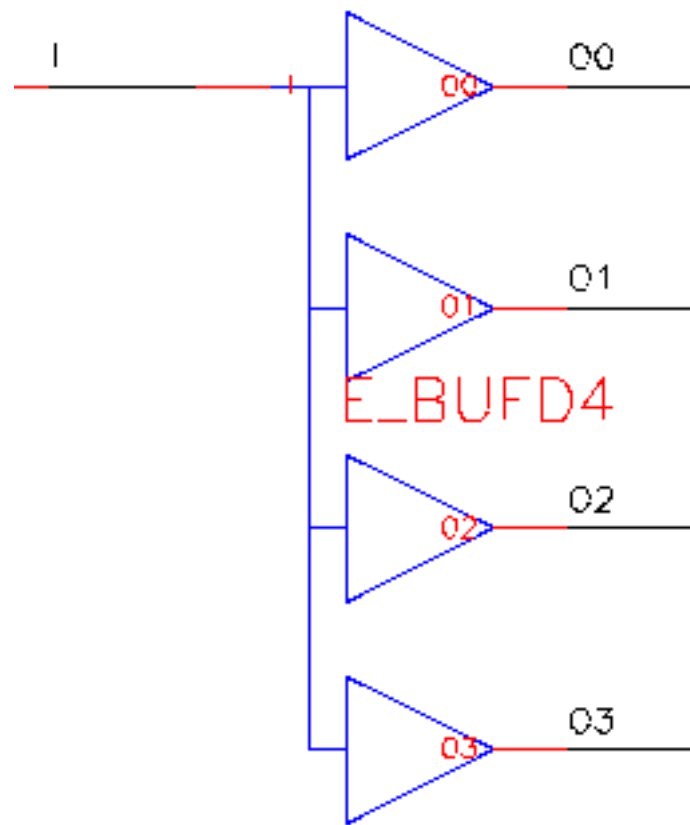
Pin	Direction	Description
I	INPUT	data
E	INPUT	output enable, active-HIGH
Z0	OUTPUT	data
Z1	OUTPUT	data

E_BUFD1 : Buffer with Complementary Outputs

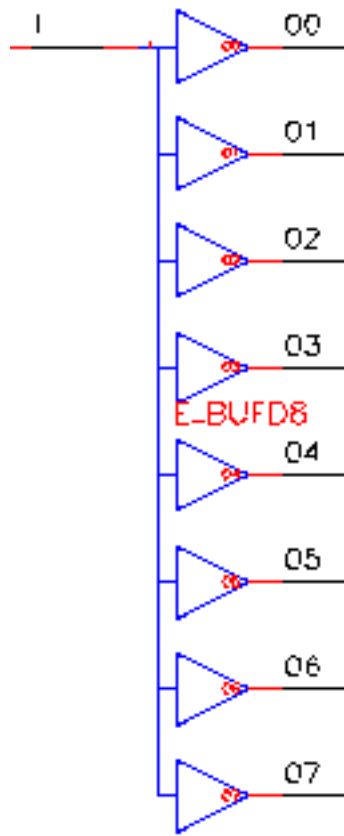
Pin	Direction	Description
I	INPUT	data
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_BUFD2 : Buffer Dual Outputs

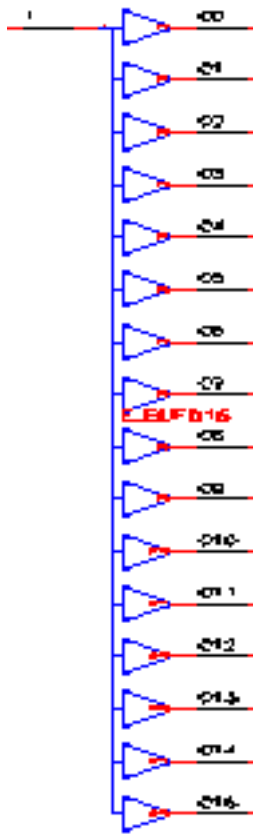
Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data

E_BUFD4 : Buffer 4 Outputs

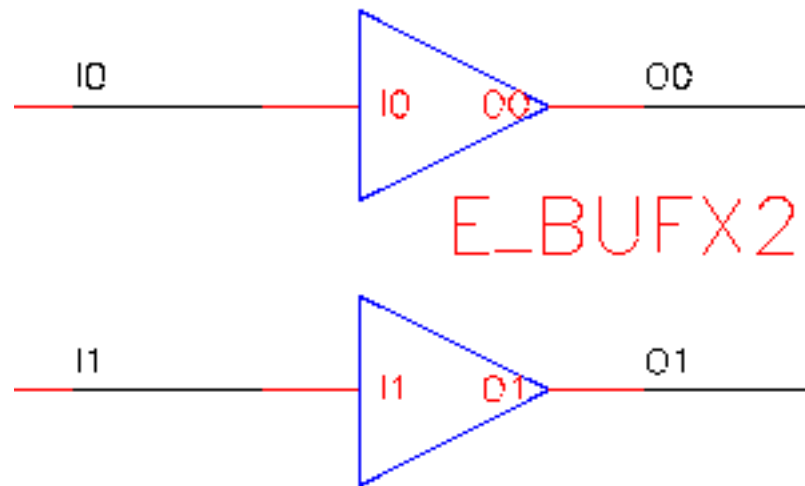
Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

E_BUFD8 : Buffer 8 Outputs

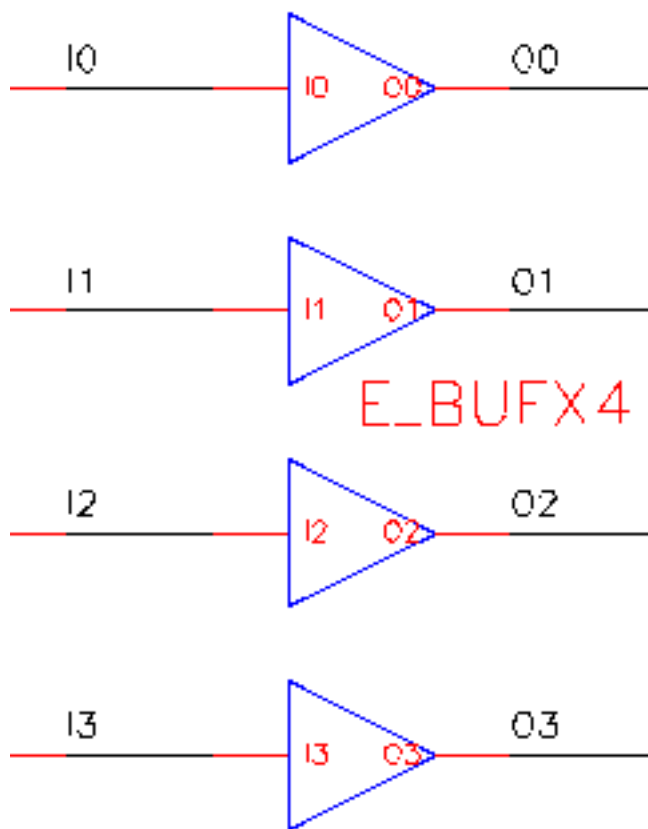
Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

E_BUF16 : Buffer 16 Outputs

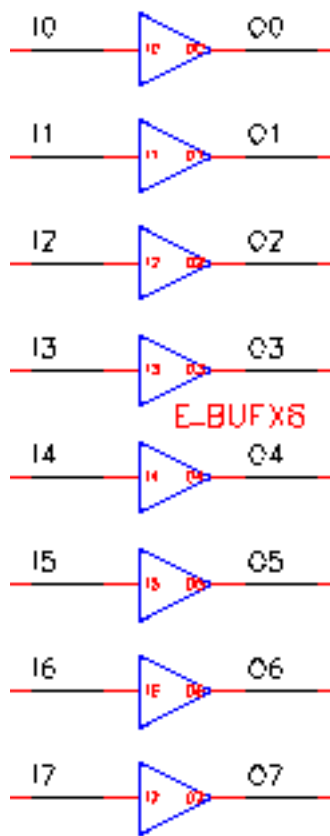
Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

E_BUF2 : 2 individual buffers

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data

E_BUF4 : 4 individual buffers

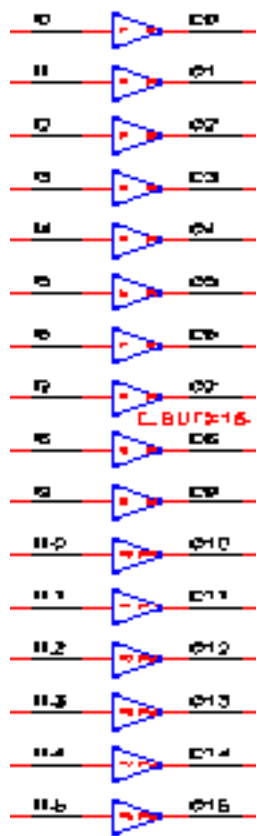
Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

E_BUFx8 : 8 individual buffers

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data

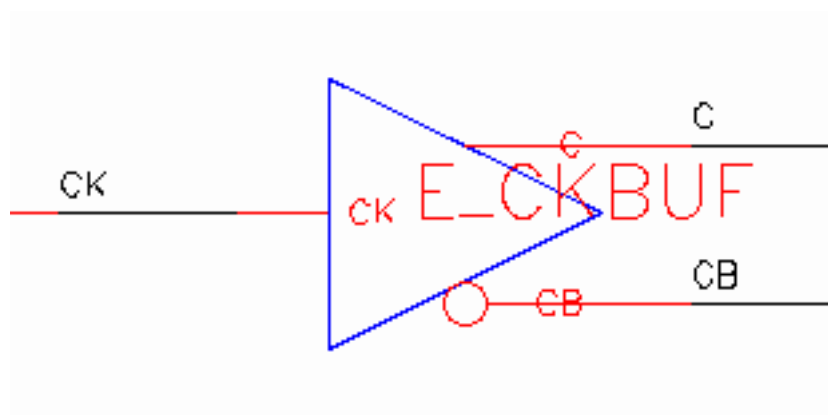
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

E_BUFX16 : 16 individual buffers



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data

I10	INPUT	data
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
I15	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data
O8	OUTPUT	data
O9	OUTPUT	data
O10	OUTPUT	data
O11	OUTPUT	data
O12	OUTPUT	data
O13	OUTPUT	data
O14	OUTPUT	data
O15	OUTPUT	data

E_CKBUF : CLOCK BUFFER

Pin	Direction	Description
CK	INPUT	data
C	OUTPUT	data
CB	OUTPUT	data, inverted

Inverters

[E_INV](#)

[E_INVT](#)

[E_INVB](#)

[E_INVD2](#)

[E_INVD4](#)

[E_INVD6](#)

[E_INVD8](#)

[E_INVD16](#)

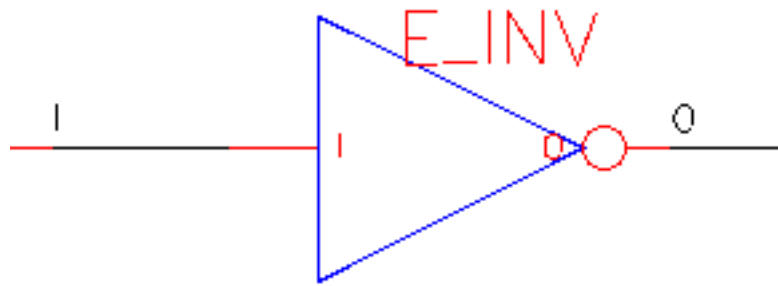
[E_INVX2](#)

[E_INVX4](#)

[E_INVX6](#)

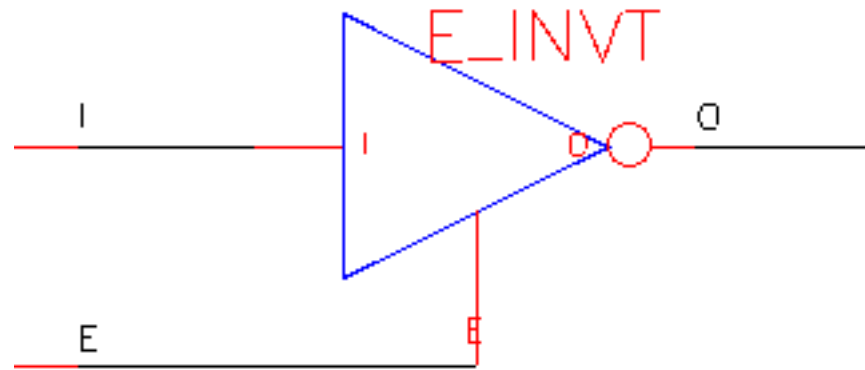
[E_INVX8](#)

[E_INVX16](#)

E_INV : Inverter

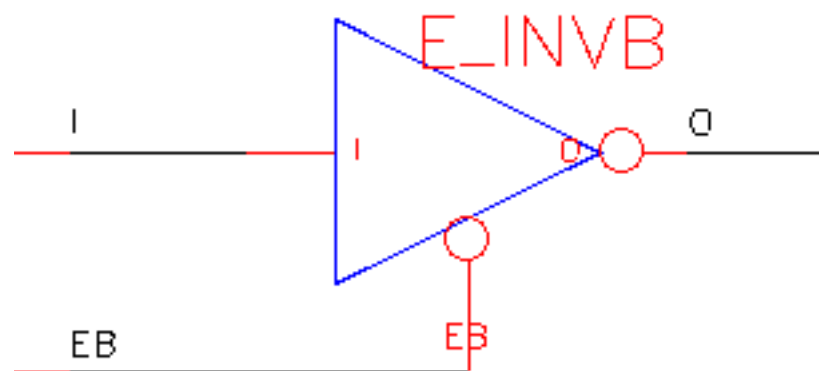
Pin	Direction	Description
I	INPUT	data
O	OUTPUT	data

E_INVT : Tri-state inverter

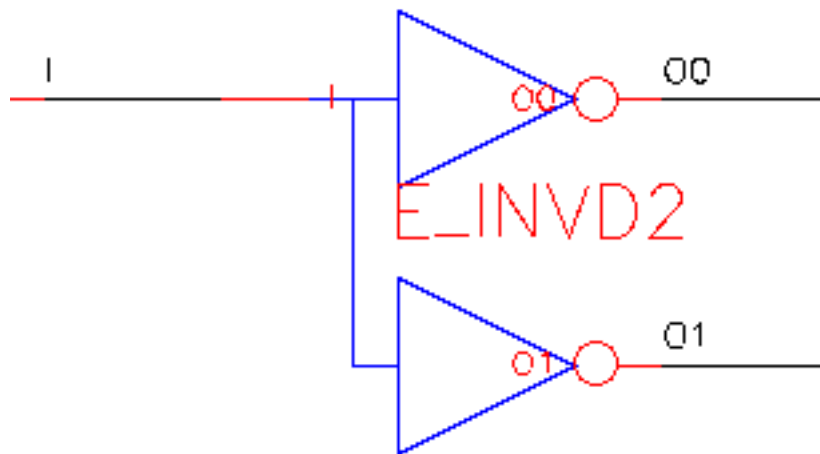


Pin	Direction	Description
I	INPUT	data
E	INPUT	output enable, active-HIGH
O	OUTPUT	data

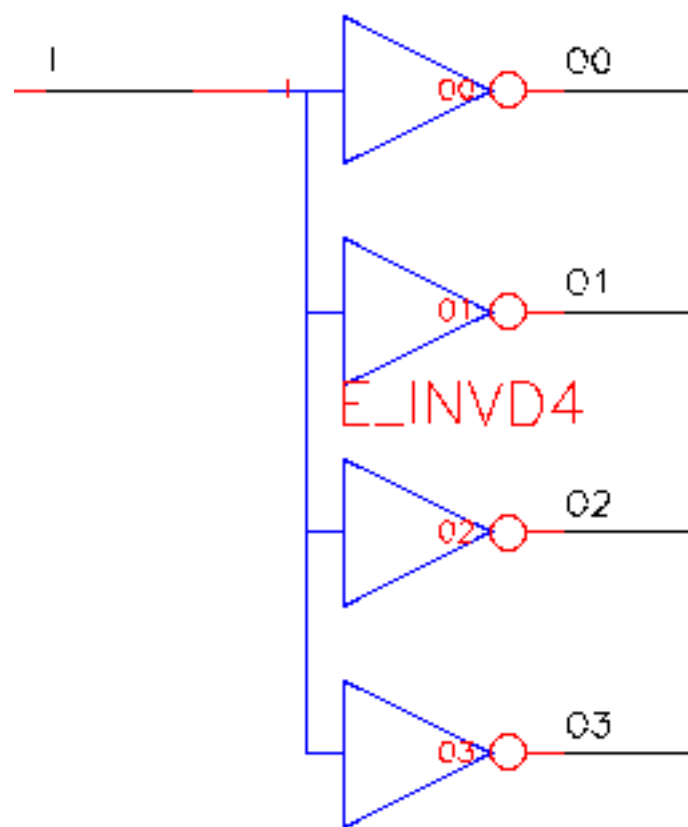
E_INVB : Tri-state inverter



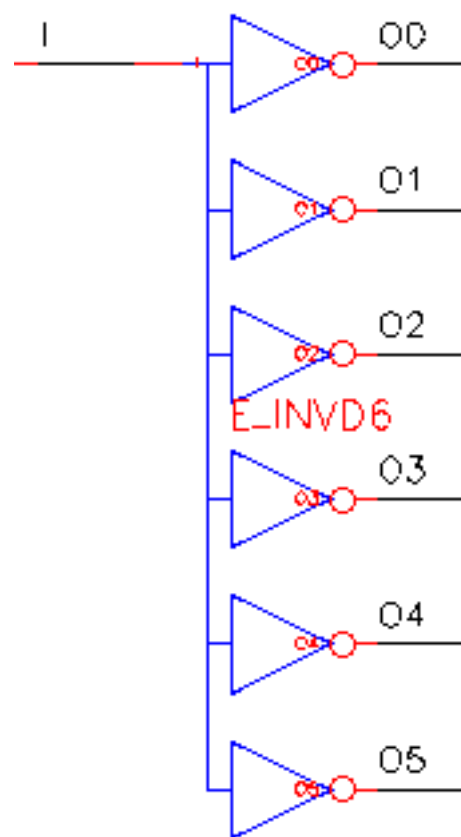
Pin	Direction	Description
I	INPUT	data
EB	INPUT	output enable, active-LOW
O	OUTPUT	data

E_INV2 : 2-output inverter

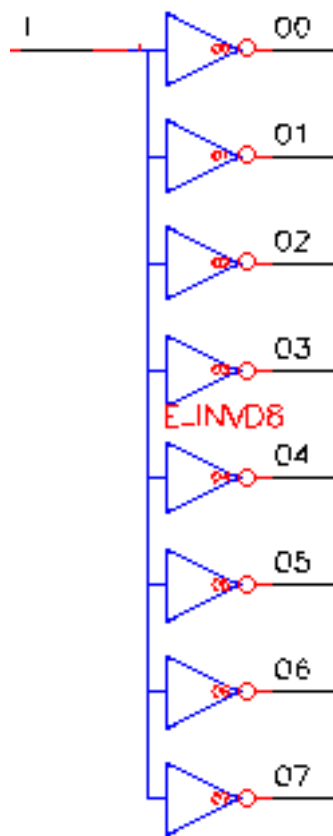
Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data

E_INV4 : 4-output inverter

Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

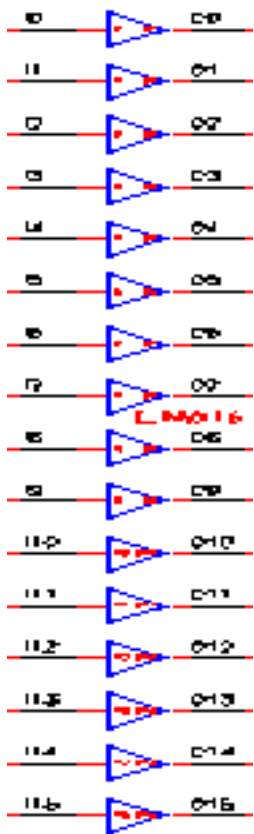
E_INV6 : 6-output inverter

Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data

E_INV8 : 8-output inverter

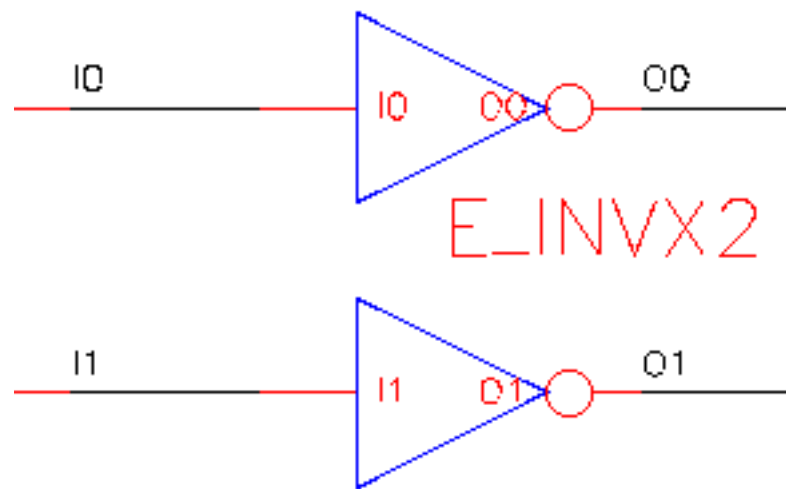
Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

E_INV16 : 16-output invertor

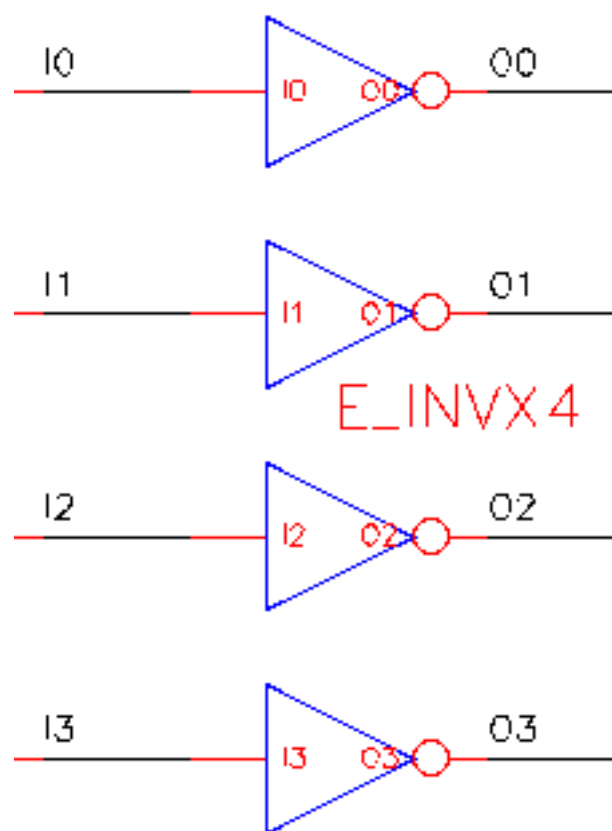


Pin	Direction	Description
I	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

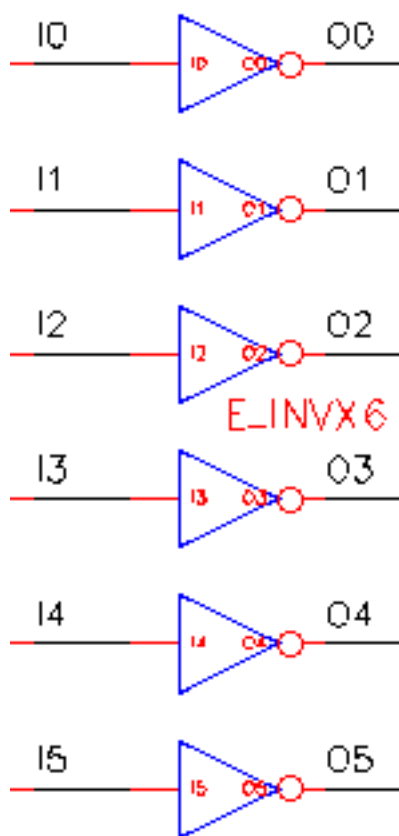
E_INVX2 : 2 individual invertors



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
O0	OUTPUT	data
Q1	OUTPUT	data

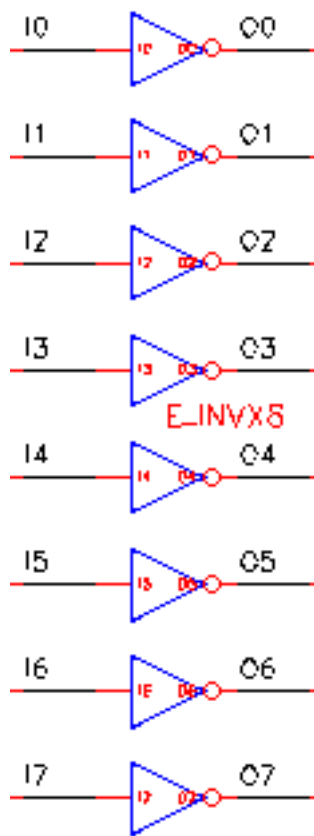
E_INVX4 : 4 individual invertors

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

E_INVX6 : 6 individual invertors

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

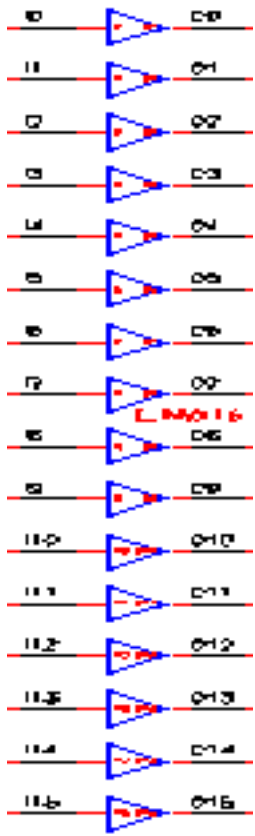
O4	OUTPUT	data
O5	OUTPUT	data

E_INVX8 : 8 individual invertors

Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data

O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

E_INVX16 : 16 individual invertors



Pin	Direction	Description
I0	INPUT	data
I1	INPUT	data
I2	INPUT	data
I3	INPUT	data
I4	INPUT	data
I5	INPUT	data
I6	INPUT	data
I7	INPUT	data
I8	INPUT	data
I9	INPUT	data

I10	INPUT	data
I11	INPUT	data
I12	INPUT	data
I13	INPUT	data
I14	INPUT	data
I15	INPUT	data
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data
O8	OUTPUT	data
O9	OUTPUT	data
O10	OUTPUT	data
O11	OUTPUT	data
O12	OUTPUT	data
O13	OUTPUT	data
O14	OUTPUT	data
O15	OUTPUT	data

Decoders

[E_DEC2](#)

[E_DEC2I](#)

[E_DEC2EI](#)

[E_DEC2ED](#)

[E_DEC2EB](#)

[E_DEC2EBI](#)

[E_DEC2EBD](#)

[E_DEC3](#)

[E_DEC3I](#)

[E_DEC3E](#)

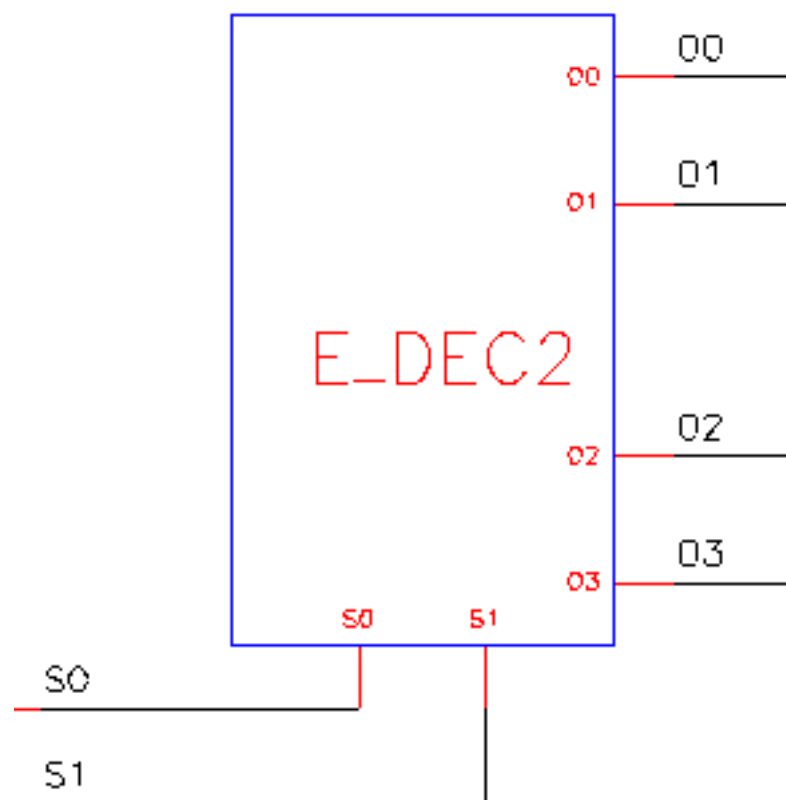
[E_DEC3EI](#)

[E_DEC3ED](#)

[E_DEC3EB](#)

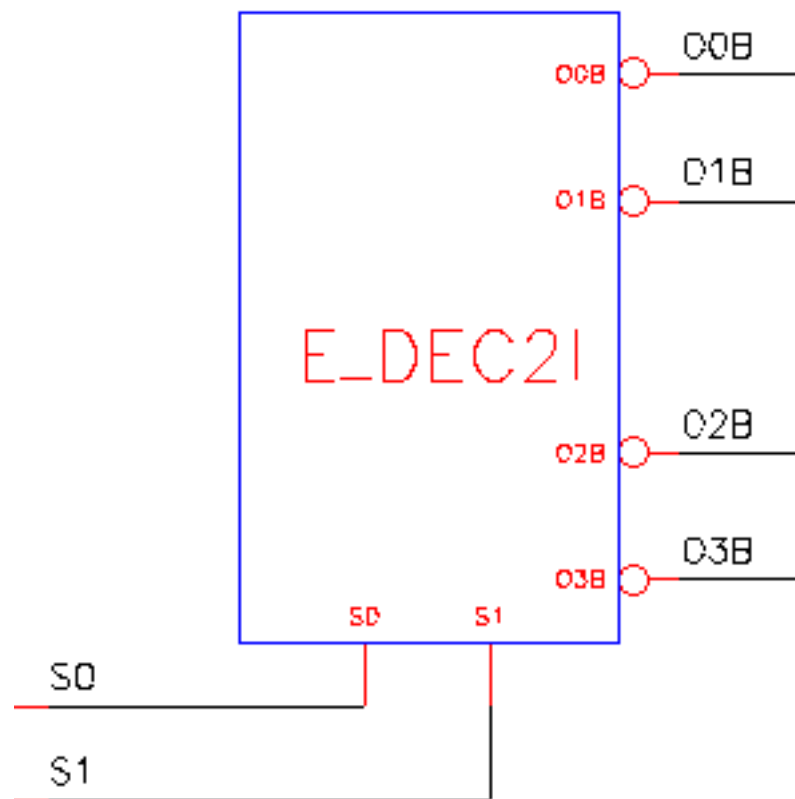
[E_DEC3EBI](#)

[E_DEC3EBD](#)

E_DEC2 : 2 to 4 decoder

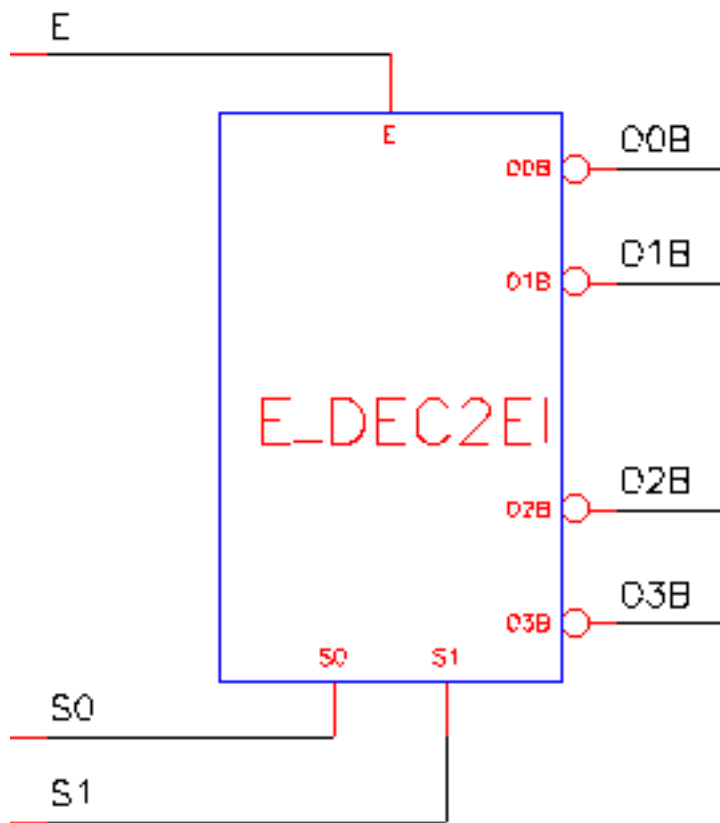
Pin	Direction	Description
I0	INPUT	data
S0	INPUT	data, LSB
S1	INPUT	data, MSB
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

E_DEC2I : 2 to 4 decoder, inverted outputs



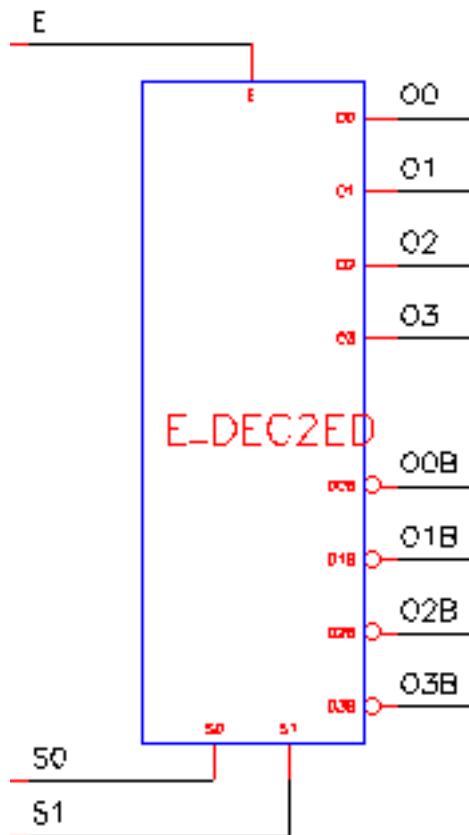
Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data, MSB
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted

E_DEC2EI : 2 to 4 decoder, active-HIGH enable, inverted outputs



Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data, MSB
E	OUTPUT	output enable, active-HIGH
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted

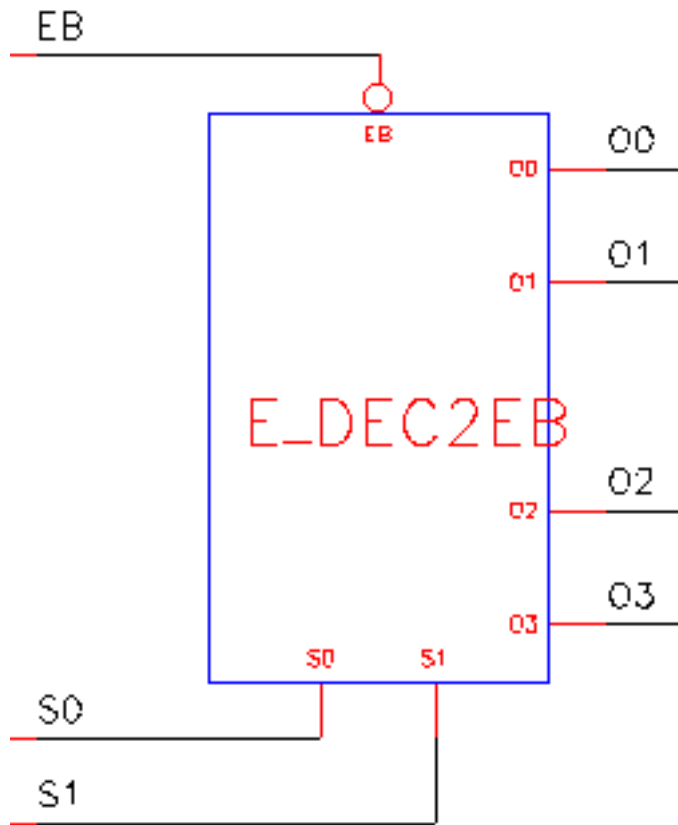
E_DEC2ED : 2 to 4 decoder, active-HIGH enable, with complementary outputs



Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data, MSB
E	OUTPUT	output enable, active-HIGH
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted

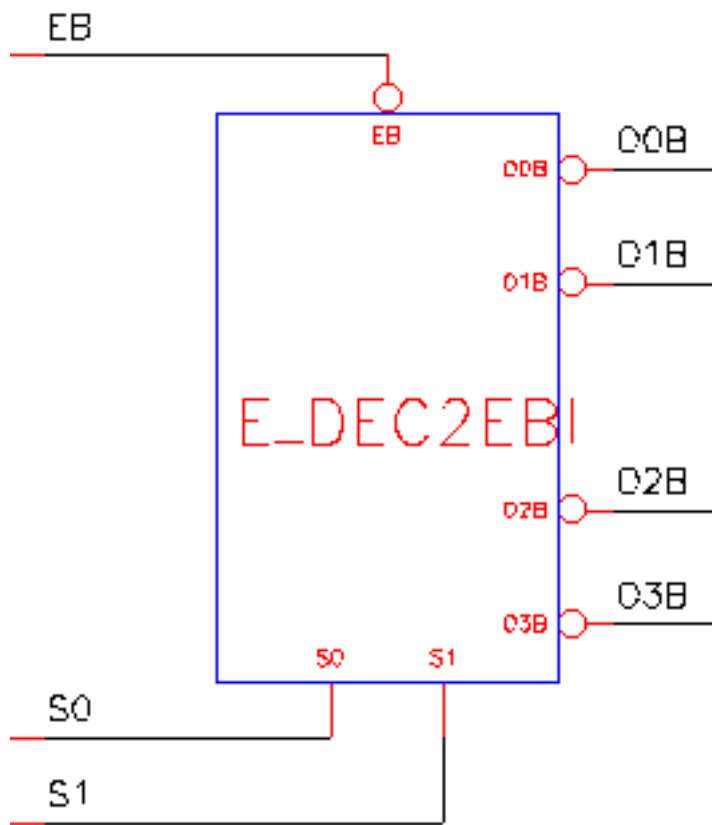
O3B	OUTPUT	data, inverted
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E_DEC2EB : 2 to 4 decoder, active-LOW enable



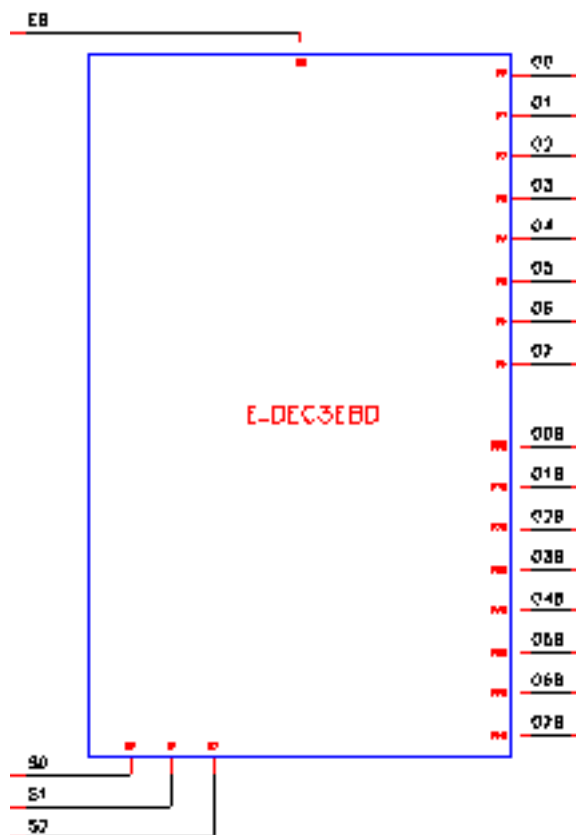
Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data, MSB
EB	INPUT	output enable, active-LOW
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data

E_DEC2EBI : 2 to 4 decoder, active-LOW enable, inverted outputs



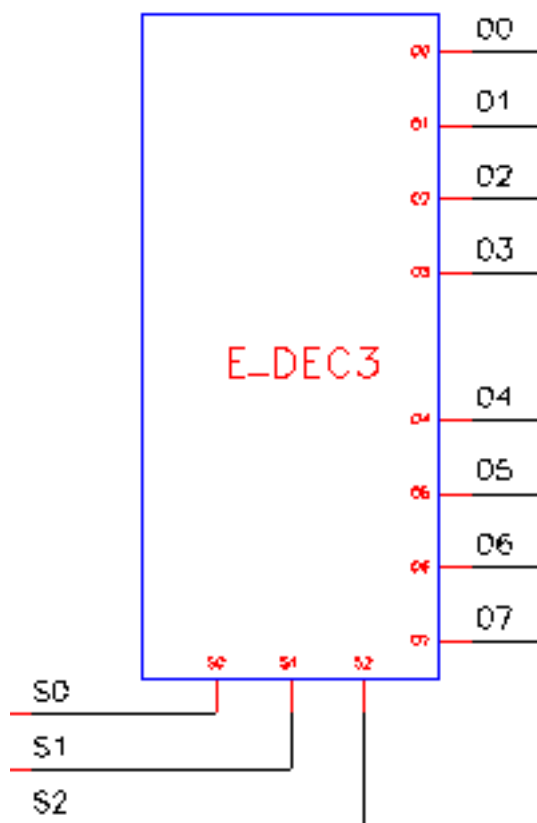
Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data, MSB
EB	INPUT	output enable, active-LOW
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted

E_DEC3EBD : 3 to 8 decoder, active-LOW enable, with complementary outputs



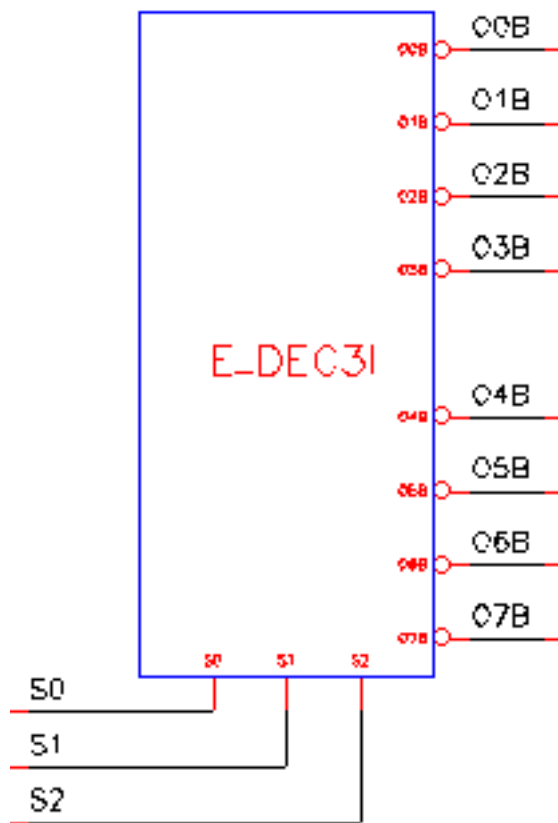
Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
EB	INPUT	output enable, active-LOW
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data
O0B	OUTPUT	data, complementary to O0
O1B	OUTPUT	data, complementary to O1
O2B	OUTPUT	data, complementary to O2
O3B	OUTPUT	data, complementary to O3
O4B	OUTPUT	data, complementary to O4
O5B	OUTPUT	data, complementary to O5
O6B	OUTPUT	data, complementary to O6
O7B	OUTPUT	data, complementary to O7

06	OUTPUT	data
07	OUTPUT	data
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted
04B	OUTPUT	data, inverted
05B	OUTPUT	data, inverted
06B	OUTPUT	data, inverted
07B	OUTPUT	data, inverted

E_DEC3 : 3 to 8 decoder

Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data

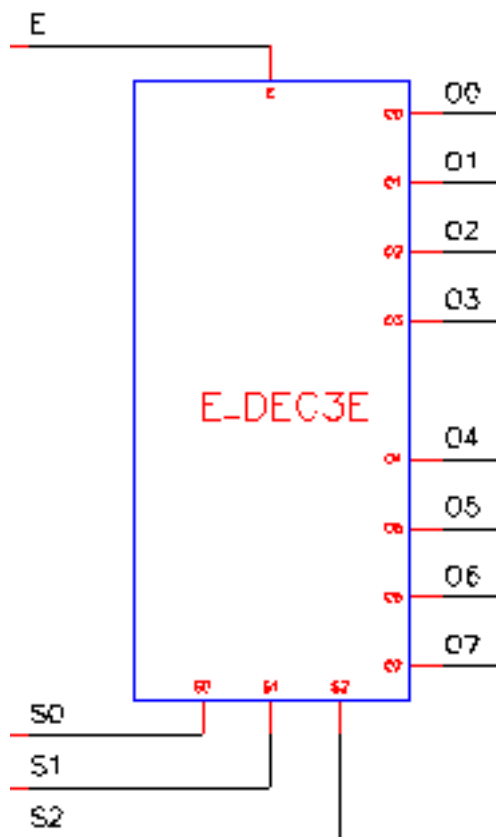
07	OUTPUT	data
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E_DEC3I : 3 to 8 decoder, inverted outputs

Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted
O4B	OUTPUT	data, inverted
O5B	OUTPUT	data, inverted
O6B	OUTPUT	data, inverted

07B	OUTPUT	data, inverted
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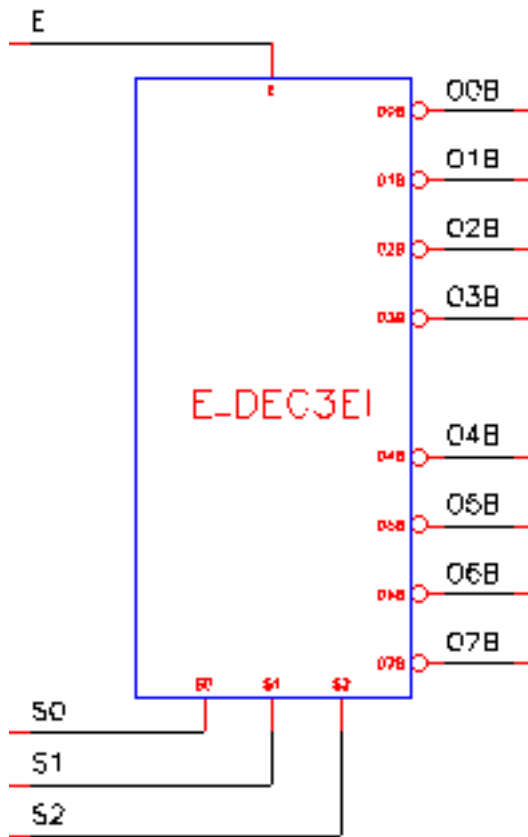
E_DEC3E : 3 to 8 decoder, active-HIGH enable



Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
E	INPUT	output enable, active-HIGH
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data
O6	OUTPUT	data
O7	OUTPUT	data

06	OUTPUT	data
07	OUTPUT	data

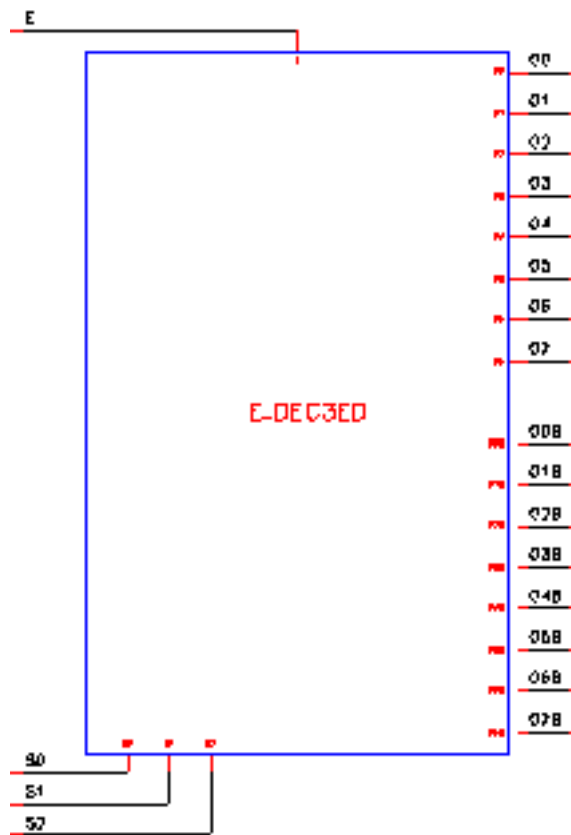
E_DEC3EI : 3 to 8 decoder, active-HIGH enable, inverted outputs



Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
E	INPUT	output enable, active-HIGH
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted
O4B	OUTPUT	data, inverted
O5B	OUTPUT	data, inverted

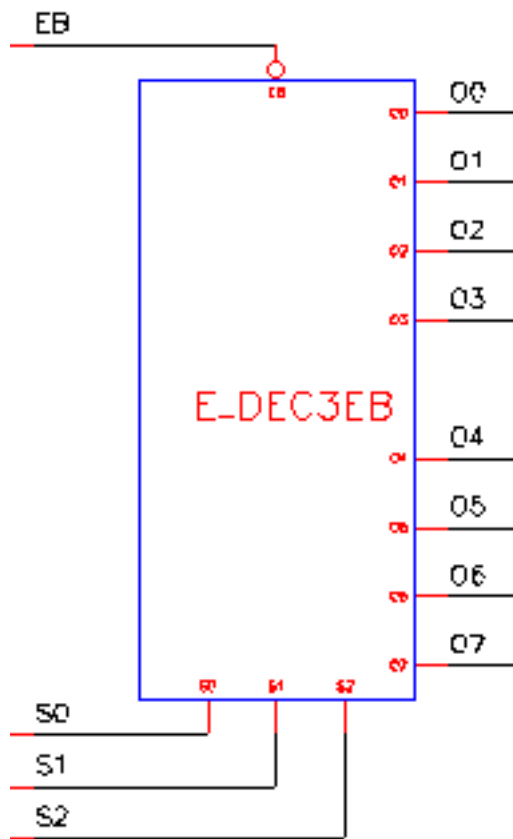
06B	OUTPUT	data, inverted
07B	OUTPUT	data, inverted

E_DEC3ED : 3 to 8 decoder, active-HIGH enable, with complementary outputs



Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
E	INPUT	output enable, active-HIGH
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
04	OUTPUT	data
05	OUTPUT	data

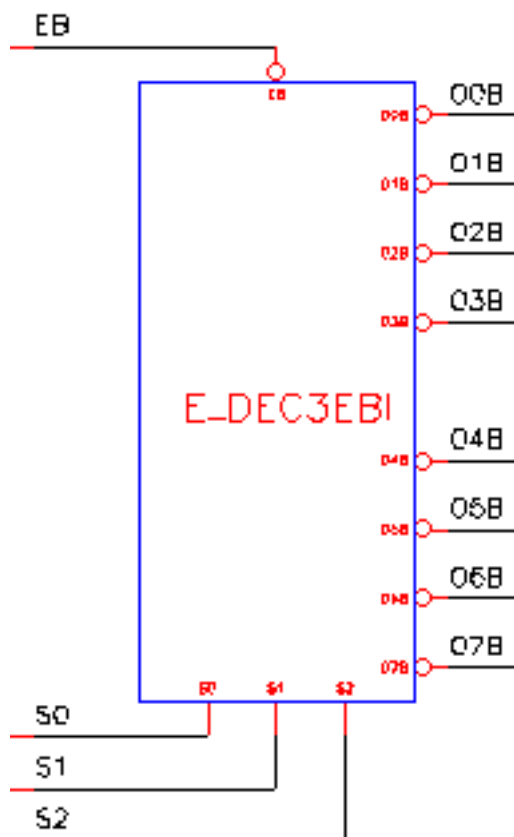
06	OUTPUT	data
07	OUTPUT	data
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted
04B	OUTPUT	data, inverted
05B	OUTPUT	data, inverted
06B	OUTPUT	data, inverted
07B	OUTPUT	data, inverted

E_DEC3EB : 3 to 8 Decoder, active-LOW enable

Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
EB	INPUT	output enable, active-LOW
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O4	OUTPUT	data
O5	OUTPUT	data

06	OUTPUT	data
07	OUTPUT	data

E_DEC3EBI : 3 to 8 decoder, active-LOW enable, inverted outputs



Pin	Direction	Description
S0	INPUT	data, LSB
S1	INPUT	data
S2	INPUT	data, MSB
EB	INPUT	output enable, active-LOW
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted
O4B	OUTPUT	data, inverted
O5B	OUTPUT	data, inverted

06B	OUTPUT	data, inverted
07B	OUTPUT	data, inverted

Adders /Substrator:

[E_FA1](#)

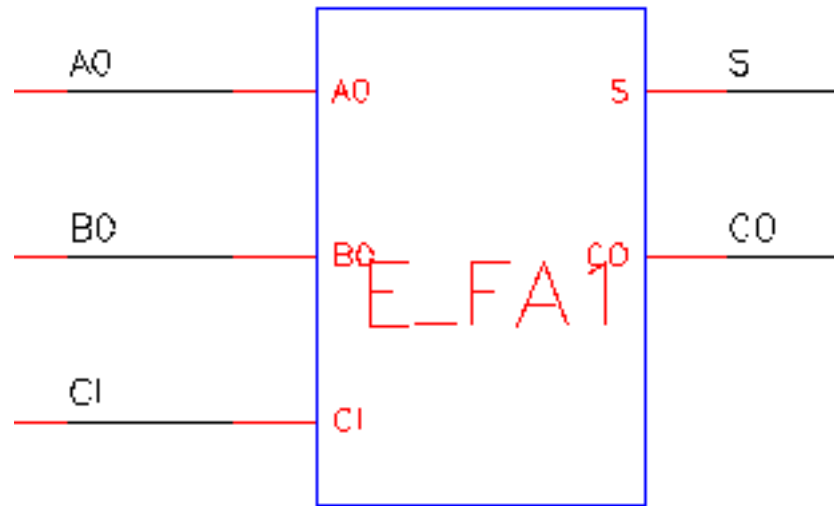
[E_FA2](#)

[E_FA4](#)

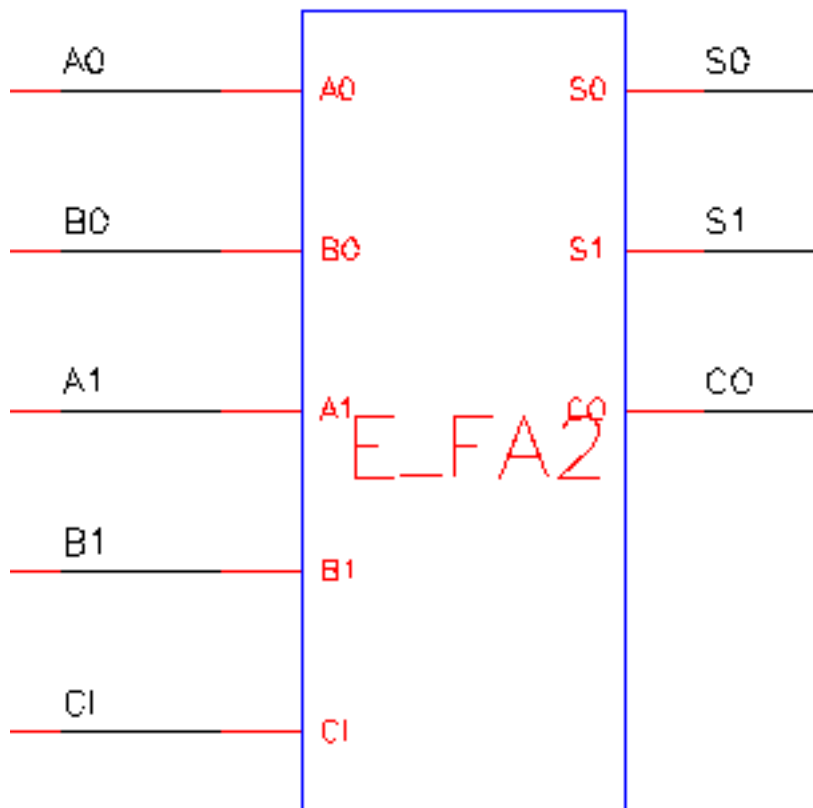
[E_HA1](#)

[E_ADDSUB](#)

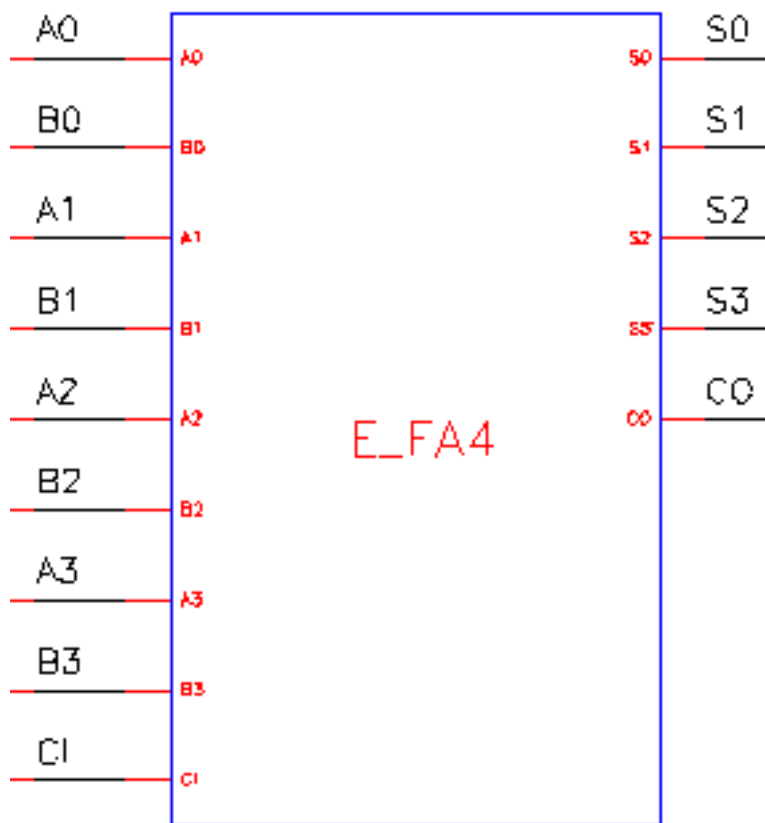
[E_ADRSUB2](#)

E_FA1 : 1-bit full adder

Pin	Direction	Description
A0	INPUT	operand
B0	INPUT	operand
CI	INPUT	carry-in
S	OUTPUT	sum
CO	OUTPUT	carry-out

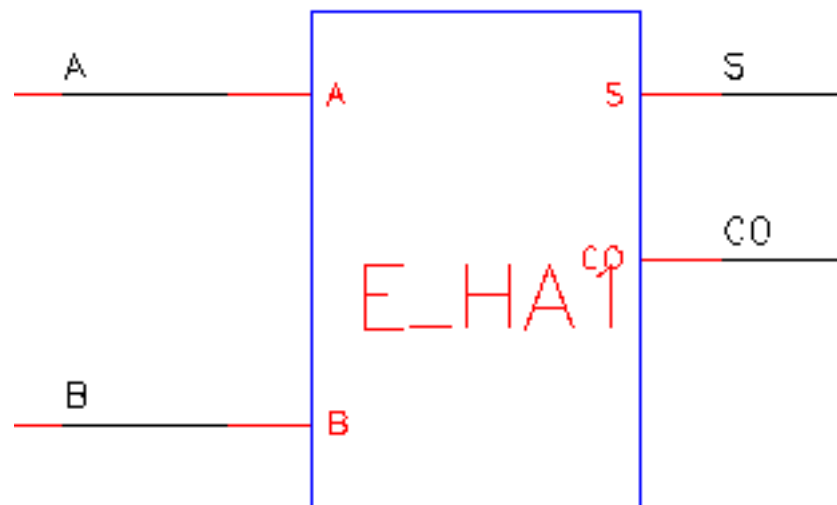
E_FA2 : 2-bit full adder

Pin	Direction	Description
A0	INPUT	operand, LSB
B0	INPUT	operand, MSB
A1	INPUT	operand, LSB
B1	INPUT	operand, MSB
CI	INPUT	carry-in
S0	OUTPUT	sum, LSB
S1	OUTPUT	sum, MSB
CO	OUTPUT	carry-out

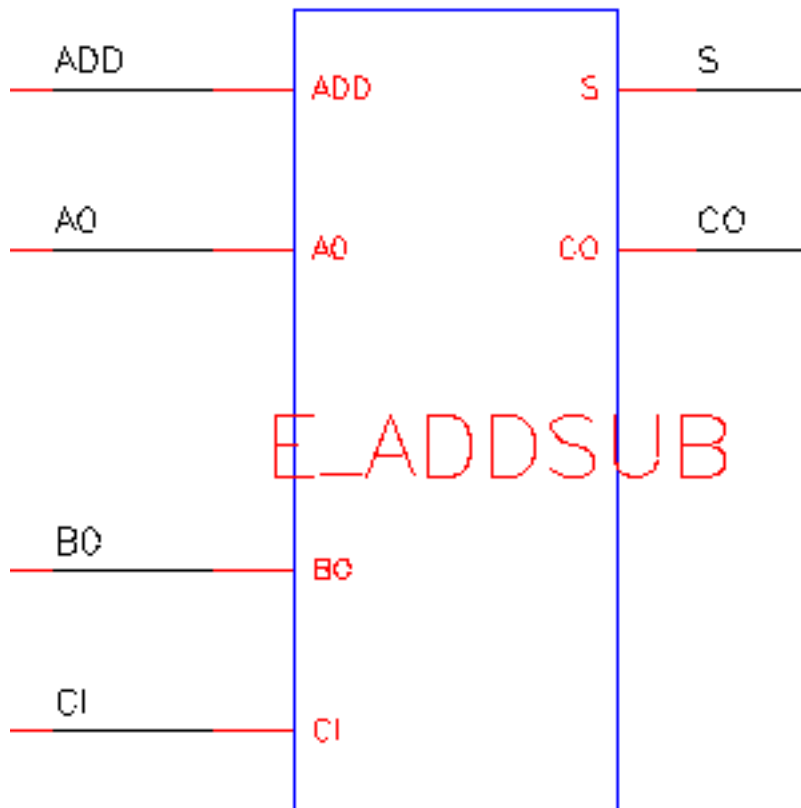
E_FA4 : 4-bit full adder

Pin	Direction	Description
A0	INPUT	operand, LSB
B0	INPUT	operand, LSB
A1	INPUT	operand
B1	INPUT	operand
A2	INPUT	operand
B2	INPUT	operand
A3	INPUT	operand, MSB
B3	INPUT	operand, MSB
CI	INPUT	carry-out
S0	OUTPUT	data

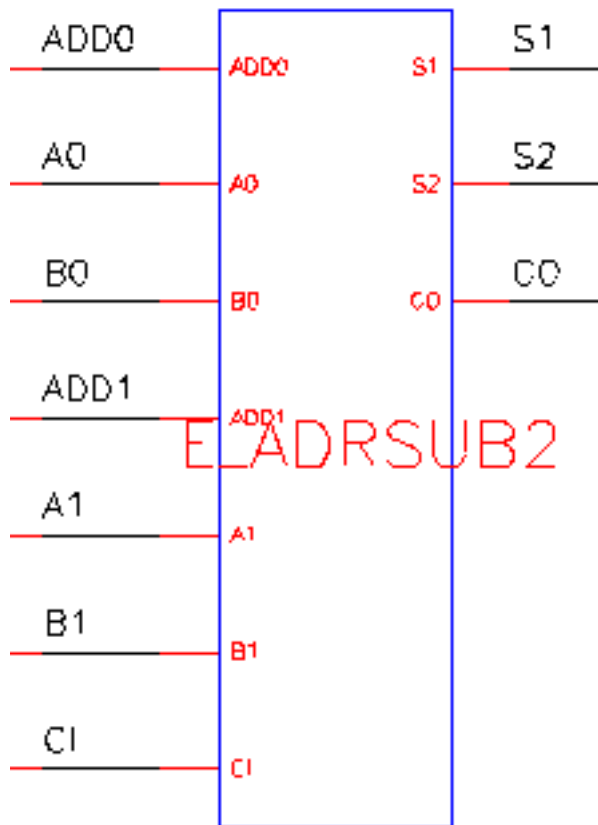
S1	OUTPUT	data
S2	OUTPUT	data
S3	OUTPUT	data
CO	OUTPUT	carry-out

E_HA1 : 1-bit half adder

Pin	Direction	Description
A	INPUT	operand
B	INPUT	operand
S	OUTPUT	carry-in
CO	OUTPUT	carry-out

E_ADDSUB : 1-bit adder/subtractor

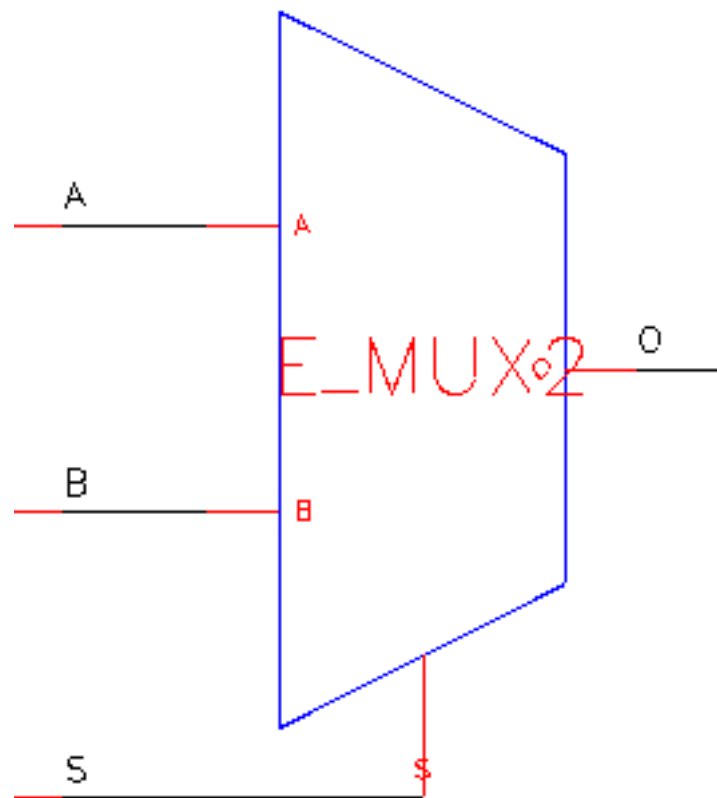
Pin	Direction	Description
ADD	INPUT	operand
A0	INPUT	operand
B0	INPUT	operand
CI	INPUT	carry-in
S	OUTPUT	sum
CO	OUTPUT	carry-out

E_ADRSUB2 : 2-bit adder/subtractor

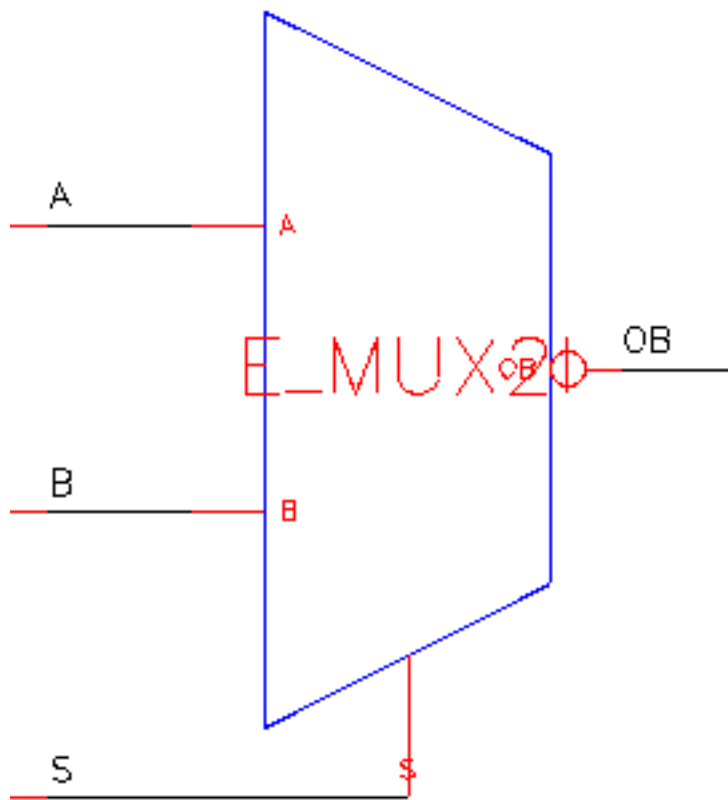
Pin	Direction	Description
ADD0	INPUT	operator, LSB
A0	INPUT	operand, LSB
B0	INPUT	operand, LSB
ADD1	INPUT	operator, MSB
A1	INPUT	operand, MSB
B1	INPUT	operand, MSB
CI	INPUT	carry-in
S1	OUTPUT	sum, LSB
S2	OUTPUT	sum, MSB
CO	OUTPUT	carry-out

Multiplexers

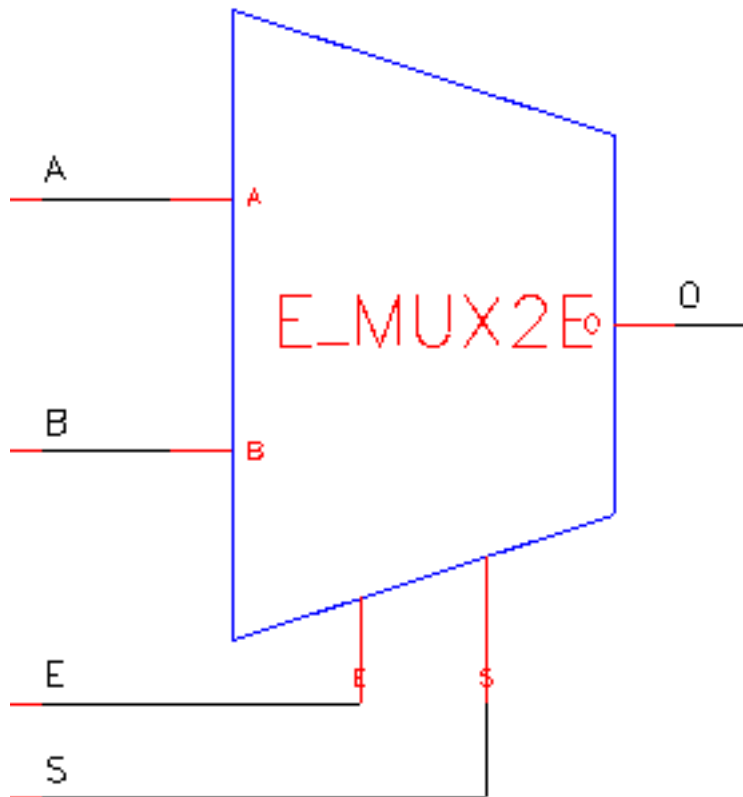
[E_MUX2](#)
[E_MUX2I](#)
[E_MUX2E](#)
[E_MUX2EI](#)
[E_MUX2ED](#)
[E_MUX2X4ED](#)
[E_MUX2EB](#)
[E_MUX2EBI](#)
[E_MUX2EBD](#)
[E_MUX2X4EBD](#)
[E_MUX3](#)
[E_MUX3I](#)
[E_MUX3E](#)
[E_MUX3EI](#)
[E_MUX3ED](#)
[E_MUX3EB](#)
[E_MUX3EBI](#)
[E_MUX3EBD](#)
[E_MUX4](#)
[E_MUX4I](#)
[E_MUX4E](#)
[E_MUX4EI](#)
[E_MUX4ED](#)
[E_MUX4EB](#)
[E_MUX4EBI](#)
[E_MUX4EBD](#)
[E_MUX5](#)
[E_MUX5I](#)
[E_MUX5E](#)
[E_MUX5EI](#)
[E_MUX5ED](#)
[E_MUX5EB](#)
[E_MUX5EBI](#)
[E_MUX5EBD](#)
[E_MUX8](#)
[E_MUX8I](#)
[E_MUX8E](#)
[E_MUX8EI](#)
[E_MUX8ED](#)
[E_MUX8EB](#)
[E_MUX8EBI](#)
[E_MUX8EBD](#)

E_MUX2 : 2-1 multiplexer

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
S	INPUT	select
O	OUTPUT	data

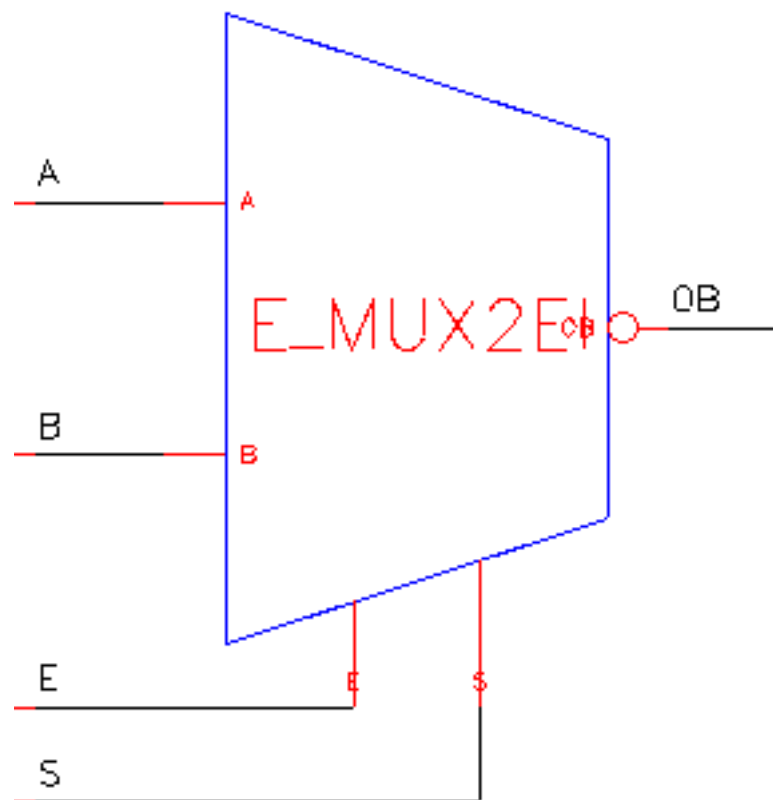
E_MUX2I : 2-1 multiplexer, inverted output

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
S	INPUT	select
OB	OUTPUT	data, inverted

E_MUX2E : 2-1 multiplexer, active-HIGH enable

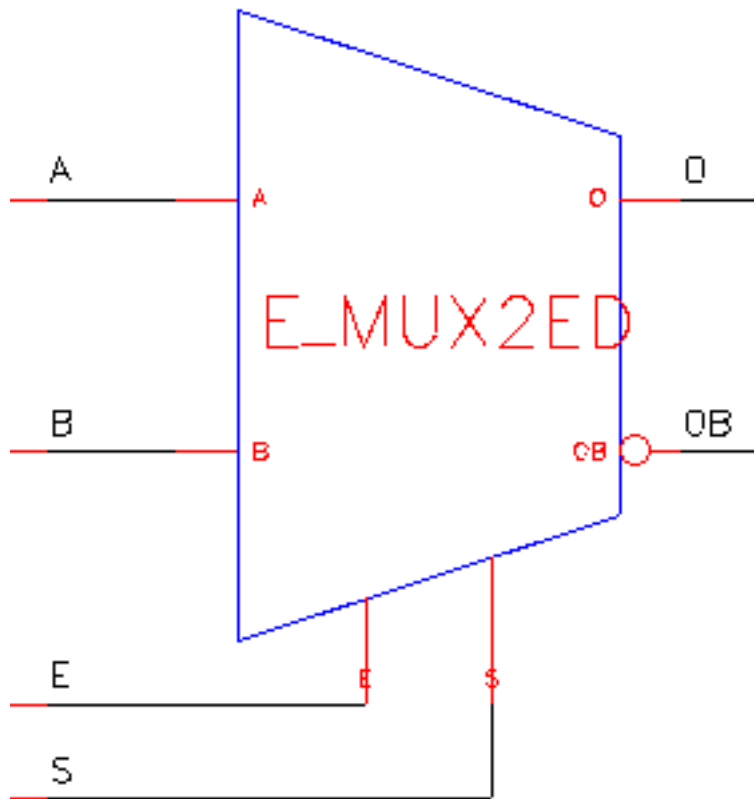
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
E	INPUT	output enable, active-HIGH
S	INPUT	select
O	OUTPUT	data

E_MUX2EI : 2-1 multiplexer, active-HIGH enable, inverted output



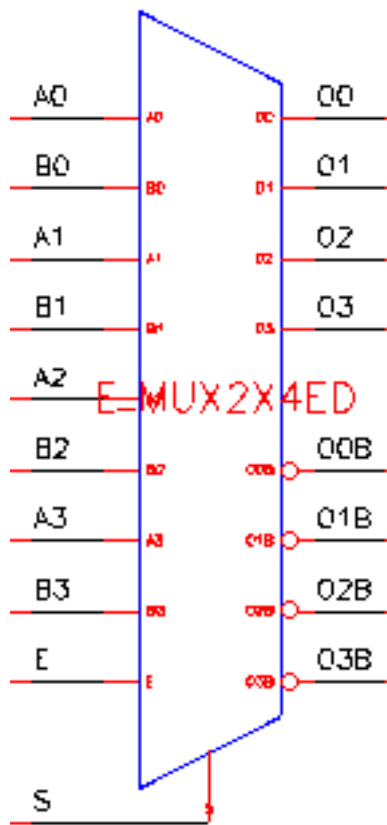
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
E	INPUT	output enable, active-HIGH
S	INPUT	select
OB	OUTPUT	data, inverted

E_MUX2ED : 2-1 multiplexer, active-HIGH enable, complementary outputs



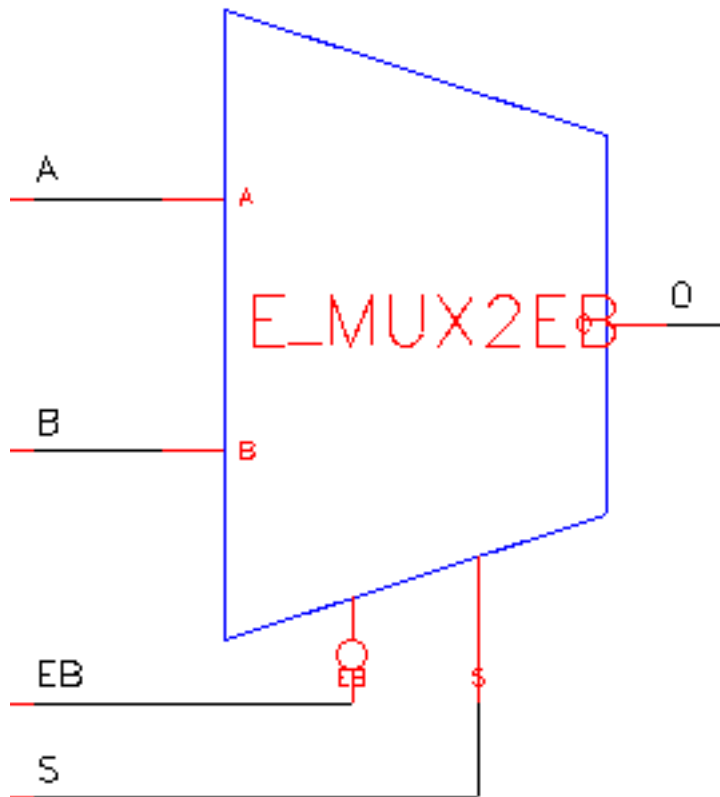
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
E	INPUT	output enable, active-HIGH
S	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_MUX2X4ED : four 2-1 multiplexers, active-HIGH enable, complementary outputs



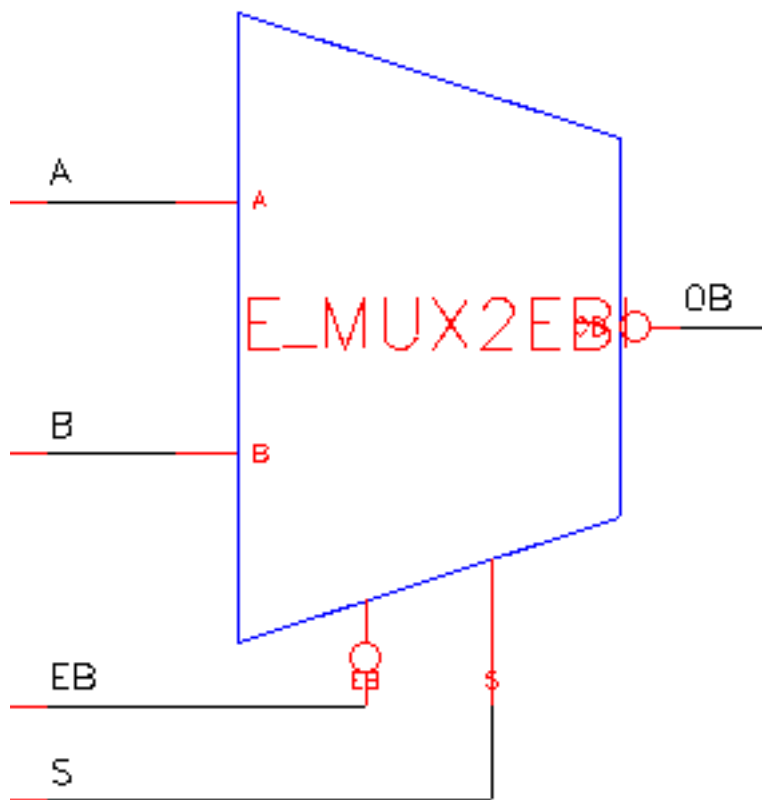
Pin	Direction	Description
A0	INPUT	data
B0	INPUT	data
A1	INPUT	data
B1	INPUT	data
A2	INPUT	data
B2	INPUT	data
A3	INPUT	data
B3	INPUT	data
E	INPUT	output enable, active-HIGH
S	INPUT	select

O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted

E_MUX2EB : 2-1 multiplexer, active-LOW enable

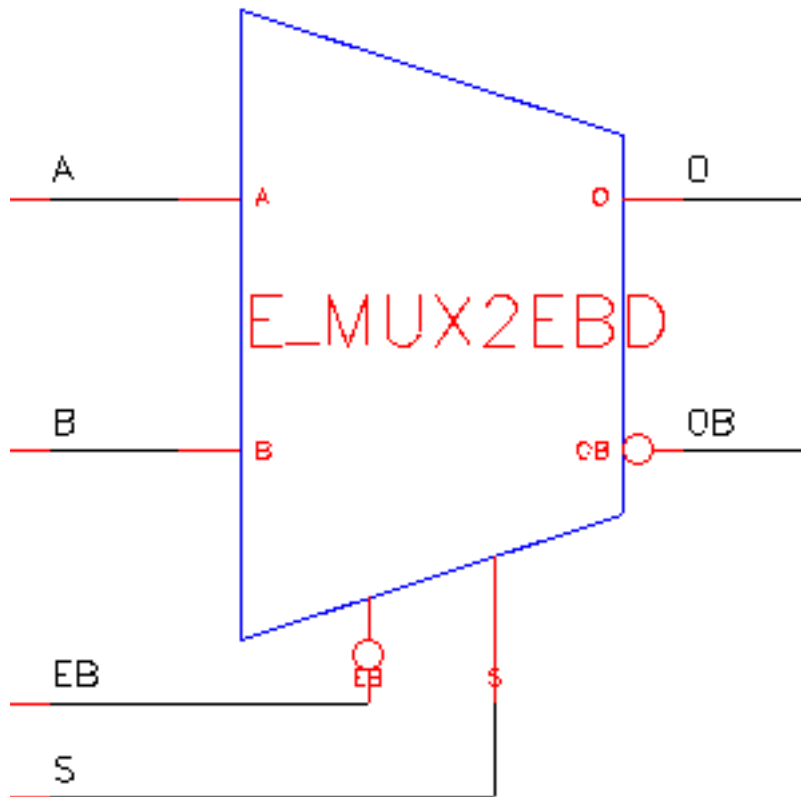
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
EB	INPUT	output enable, active-LOW
S	INPUT	select
O	OUTPUT	data

E_MUX2EBI : 2-1 multiplexer, active-LOW enable, inverted output



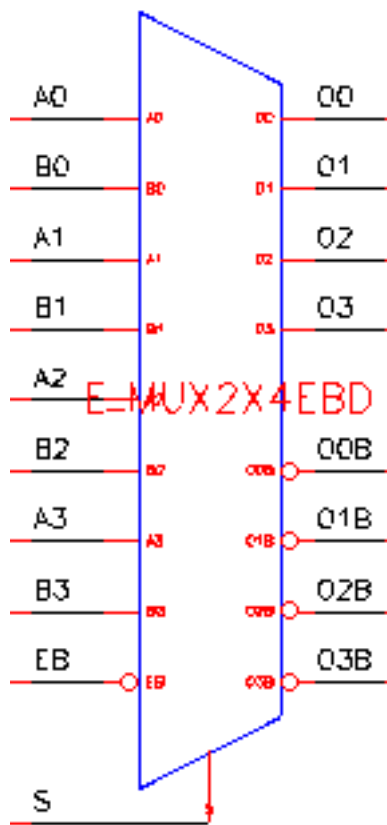
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
EB	INPUT	output enable, active-LOW
S	INPUT	select
OB	OUTPUT	data, inverted

E_MUX2EBD : 2-1 multiplexer, active-LOW enable, complementary outputs



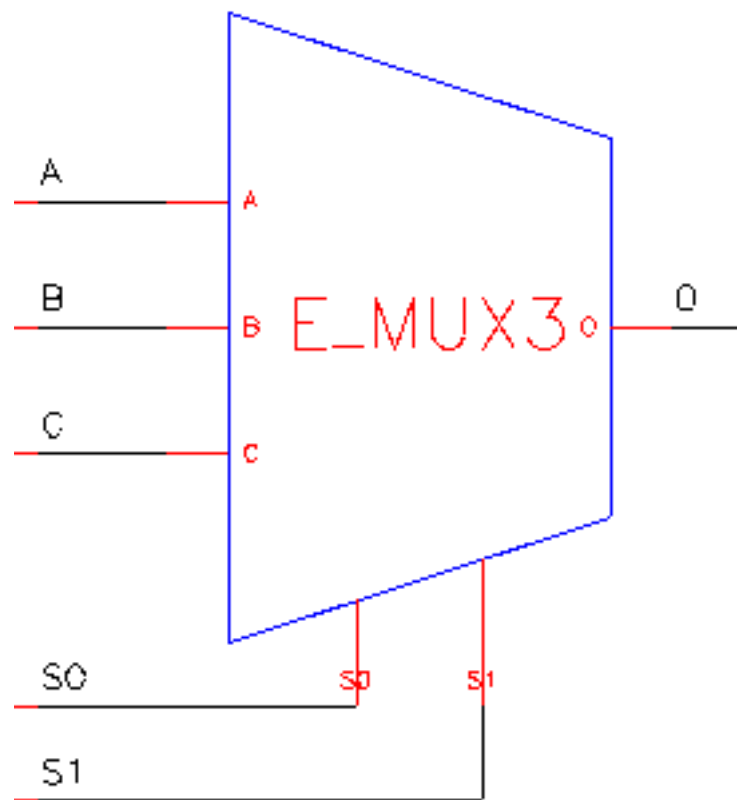
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
EB	INPUT	output enable, active-LOW
S	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_MUX2X4EBD : four 2-1 multiplexers, active-LOW enable, complementary outputs

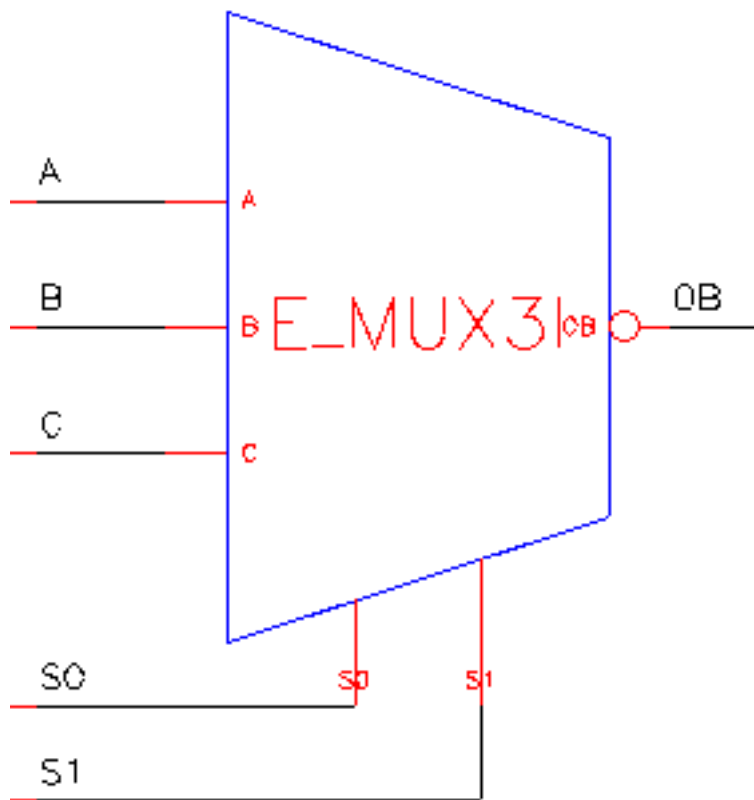


Pin	Direction	Description
A0	INPUT	data
B0	INPUT	data
A1	INPUT	data
B1	INPUT	data
A2	INPUT	data
B2	INPUT	data
A3	INPUT	data
B3	INPUT	data
EB	INPUT	output enable, active-LOW
S	INPUT	select

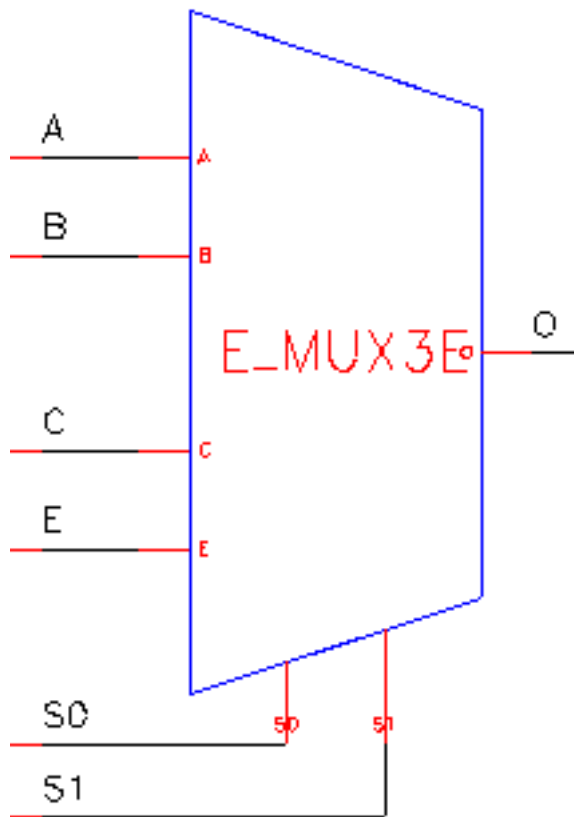
O0	OUTPUT	data
O1	OUTPUT	data
O2	OUTPUT	data
O3	OUTPUT	data
O0B	OUTPUT	data, inverted
O1B	OUTPUT	data, inverted
O2B	OUTPUT	data, inverted
O3B	OUTPUT	data, inverted

E_MUX3 : 3-1 multiplexer

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data

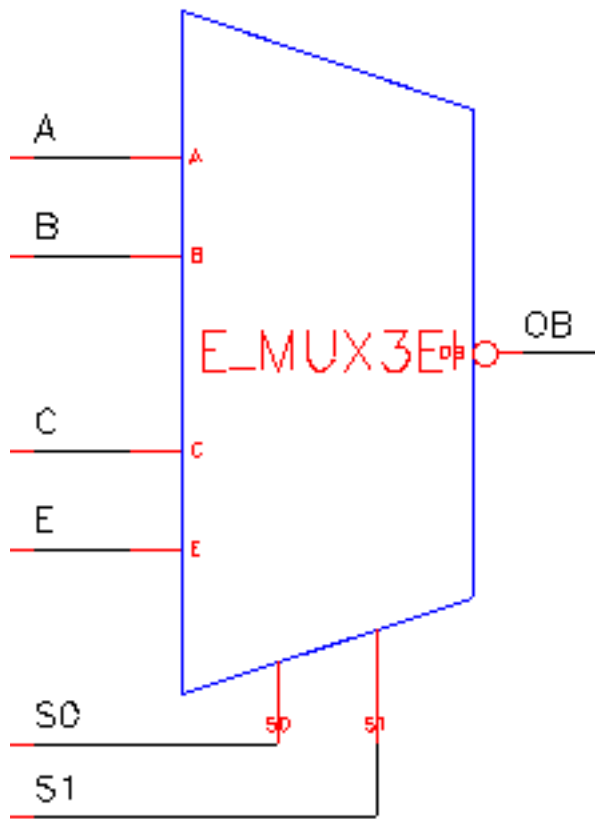
E_MUX3I : 3-1 multiplexer, inverted output

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
S0	INPUT	select
S1	INPUT	select
OB	OUTPUT	data, inverted

E_MUX3E : 3-1 multiplexer, active-HIGH enable

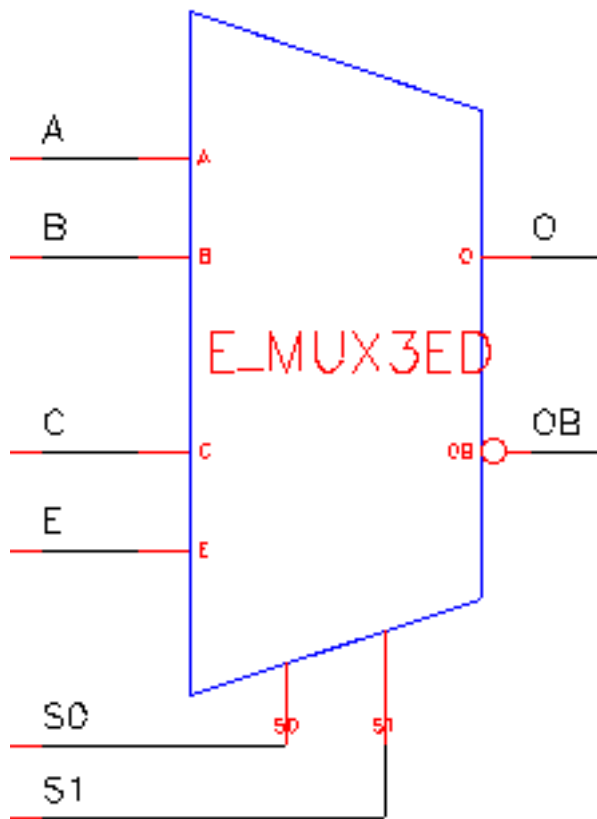
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
E	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data

E_MUX3EI : 3-1 multiplexer, active-HIGH enable, inverted output



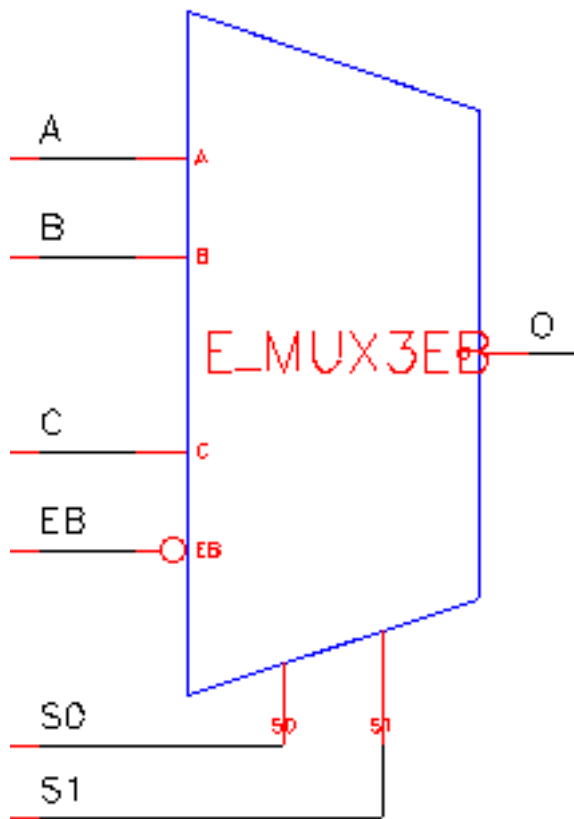
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
E	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
OB	OUTPUT	data, inverted

E_MUX3ED : 3-1 multiplexer, active-HIGH enable, complementary outputs



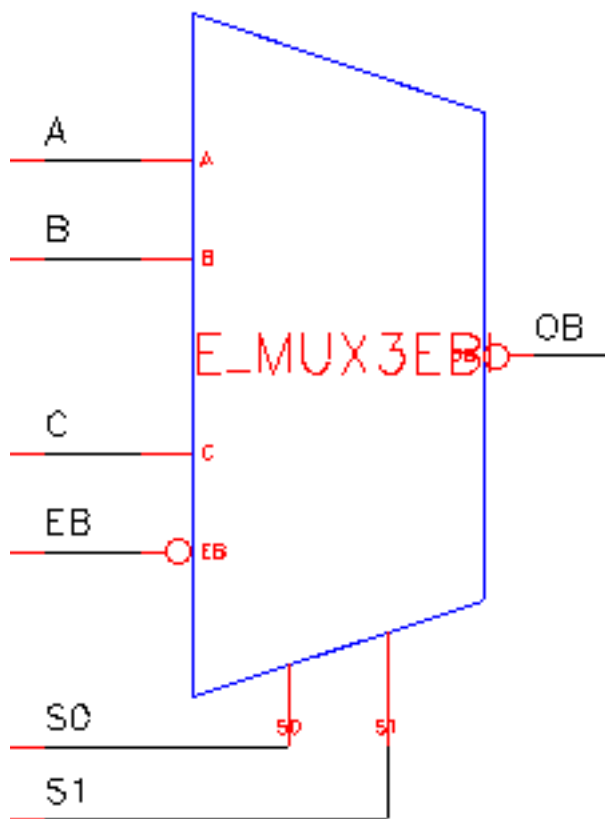
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
E	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_MUX3EB : 3-1 multiplexer, active-LOW enable



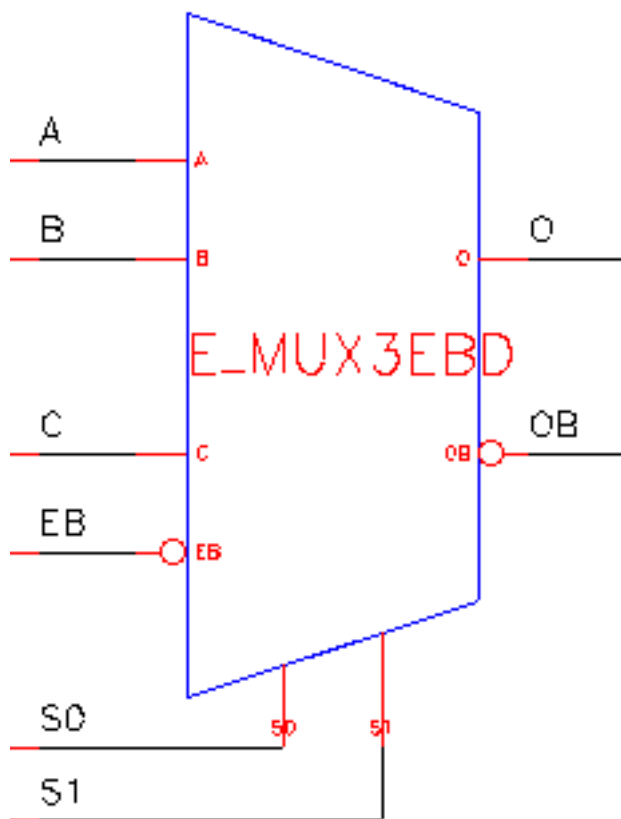
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
EB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data

E_MUX3EBI : 3-1 multiplexer, active-LOW enable, inverted output

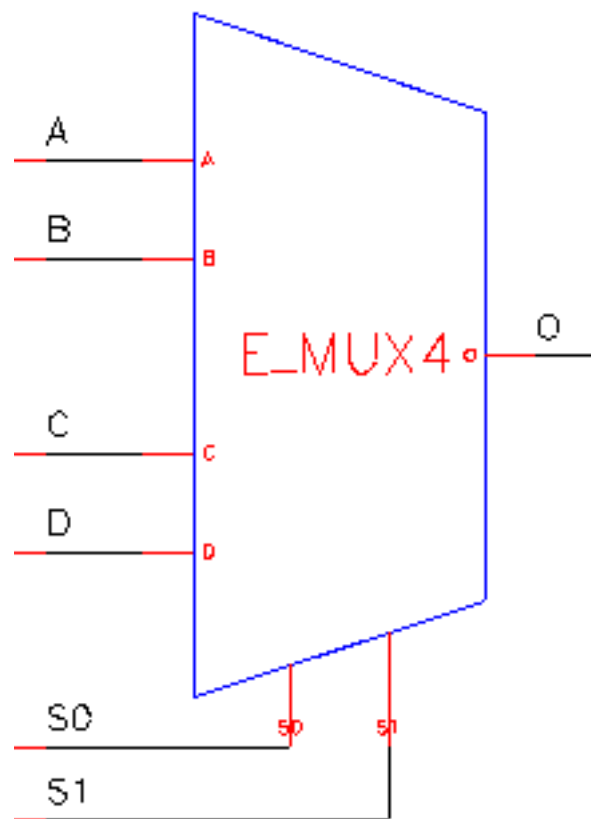


Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
EB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
OB	OUTPUT	data, inverted

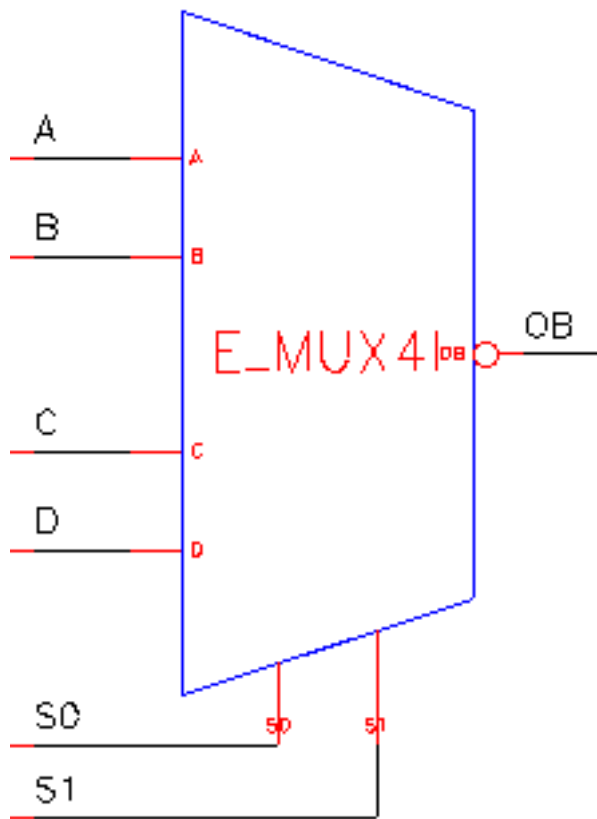
E_MUX3EBD : 3-1 multiplexer, active-LOW enable, complementary outputs



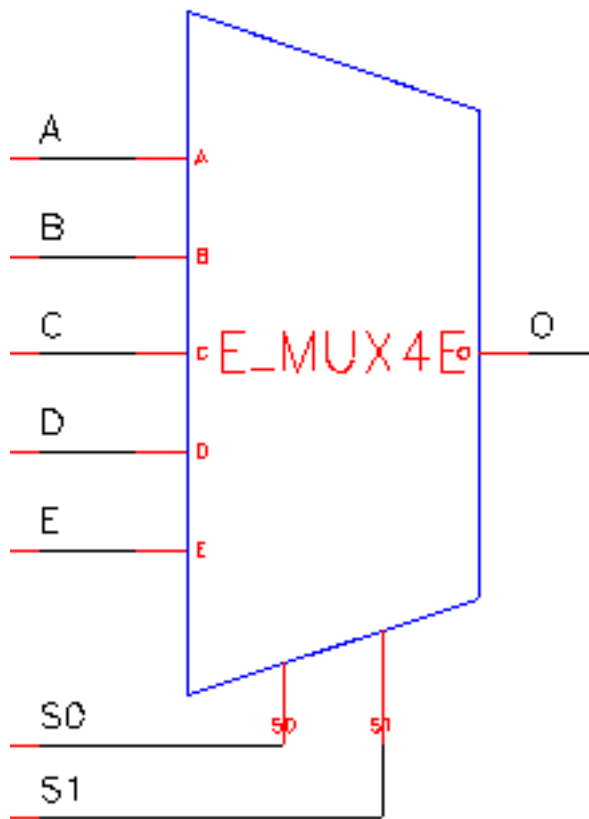
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
EB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_MUX4 : 4-1 multiplexer

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data

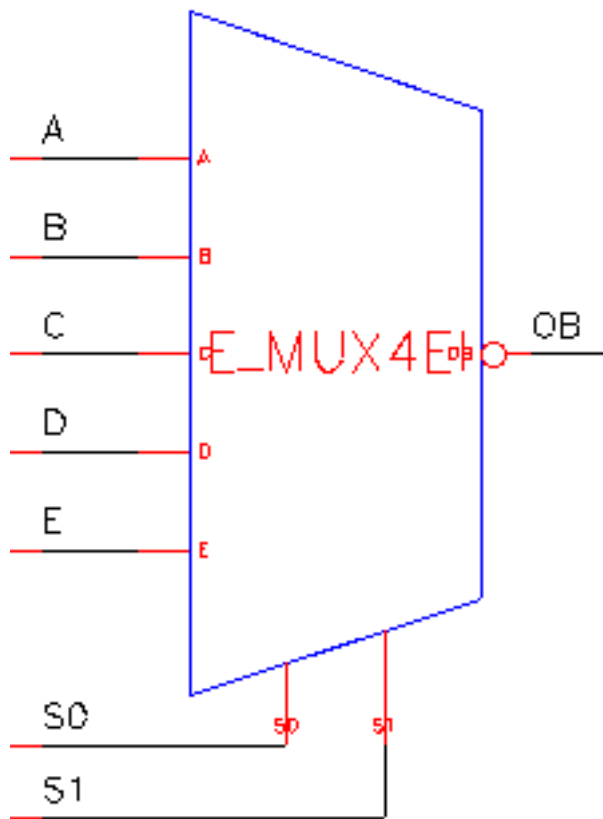
E_MUX4I : 4-1 multiplexer, inverted output

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
S0	INPUT	select
S1	INPUT	select
OB	OUTPUT	data, inverted

E_MUX4E : 4-1 multiplexer, active-HIGH enable

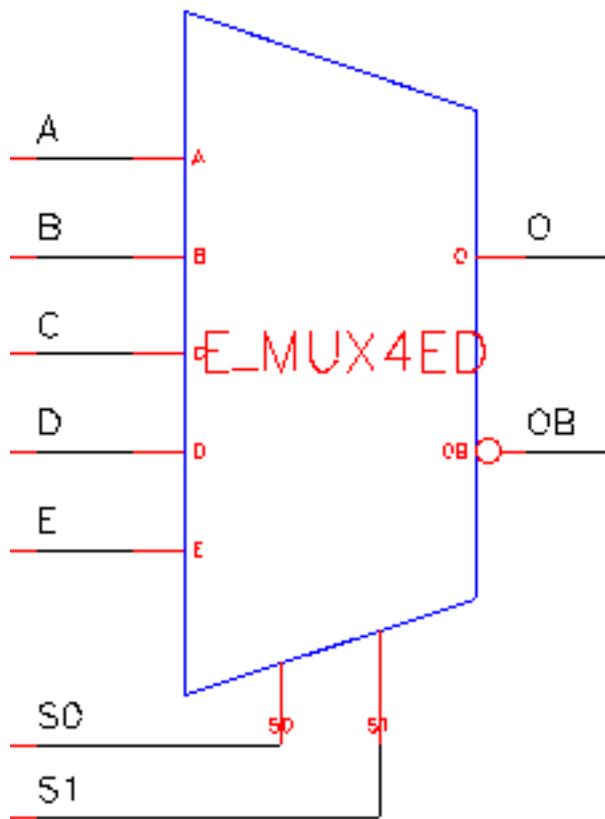
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data

E_MUX4EI : 4-1 multiplexer, active-HIGH enable, inverted output



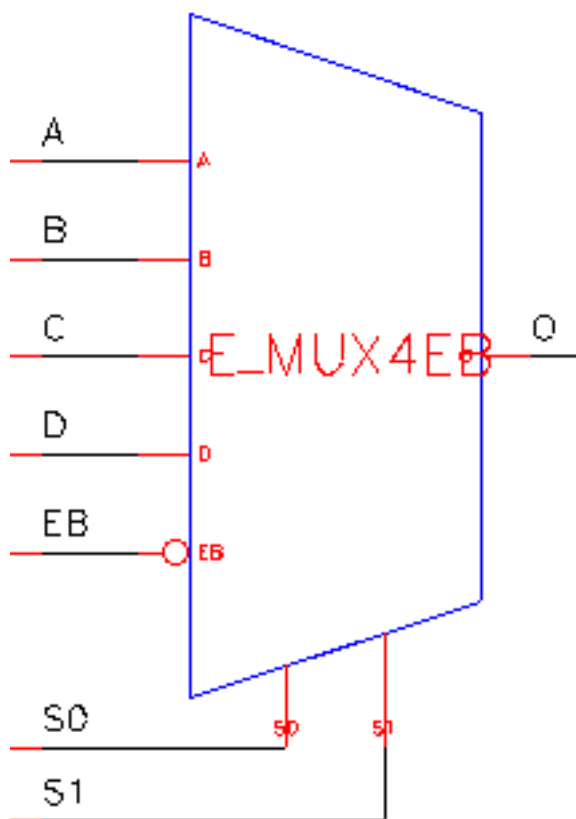
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
OB	OUTPUT	data, inverted

E_MUX4ED : 4-1 multiplexer, active-HIGH enable, complementary outputs

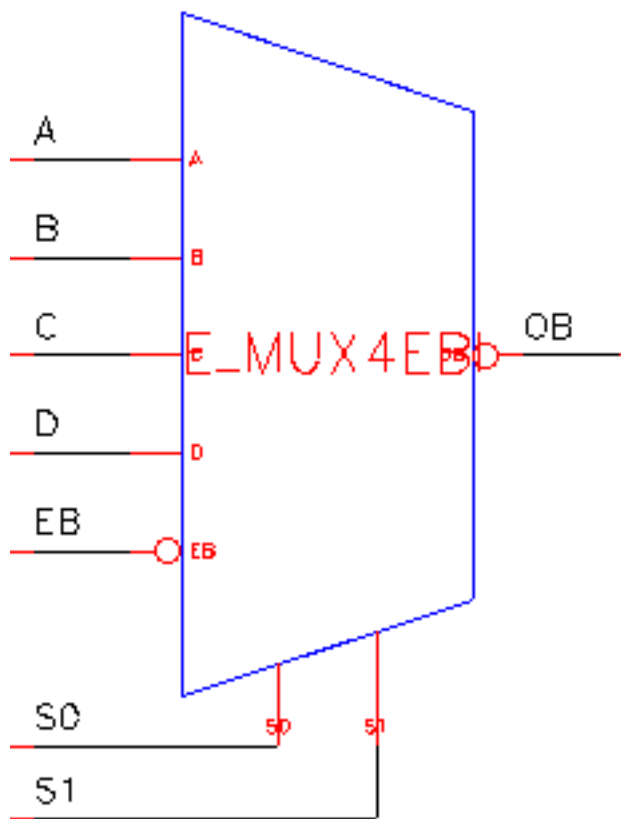


Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_MUX4EB : 4-1 multiplexer, active-LOW enable

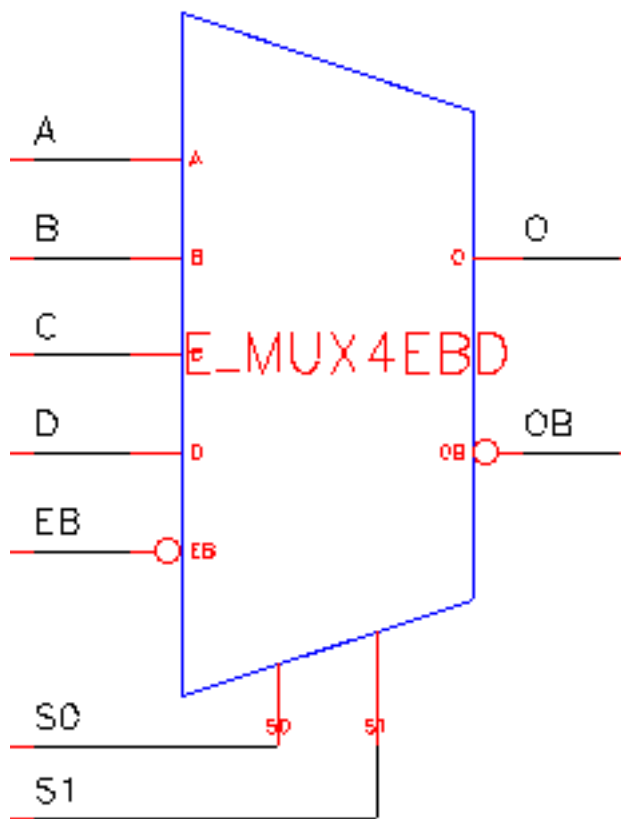


Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
EB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data

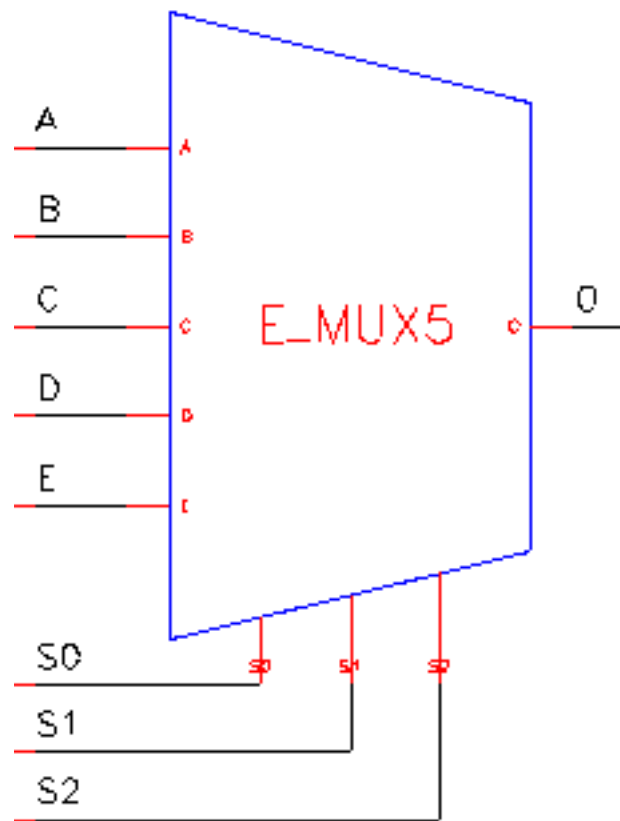
E_MUX4EBI : 4-1 multiplexer, active-LOW enable, inverted output

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
EB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
OB	OUTPUT	data, inverted

E_MUX4EBD : 4-1 multiplexer, active-LOW enable, complementary outputs

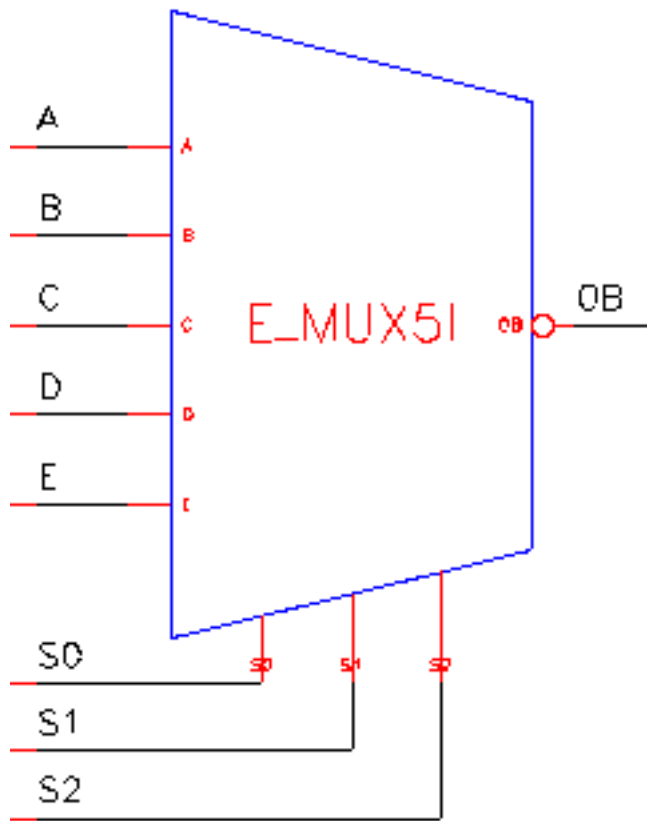


Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
EB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

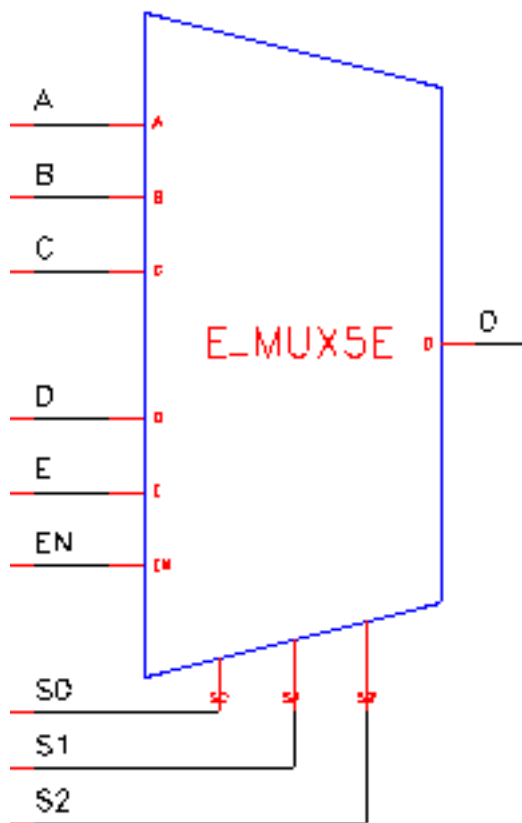
E_MUX5 : 5-1 multiplexer

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

E_MUX5I : 5-1 multiplexer, inverted output

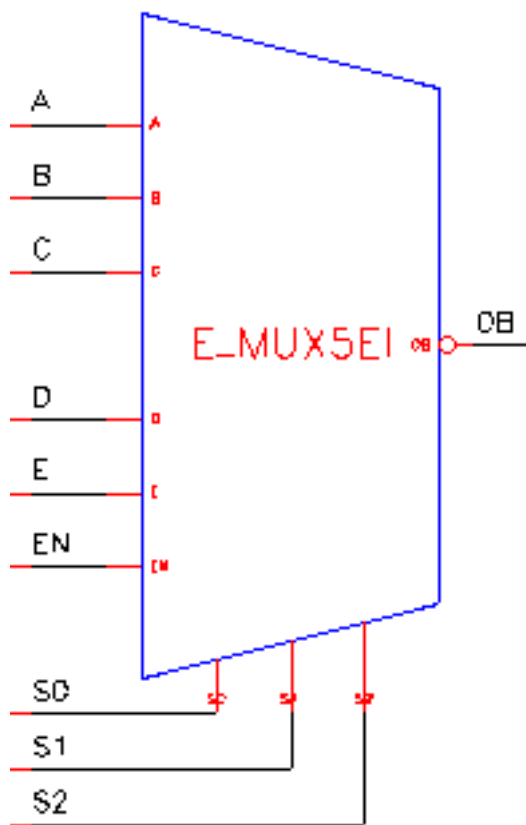


Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
OB	OUTPUT	data, inverted

E_MUX5E : 5-1 multiplexer, active-HIGH enable

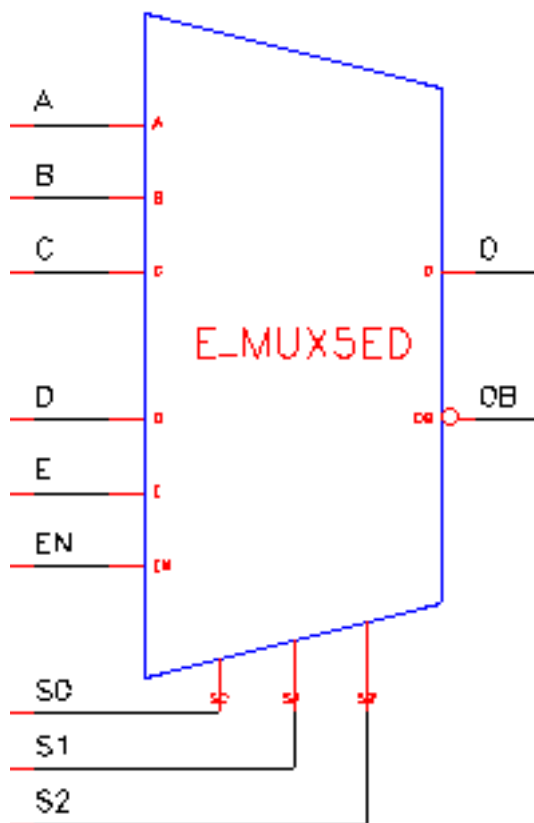
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
EN	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

E_MUX5EI : 5-1 multiplexer, active-HIGH enable, inverted output



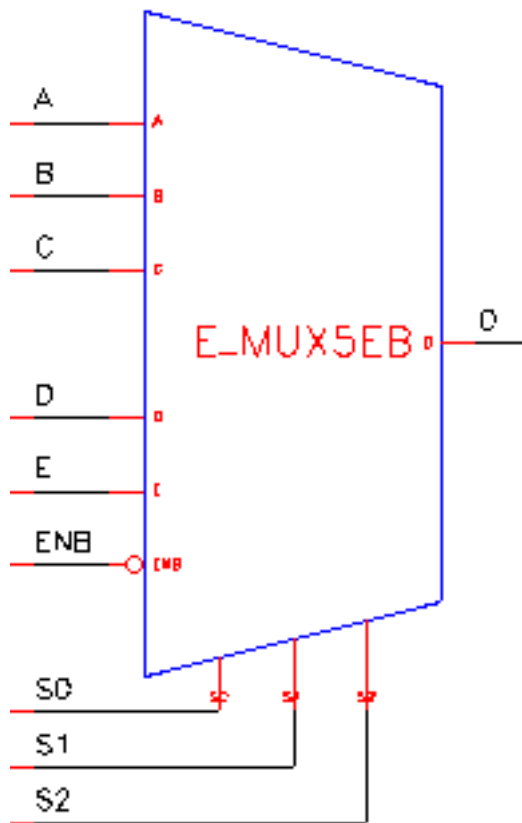
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
EN	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
OB	OUTPUT	data, inverted

E_MUX5ED : 5-1 multiplexer, active-HIGH enable, complementary outputs



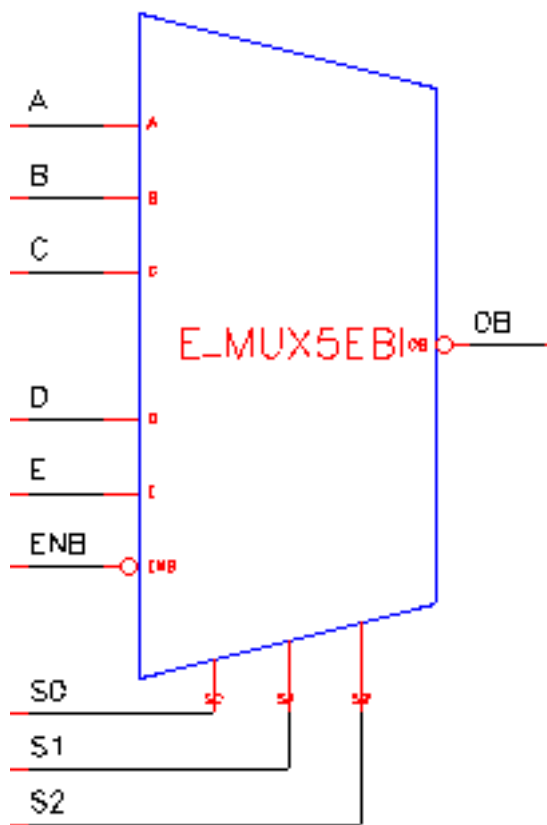
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
EN	INPUT	output enable, active-HIGH
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

OB	OUTPUT	data, inverted
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E_MUX5EB : 5-1 multiplexer, active-LOW enable

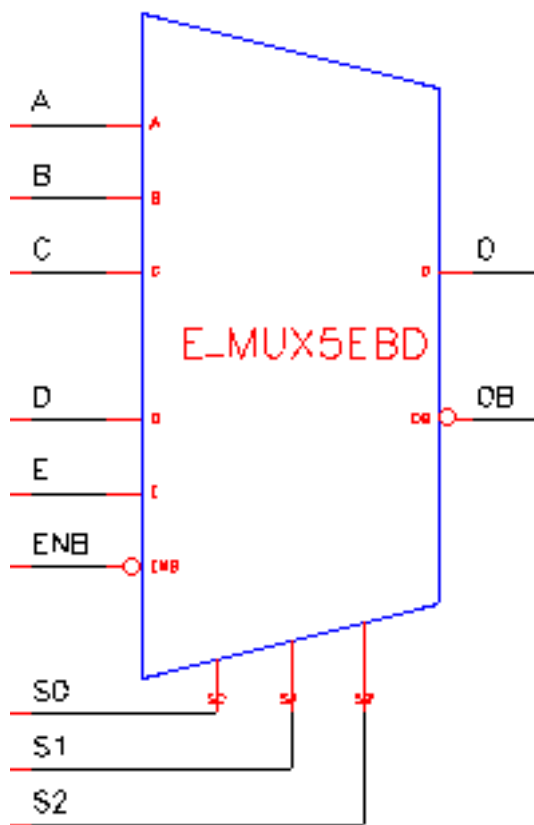
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
ENB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

E_MUX5EBI : 5-1 multiplexer, active-LOW enable, inverted output



Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
ENB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
OB	OUTPUT	data, inverted

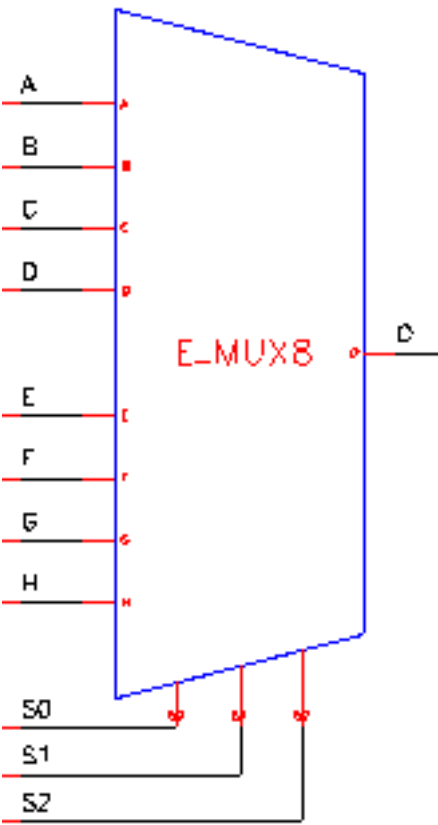
E_MUX5EBD : 5-1 multiplexer, active-LOW enable, complementary outputs



Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
ENB	INPUT	output enable, active-LOW
S0	INPUT	select
S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

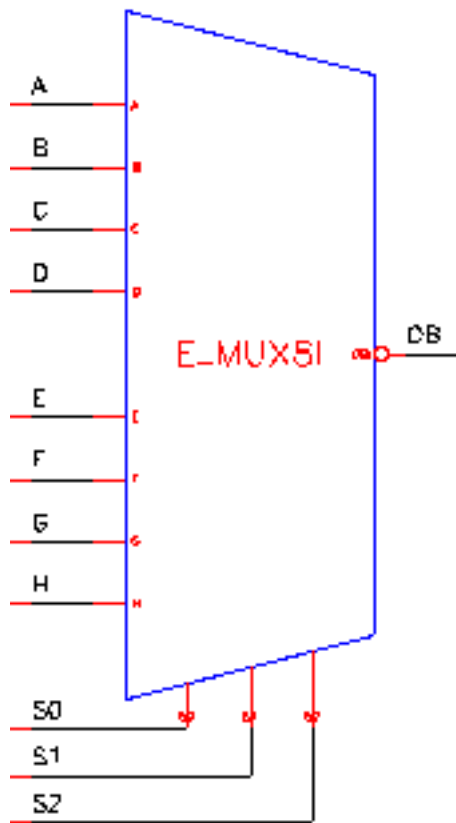
OB	OUTPUT	data, inverted
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E_MUX8 : 8-1 multiplexer



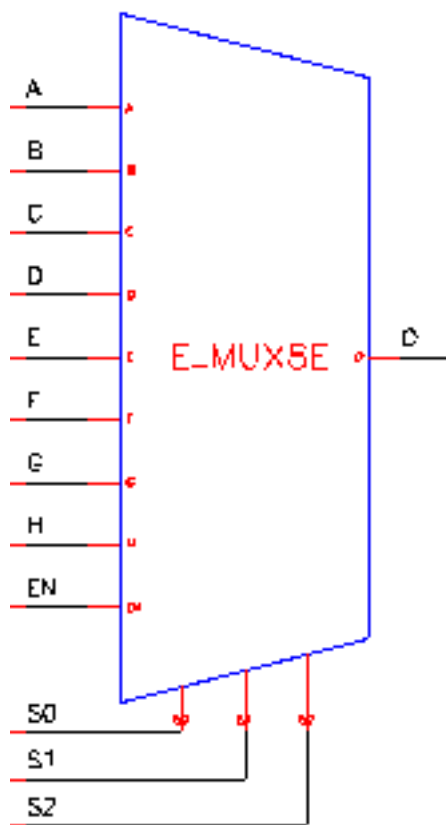
Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
S0	INPUT	select
S1	INPUT	select

S2	INPUT	select
O	OUTPUT	data

E_MUX8I : 8-1 multiplexer, inverted output

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
S0	INPUT	select
S1	INPUT	select

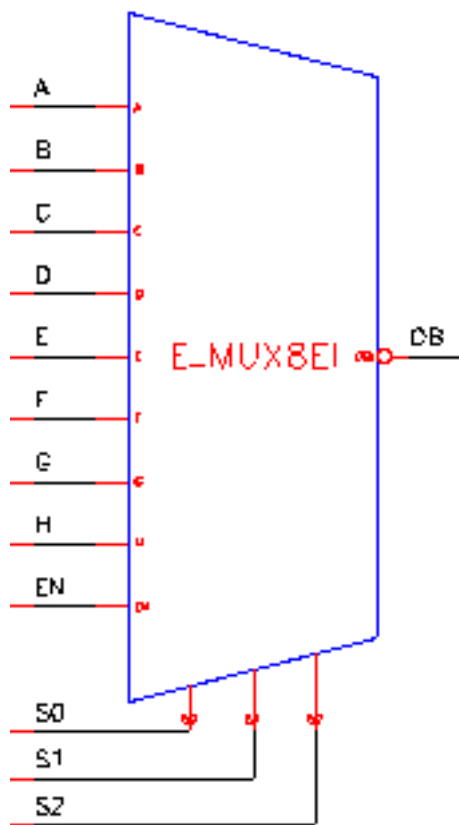
S2	INPUT	select
OB	OUTPUT	data, inverted

E_MUX8E : 8-1 multiplexer, active-HIGH enable

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
EN	INPUT	output enable, active-HIGH
S0	INPUT	select

S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

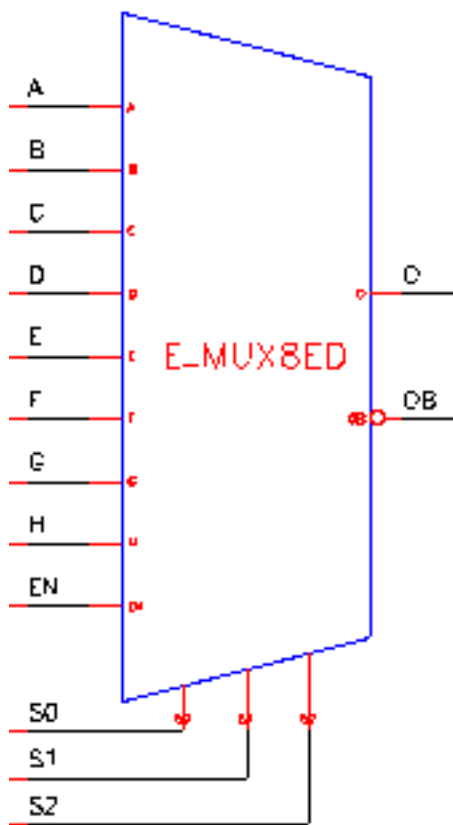
E_MUX8EI : 8-1 multiplexer, active-HIGH enable, inverted output



Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
EN	INPUT	output enable, active-HIGH
S0	INPUT	select

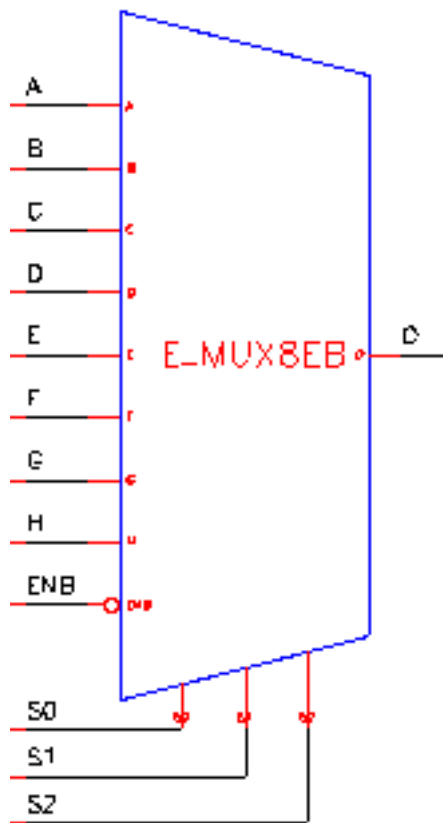
S1	INPUT	select
S2	INPUT	select
OB	OUTPUT	data, inverted

E_MUX8ED : 8-1 multiplexer, active-HIGH enable, complementary outputs



Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
EN	INPUT	output enable, active-HIGH
S0	INPUT	select

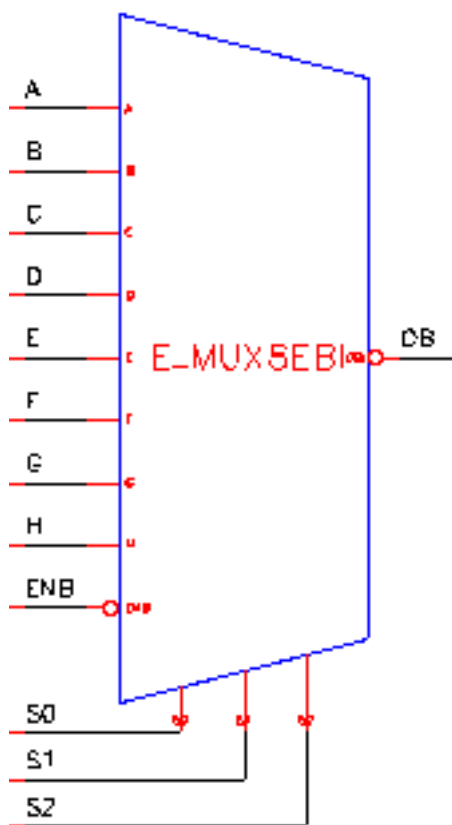
S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

E_MUX8EB : 8-1 multiplexer, active-LOW enable

Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
ENB	INPUT	output enable, active-LOW
S0	INPUT	select

S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data

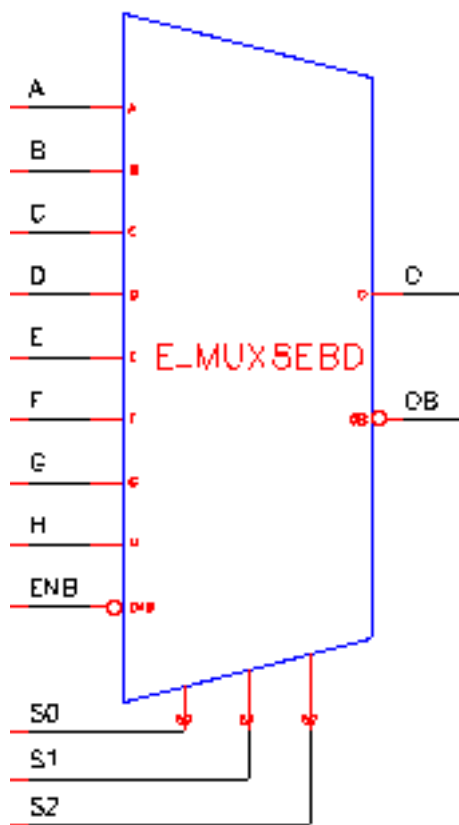
E_MUX8EBI : 8-1 multiplexer, active-LOW enable, inverted output



Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
ENB	INPUT	output enable, active-LOW
S0	INPUT	select

S1	INPUT	select
S2	INPUT	select
OB	OUTPUT	data, inverted

E_MUX8EBD : 8-1 multiplexer, active-LOW enable, complementary outputs



Pin	Direction	Description
A	INPUT	data
B	INPUT	data
C	INPUT	data
D	INPUT	data
E	INPUT	data
F	INPUT	data
G	INPUT	data
H	INPUT	data
ENB	INPUT	output enable, active-LOW
S0	INPUT	select

S1	INPUT	select
S2	INPUT	select
O	OUTPUT	data
OB	OUTPUT	data, inverted

Memories

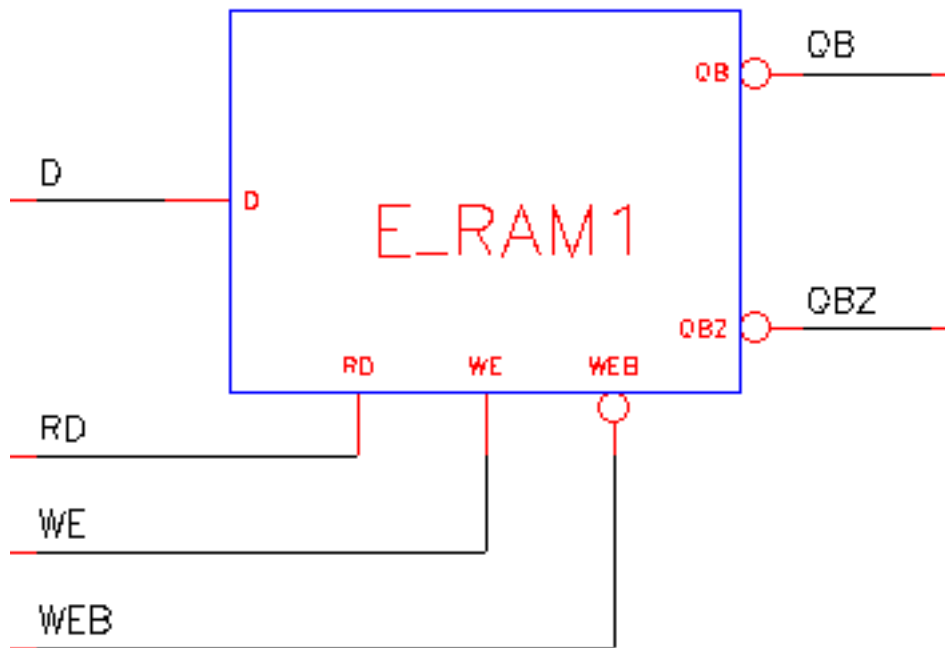
[E_RAM1](#)

[E_RAM16](#)

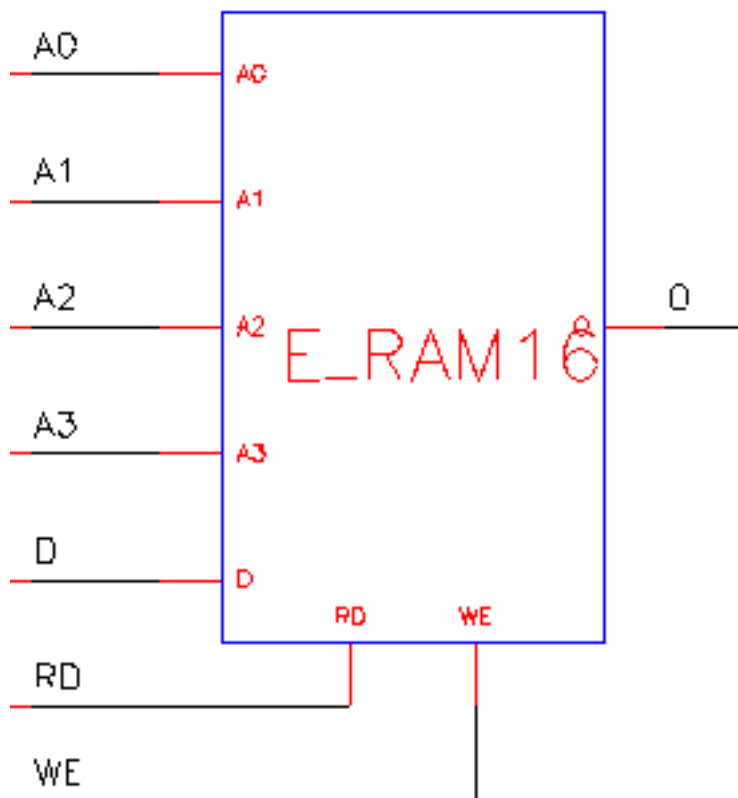
[E_RAM32](#)

[E_ROM16](#)

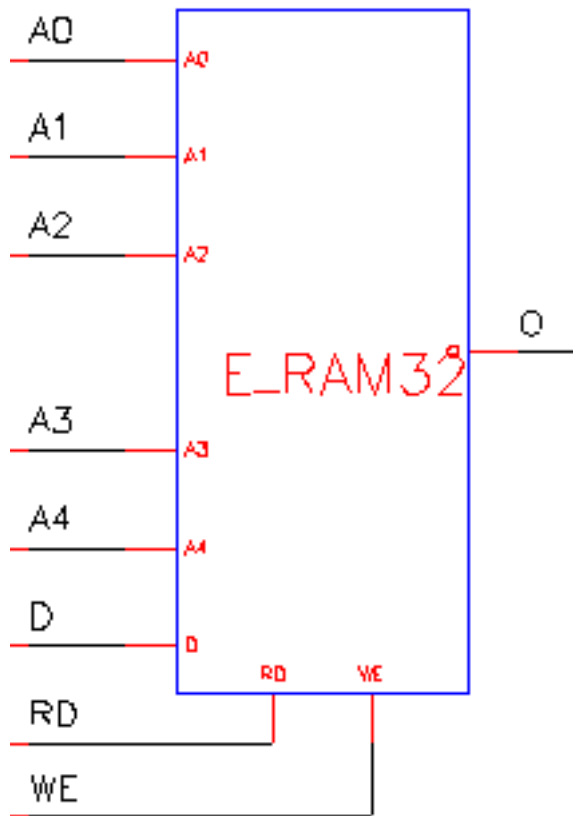
[E_ROM32](#)

E_RAM1 : 1-bit RAM Cell with inverting and tri-state inverting output

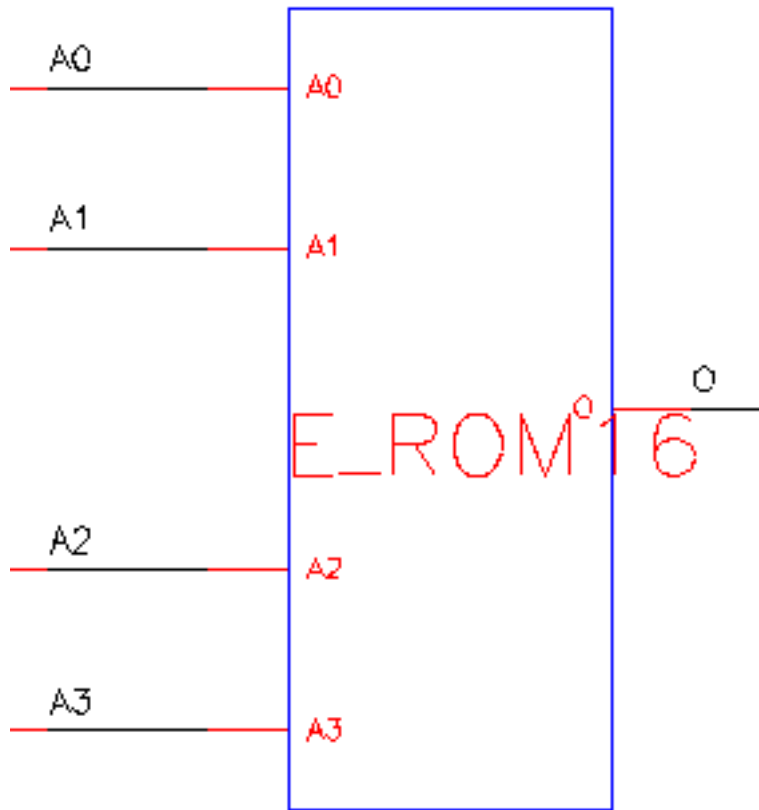
Pin	Direction	Description
D	INPUT	data
RD	INPUT	read-enable
WE	INPUT	write-enable
WEB	INPUT	write-enable, active-LOW
QB	OUTPUT	data
QBZ	OUTPUT	data, tri-state

E_RAM16 : 16-bit RAM Cell

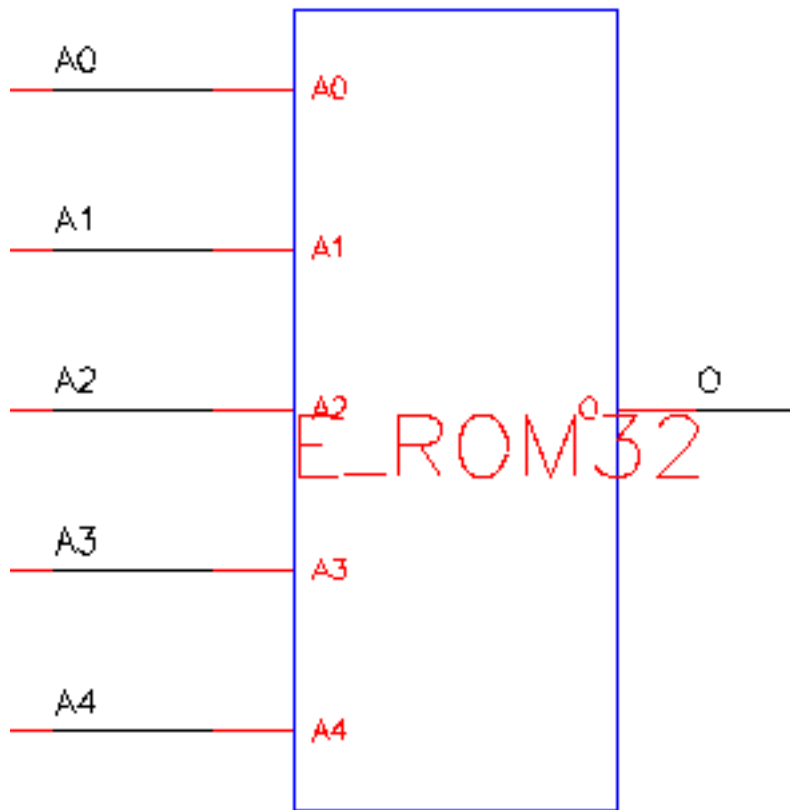
Pin	Direction	Description
A0	INPUT	address, LSB
A1	INPUT	address
A2	INPUT	address
A3	INPUT	address, MSB
D	INPUT	data
RD	INPUT	read-enable
WE	INPUT	write-enable
O	OUTPUT	data

E_RAM32 : 32-bit RAM Cell

Pin	Direction	Description
A0	INPUT	address, LSB
A1	INPUT	address
A2	INPUT	address
A3	INPUT	address
A4	INPUT	address, MSB
D	INPUT	data
RD	INPUT	read-enable
WE	INPUT	write-enable
O	OUTPUT	data

E_ROM16 : 16-bit ROM Cell

Pin	Direction	Description
A0	INPUT	address, LSB
A1	INPUT	address
A2	INPUT	address
A3	INPUT	address, MSB
O	OUTPUT	data

E_ROM32 : 32-bit ROM Cell

Pin	Direction	Description
A0	INPUT	address, LSB
A1	INPUT	address
A2	INPUT	address
A3	INPUT	address
A4	INPUT	address, MSB
O	OUTPUT	data

Latches

[E_LDN](#)

[E_LDNRB](#)

[E_LDNRSB](#)

[E_LDNR](#)

[E_LDNS](#)

[E_LDNRS](#)

[E_LDP](#)

[E_LDPRB](#)

[E_LDPSB](#)

[E_LDPRSB](#)

[E_LDPR](#)

[E_LDPS](#)

[E_LDPRS](#)

[E_LDN4](#)

[E_LDN4R](#)

[E_LDN4RB](#)

[E_LDN4SB](#)

[E_LDN4RSB](#)

[E_LDN4S](#)

[E_LDN4RS](#)

[E_LDN8](#)

[E_LDN8RB](#)

[E_LDN8SB](#)

[E_LDN8RSB](#)

[E_LDN8R](#)

[E_LDN8S](#)

[E_LDN8RS](#)

[E_LDP4](#)

[E_LDP4RB](#)

[E_LDP4SB](#)

[E_LDP4RSB](#)

[E_LDP4R](#)

[E_LDP4S](#)

[E_LDP4RS](#)

[E_LDP8](#)

[E_LDP8RB](#)

[E_LDP8SB](#)

[E_LDP8RSB](#)

[E_LDP8R](#)

[E_LDP8S](#)

[E_LDP8RS](#)

[E_LDNT](#)

[E_LDNTRB](#)

[E_LDNTSB](#)

[E_LDNTRSB](#)

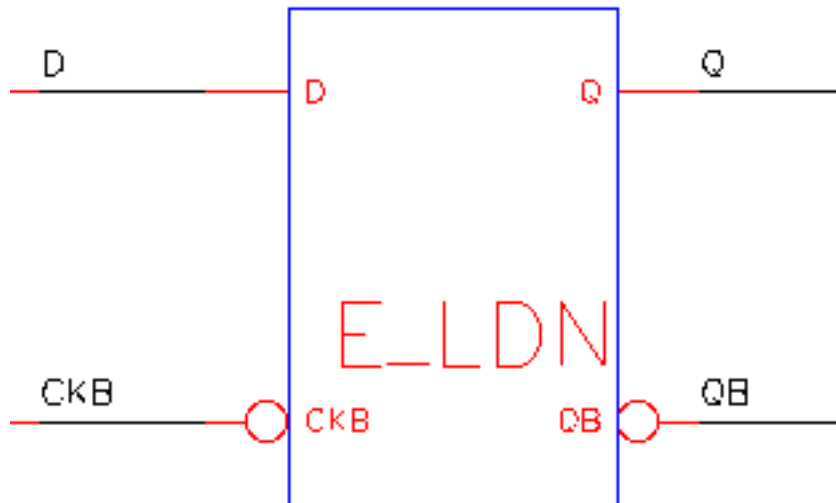
[E_LDNTR](#)

[E_LDNTS](#)

[E_LDNTRS](#)

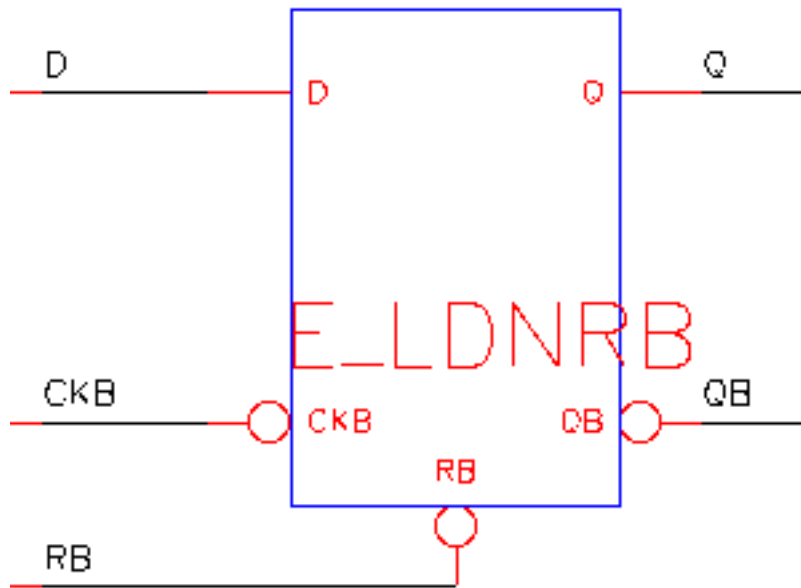
[E LDPT](#)
[E LDPTRB](#)
[E LDPTSB](#)
[E LDPTRSB](#)
[E LDPTR](#)
[E LDPTS](#)
[E LDPTRS](#)
[E LDNM](#)
[E LDNMRB](#)
[E LDNMSB](#)
[E LDNMRSB](#)
[E LDNMR](#)
[E LDNMS](#)
[E LDNMRS](#)
[E LDNW2](#)
[E LDNST](#)
[E LDNSTRB](#)
[E LDNSTRSB](#)
[E LDNSTR](#)
[E LDNSTS](#)
[E LDNSTRS](#)
[E LDPM](#)
[E LDPMRB](#)
[E LDPMSB](#)
[E LDPMRSB](#)
[E LDPMR](#)
[E LDPMS](#)
[E LDPMRS](#)
[E LDPW2](#)
[E LDPST](#)
[E LDPSTRB](#)
[E LDPSTSB](#)
[E LDPSTRSB](#)
[E LDPSTR](#)
[E LDPSTS](#)
[E LDPSTRS](#)
[E DBHD](#)
[E DLHMP](#)
[E DLHMPW2](#)
[E LSRG](#)
[E LSRGB](#)
[E NRL](#)
[E NDL](#)

E_LDN : D-Latch, active-LOW enable

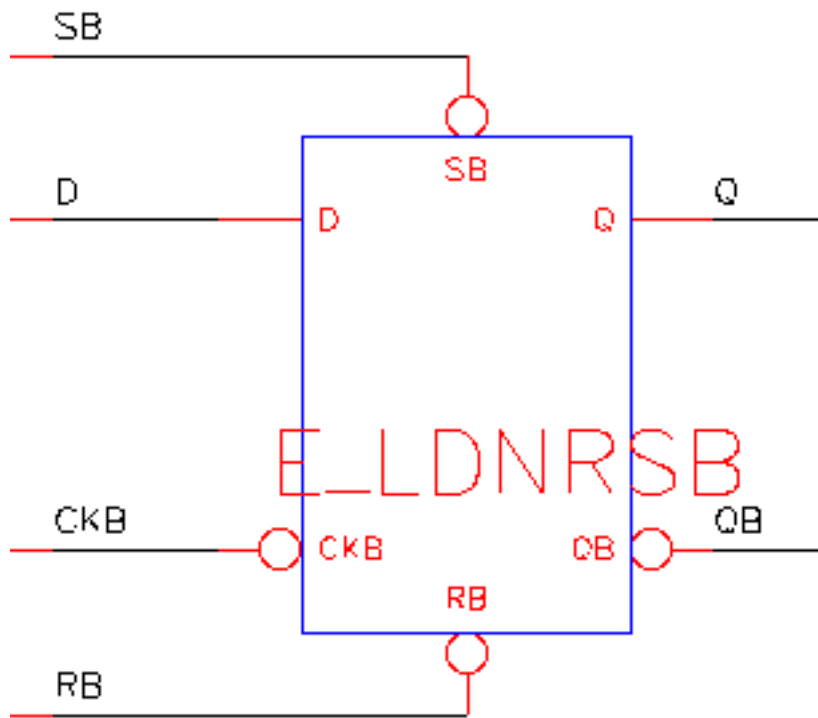


Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNRB : D-Latch, active-LOW enable, active-LOW reset

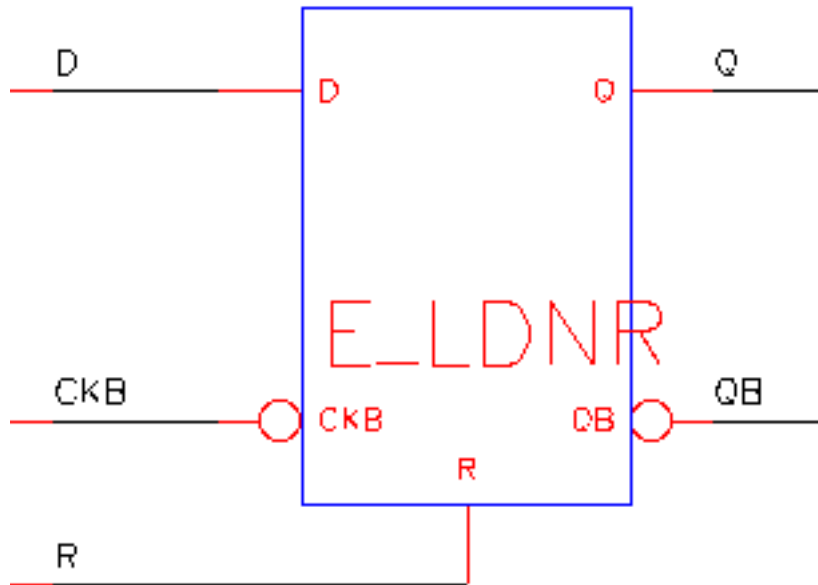


Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNRSB : D-Latch, active-LOW enable, active-LOW set & reset

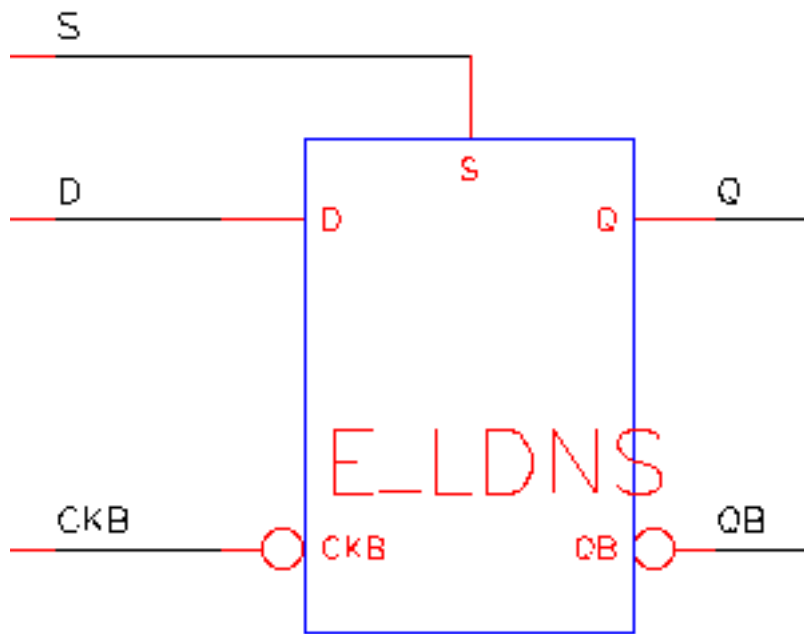
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNR : D-Latch, active-LOW enable, active-HIGH reset

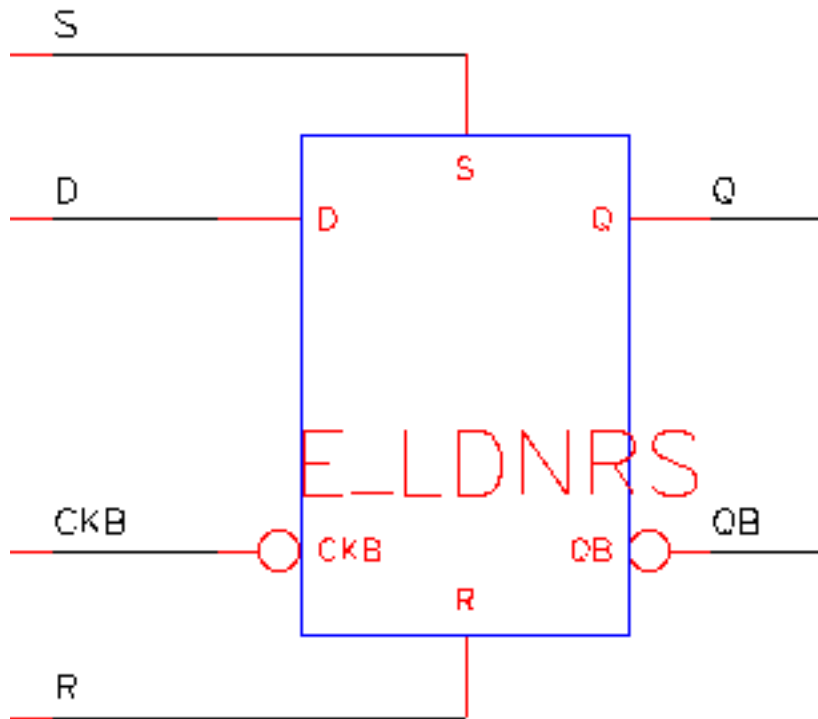


Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNS : D-Latch, active-LOW enable, active-HIGH set

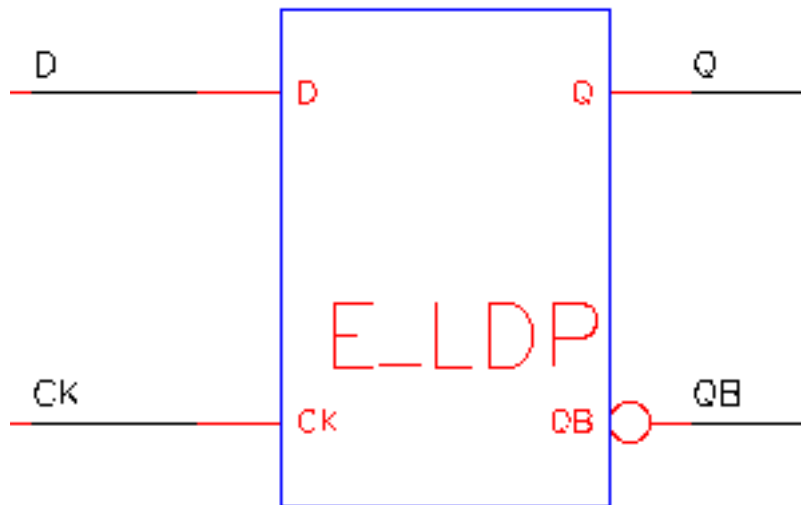


Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNRS : D-Latch, active-LOW enable, active-HIGH set & reset

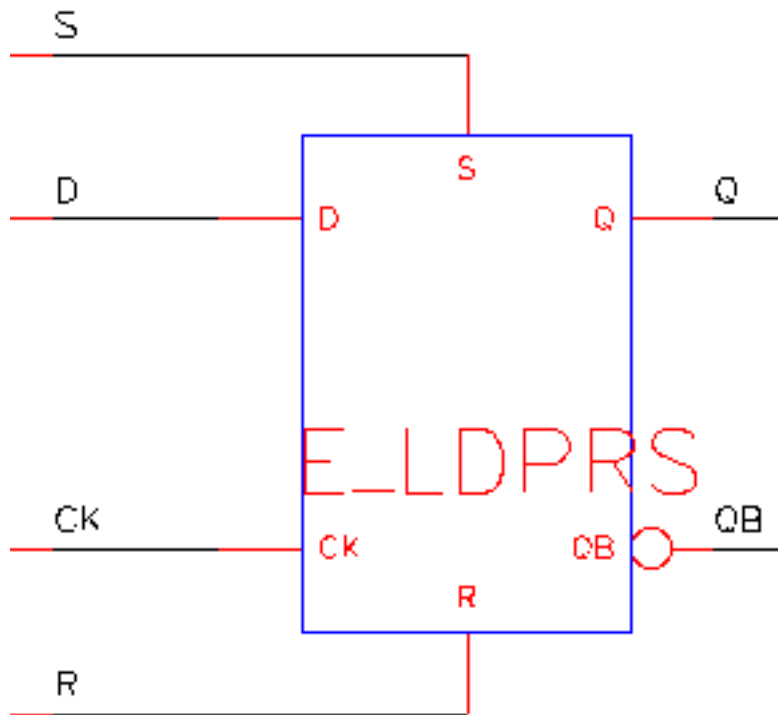
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDP : D-Latch, active-HIGH enable



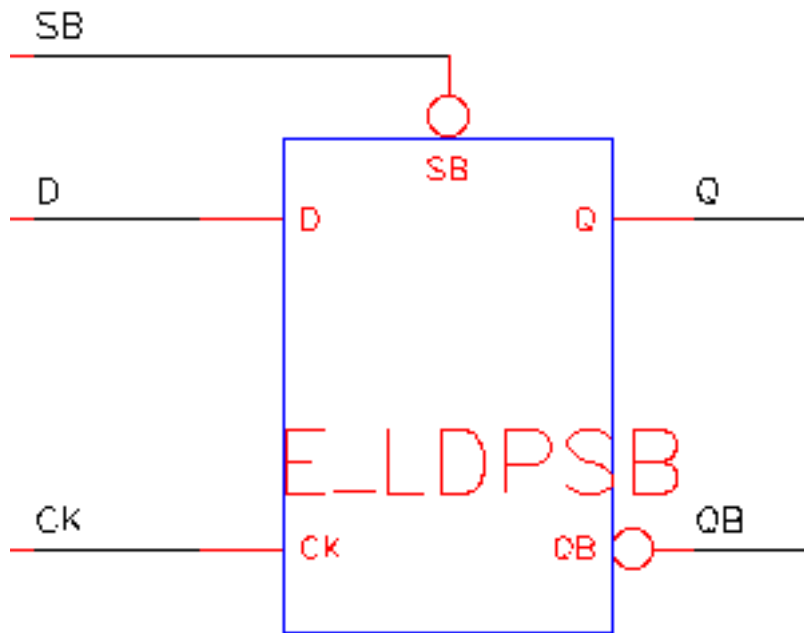
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPRB "": D-Latch, active-HIGH enable, active-LOW reset



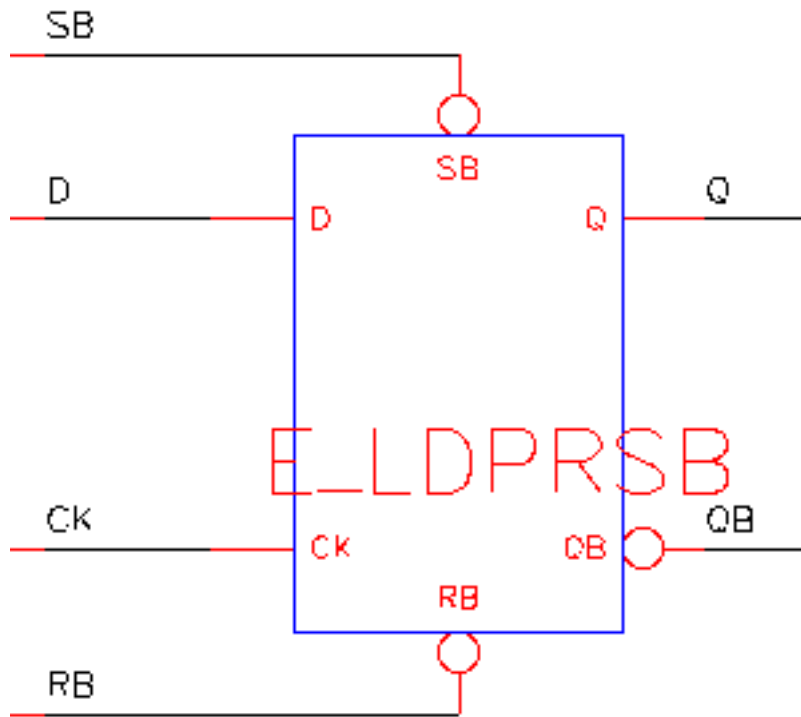
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPSB : D-Latch, active-HIGH enable, active-LOW set



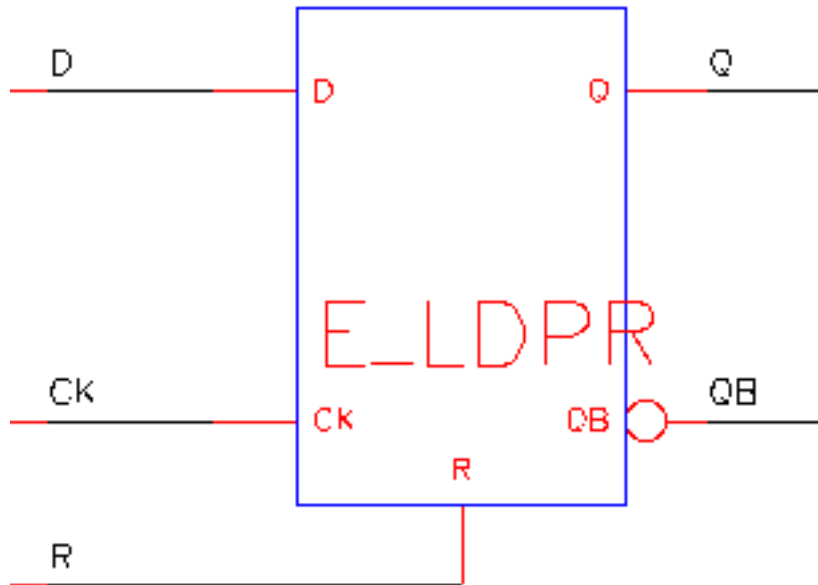
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPRSB : D-Latch, active-HIGH enable, active-LOW set & reset



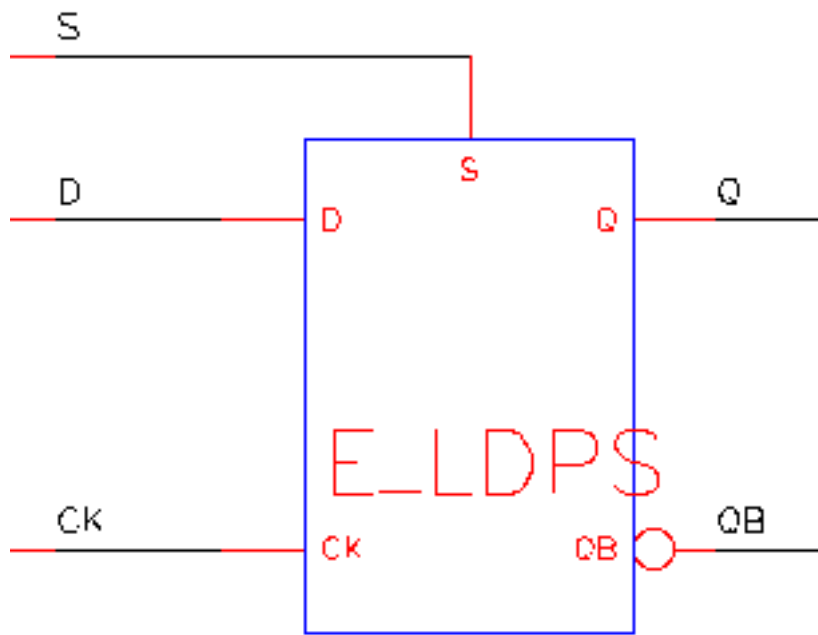
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPR : D-Latch, active-HIGH enable, active-HIGH reset

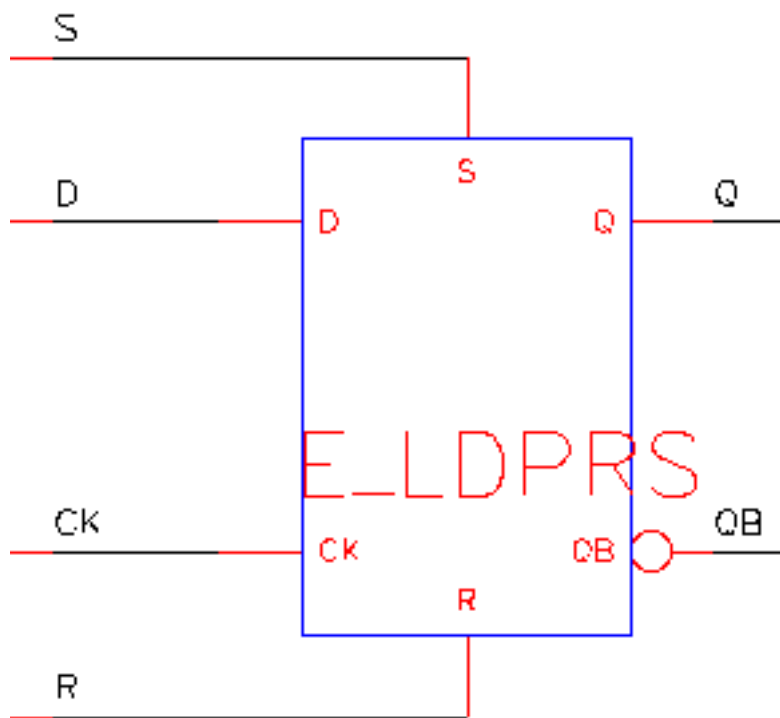


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
R	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPS : D-Latch, active-HIGH enable, active-HIGH set

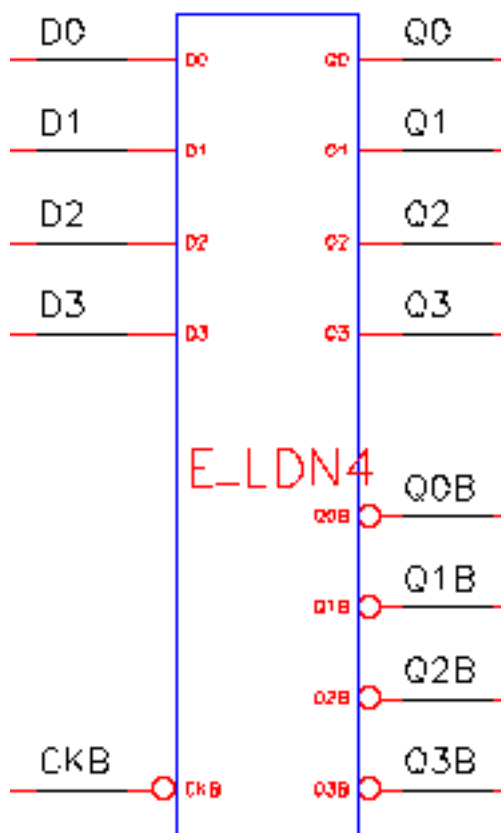


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPRS : D-L

Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

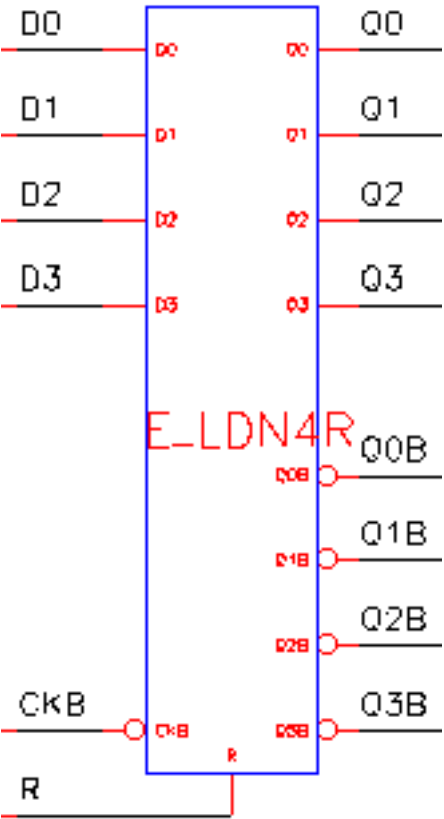
E_LDN4 : 4 bits D-Latch, active-LOW enable



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted

Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

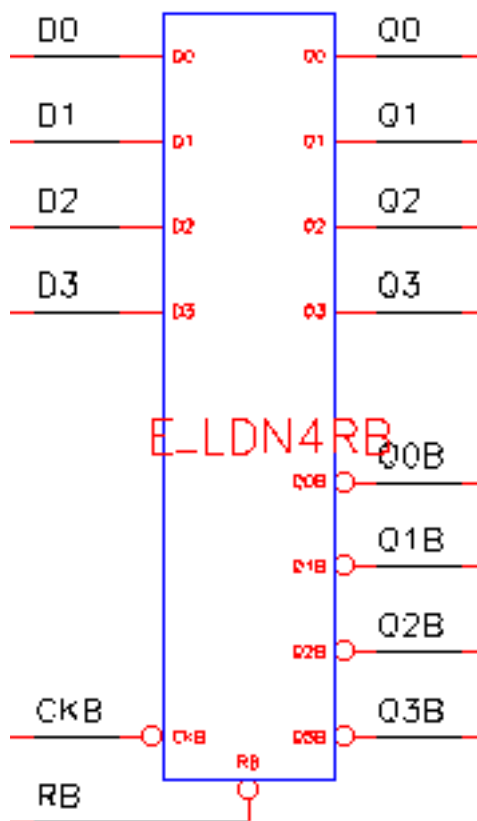
E_LDN4R : 4 bits D-Latch, active-LOW enable, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
R	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

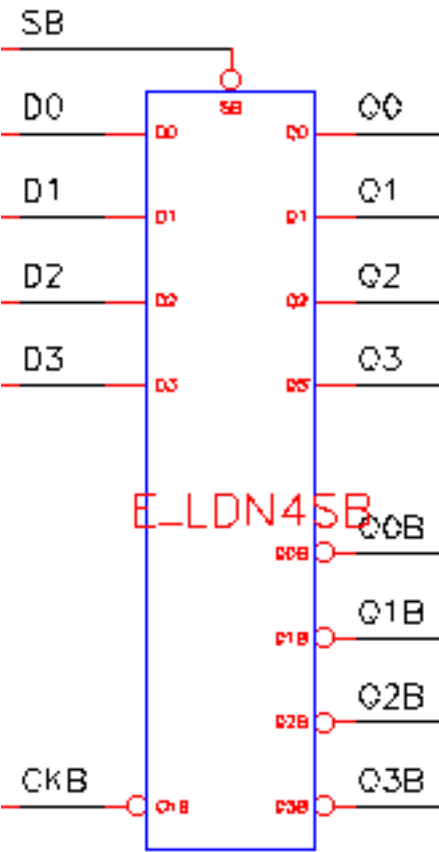
E_LDN4RB : 4 bits D-Latch, active-LOW enable, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

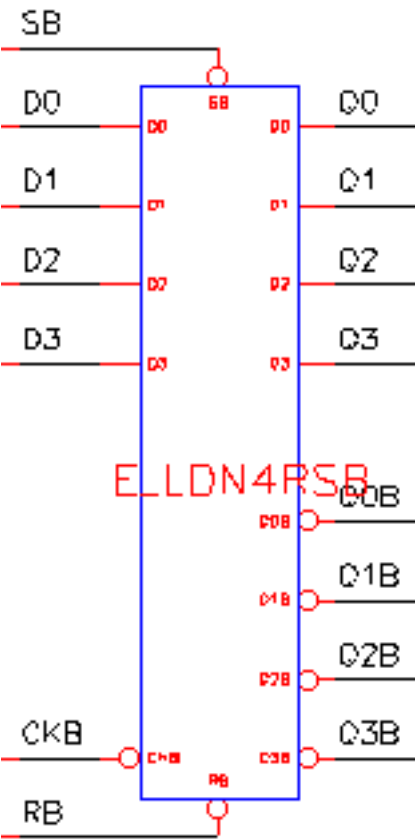
E_LDN4SB : 4 bits D-Latch, active-LOW enable, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

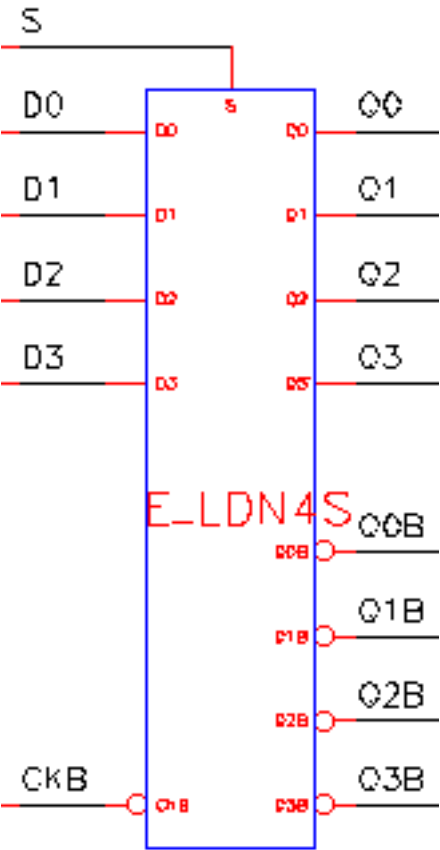
E_LDN4RSB : 4 bits D-Latch, active-LOW enable, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

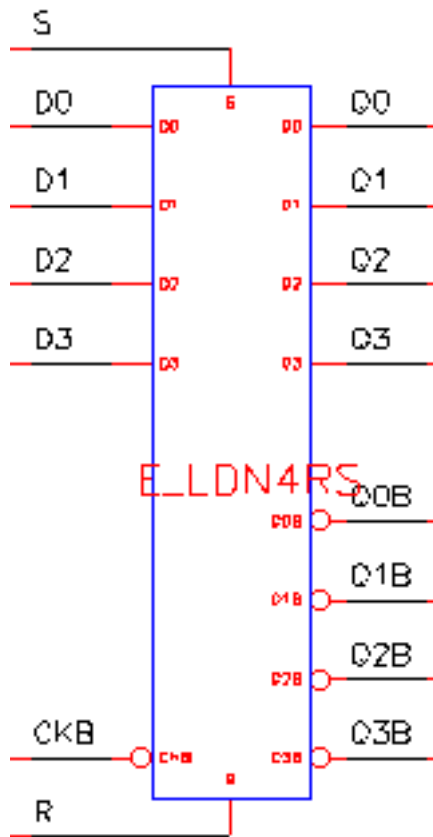
E_LDN4S : 4 bits D-Latch, active-LOW enable, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

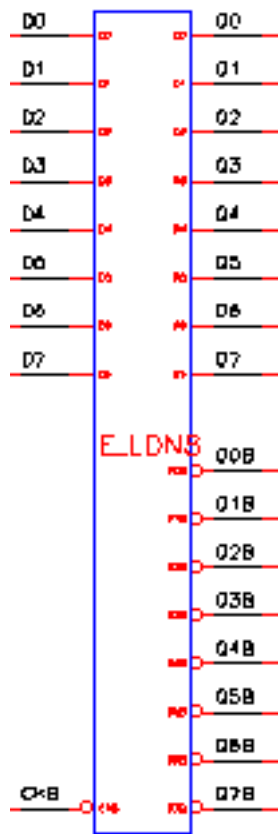
E_LDN4RS : 4 bits D-Latch, active-LOW enable, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, active-LOW
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_LDN8 : 8 bits D-Latch, active-LOW enable



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
Q0	OUTPUT	data

Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDN8RB : 8 bits D-Latch, active-LOW enable, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
RB	INPUT	async reset, active-LOW

Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDN8SB : 8 bits D-Latch, active-LOW enable, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
SB	INPUT	async set, active-LOW

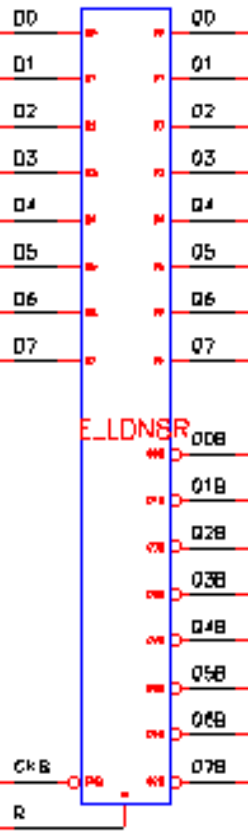
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDN8RSB : 8 bits D-Latch, active-LOW enable, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
RB	INPUT	async reset, active-LOW

E_LDN8R : 8 bits D-Latch, active-LOW enable, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
R	INPUT	async reset, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data

Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

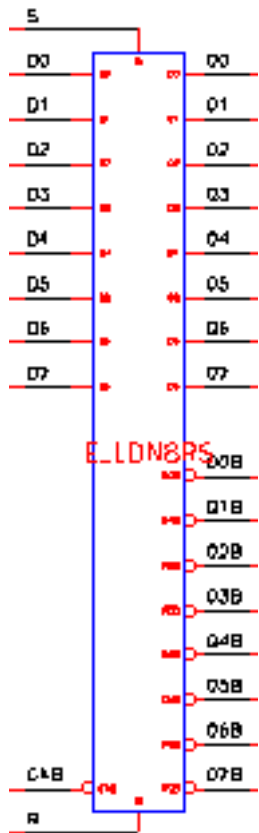
E_LDN8S : 8 bits D-Latch, active-LOW enable, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
S	INPUT	async set, active-HIGH

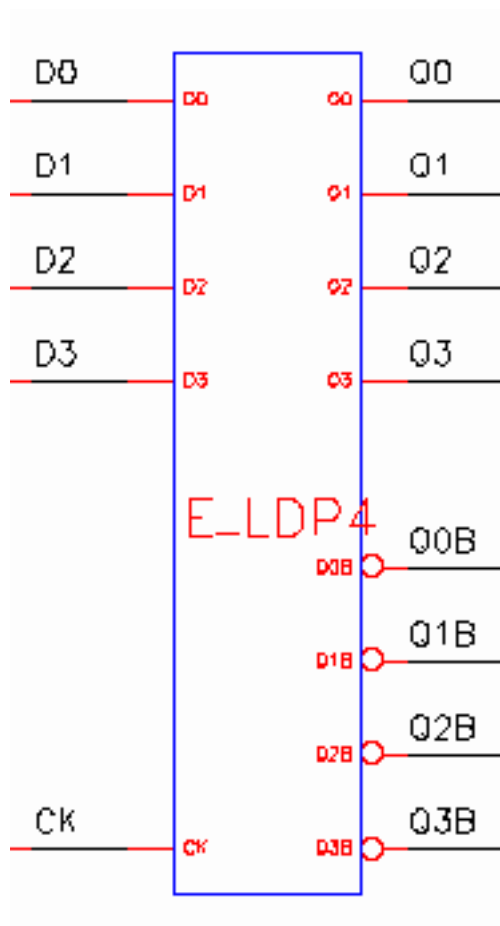
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDN8RS : 8 bits D-Latch, active-LOW enable, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, active-LOW
R	INPUT	async reset, active-HIGH

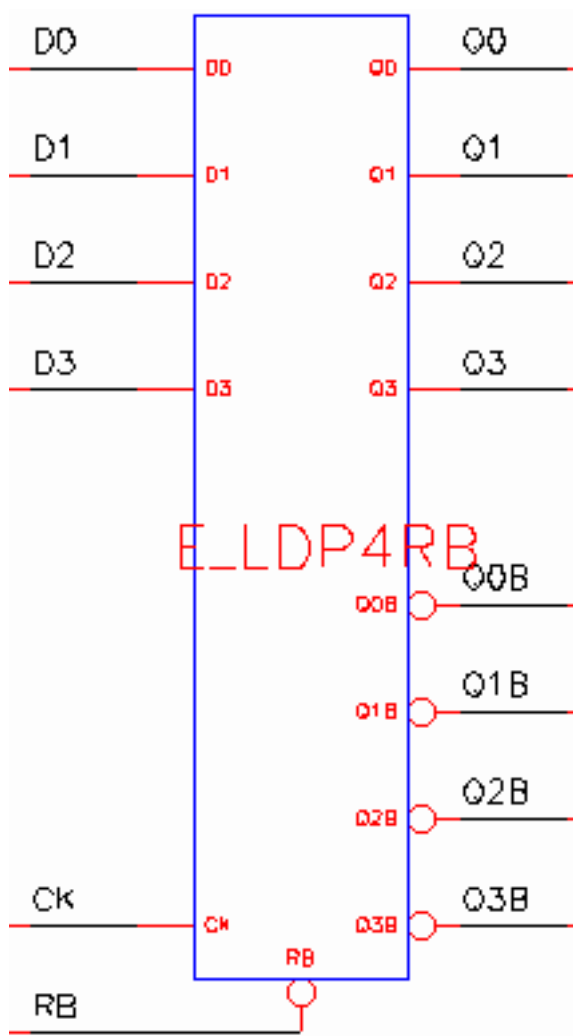
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDP4 : 4 bits D-Latch, active-HIGH enable

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

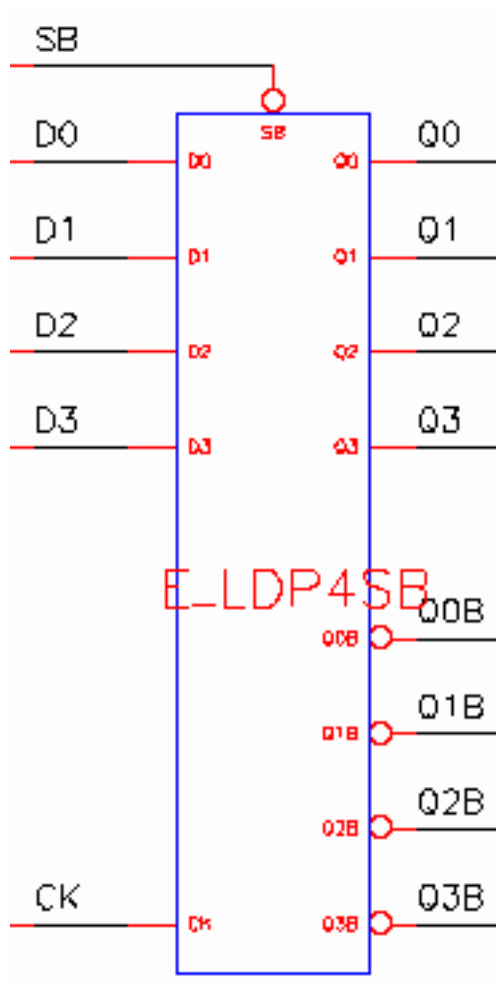
E_LDP4RB : 4 bits D-Latch, active-HIGH enable, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data

Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

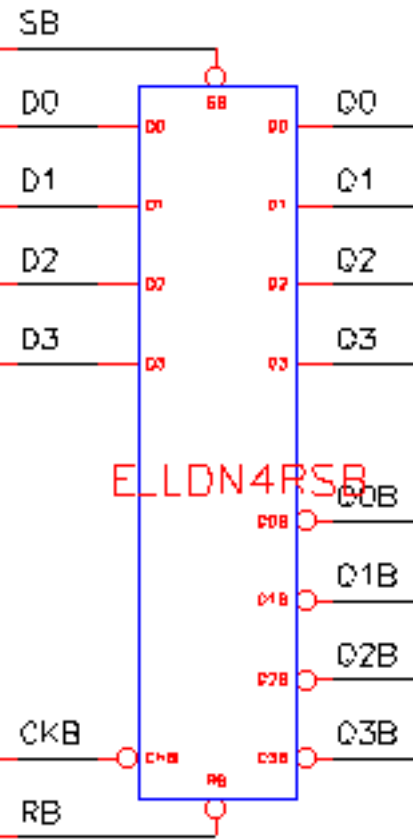
E_LDP4SB : 4 bits D-Latch, active-HIGH enable, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

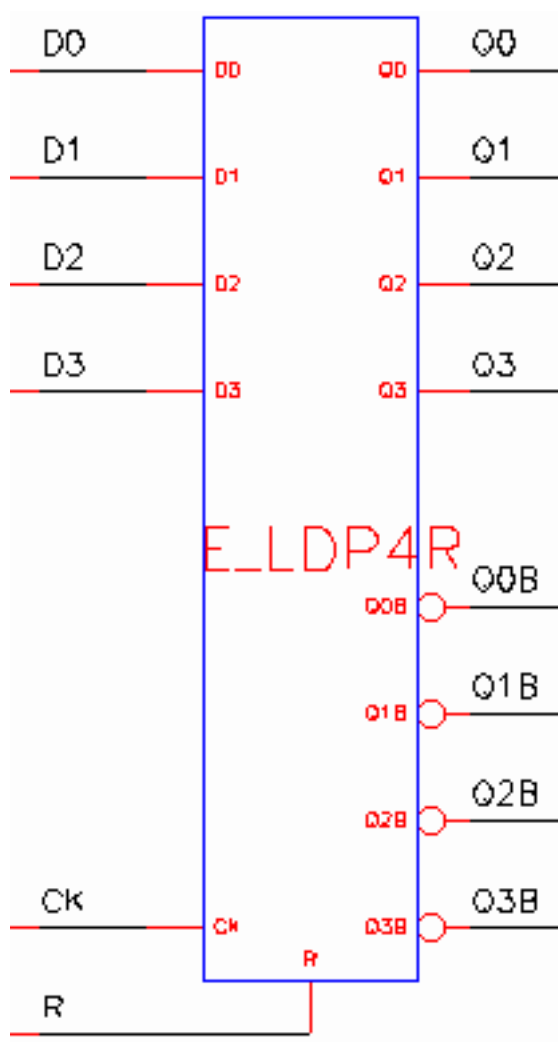
E_LDP4RSB : 4 bits D-Latch, active-HIGH enable, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

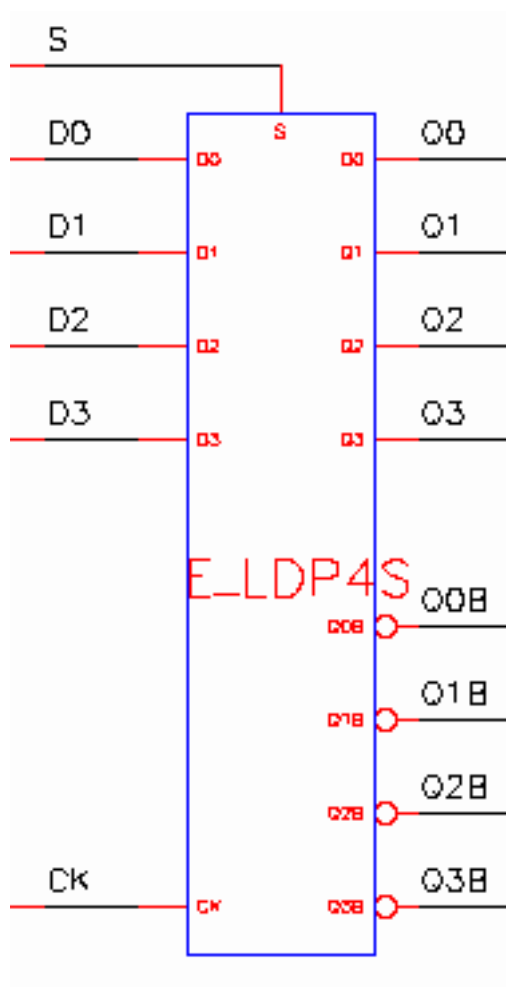
E_LDP4R : 4 bits D-Latch, active-HIGH enable, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
R	INPUT	async reset, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data

Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

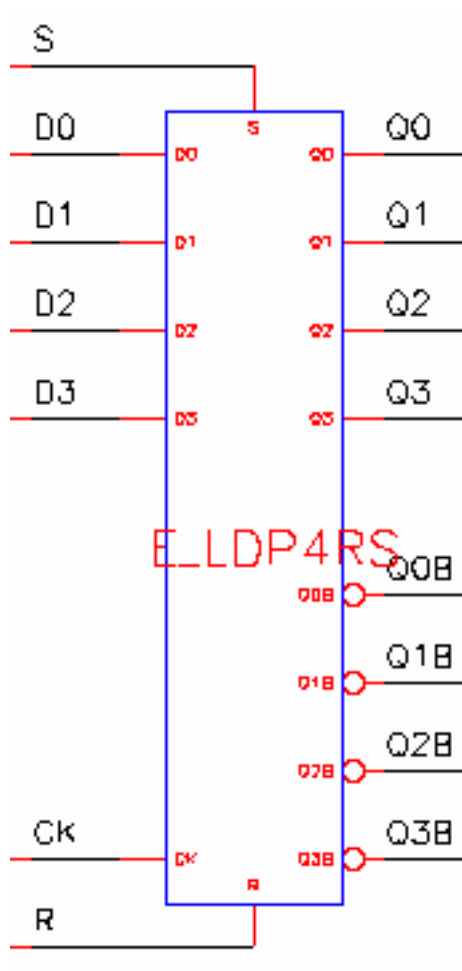
E_LDP4S : 4 bits D-Latch, active-HIGH enable, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

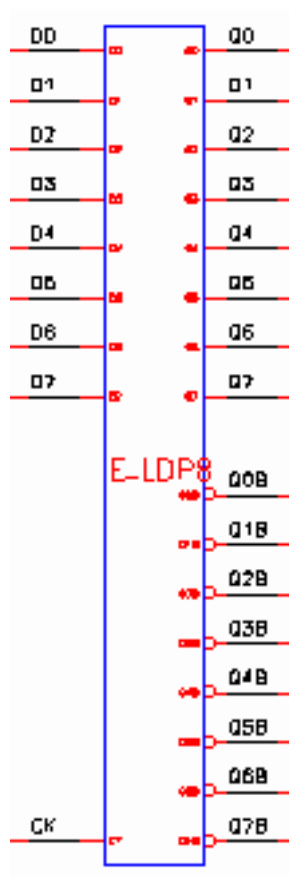
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_LDP4RS : 4 bits D-Latch, active-HIGH enable, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, active-HIGH
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data

Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_LDP8 : 8 bits D-Latch, active-HIGH enable

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CK	INPUT	clock, active-HIGH
Q0	OUTPUT	data

Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDP8RB : 8 bits D-Latch, active-HIGH enable, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data

CK	INPUT	clock, active-HIGH
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDP8SB : 8 bits D-Latch, active-HIGH enable, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data

CK	INPUT	clock, active-HIGH
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDP8RSB : 8 bits D-Latch, active-HIGH enable, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data

CK	INPUT	clock, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

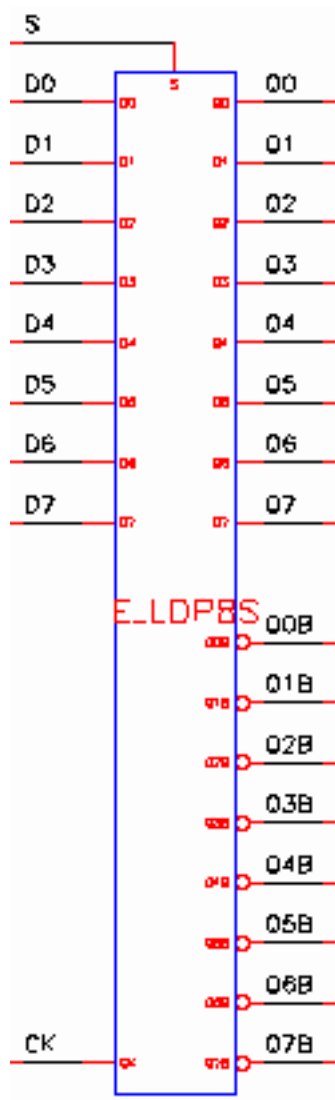
E_LDP8R : 8 bits D-Latch, active-HIGH enable, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, active-HIGH
R	INPUT	async reset, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDP8S : 8 bits D-Latch, active-HIGH enable, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

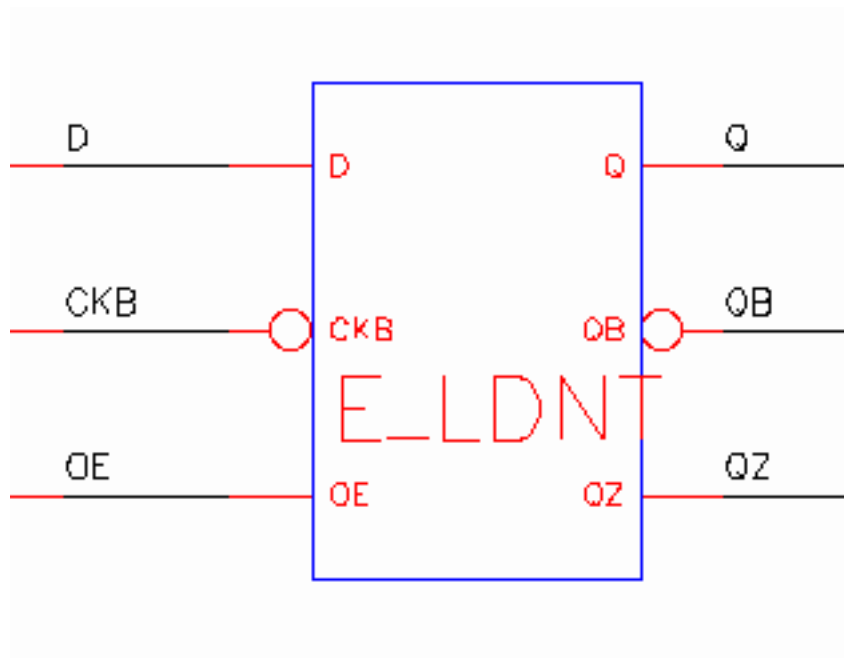
E_LDP8RS : 8 bits D-Latch, active-HIGH enable, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

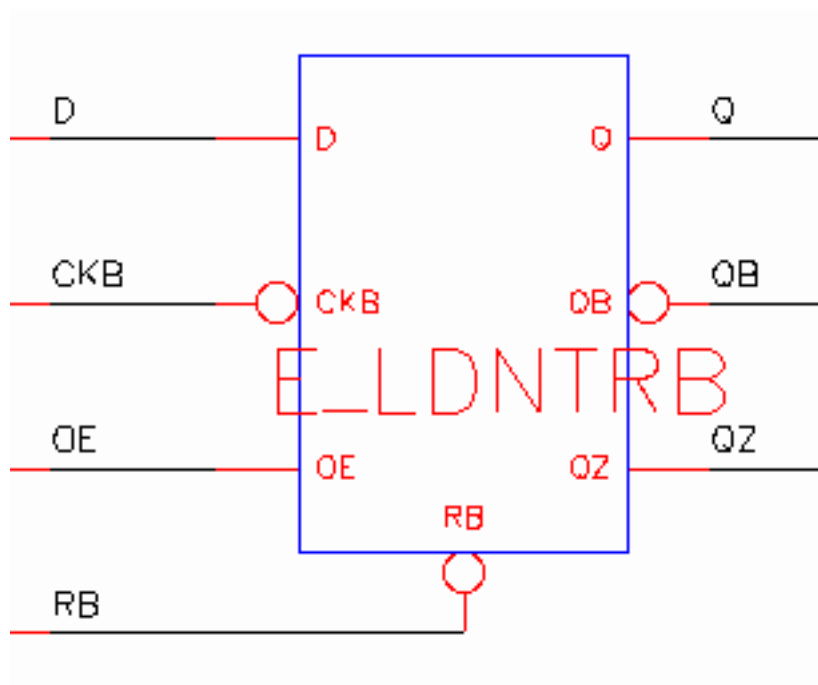
D6	INPUT	data
D7	INPUT	data
CK	INPUT	clock, active-HIGH
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_LDNT : D-Latch, active-LOW enable, active-HIGH enable tri-state output



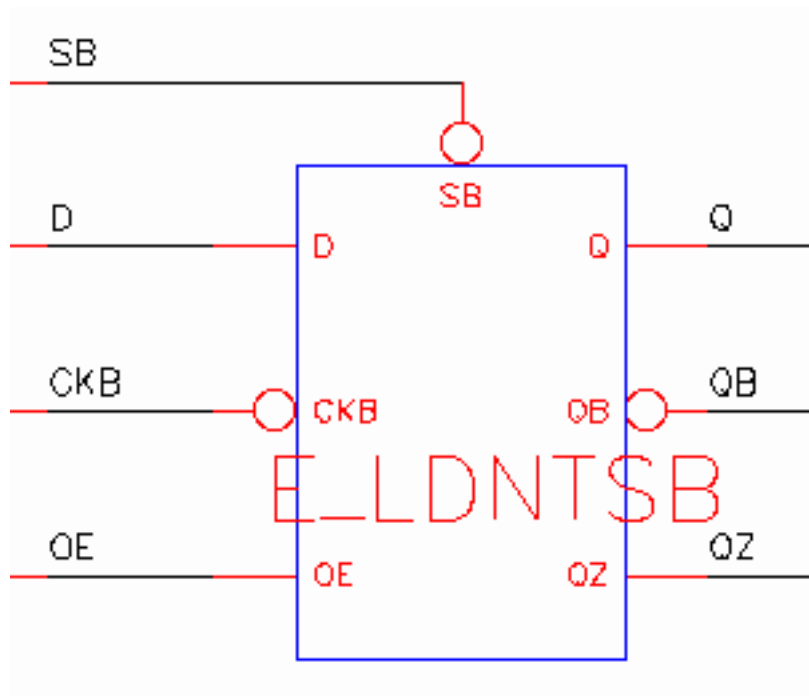
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNTRB : D-Latch, active-LOW enable, active-LOW reset, active-HIGH enable tri-state output



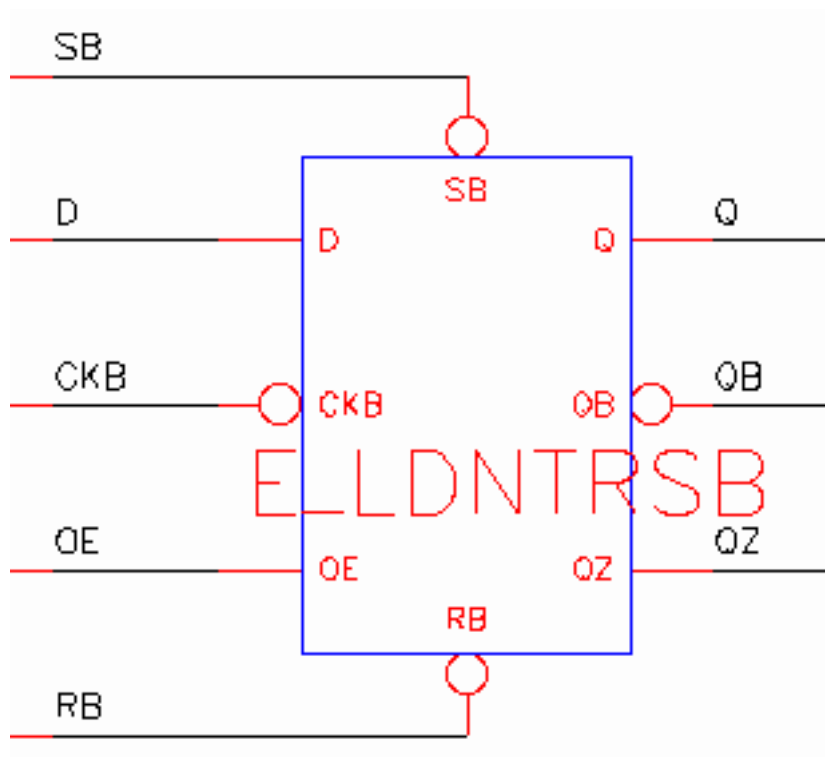
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNTSB : D-Latch, active-LOW enable, active-LOW set, active-HIGH enable tri-state output



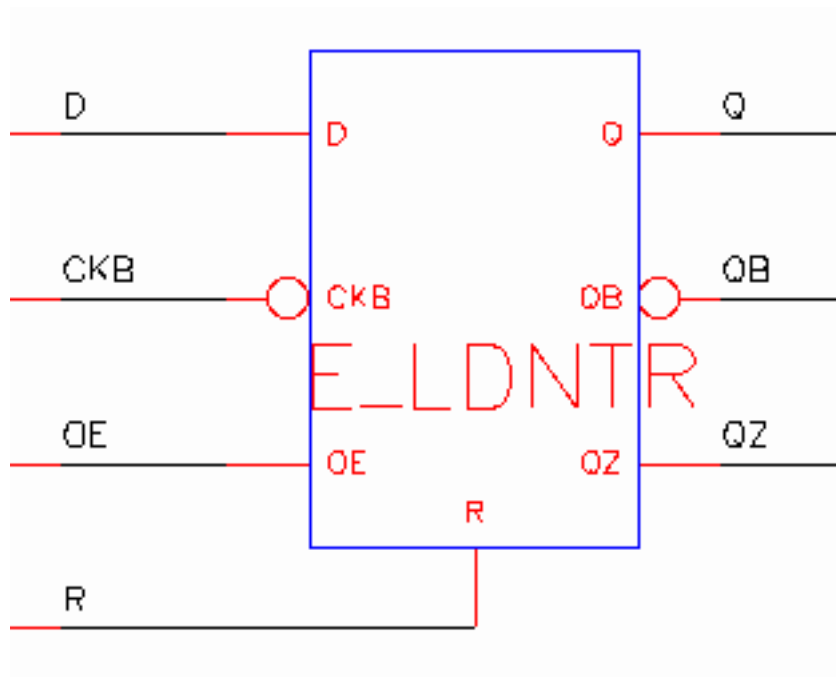
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNTRSB : D-Latch, active-LOW enable, active-LOW set & reset, active-HIGH enable tri-state output



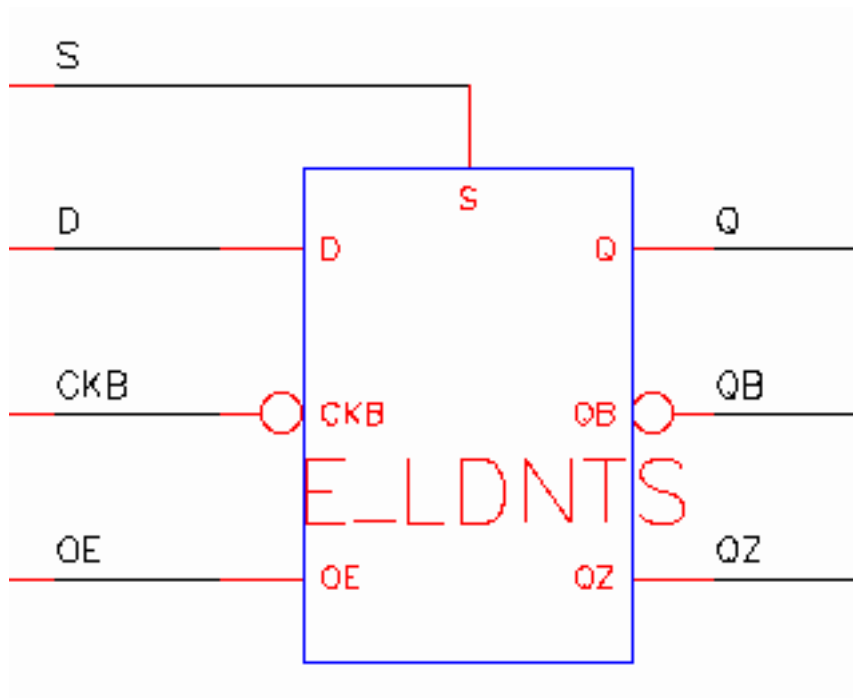
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNTNTR : D-Latch, active-LOW enable, active-HIGH reset, active-HIGH enable tri-state output



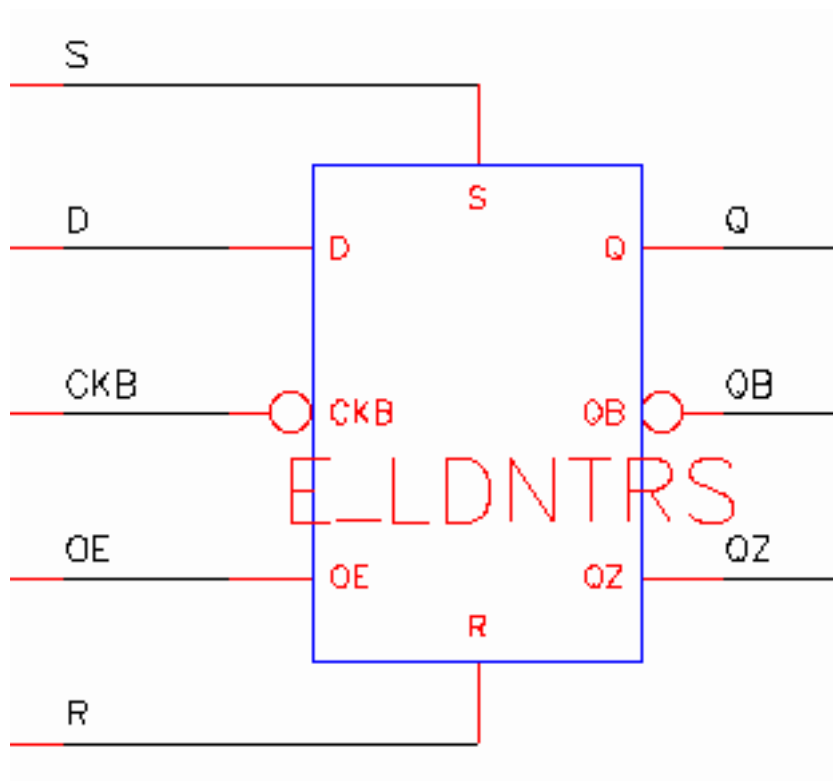
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNTS : D-Latch, active-LOW enable, active-HIGH set, active-HIGH enable tri-state output



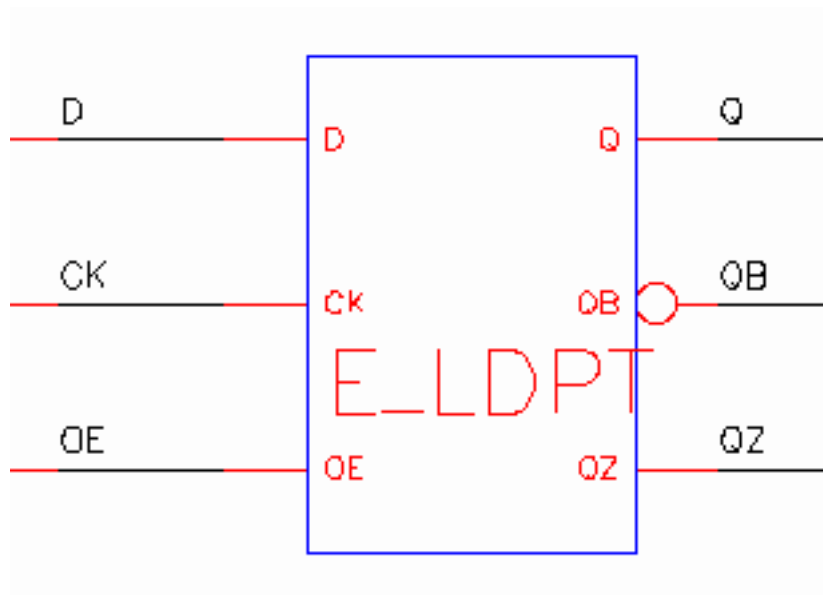
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNTRS : D-Latch, active-LOW enable, active-HIGH set & reset, active-HIGH enable tri-state output



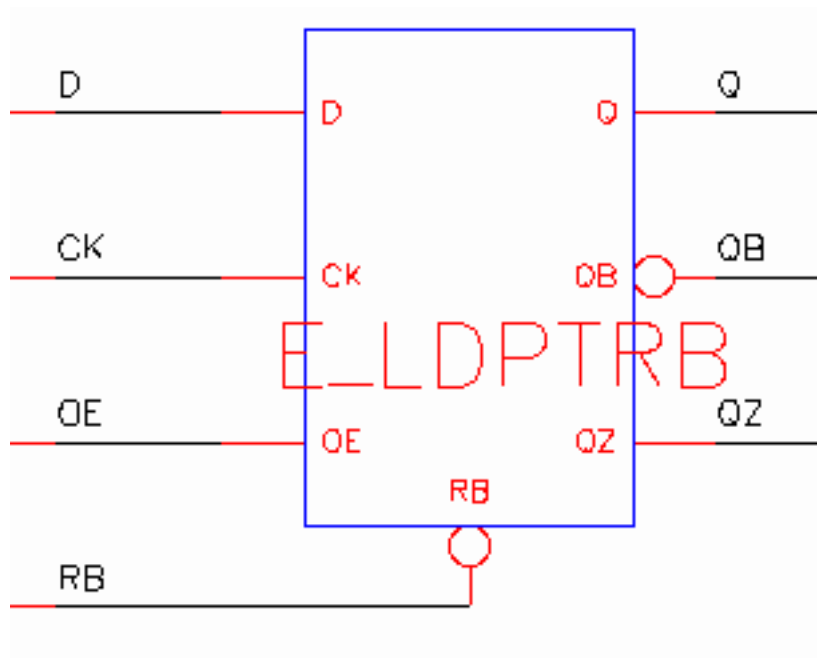
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	enable, active-LOW
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPT : D-Latch, active-HIGH enable, active-HIGH enable tri-state output



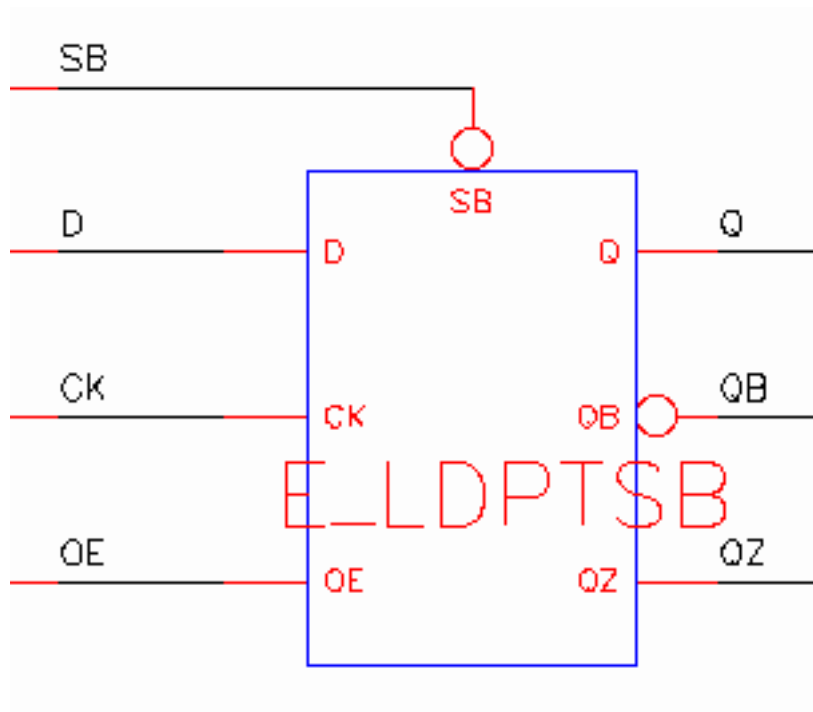
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPTRB : D-Latch, active-HIGH enable, active-LOW reset, active-HIGH enable tri-state output



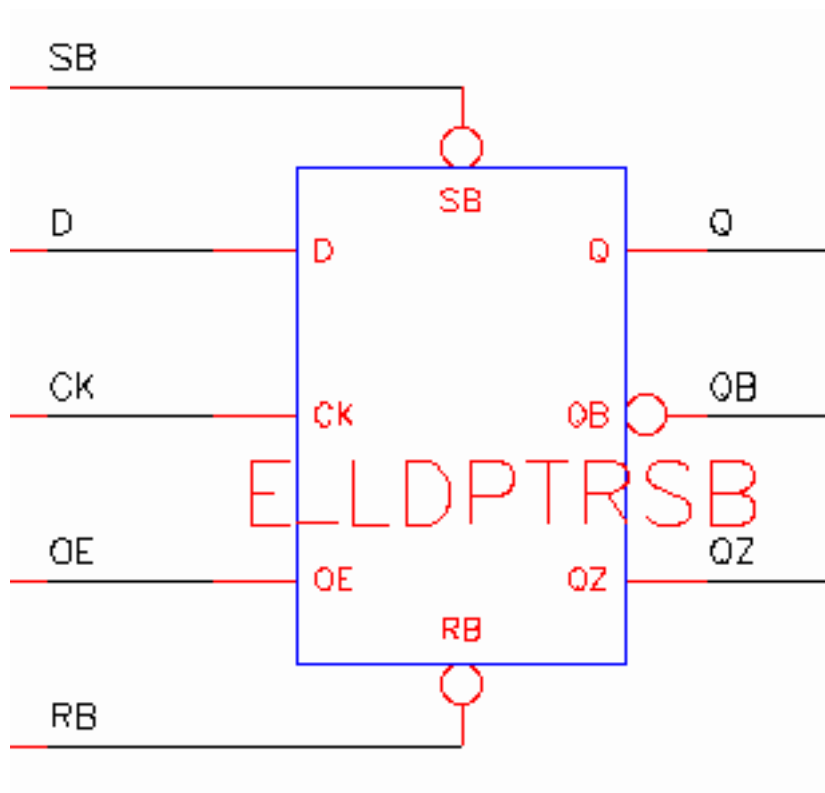
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPTSB : D-Latch, active-HIGH enable, active-LOW set, active-HIGH enable tri-state output



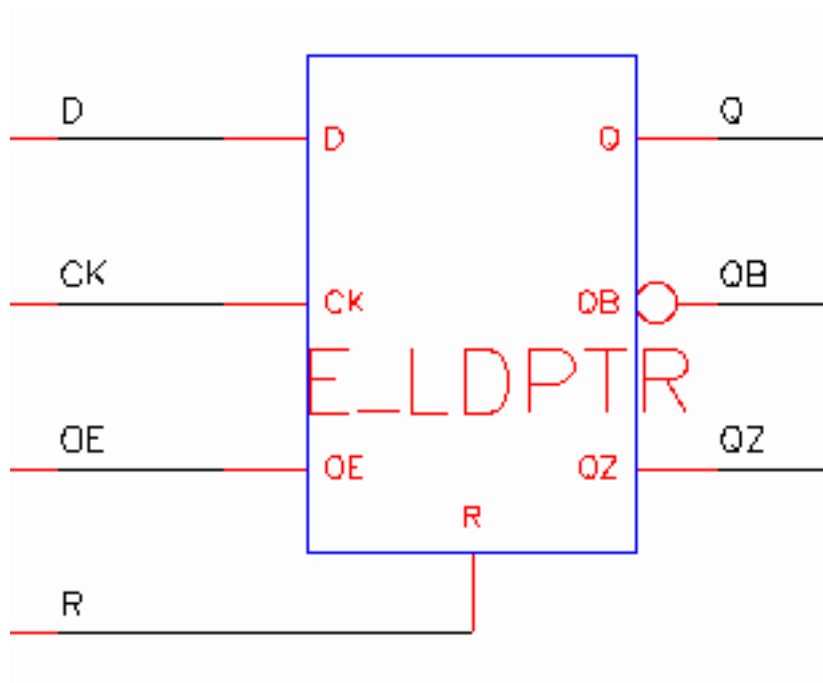
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPTRSB : D-Latch, active-HIGH enable, active-LOW set & reset, active-HIGH enable tri-state output



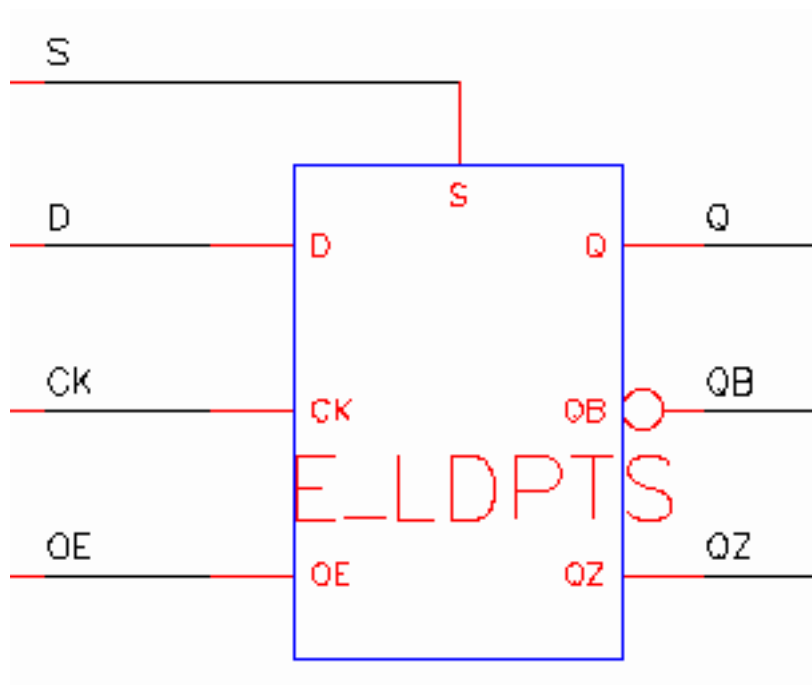
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPTTR : D-Latch, active-HIGH enable, active-HIGH reset, active-HIGH enable tri-state output



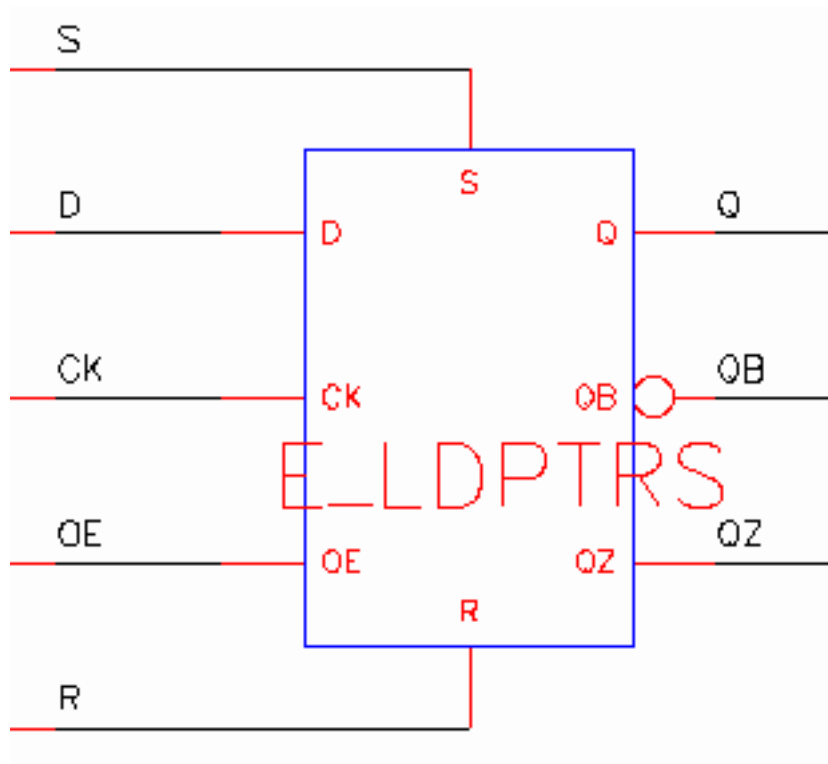
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPTS : D-Latch, active-HIGH enable, active-HIGH set, active-HIGH enable tri-state output



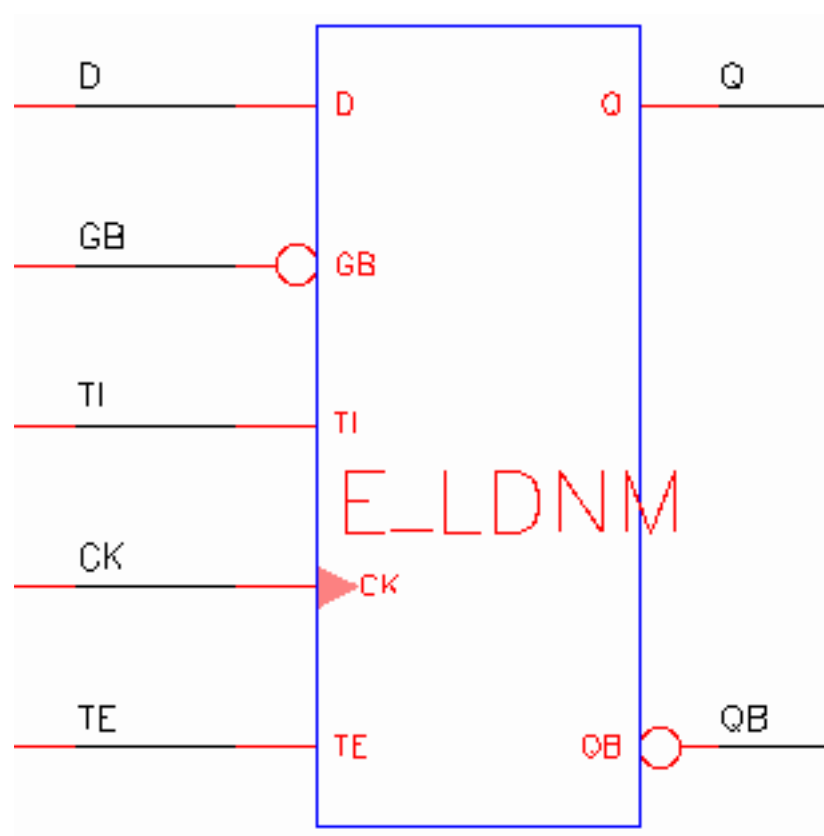
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDPTRS : D-Latch, active-HIGH enable, active-HIGH set & reset, active-HIGH enable tri-state output



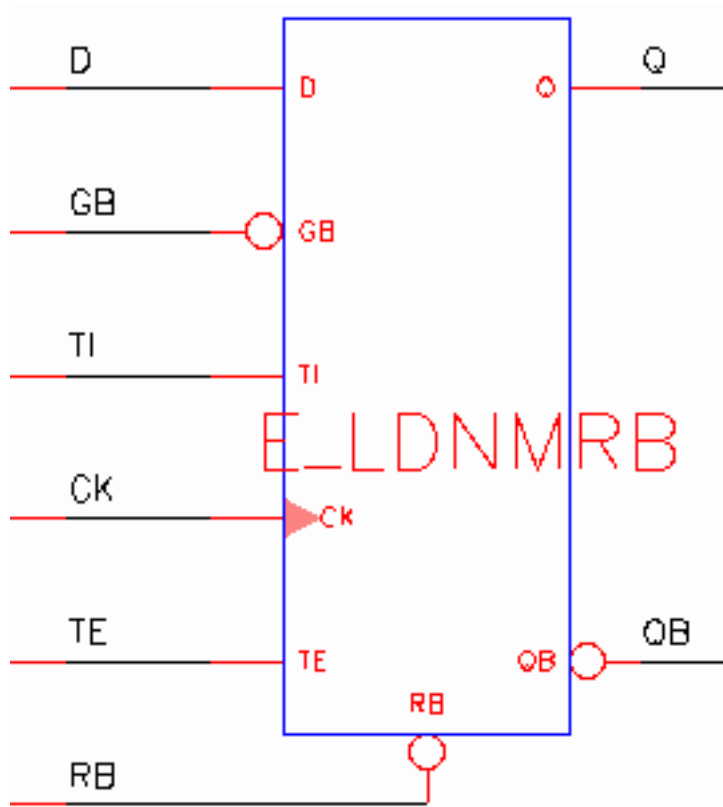
Pin	Direction	Description
D	INPUT	data
CK	INPUT	enable, active HIGH
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_LDNM : D-Latch, active-LOW enable, multiplexed with D-type flip-flop, positive-edge-triggered



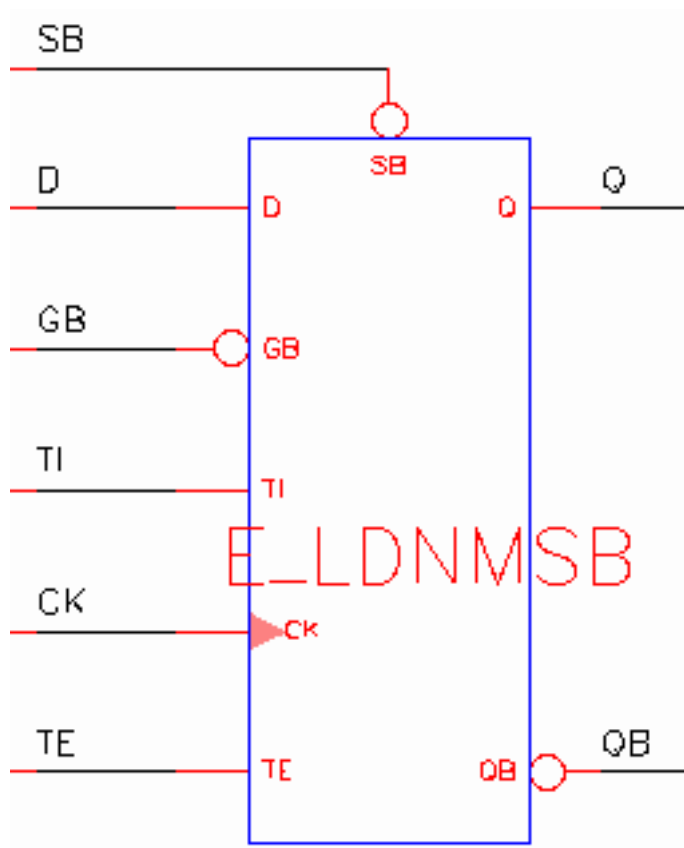
Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNMRB : D-Latch, active-LOW enable, active-LOW reset, multiplexed with D-type flip-flop, positive-edge-triggered



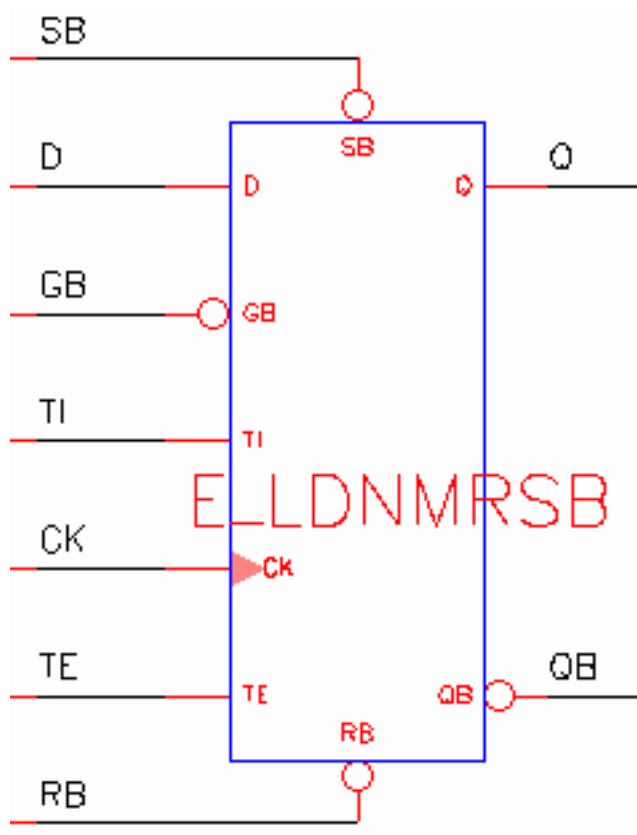
Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNMSB : D-Latch, active-LOW enable, active-LOW set, multiplexed with D-type flip-flop, positive-edge-triggered



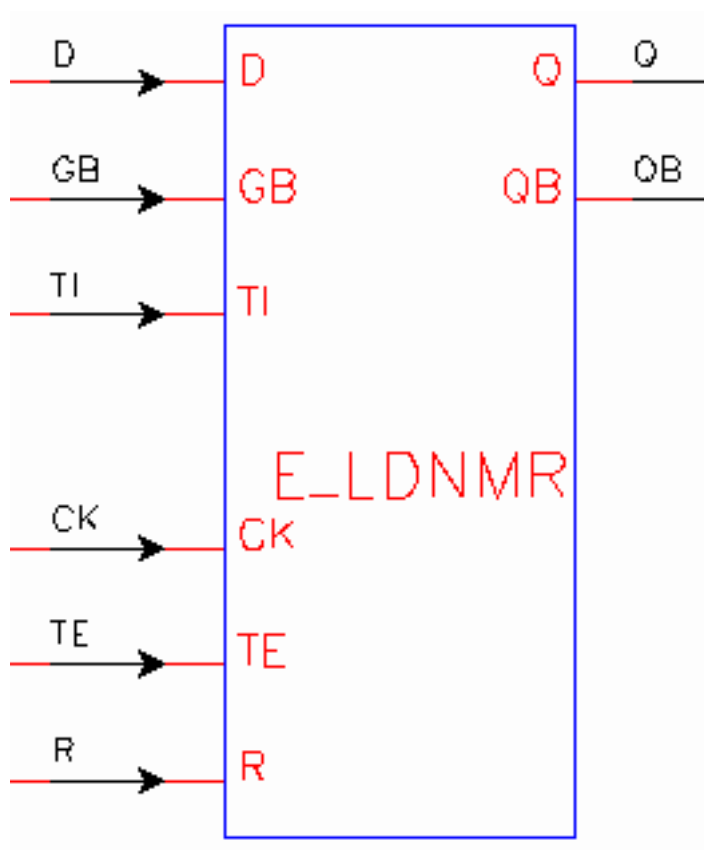
Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNMRSB : D-Latch, active-LOW enable, active-LOW set & reset, multiplexed with D-type flip-flop, positive-edge-triggered



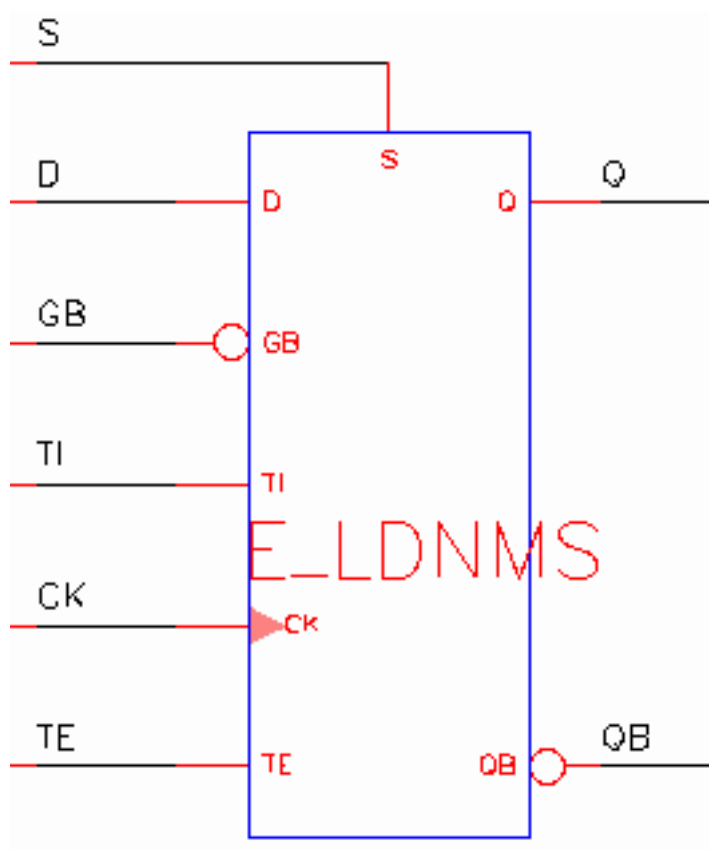
Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNMR : D-Latch, active-LOW enable, active-HIGH reset, multiplexed with D-type flip-flop, positive-edge-triggered



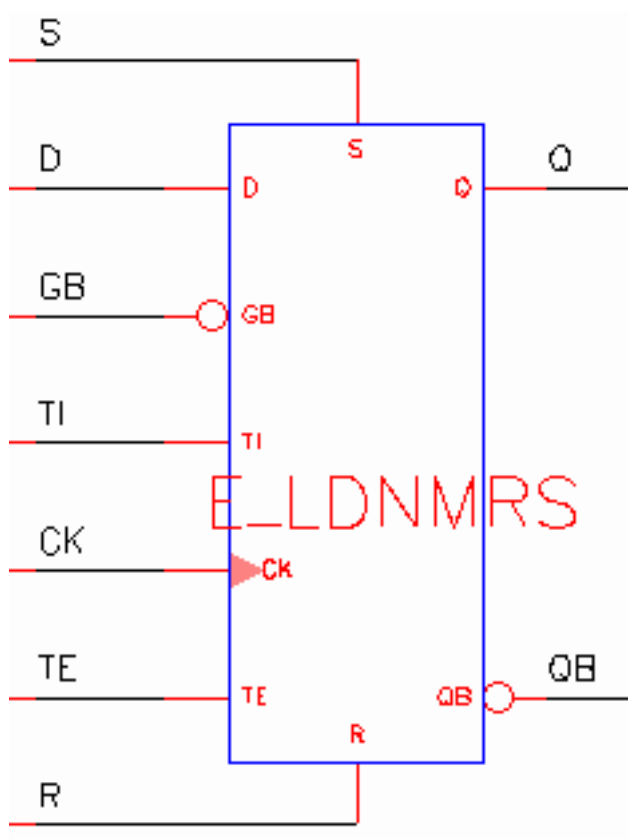
Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNMS : D-Latch, active-LOW enable, active-HIGH set, multiplexed with D-type flip-flop, positive-edge-triggered



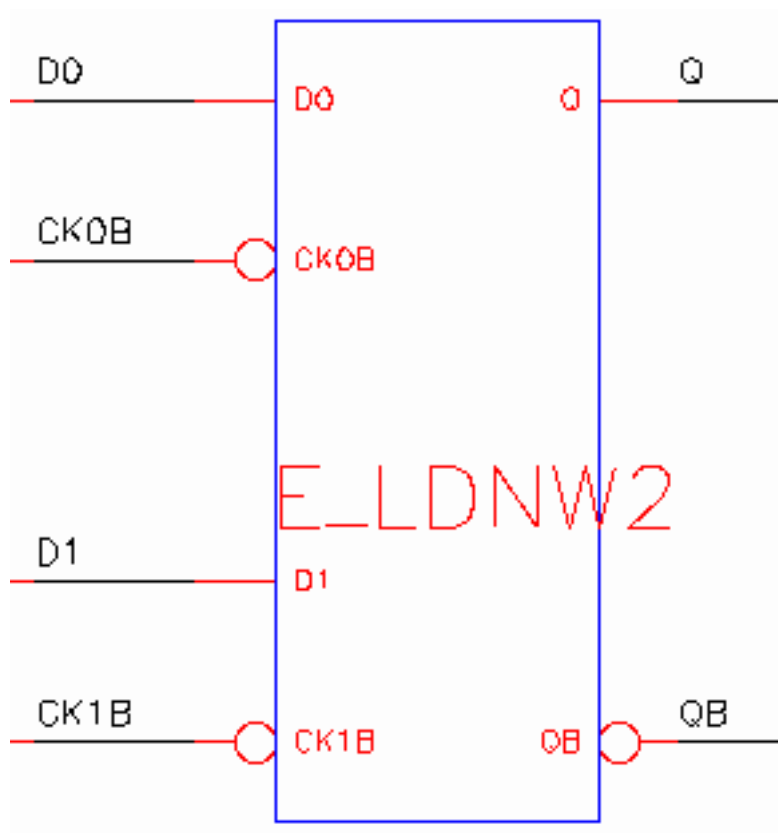
Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNMRS : D-Latch, active-LOW enable, active-HIGH set & reset, multiplexed with D-type flip-flop, positive-edge-triggered

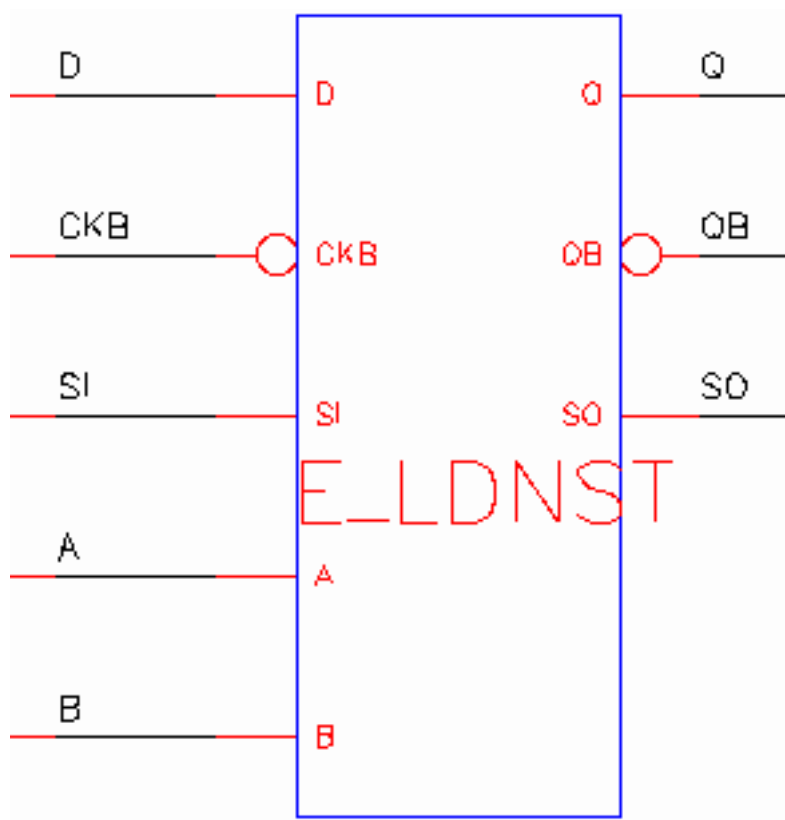


Pin	Direction	Description
D	INPUT	data
GB	INPUT	enable, active-LOW
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNW2 : D-Latch, active-LOW enable, 2 write ports

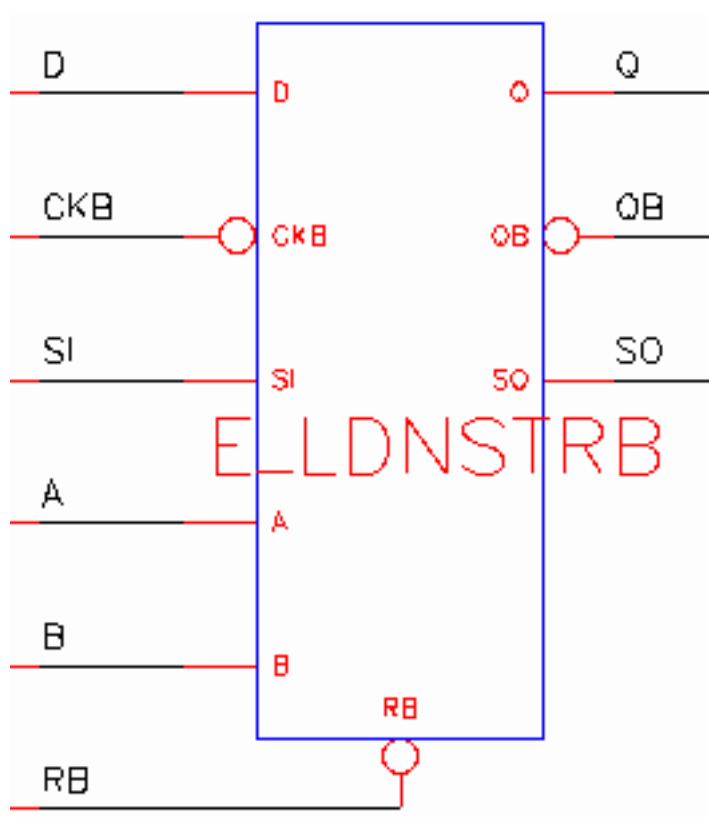


Pin	Direction	Description
D0	INPUT	data 0
CK0B	INPUT	clock 0, active-LOW
D1	INPUT	data 1
CK1B	INPUT	clock 1, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDNST : D-Latch with Scan Test Input, active-LOW enable

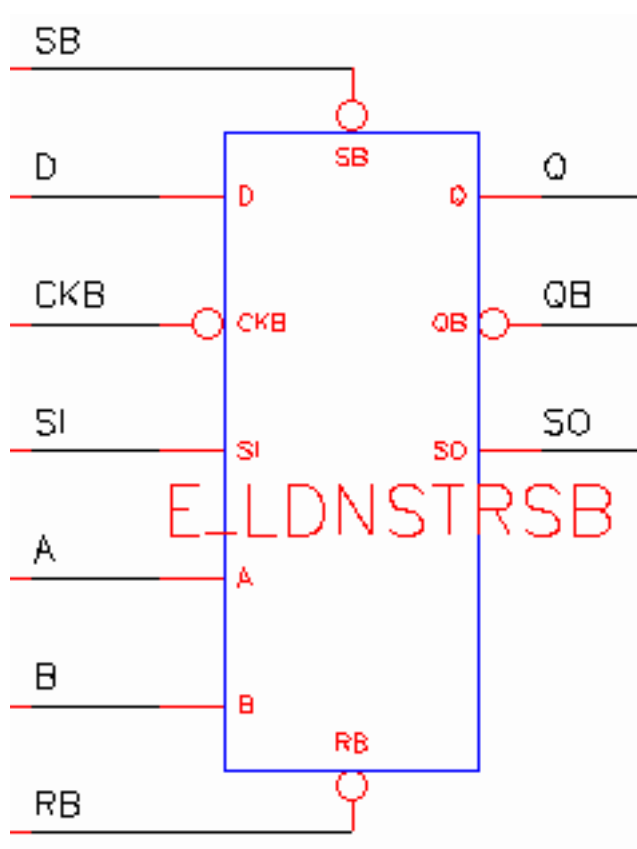
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDNSTRB : D-Latch with Scan Test Input, active-LOW enable, active-LOW reset



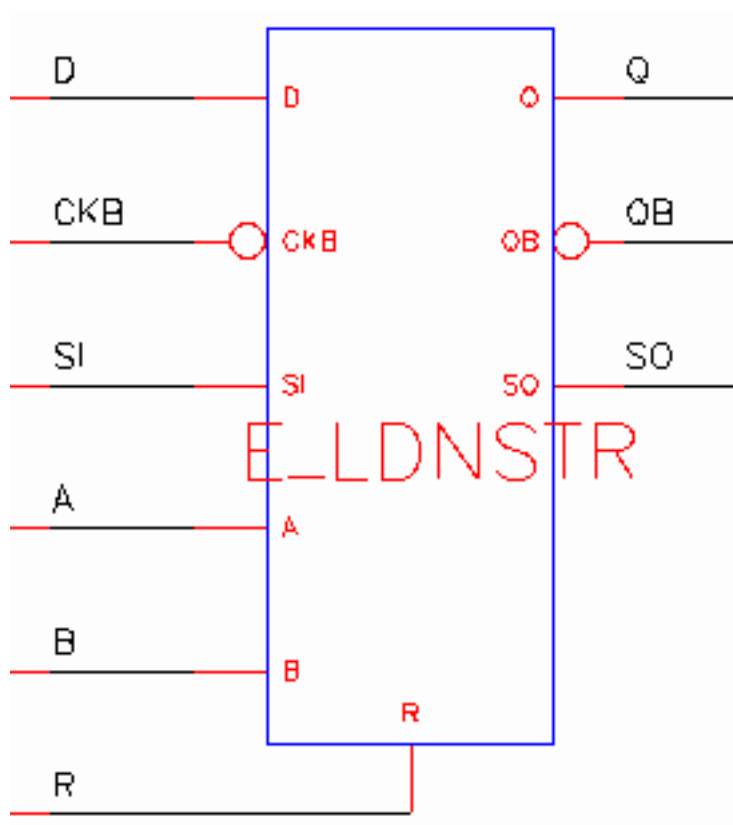
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDNSTRSB : D-Latch with Scan Test Input, active-LOW enable, active-LOW set & reset



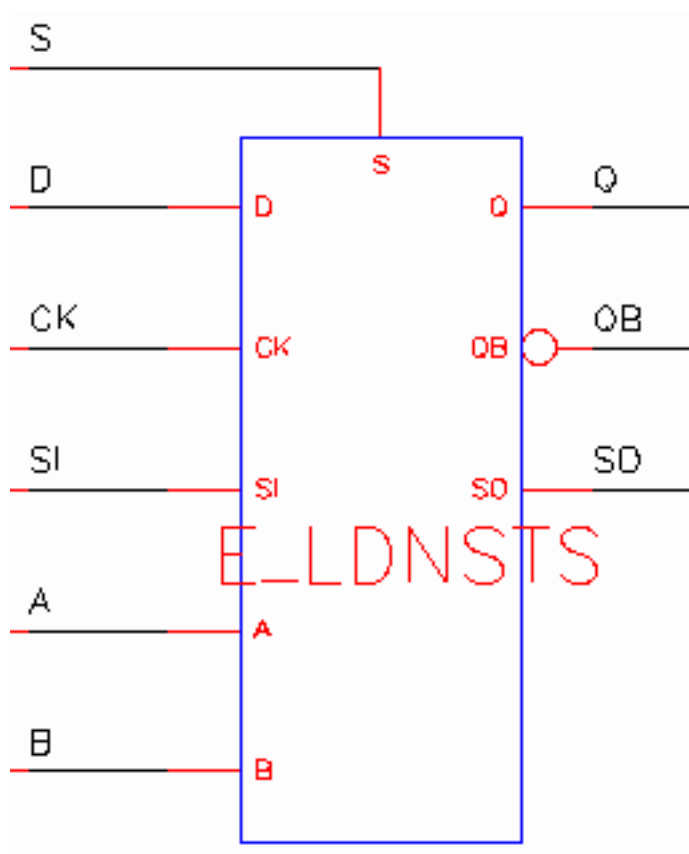
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDNSTR : D-Latch with Scan Test Input, active-LOW enable, active-HIGH reset



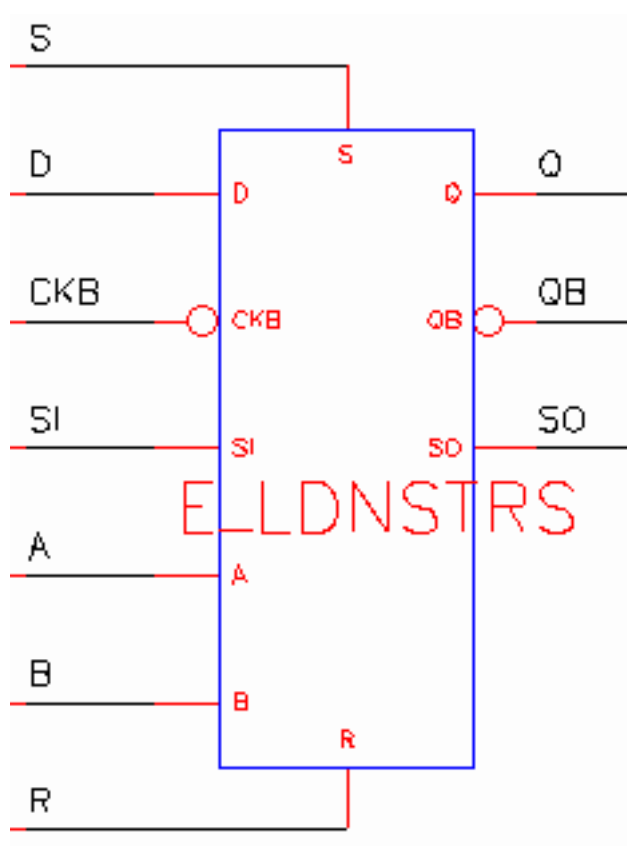
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDNSTS : D-Latch with Scan Test Input, active-LOW enable, active-HIGH set



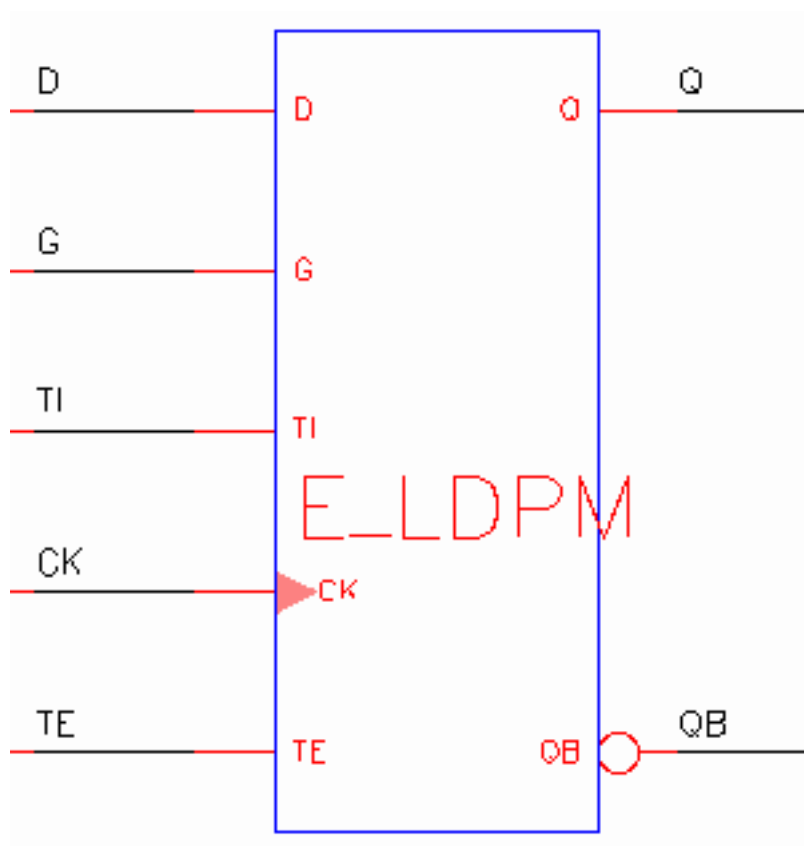
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDNSTRS : D-Latch with Scan Test Input, active-LOW enable, active-HIGH set & reset



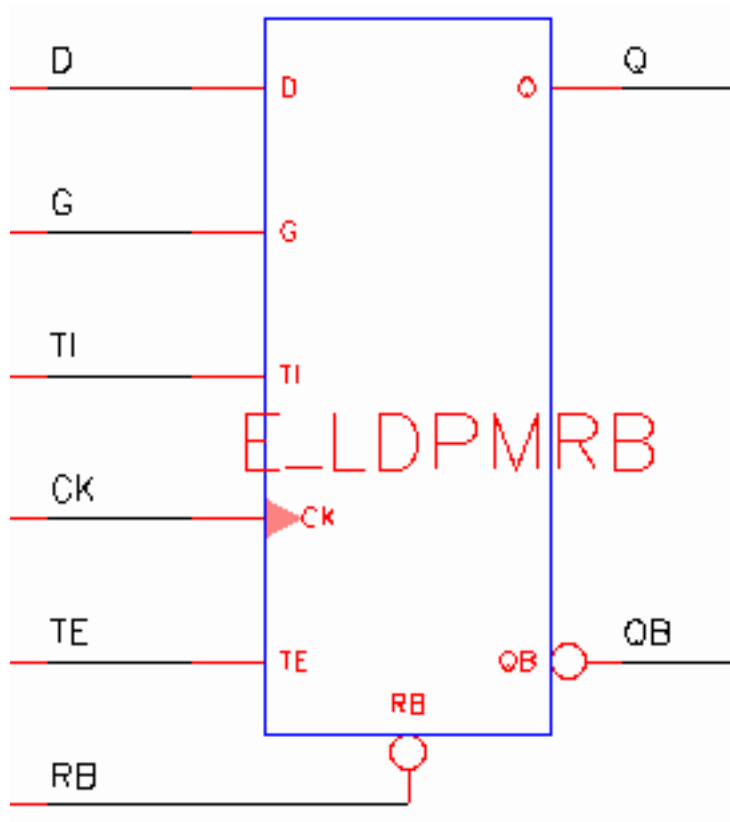
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, active-LOW
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPM : D-Latch, active-HIGH enable, multiplexed with D-type flip-flop, positive-edge-triggered



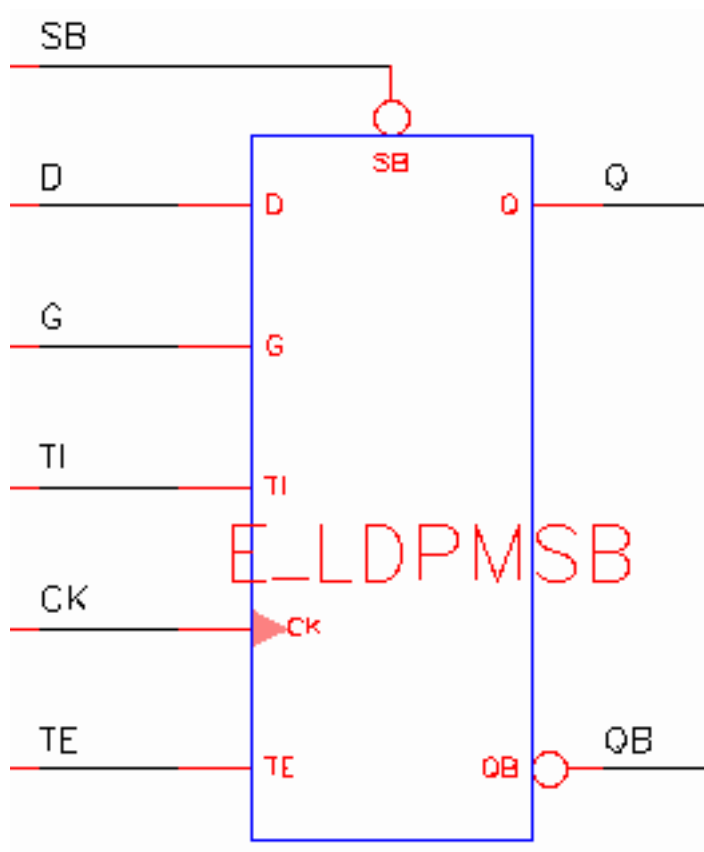
Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPMRB : D-Latch, active-HIGH enable, active-LOW reset, multiplexed with D-type flip-flop, positive-edge-triggered



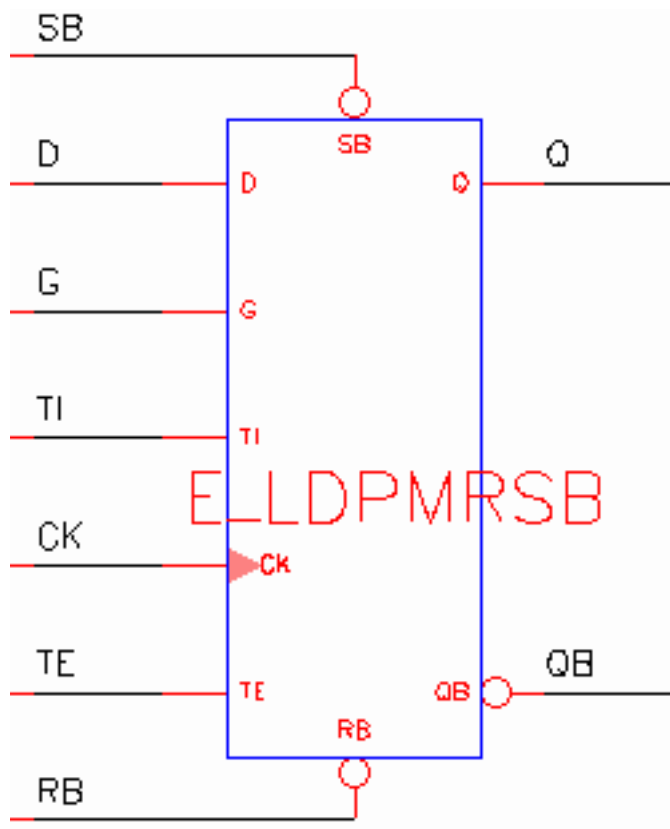
Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPMsb : D-Latch, active-HIGH enable, active-LOW set, multiplexed with D-type flip-flop, positive-edge-triggered



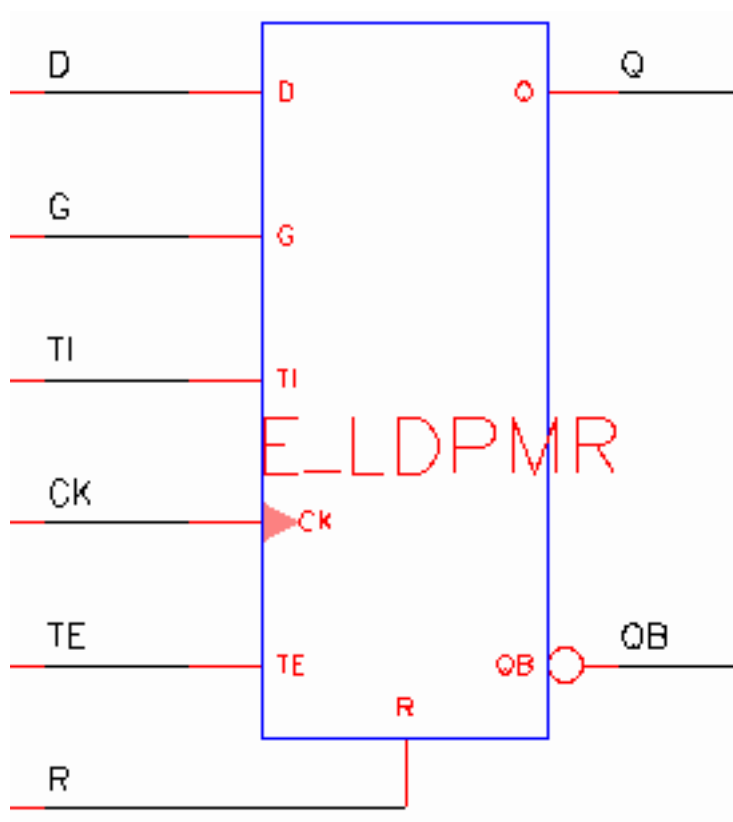
Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPMRSB : D-Latch, active-HIGH enable, active-LOW set & reset, multiplexed with D-type flip-flop, positive-edge-triggered



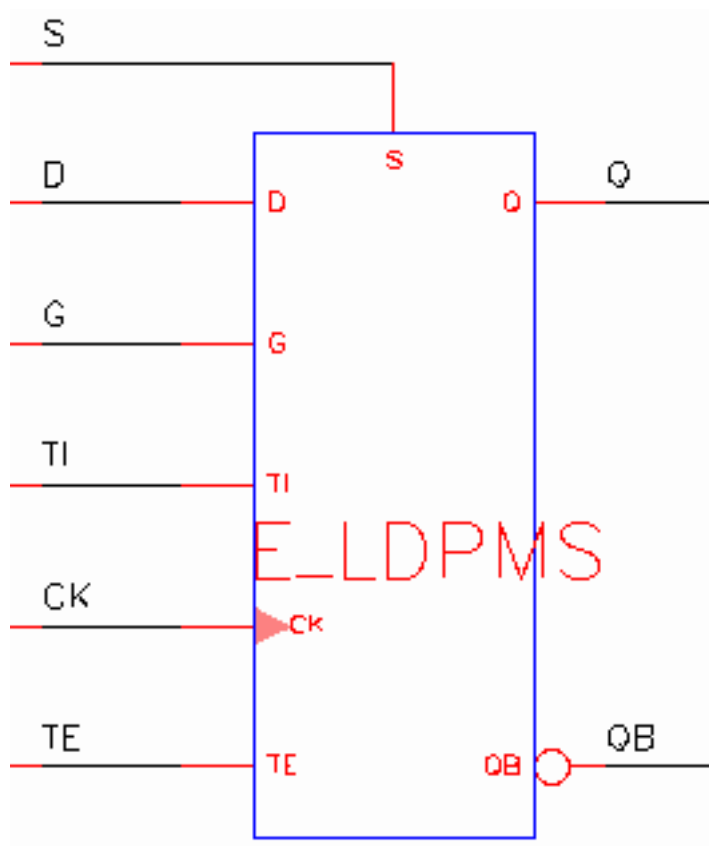
Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPMR : D-Latch, active-HIGH enable, active-HIGH reset, multiplexed with D-type flip-flop, positive-edge-triggered



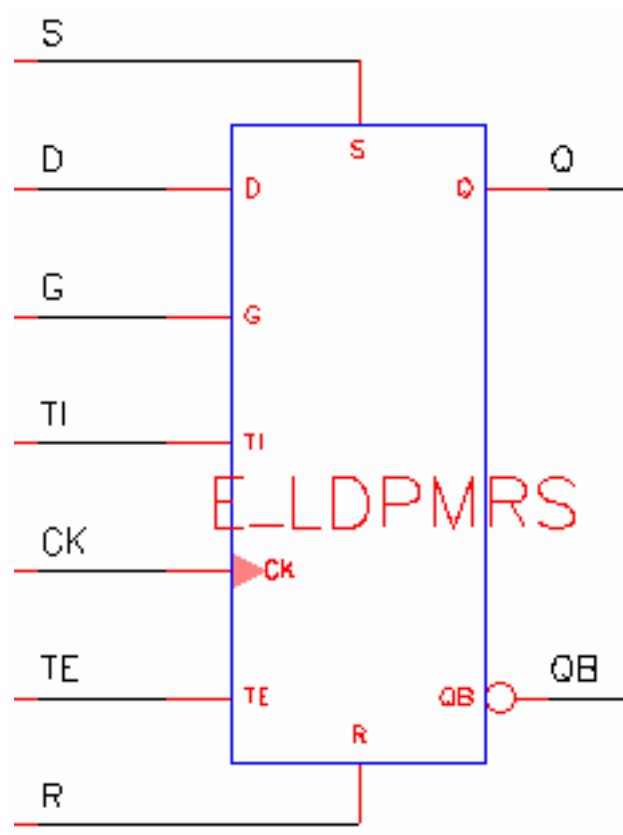
Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPMs : D-Latch, active-HIGH enable, active-HIGH set, multiplexed with D-type flip-flop, positive-edge-triggered



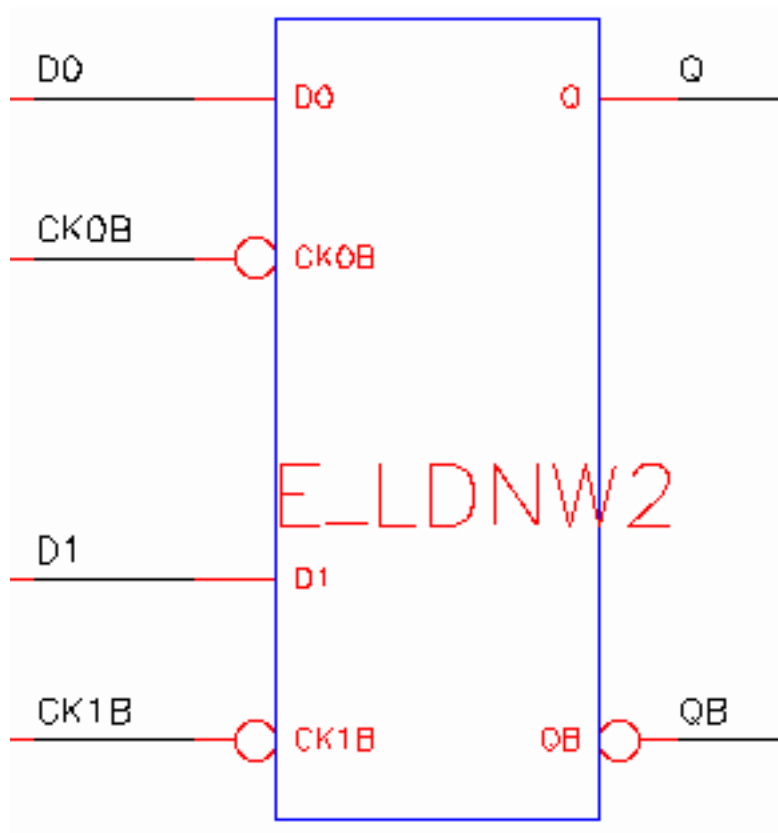
Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPMRS : D-Latch, active-HIGH enable, active-HIGH set & reset, multiplexed with D-type flip-flop, positive-edge-triggered

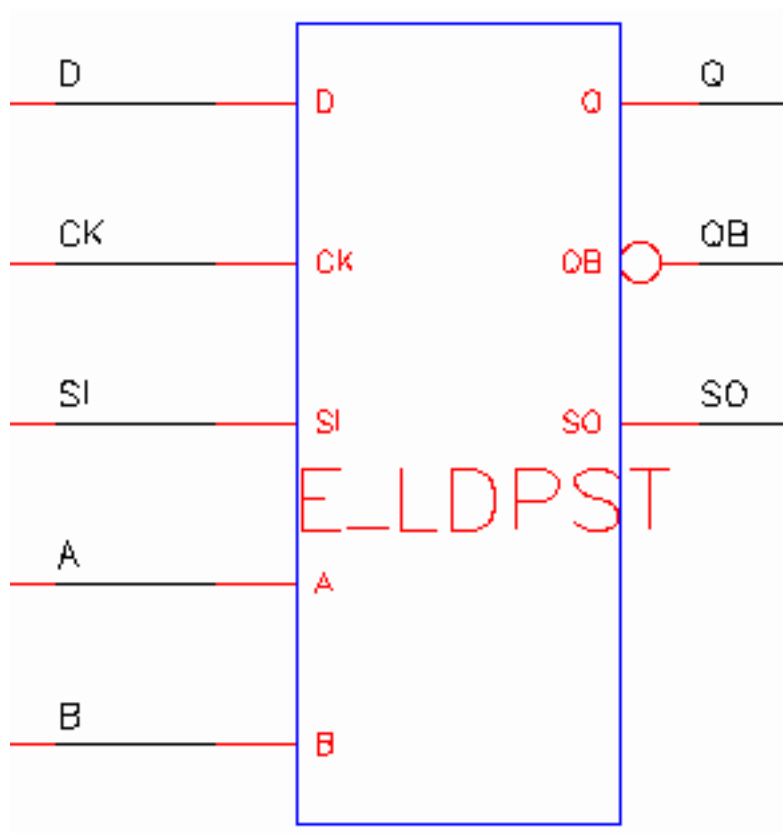


Pin	Direction	Description
D	INPUT	data
G	INPUT	enable, active-HIGH
TI	INPUT	scan data
CK	INPUT	clock, active-HIGH
TE	INPUT	scan enable
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPW2 : D-Latch, active-HIGH enable, 2 write ports

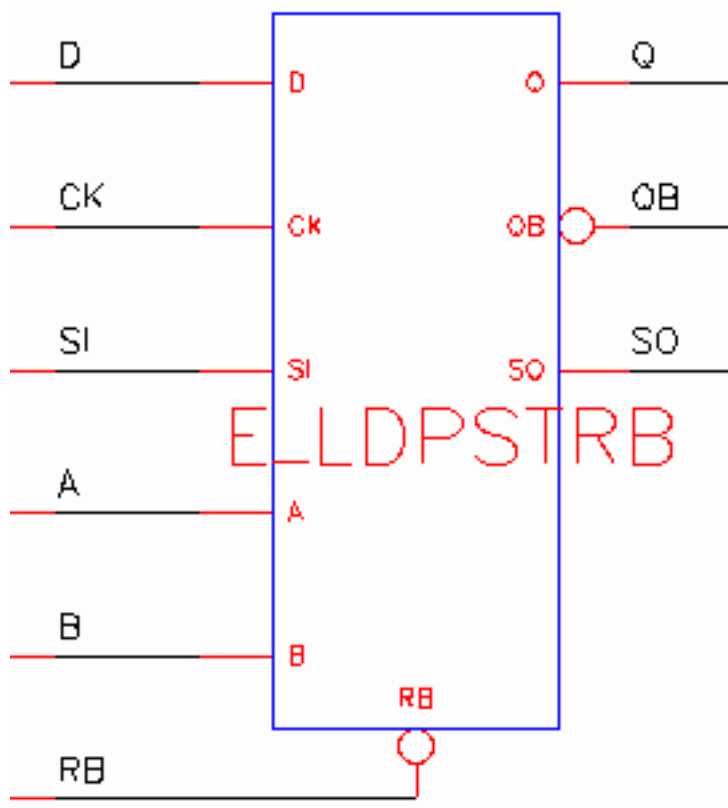


Pin	Direction	Description
D0	INPUT	data 0
CK0	INPUT	clock 0, active-HIGH
D1	INPUT	data 1
CK1	INPUT	clock 1, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LDPST : D-Latch with Scan Test Input, active-HIGH enable

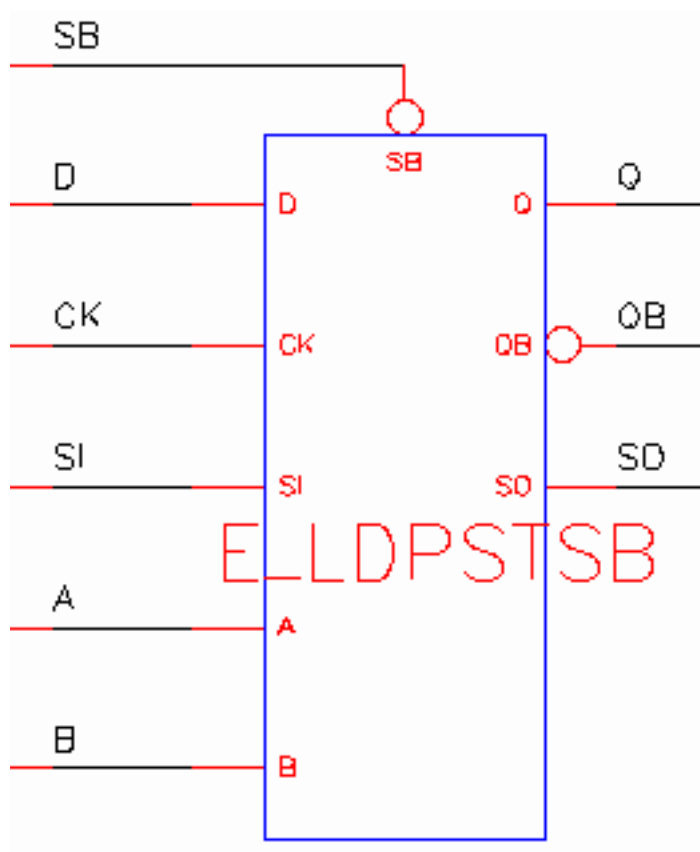
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPSTRB : D-Latch with Scan Test Input, active-HIGH enable, active-LOW reset

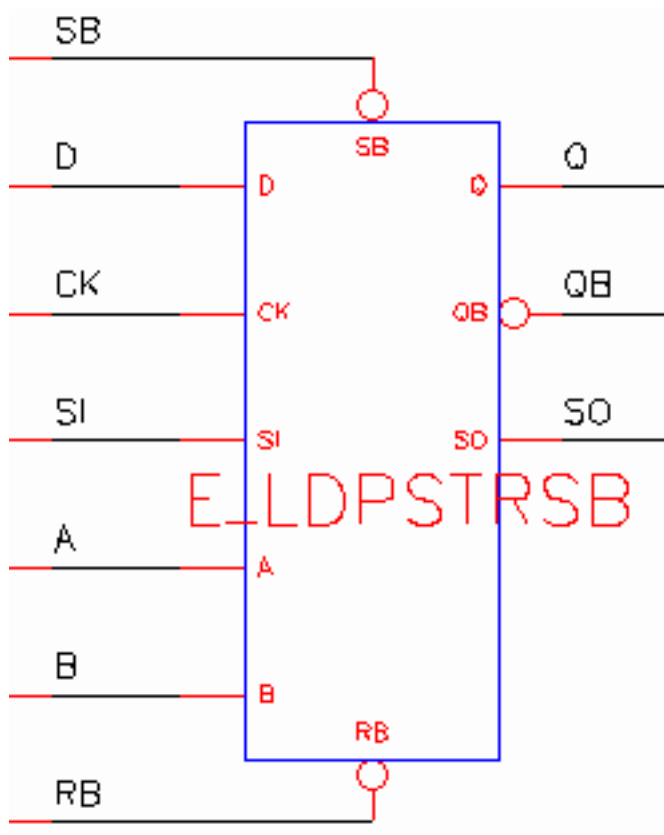


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
RB	INPUT	async reset, active-LOW
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPSTSB : D-Latch with Scan Test Input, active-HIGH enable, active-LOW set

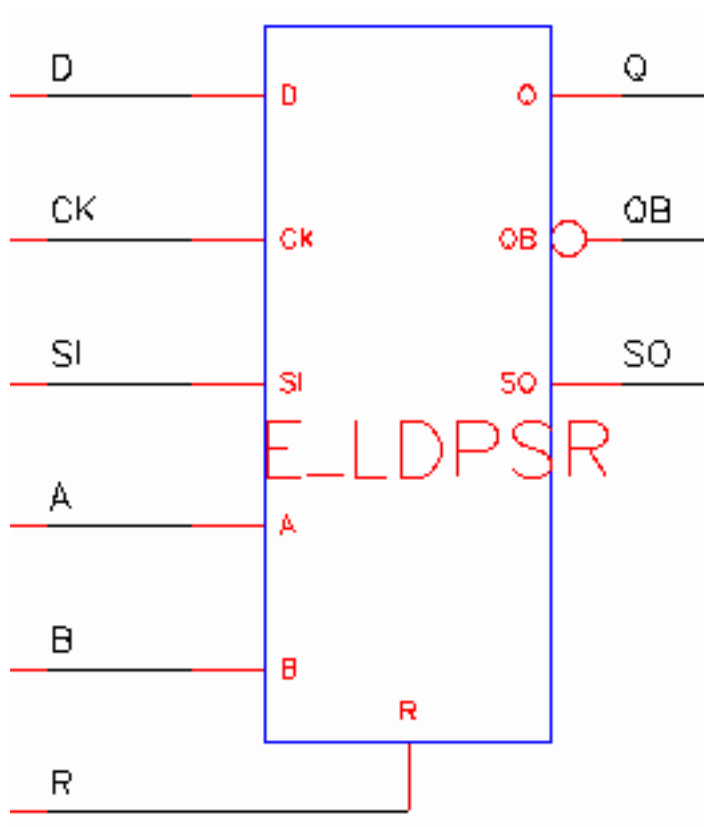


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPSTRSB : D-Latch with Scan Test Input, active-HIGH enable, active-LOW set & reset

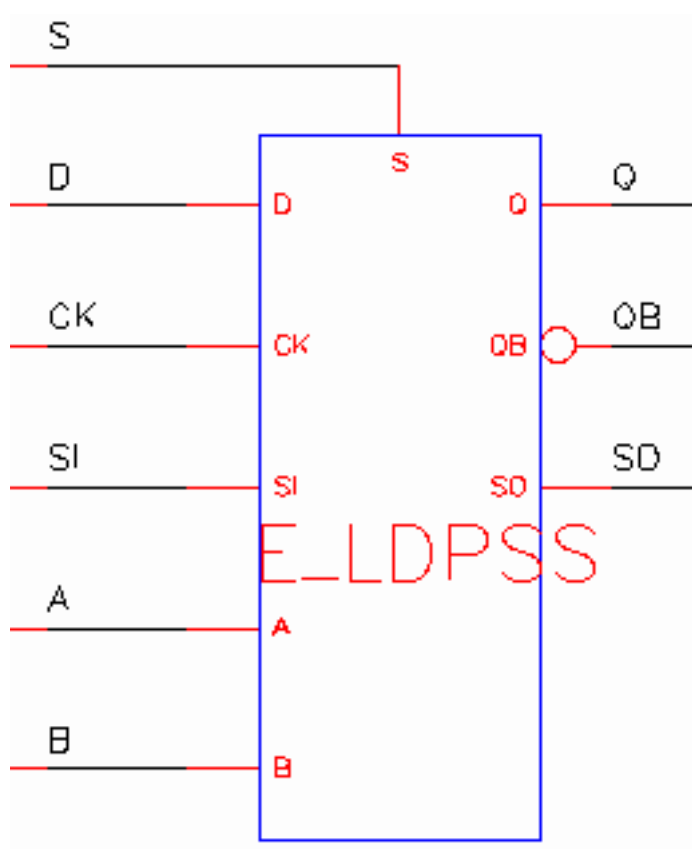
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPSTR : D-Latch with Scan Test Input, active-HIGH enable, active-HIGH reset

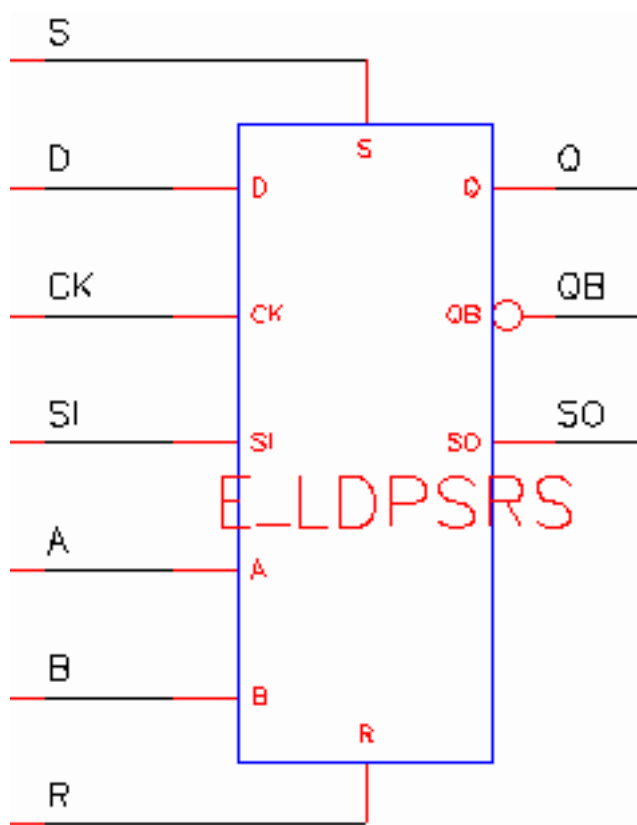


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
R	INPUT	async reset, active HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPSTS : D-Latch with Scan Test Input, active-HIGH enable, active-HIGH set

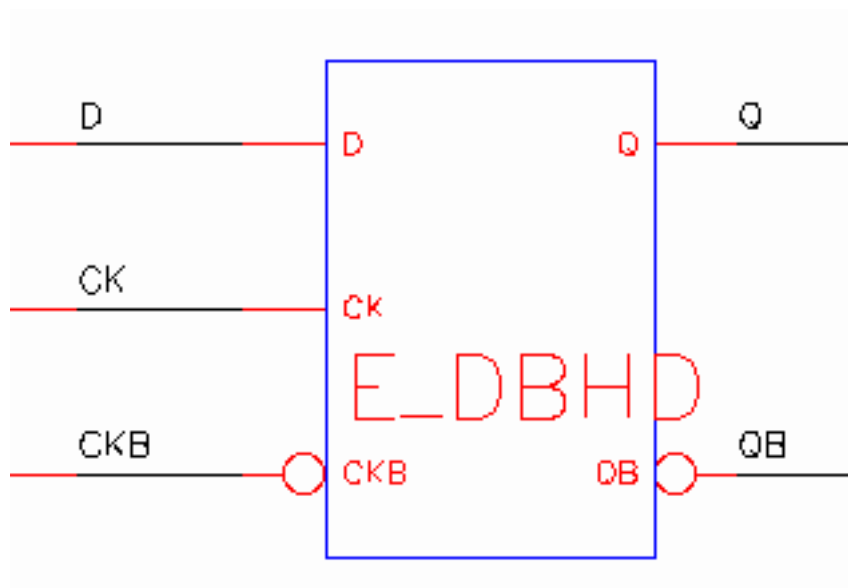


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
S	INPUT	async set, active HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_LDPSTRS : D-Latch with Scan Test Input, active-HIGH enable, active-HIGH set & reset

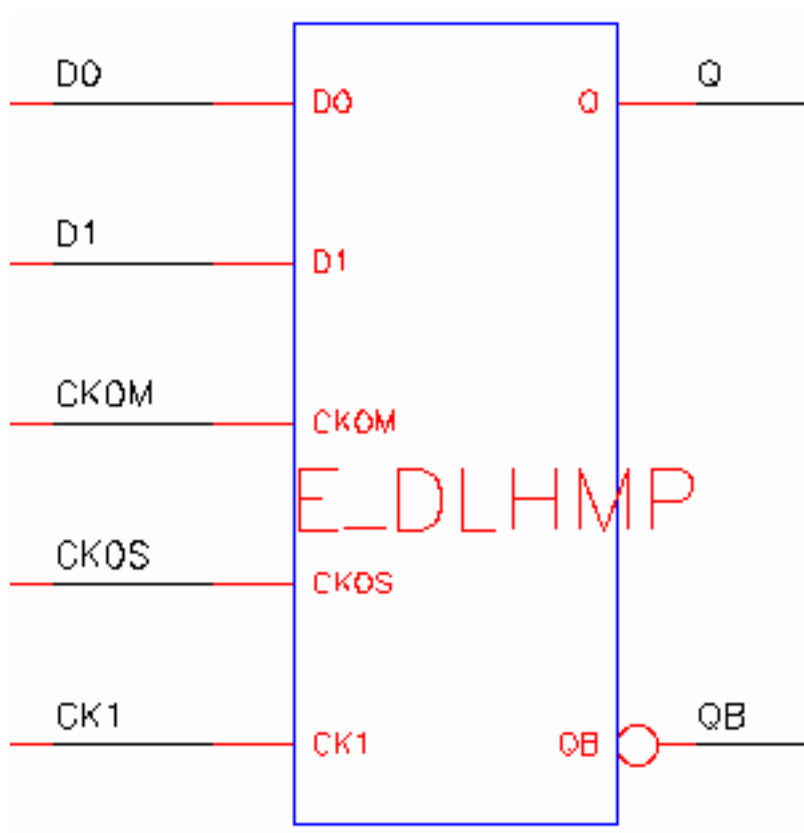
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
SI	INPUT	serial data in
A	INPUT	data A
B	INPUT	data B
R	INPUT	async reset, active HIGH
S	INPUT	async set, active HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
SO	OUTPUT	serial data out

E_DBHD : D-Latch, complementary enables

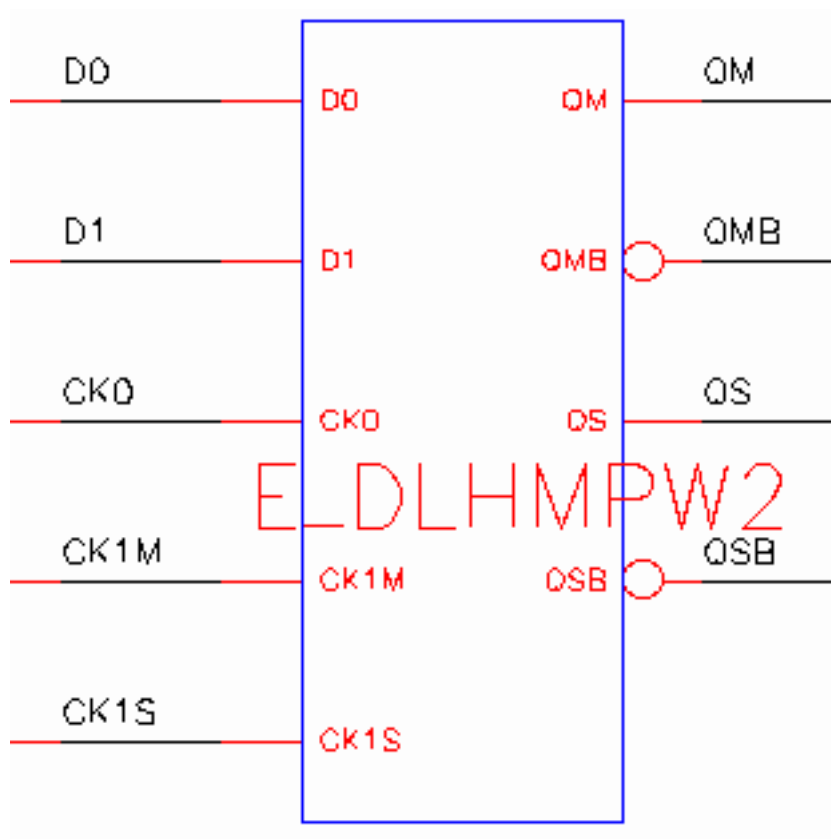


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, active-HIGH
CKB	INPUT	clock, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DLHMP : Master-slave latch, with additional D-input for slave stage

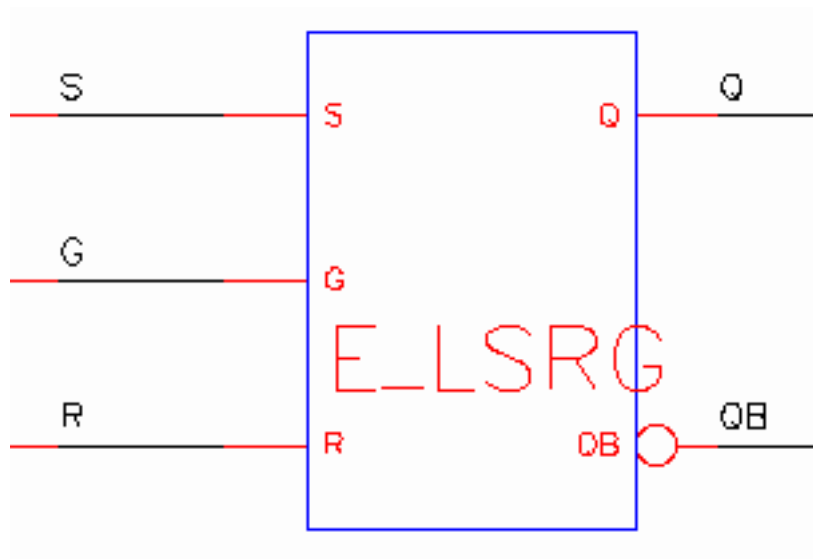


Pin	Direction	Description
D0	INPUT	data 0
D1	INPUT	data 1
CK0M	INPUT	clock 0, master, active-HIGH
CK0S	INPUT	clock 0, slave, active-HIGH
CK1	INPUT	clock 1, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DLHMPW2 : Master-slave latch, 2 write ports

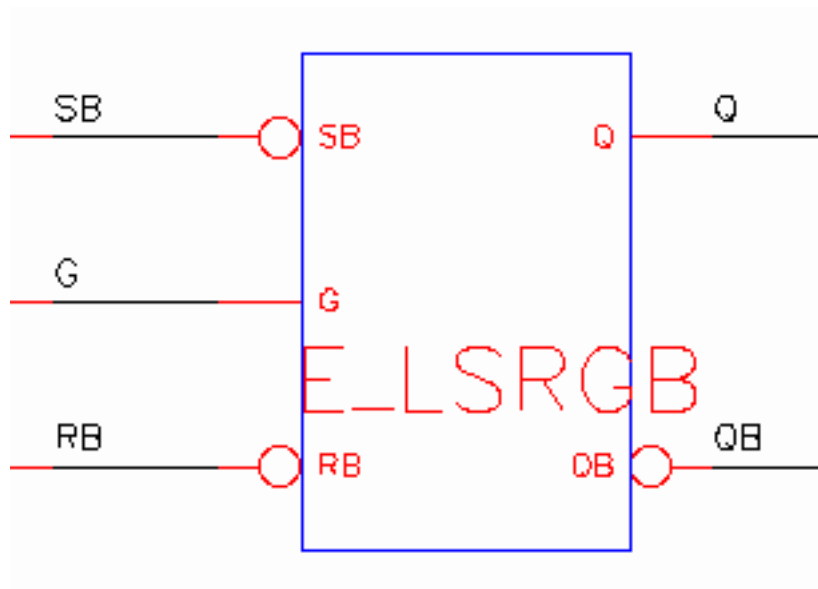
Pin	Direction	Description
D0	INPUT	data 0
D1	INPUT	data 1
CKD	INPUT	clock 0, active-HIGH
CK1M	INPUT	clock 1, master, active-HIGH
CK1S	INPUT	clock 1, slave, active-HIGH
QM	OUTPUT	data, master
QMB	OUTPUT	data, inverted, master
QS	OUTPUT	data, slave
QSB	OUTPUT	data, inverted,slave

E_LSRG : SR with COMMON GATED, active-HIGH set & reste

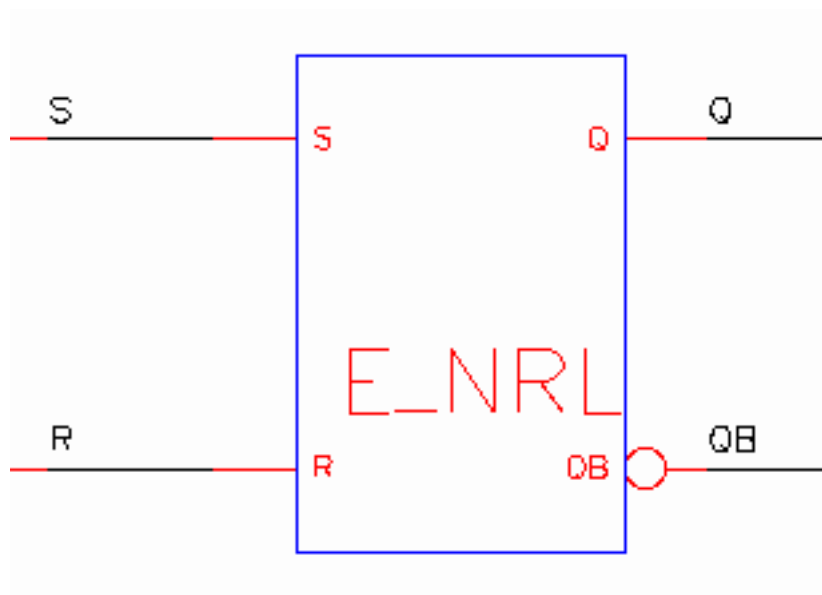


Pin	Direction	Description
S	INPUT	async set, active-HIGH
G	INPUT	enable, active-HIGH
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_LSRGB : SR with COMMON GATED, active-LOW set & reset

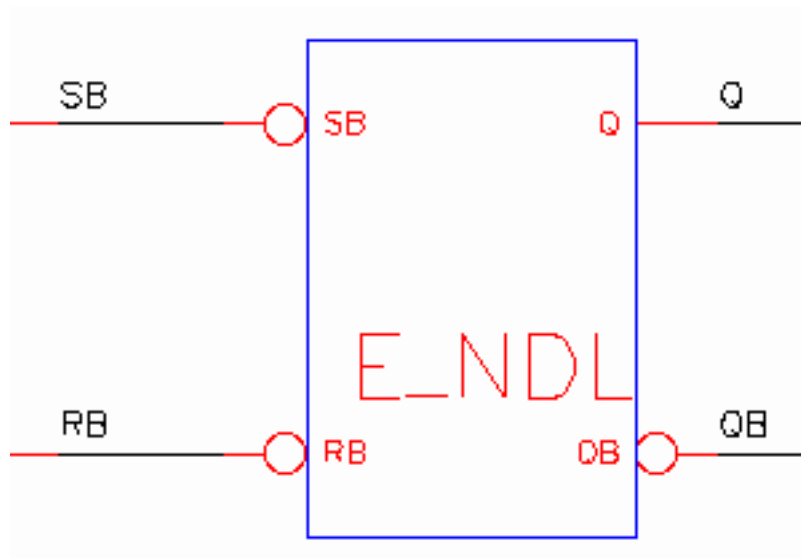


Pin	Direction	Description
SB	INPUT	async set, active-LOW
G	INPUT	enable, active-HIGH
RB	INPUT	async reset, active LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_NRL : SR latch, NOR type

Pin	Direction	Description
S	INPUT	async set, active-LOW
R	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_NDL : SR latch, NAND type



Pin	Direction	Description
SB	INPUT	async set, active-LOW
RB	INPUT	async reset, active LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

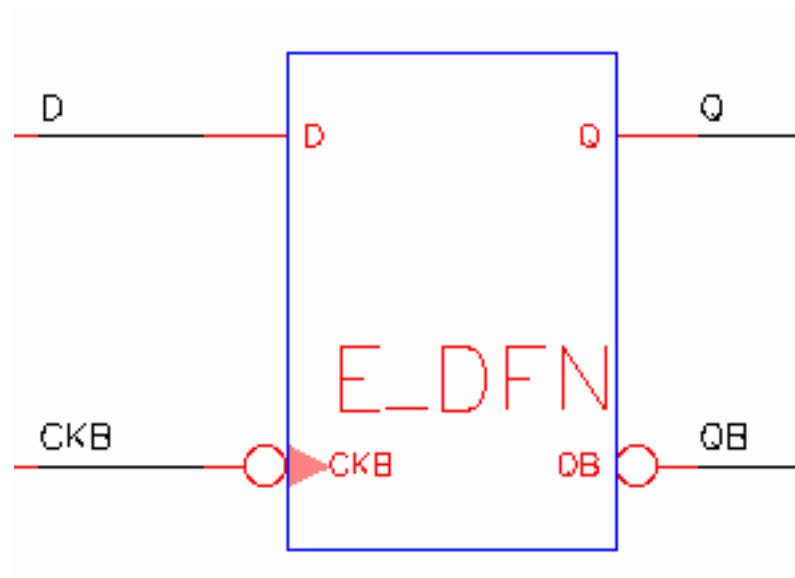
FlipFlops

[E_DFN](#)
[E_DFNRB](#)
[E_DFNSB](#)
[E_DFNRSB](#)
[E_DFNR](#)
[E_DFNS](#)
[E_DFNRS](#)
[E_DFP](#)
[E_DFPRB](#)
[E_DFPSB](#)
[E_DFPRSB](#)
[E_DFPR](#)
[E_DFPS](#)
[E_DFPRS](#)
[E_DFNCL](#)
[E_DFNCLRB](#)
[E_DFNCLSB](#)
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[E_DFNCLR](#)
[E_DFNCLS](#)
[E_DFNCLRS](#)
[E_DFPCL](#)
[E_DFPCLRB](#)
[E_DFPCLSB](#)
[E_DFPCLRSE](#)
[E_DFPCLR](#)
[E_DFPCLS](#)
[E_DFPCLRS](#)
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[E_DFPTS](#)
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[E_DFN4](#)
[E_DFN4RB](#)
[E_DFN4SB](#)
[E_DFN4RSB](#)
[E_DFN4R](#)

[E_DFN4S](#)
[E_DFN4RS](#)
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[E_DFNZT](#)
[E_DFNZTRB](#)
[E_DFNZTSB](#)
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[E_DFPZRSB](#)
[E_DFPZT](#)
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[E_DDBTNB](#)
[E_DFFZRSB](#)
[E_DFFZRSBLD](#)
[E_DFFNLRSB](#)
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[E_JKFNR](#)
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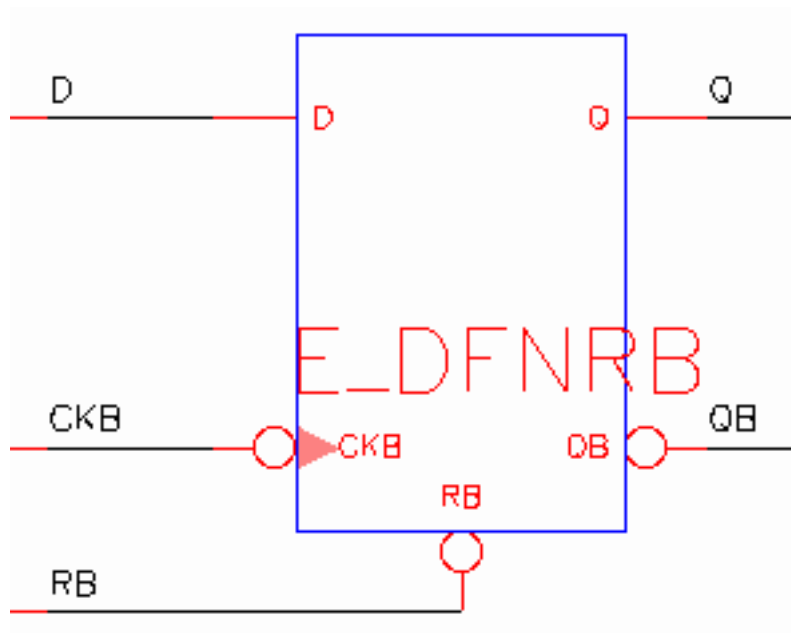
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[E_JKFP](#)
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[E_JKFPSB](#)
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[E_JKFNSCRSB](#)
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[E_TFFNRB](#)
[E_TFFNSB](#)
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[E_TFFNRS](#)
[E_TFFP](#)
[E_TFFPRB](#)
[E_TFFPSB](#)
[E_TFFPRSB](#)
[E_TFFPR](#)
[E_TFFPS](#)
[E_TFFPRS](#)

E_MFFRSB
E_MBNFNQ

E_DFN : DFF, negative-edge-trigger

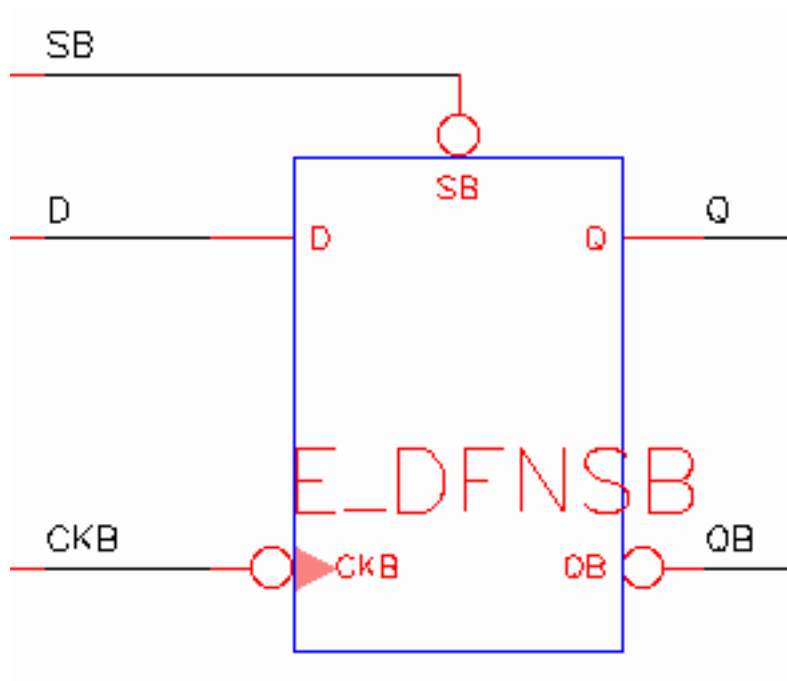
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNRB : DFF, negative-edge-trigger, active-LOW reset



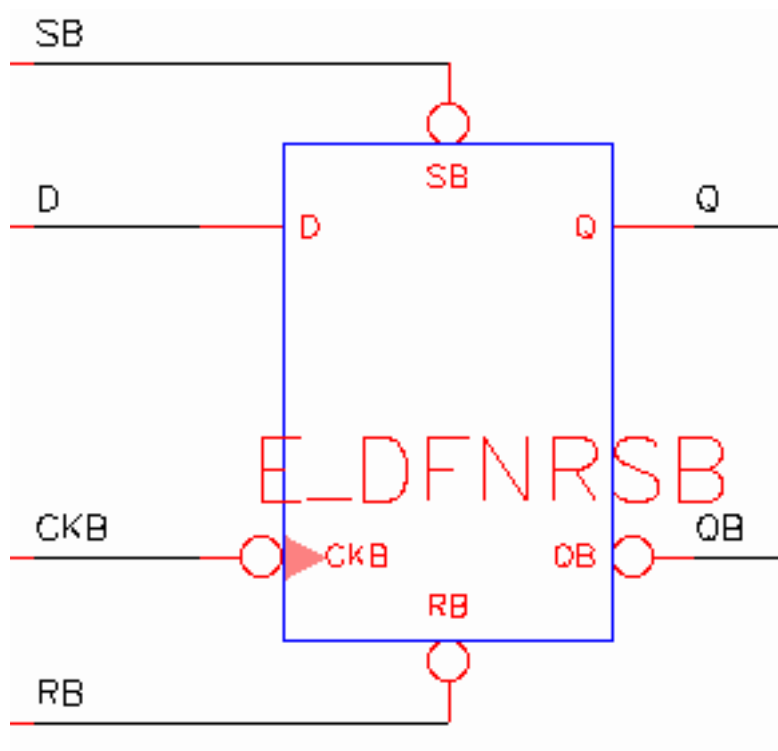
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNSB : DFF, negative-edge-trigger, active-LOW set



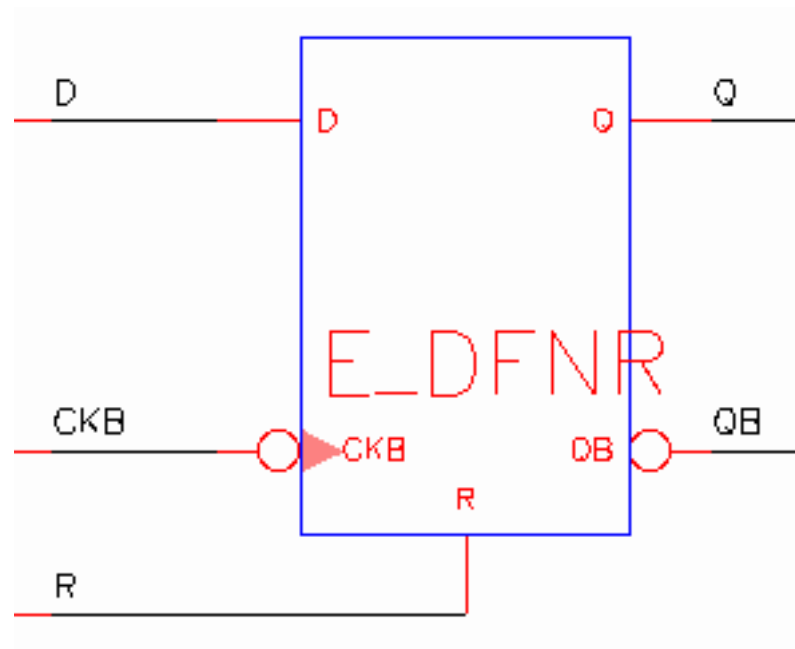
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNRSB ; DFF, negative-edge-trigger, active-LOW set & reset



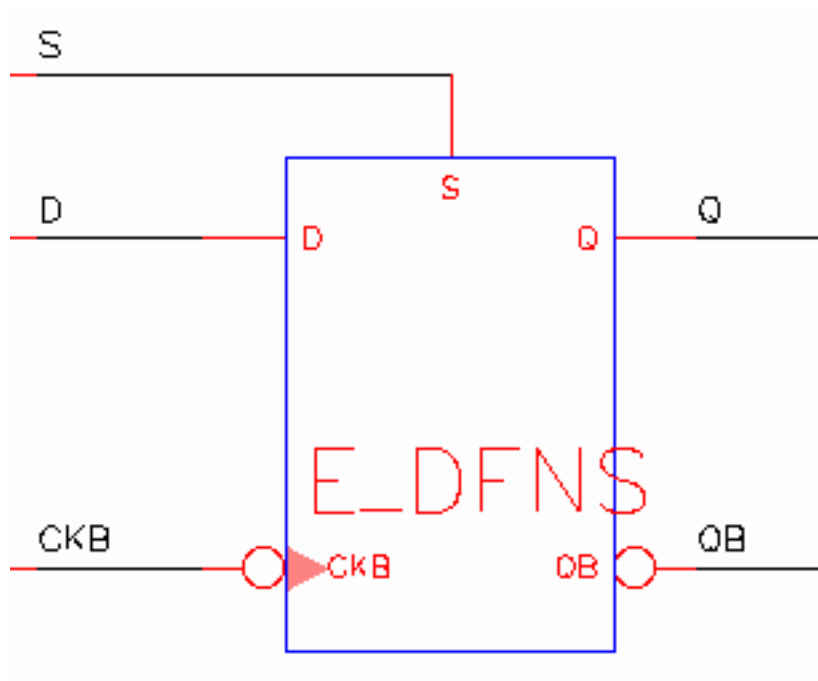
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNR : DFF, negative-edge-trigger, active-HIGH reset



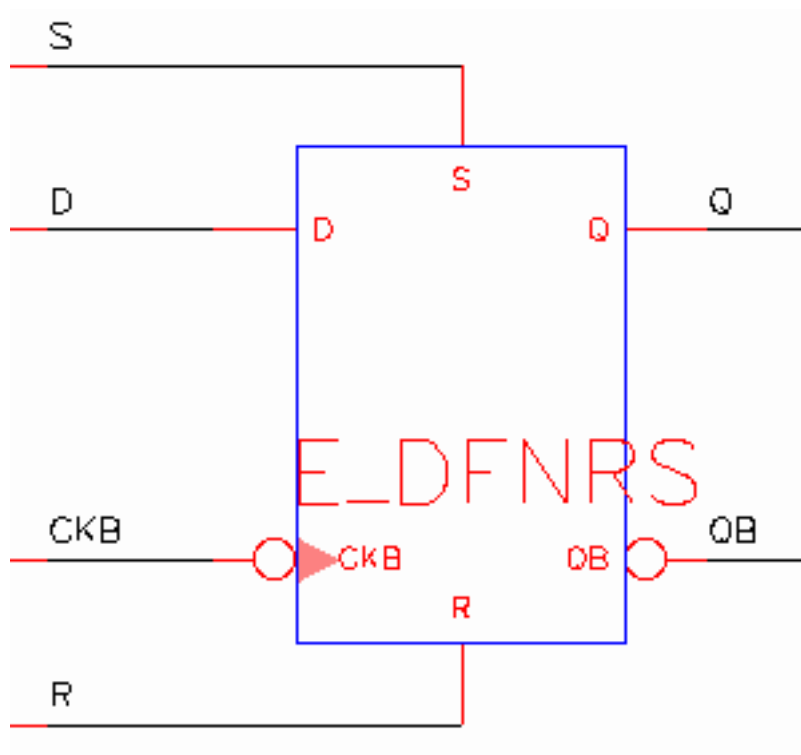
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNS : DFF, negative-edge-trigger, active-HIGH set



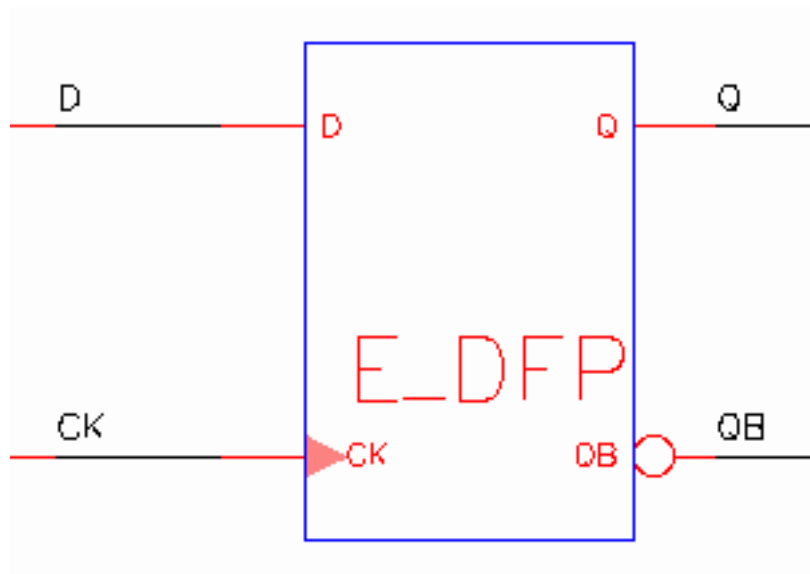
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNRS : DFF, negative-edge-trigger, active-HIGH set & reset



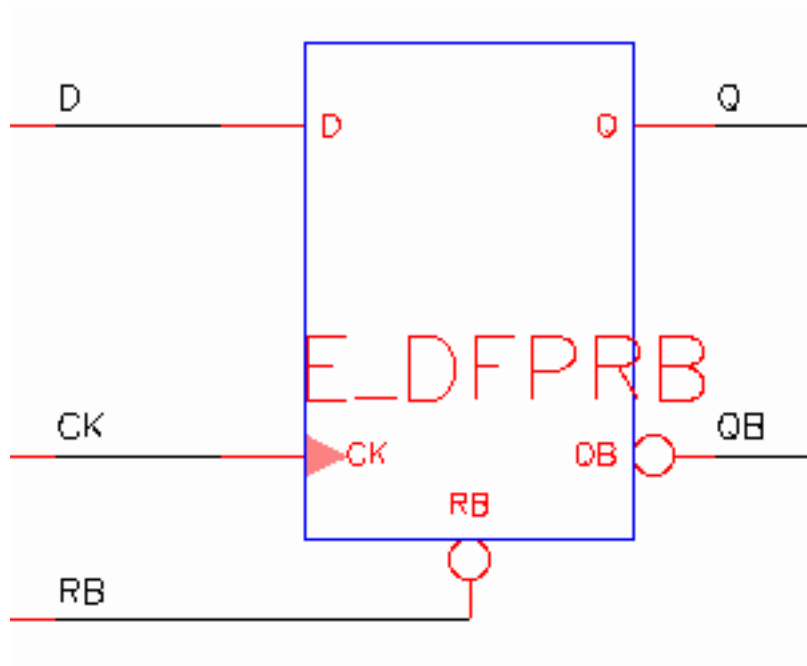
Pin	Direction	Description
D	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFP : DFF, positive-edge-trigger



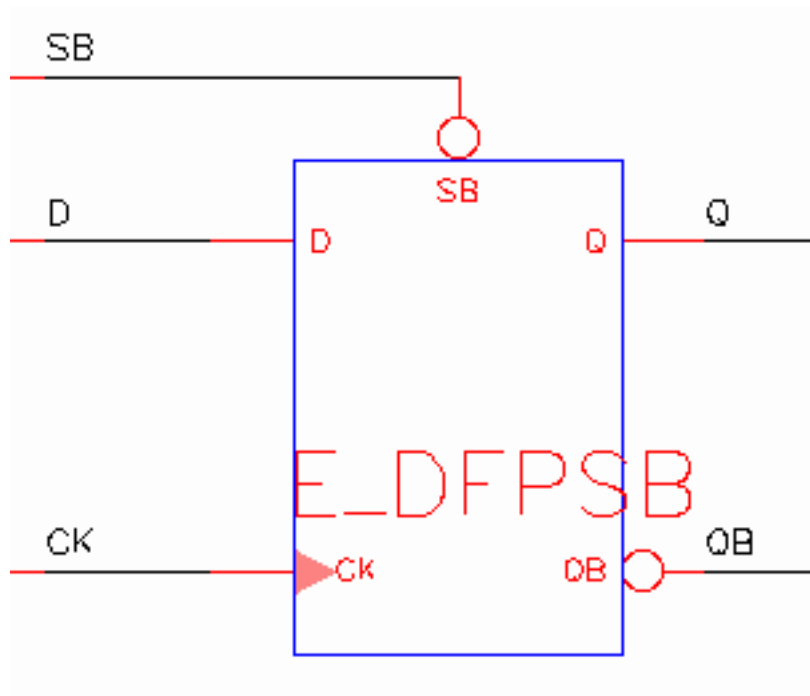
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPRB : DFF, positive-edge-trigger, active-LOW reset



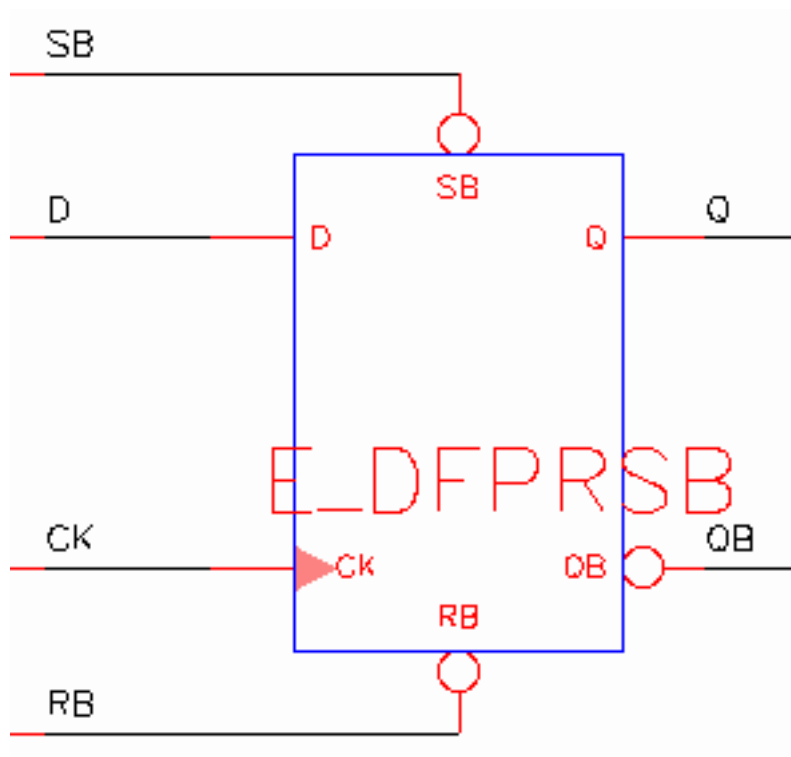
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPSB : DFF, positive-edge-trigger, active-LOW set



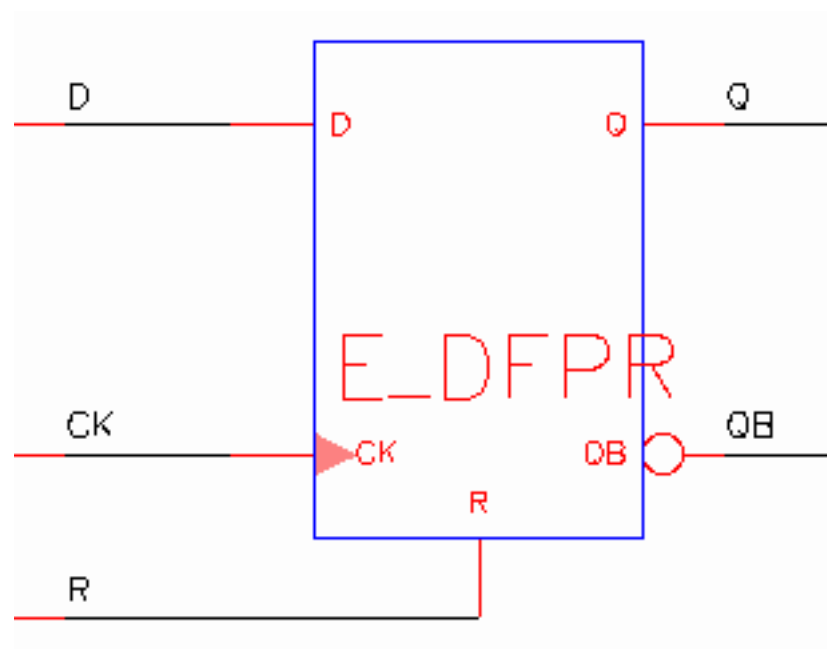
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPRSB : DFF, positive-edge-trigger, active-LOW set & reset



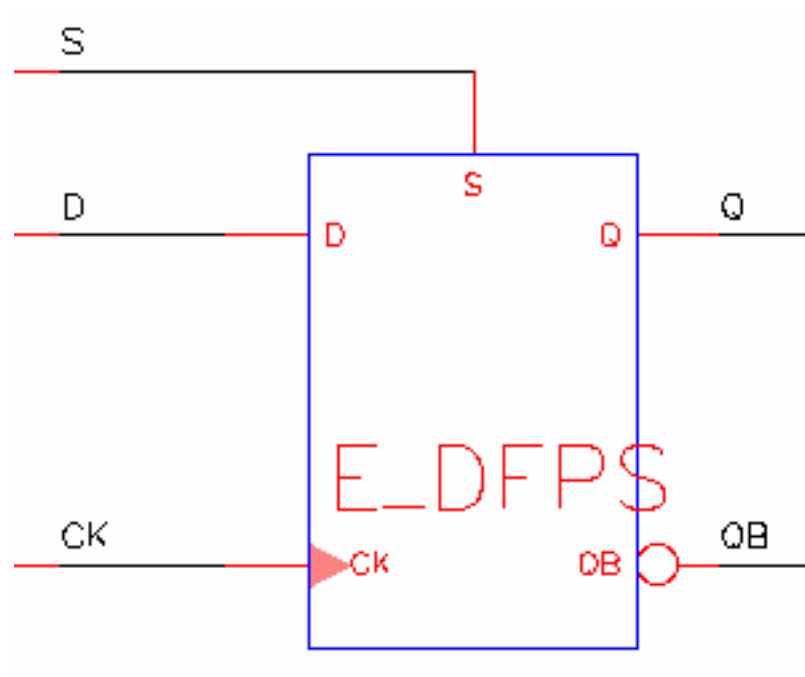
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPR : DFF, positive-edge-trigger, active-HIGH reset



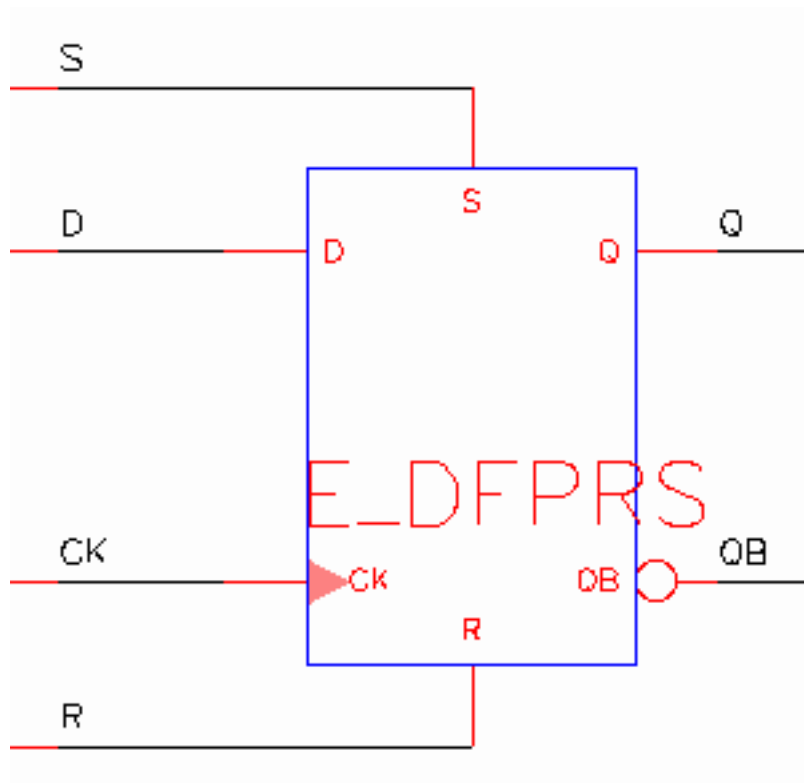
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPS : DFF, positive-edge-trigger, active-HIGH set



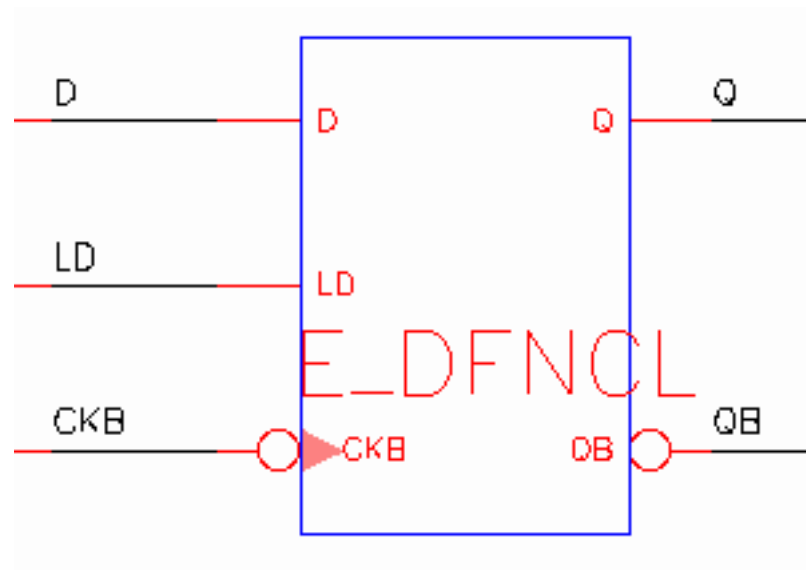
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPRS : DFF, positive-edge-trigger, active-HIGH set & reset



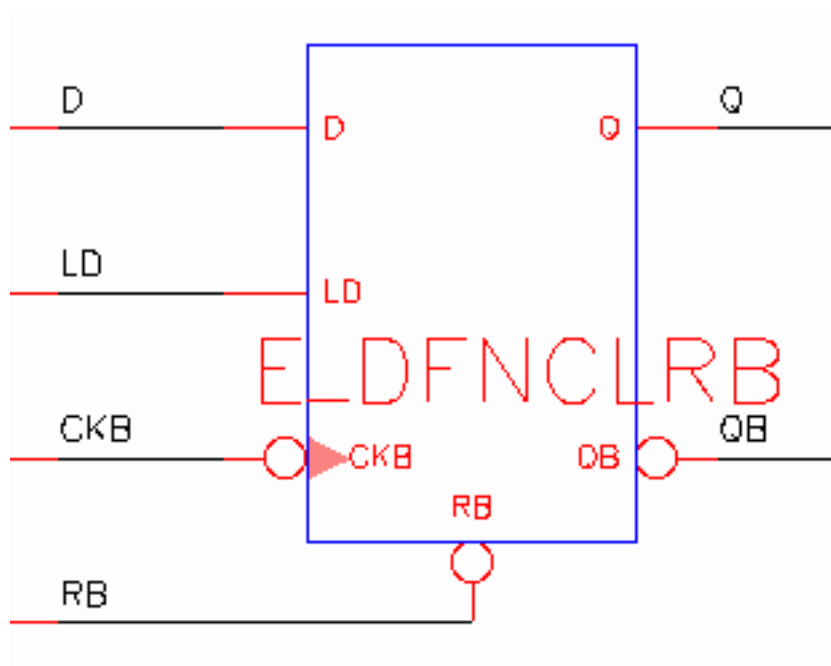
Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCL : DFF, negative-edge-trigger, Load



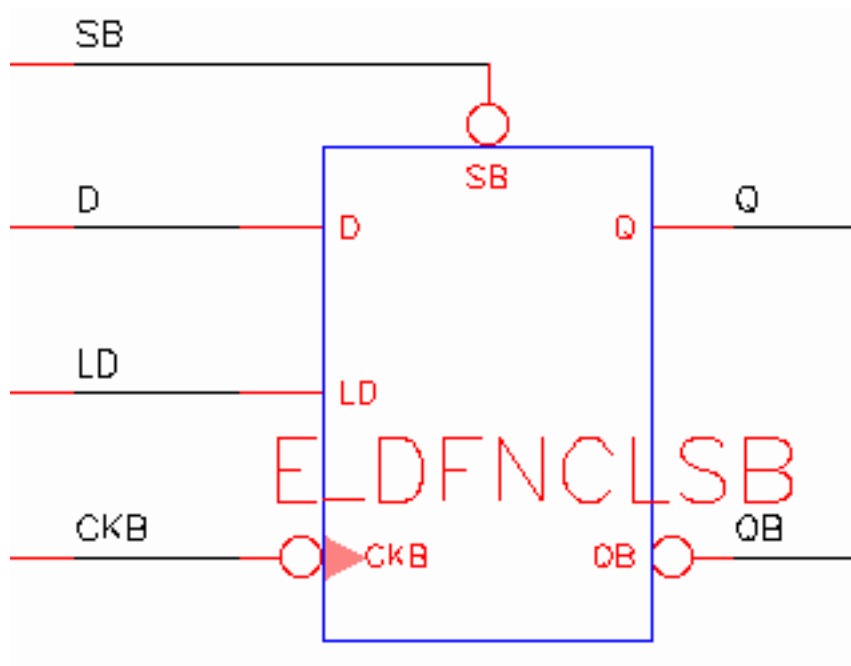
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCRLB : DFF, negative-edge-trigger, active-LOW reset, Load



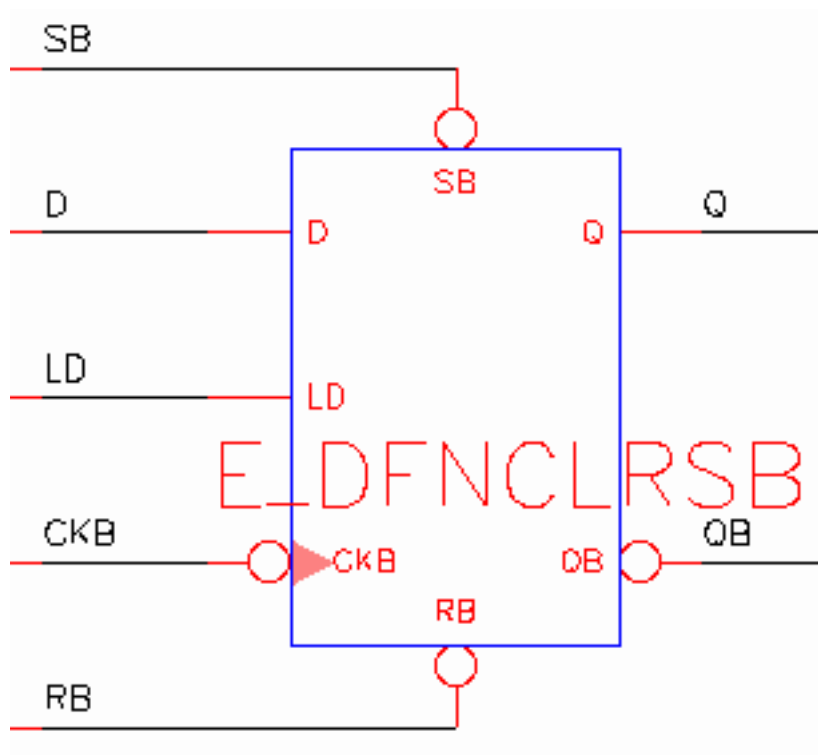
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCLSB : DFF, negative-edge-trigger, active-LOW set, Load



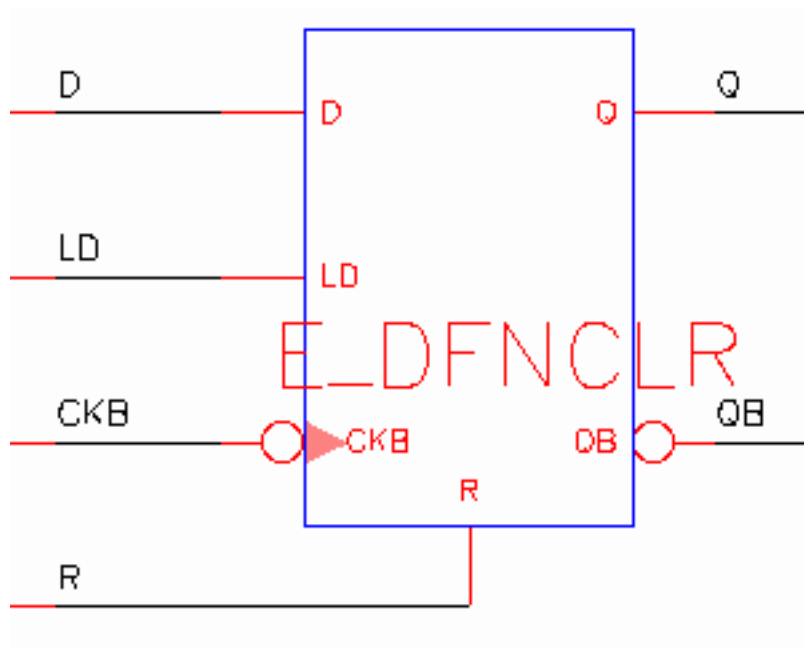
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCLRSB : DFF, negative-edge-trigger, active-LOW set & reset, Load



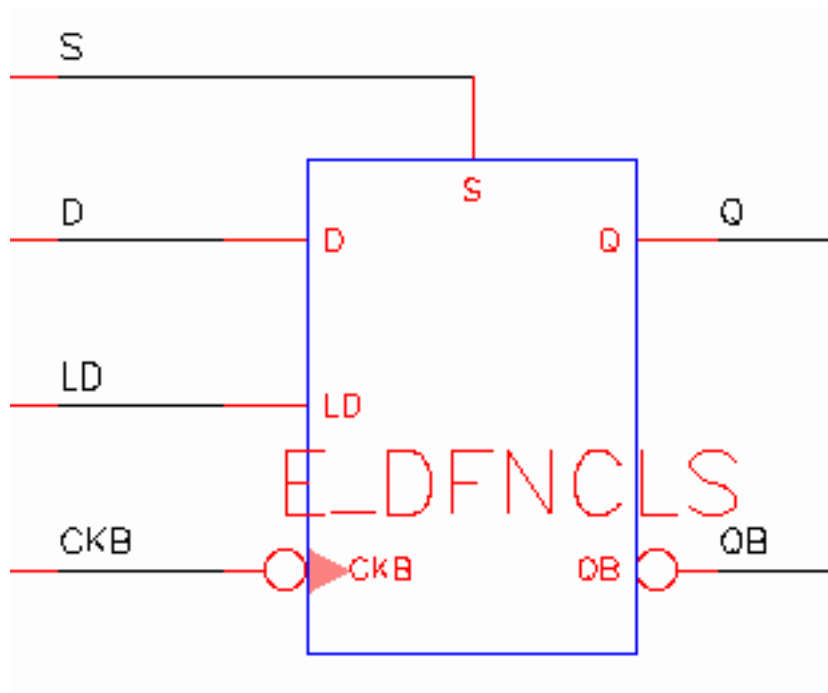
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCLR : DFF, negative-edge-trigger, active-HIGH reset, Load



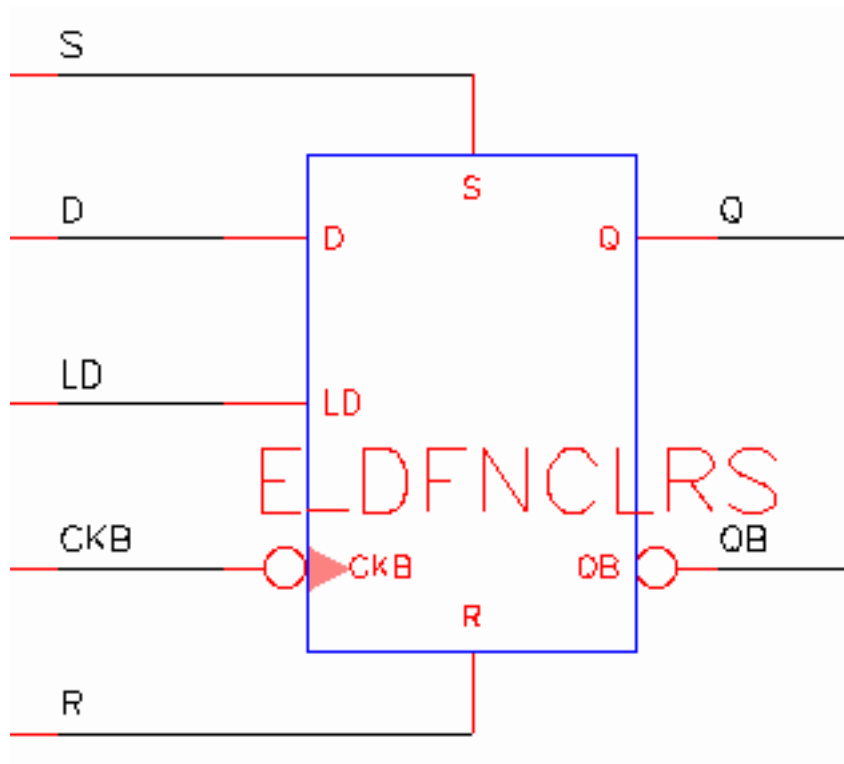
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCLS : DFF, negative-edge-trigger, active-HIGH set, Load



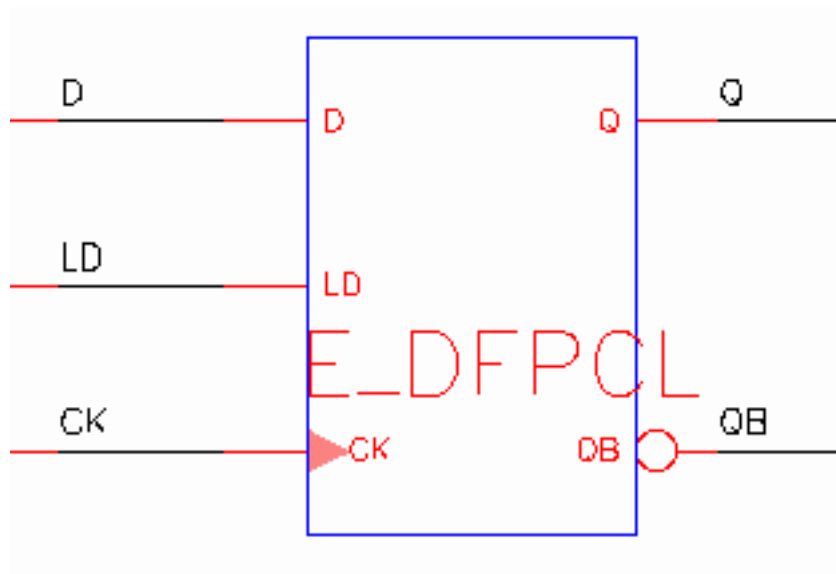
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNCLRS : DFF, negative-edge-trigger, active-HIGH set & reset, Load



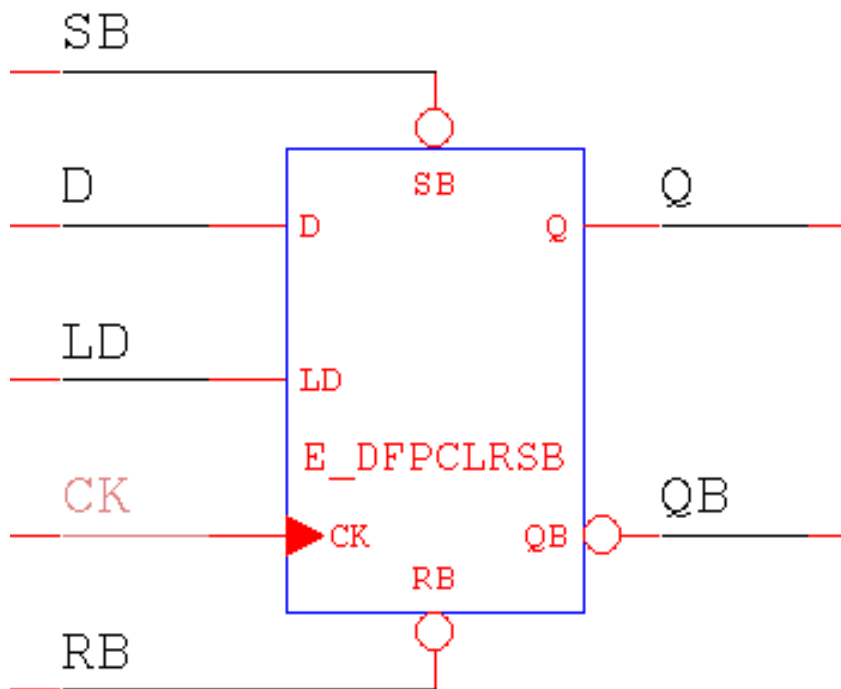
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPCL : DFF, positive-edge-trigger, Load



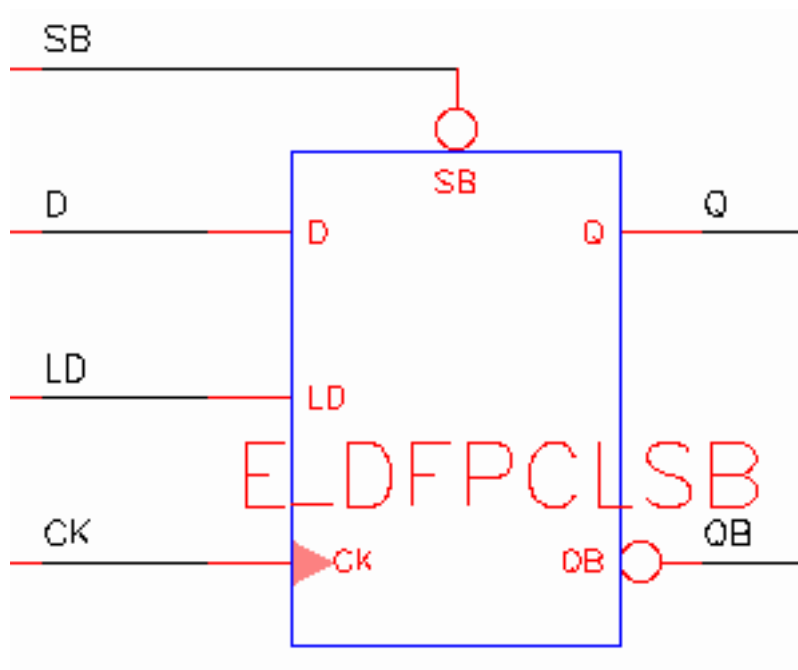
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPCLRSB : DFF, positive-edge-trigger, active-LOW set & reset, Load



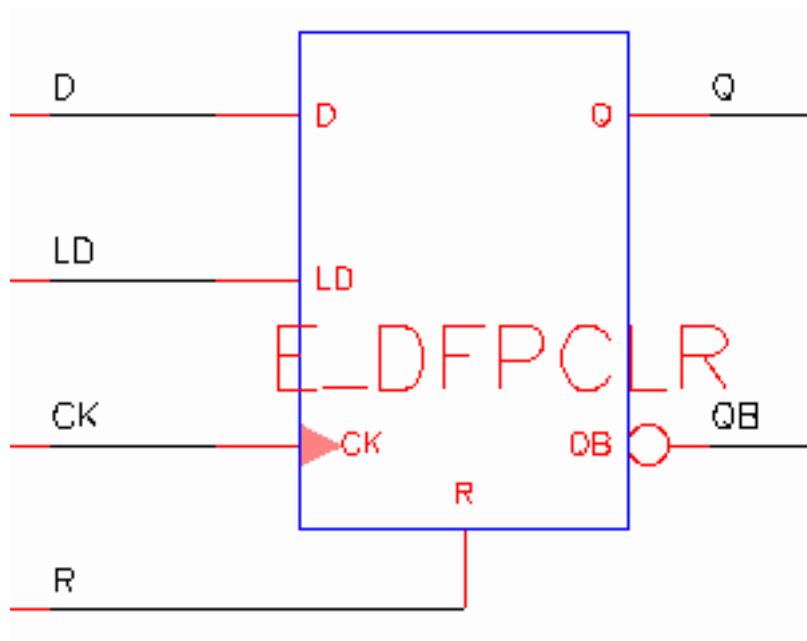
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPCLSB : DFF, positive-edge-trigger, active-LOW set, Load



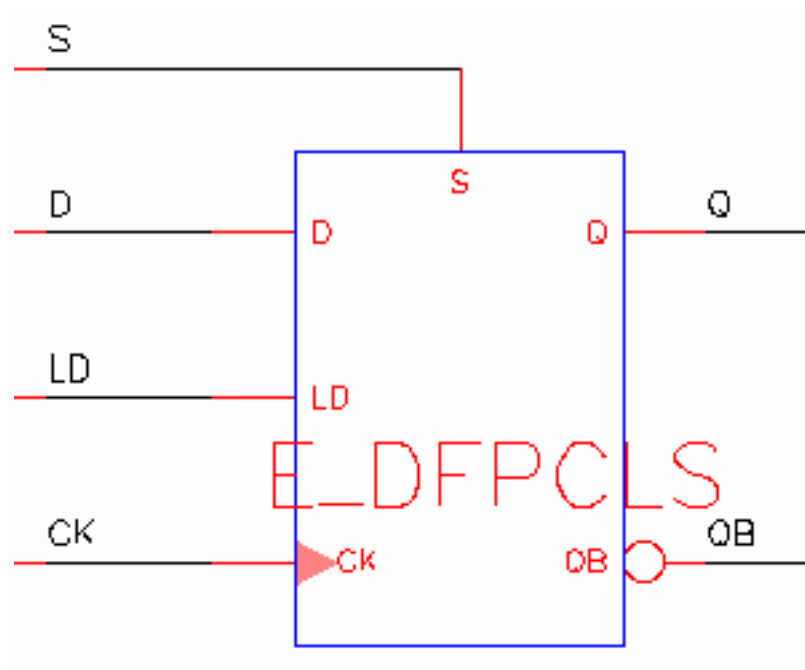
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPCLR : DFF, positive-edge-trigger, active-HIGH reset, Load



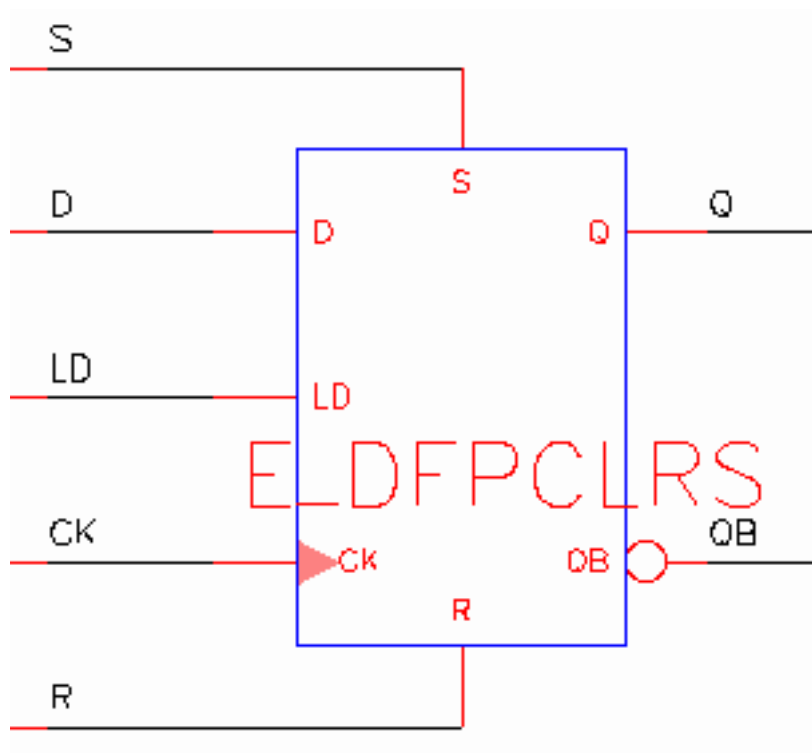
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPCLS : DFF, positive-edge-trigger, active-HIGH set, Load



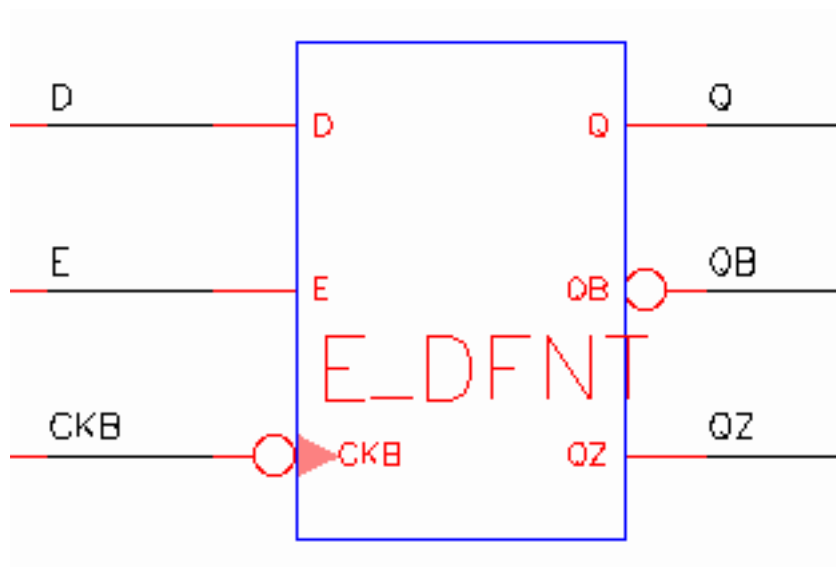
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPCLRS : DFF, positive-edge-trigger, active-HIGH set & reset, Load



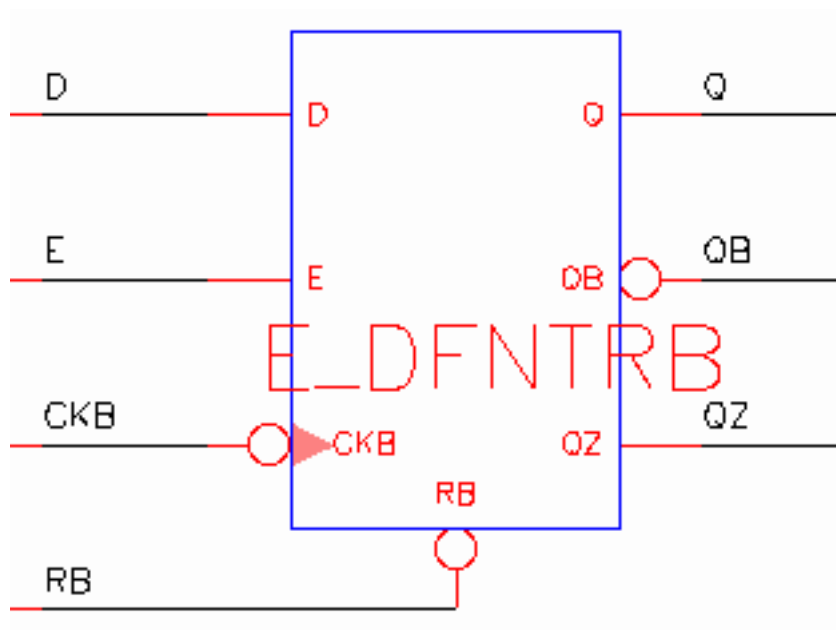
Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNT : DFF, negative-edge-trigger with Tri-State



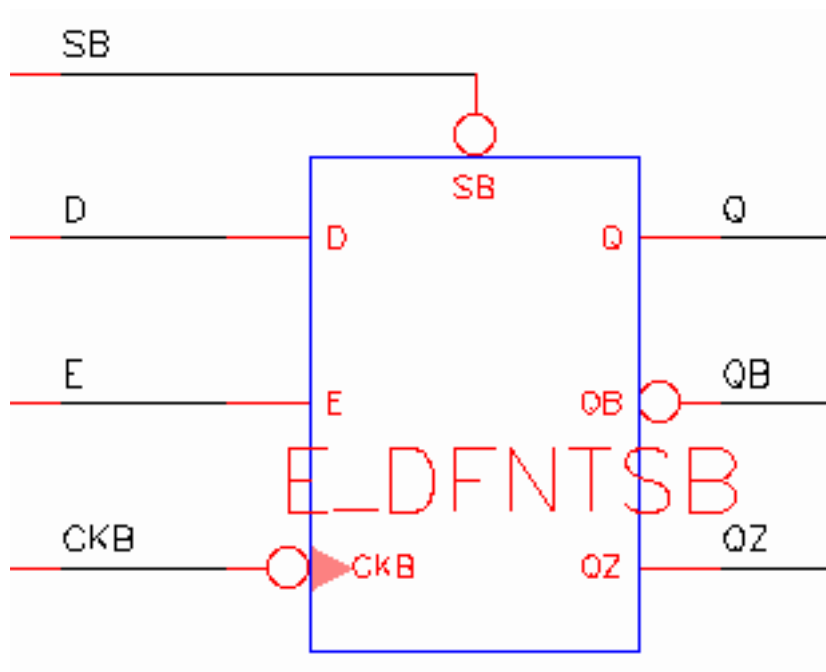
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFNRB : DFF, negative-edge-trigger with Tri-State, active-LOW reset



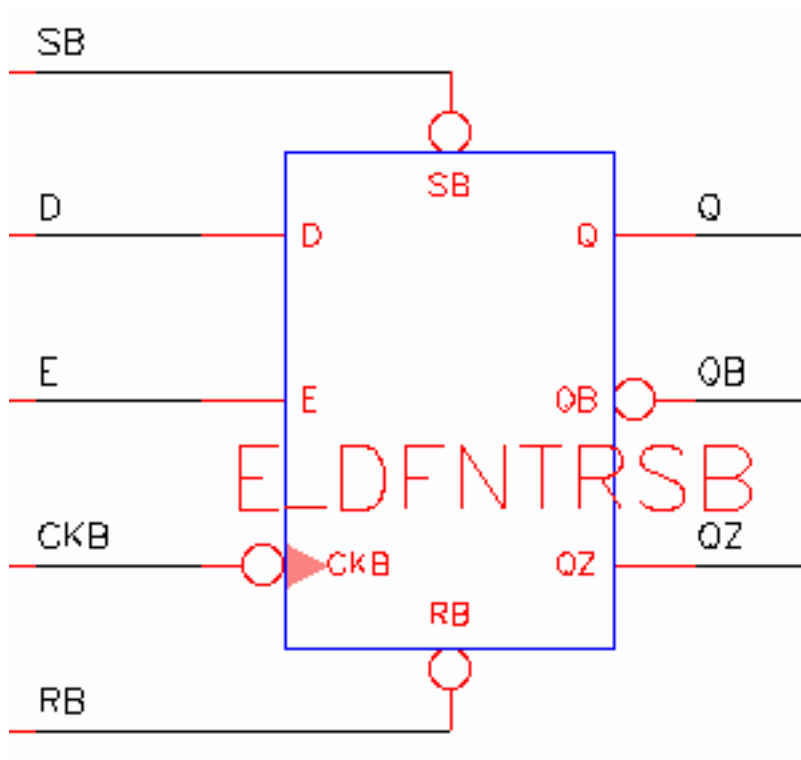
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFNTSB : DFF, negative-edge-trigger with Tri-State, active-LOW set



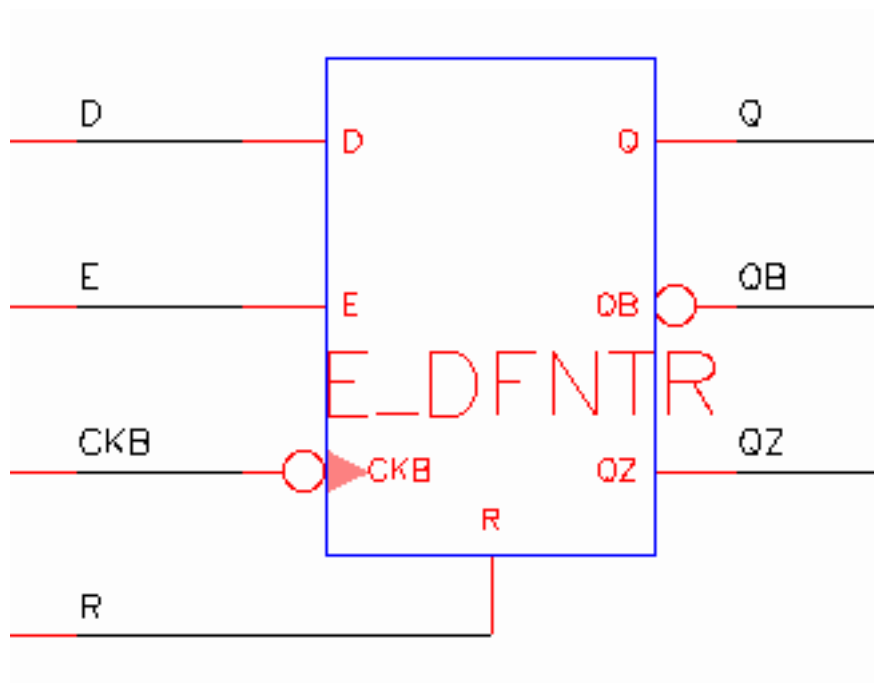
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFNRBSB : DFF, negative-edge-trigger with Tri-State, active-LOW set & reset



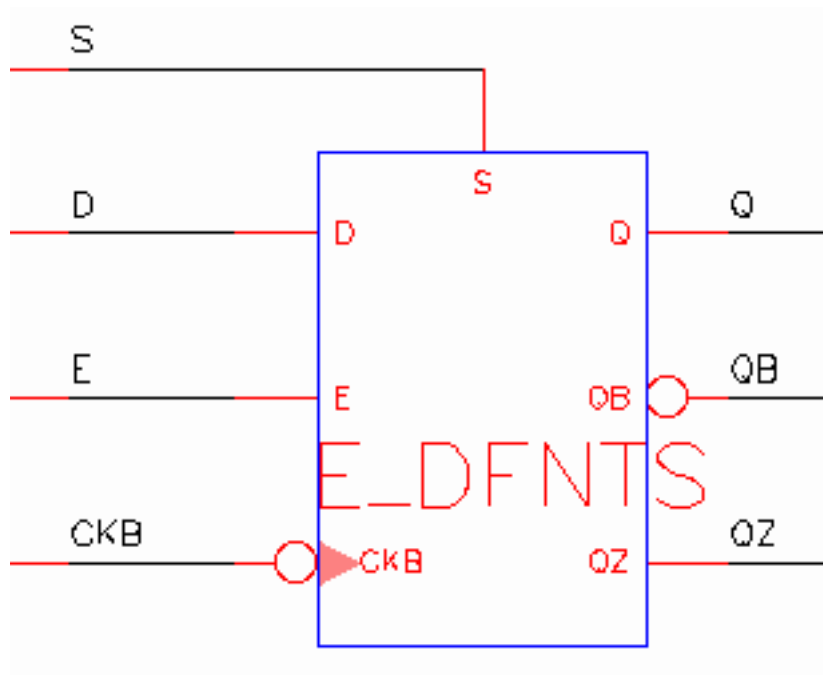
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFNTTR : DFF, negative-edge-trigger with Tri-State, active-HIGH reset



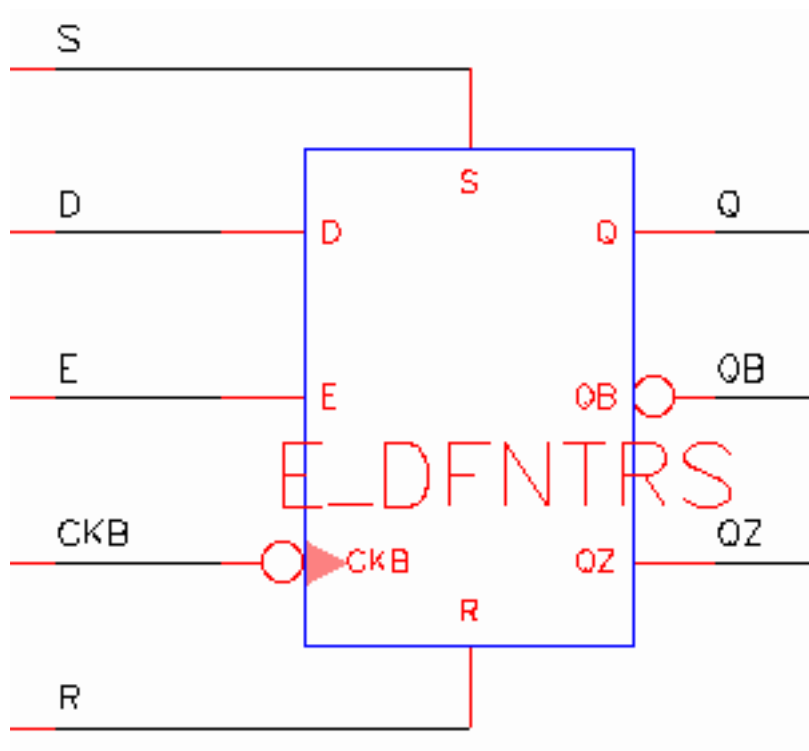
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFNTS : DFF, negative-edge-trigger with Tri-State, active-HIGH set



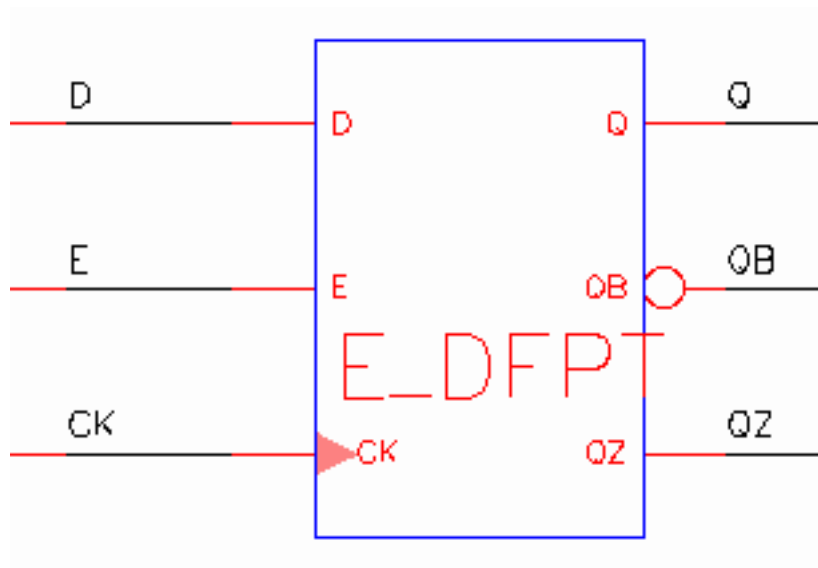
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFNTRS : DFF, negative-edge-trigger with Tri-State, active-HIGH set & reset



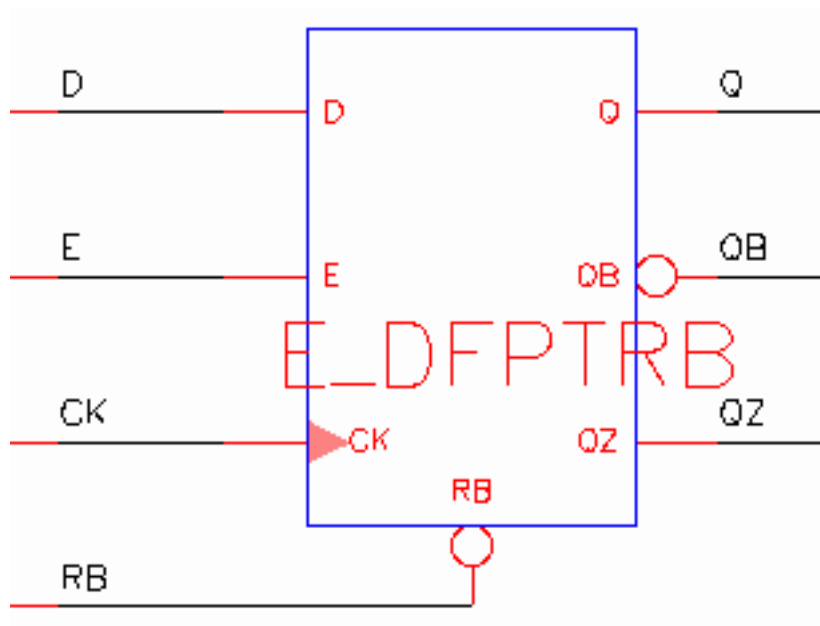
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-HIGH
SB	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPT : DFF, positive-edge-trigger with Tri-State



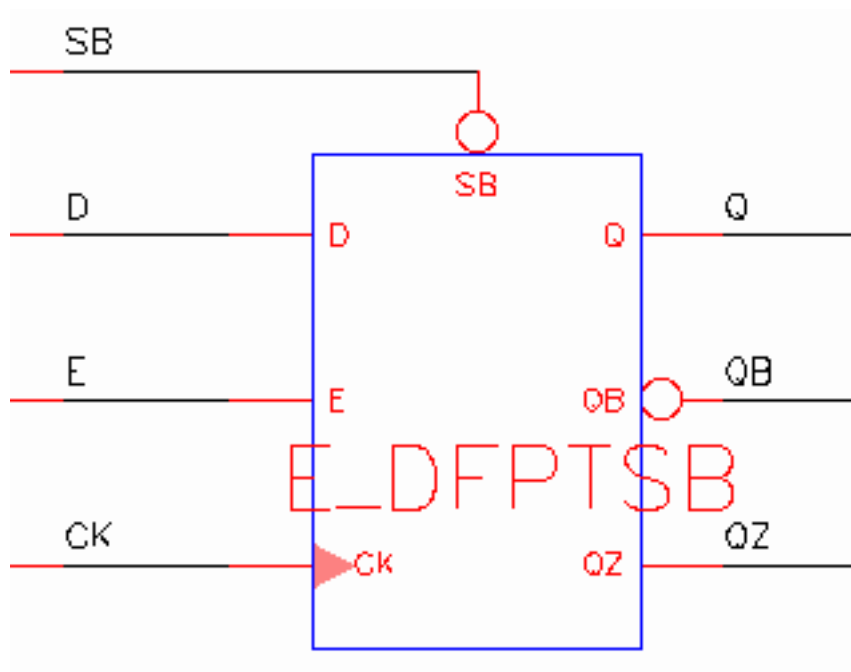
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPTRB : DFF, positive-edge-trigger with Tri-State, active-LOW reset



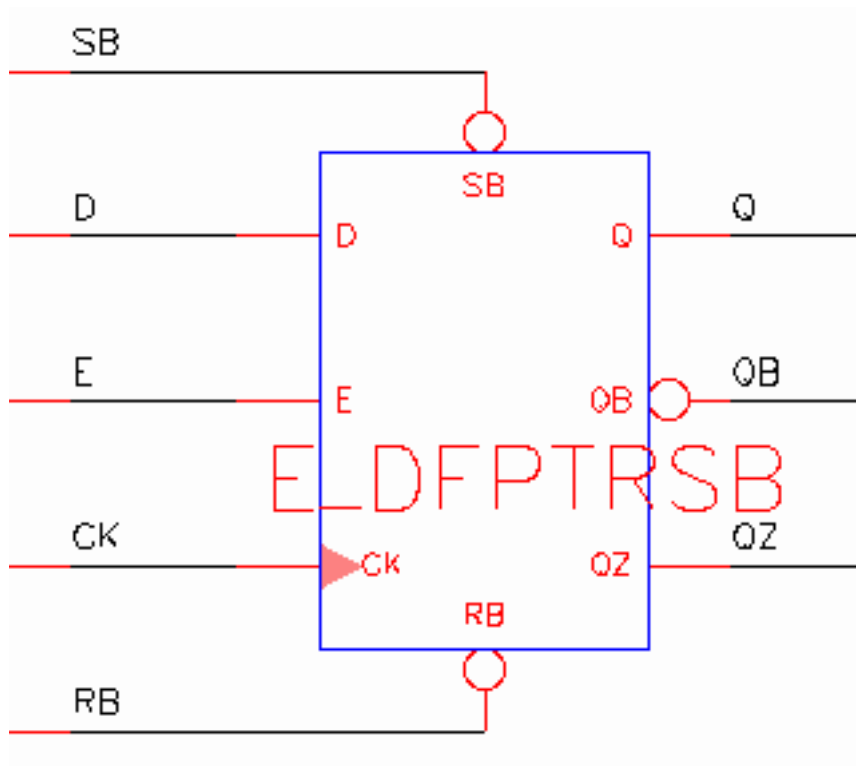
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPTSB : DFF, positive-edge-trigger with Tri-State, active-LOW set



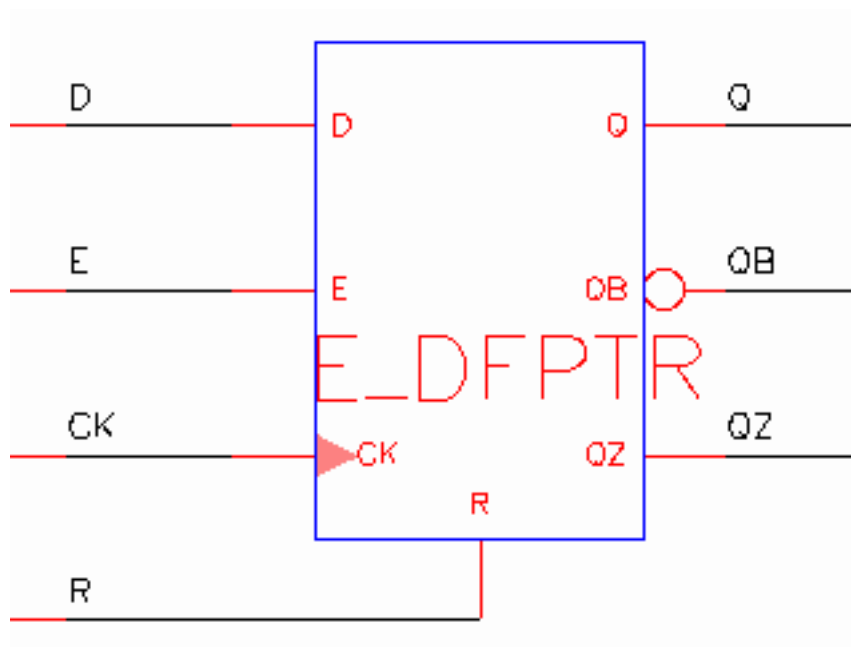
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPTRSB : DFF, positive-edge-trigger with Tri-State, active-LOW set & reset



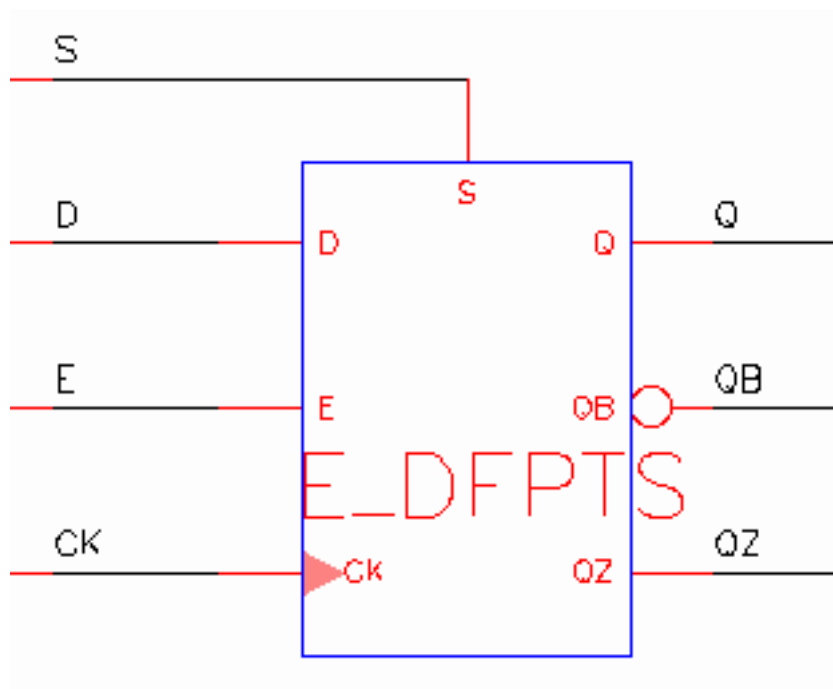
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPTR : DFF, positive-edge-trigger with Tri-State, active-HIGH reset



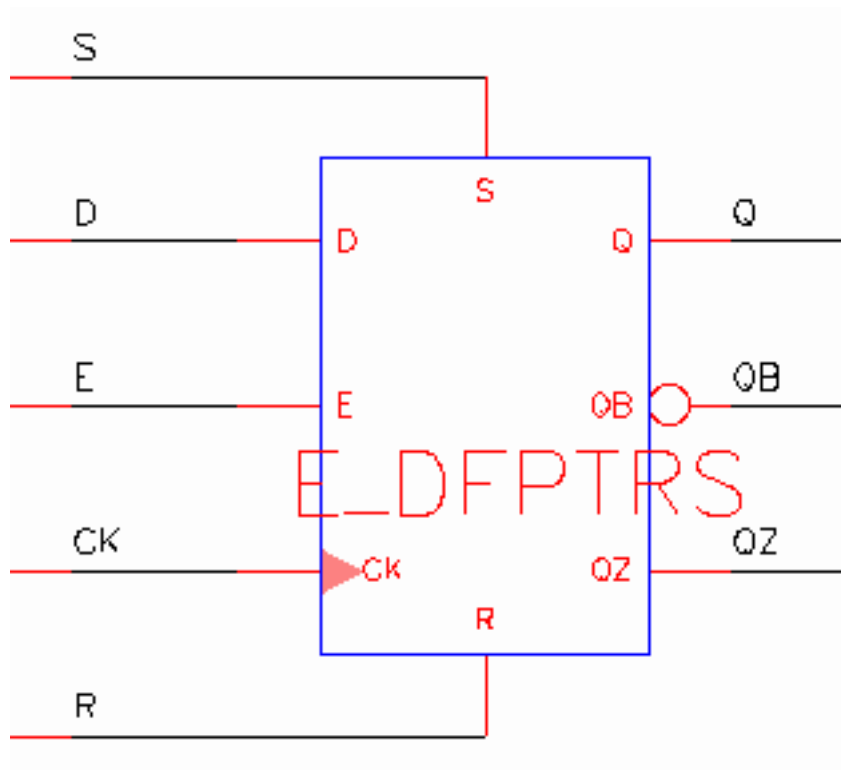
Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPTS : DFF, positive-edge-trigger with Tri-State, active-HIGH set

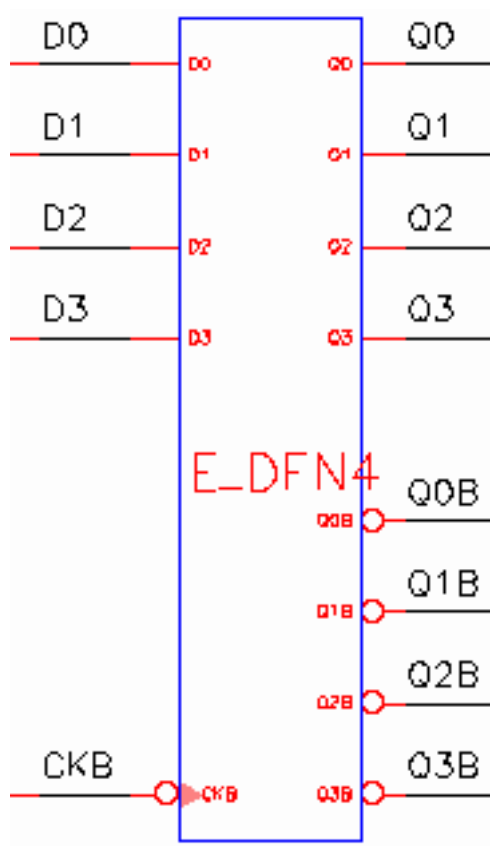


Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFPTRS : DFF, positive-edge-trigger with Tri-State, active-HIGH set & reset

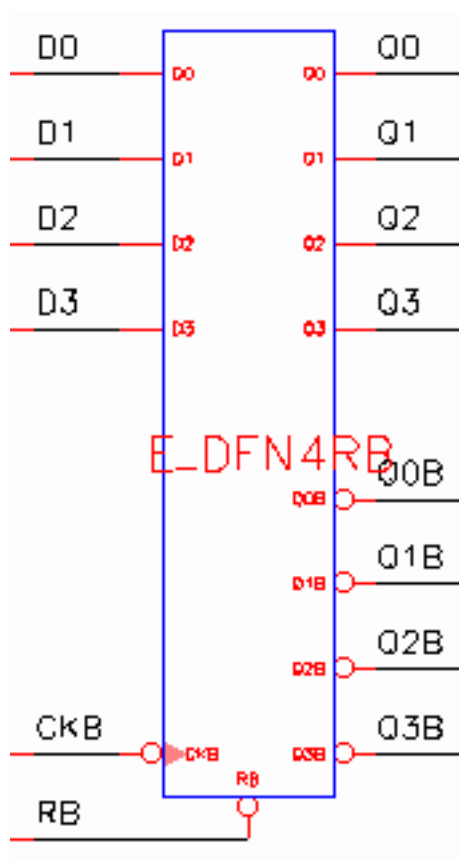


Pin	Direction	Description
D	INPUT	data
E	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-HIGH
SB	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_DFN4 : 4Bit DFF, negative-edge-trigger

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted

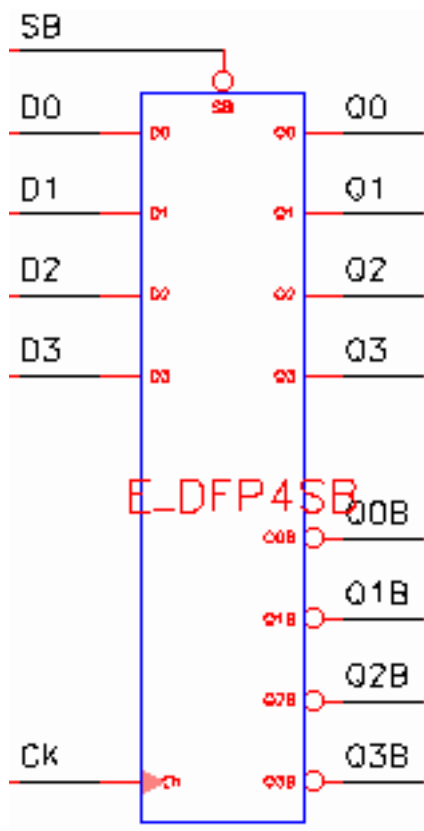
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_DFN4RB : 4Bit DFF, negative-edge-trigger, active-LOW reset

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

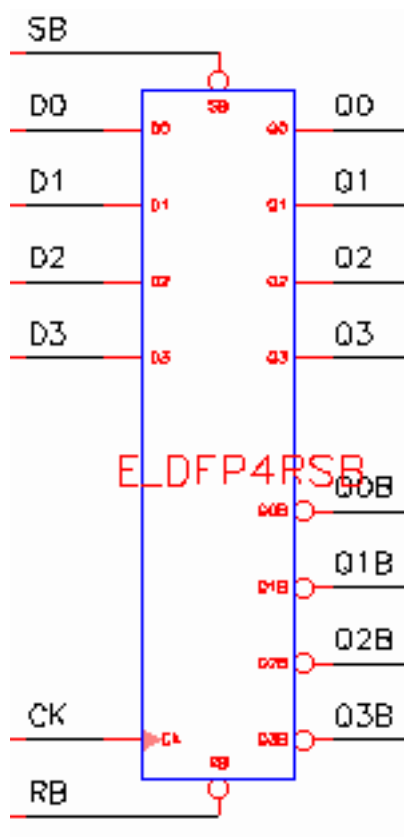
E_DFN4SB : 4Bit DFF, negative-edge-trigger, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted

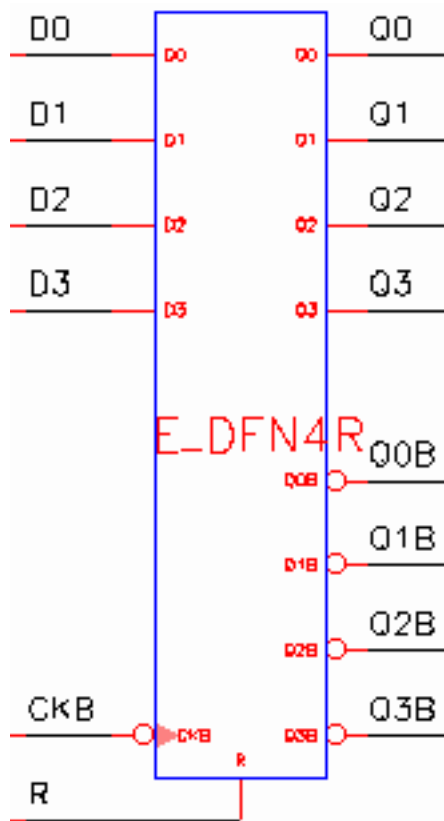
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_DFN4RSB : 4Bit DFF, negative-edge-trigger, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

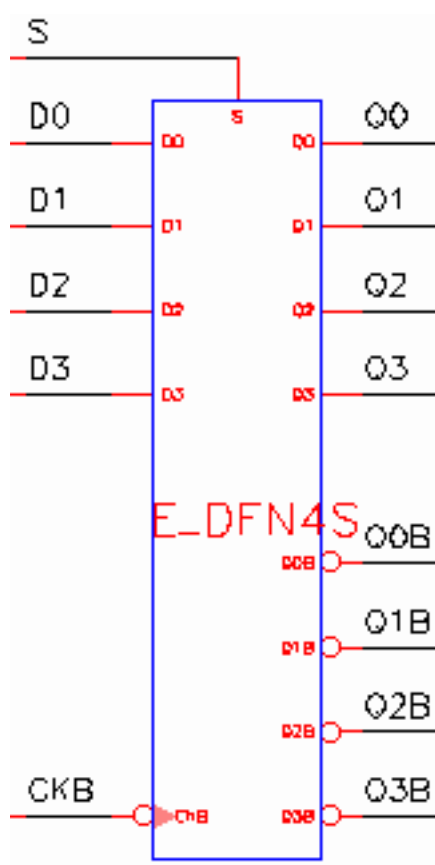
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_DFN4R : 4Bit DFF, negative-edge-trigger, active-HIGH reset

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

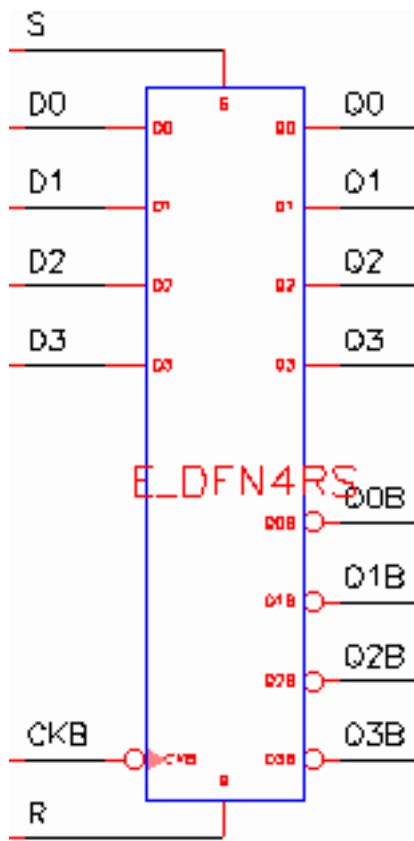
E_DFN4S : 4Bit DFF, negative-edge-trigger, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

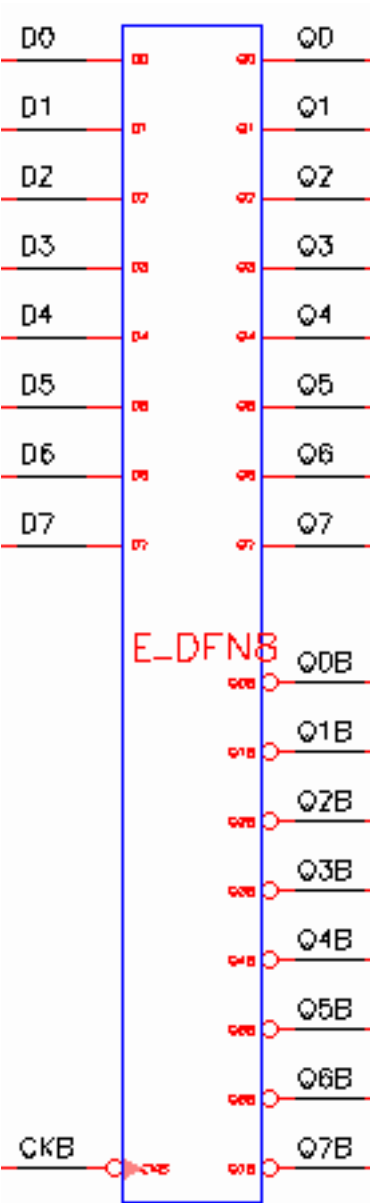
E_DFN4RS : 4Bit DFF, negative-edge-trigger, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

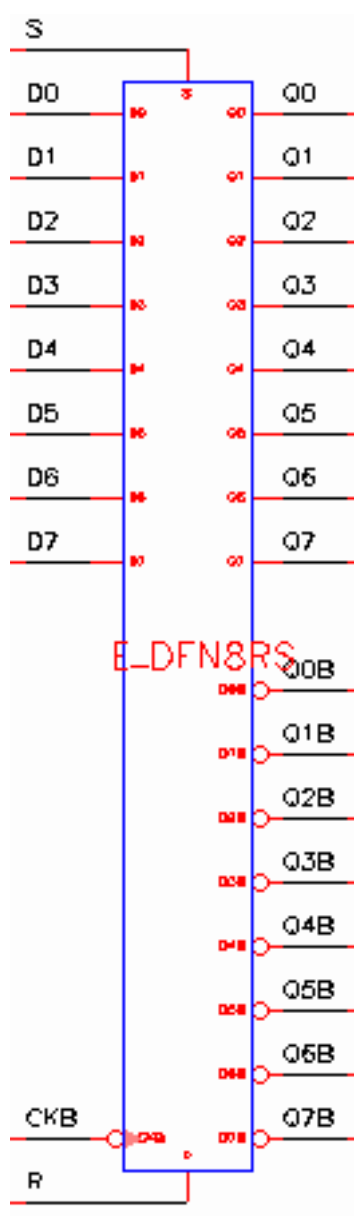
E_DFN8 : 8Bit DFF, negative-edge-trigger



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

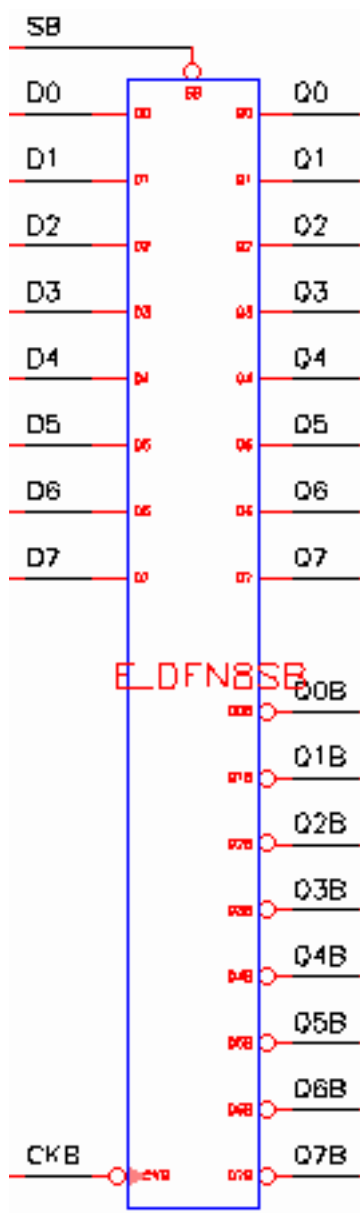
E_DFN8RB : 8Bit DFF, negative-edge-trigger, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

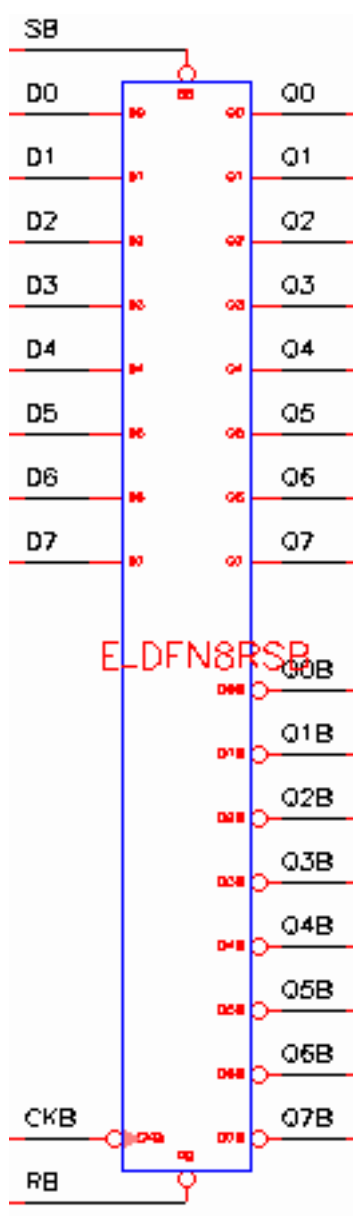
E_DFN8SB : 8Bit DFF, negative-edge-trigger, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

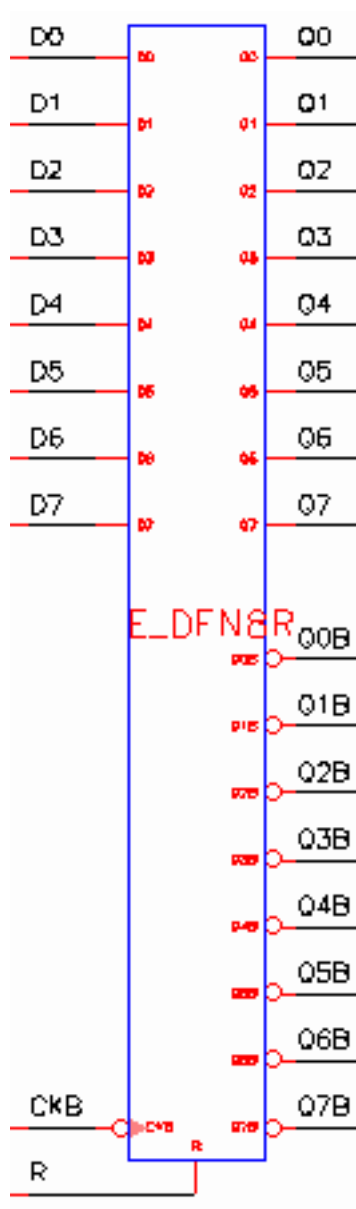
E_DFN8RSB : 8Bit DFF, negative-edge-trigger, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

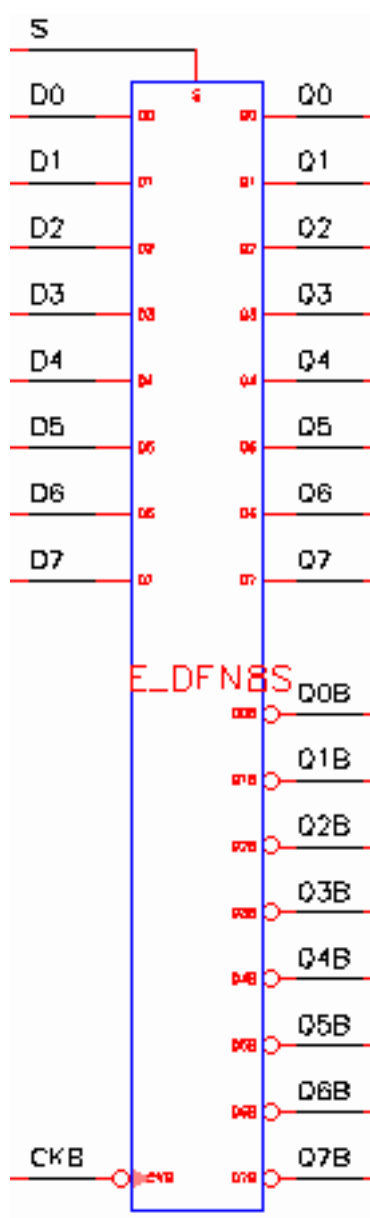
E_DFN8R : 8Bit DFF, negative-edge-trigger, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
R	INPUT	reset

D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

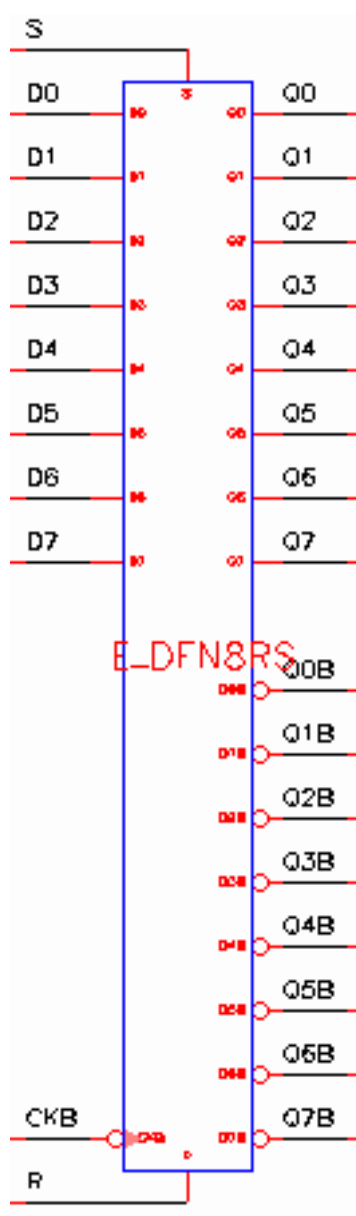
E_DFN8S : 8Bit DFF, negative-edge-trigger, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

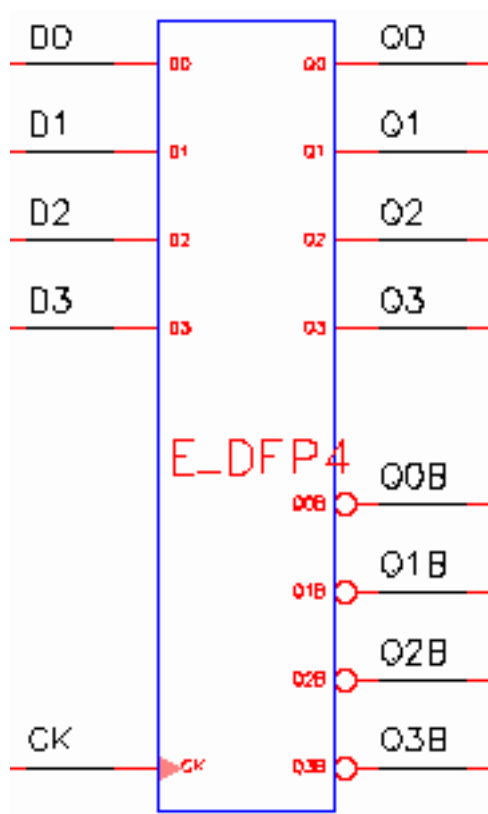
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_DFN8RS : 8Bit DFF, negative-edge-trigger, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data

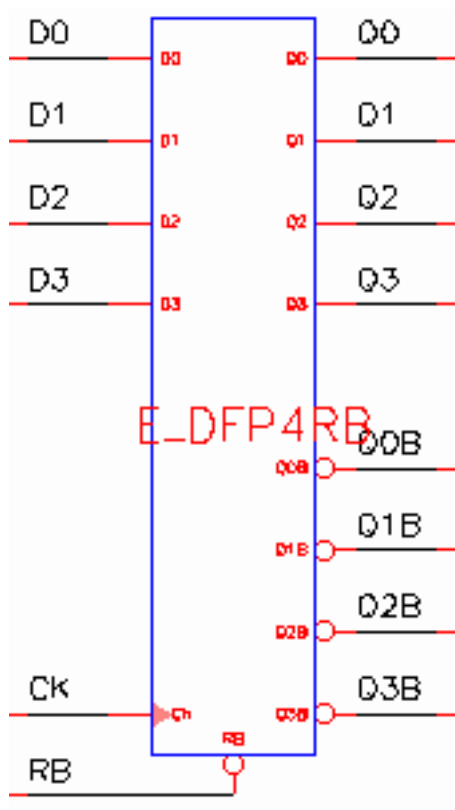
D6	INPUT	data
D7	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_DFP4 : 4Bit DFF, positive-edge-trigger

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted

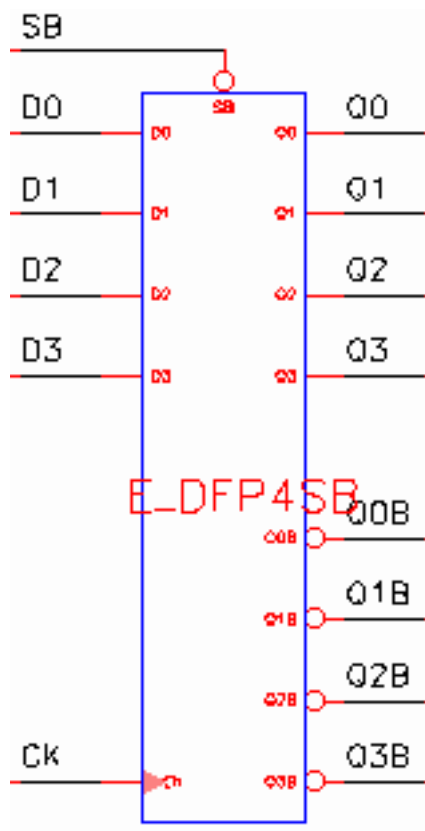
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_DFP4RB : 4Bit DFF, positive-edge-trigger, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted

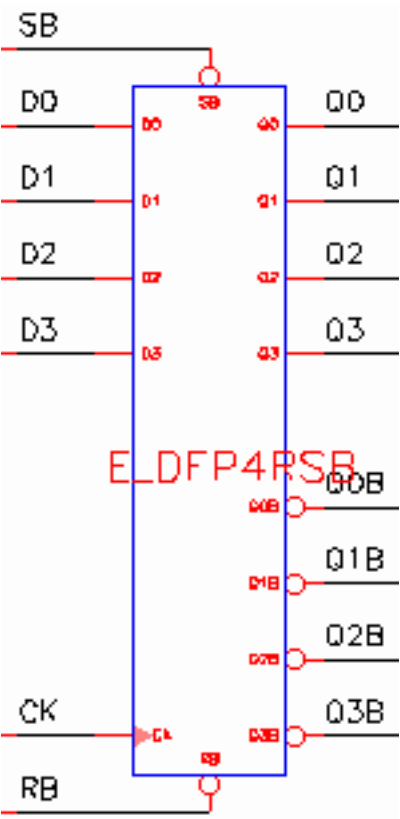
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_DFP4SB : 4Bit DFF, positive-edge-trigger, active-LOW set

Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted

Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

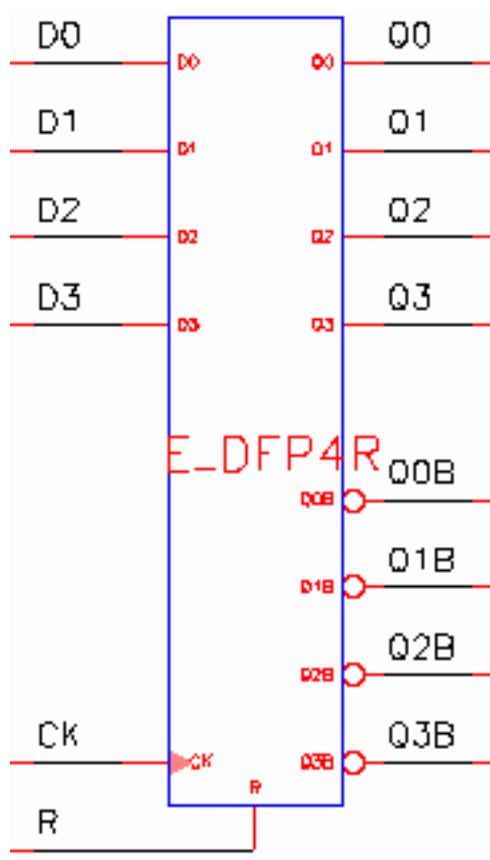
E_DFP4RSB : 4Bit DFF, positive-edge-trigger, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

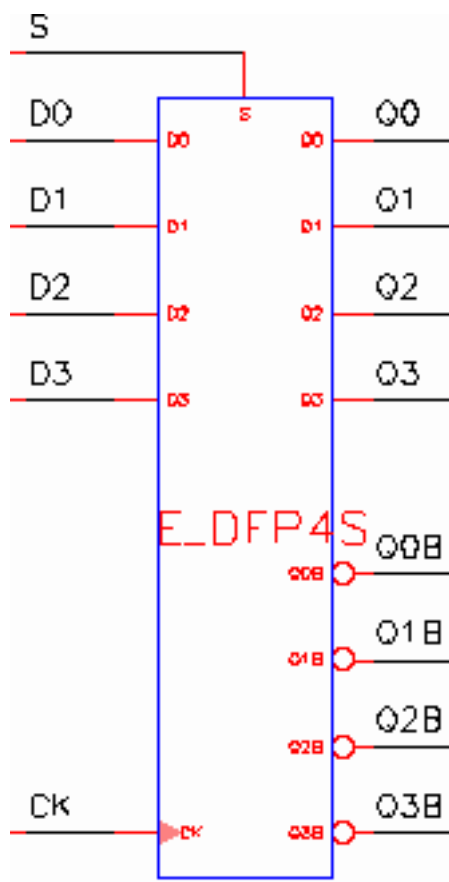
E_DFP4R : 4Bit DFF, positive-edge-trigger, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

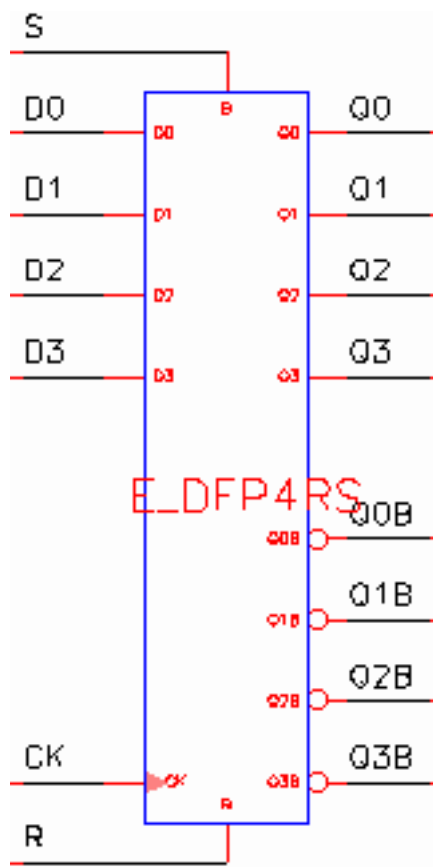
E_DFP4S : 4Bit DFF, positive-edge-trigger, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data

Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

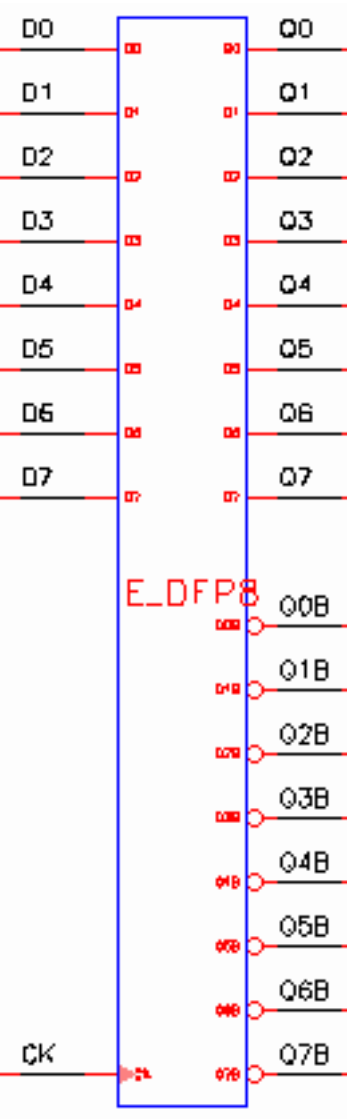
E_DFP4RS : 4Bit DFF, positive-edge-trigger, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data

Q3	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted

E_DFP8 : 8Bit DFF, positive-edge-trigger



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

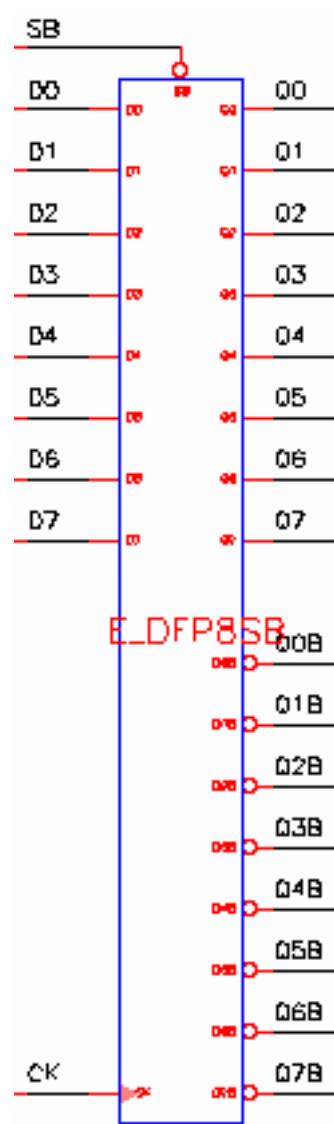
E_DFP8RB : 8Bit DFF, positive-edge-trigger, active-LOW reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_DFP8SB : 8Bit DFF, positive-edge-trigger, active-LOW set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data
D7	INPUT	data

D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

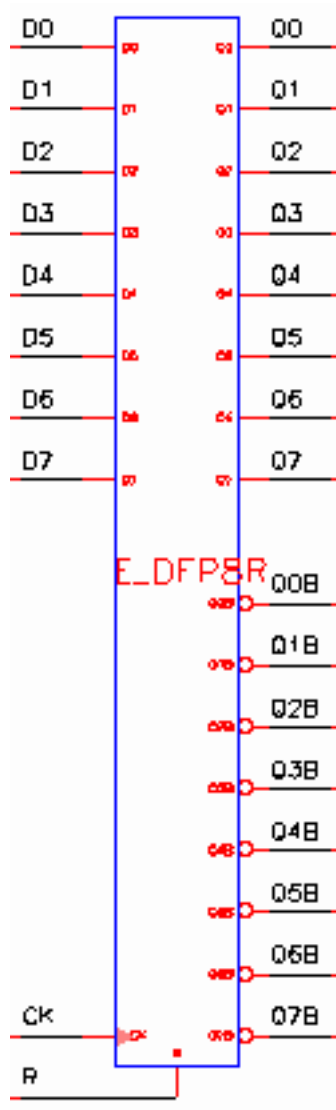
E_DFP8RSB : 8Bit DFF, positive-edge-trigger, active-LOW set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

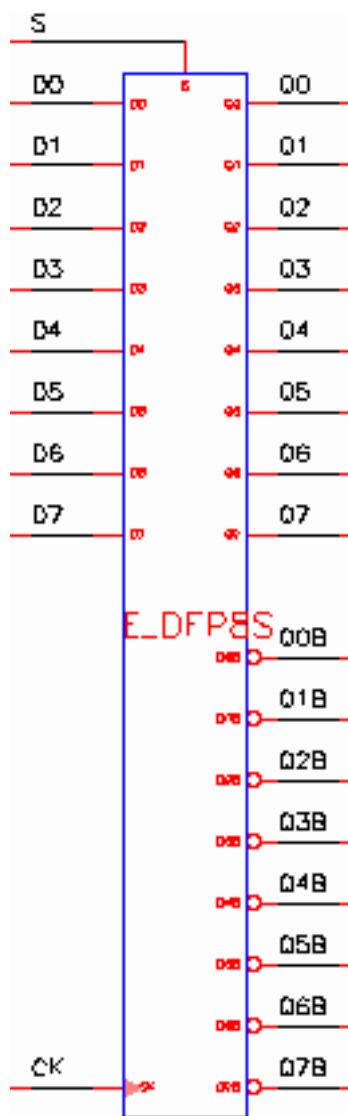
E_DFP8R : 8Bit DFF, positive-edge-trigger, active-HIGH reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_DFP8S : 8Bit DFF, positive-edge-trigger, active-HIGH set



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

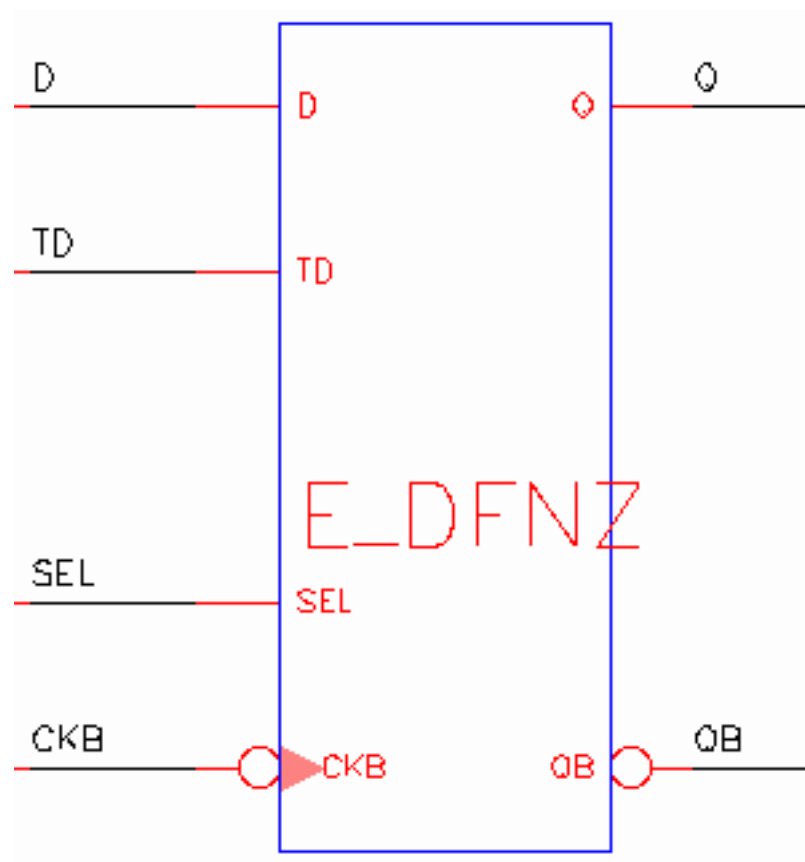
E_DFP8RS : 8Bit DFF, positive-edge-trigger, active-HIGH set & reset



Pin	Direction	Description
D0	INPUT	data
D1	INPUT	data
D2	INPUT	data
D3	INPUT	data
D4	INPUT	data
D5	INPUT	data
D6	INPUT	data

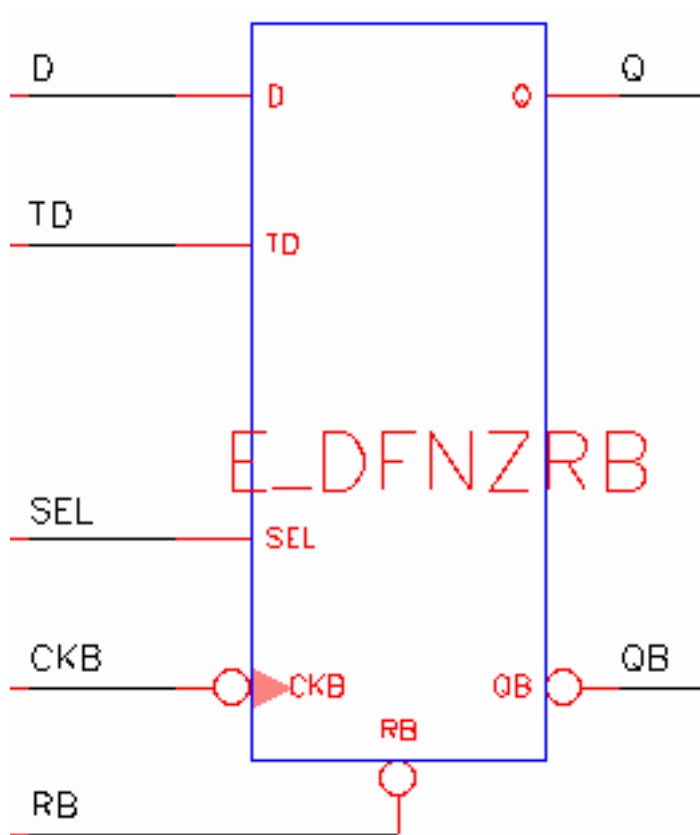
D7	INPUT	data
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q0	OUTPUT	data
Q1	OUTPUT	data
Q2	OUTPUT	data
Q3	OUTPUT	data
Q4	OUTPUT	data
Q5	OUTPUT	data
Q6	OUTPUT	data
Q7	OUTPUT	data
Q0B	OUTPUT	data, inverted
Q1B	OUTPUT	data, inverted
Q2B	OUTPUT	data, inverted
Q3B	OUTPUT	data, inverted
Q4B	OUTPUT	data, inverted
Q5B	OUTPUT	data, inverted
Q6B	OUTPUT	data, inverted
Q7B	OUTPUT	data, inverted

E_DFNZ : DFF with SCAN, negative-edge-trigger



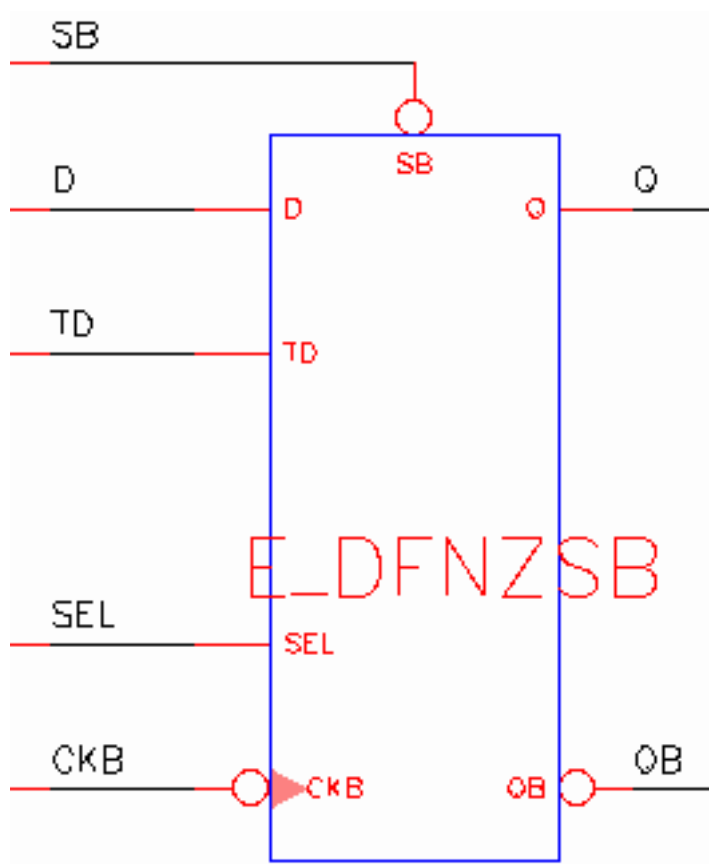
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZRB : DFF with SCAN, negative-edge-trigger, active-LOW reset



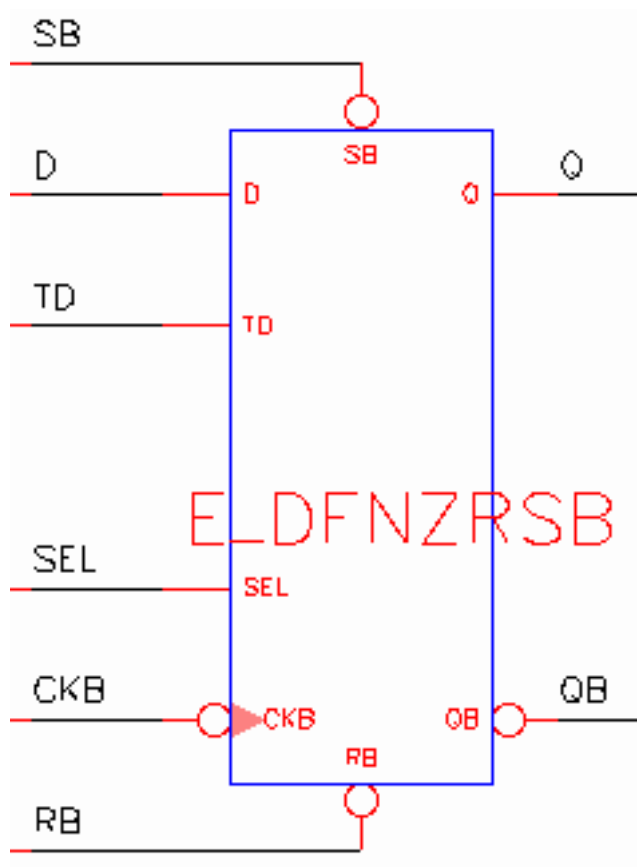
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZSB : DFF with SCAN, negative-edge-trigger, active-LOW set



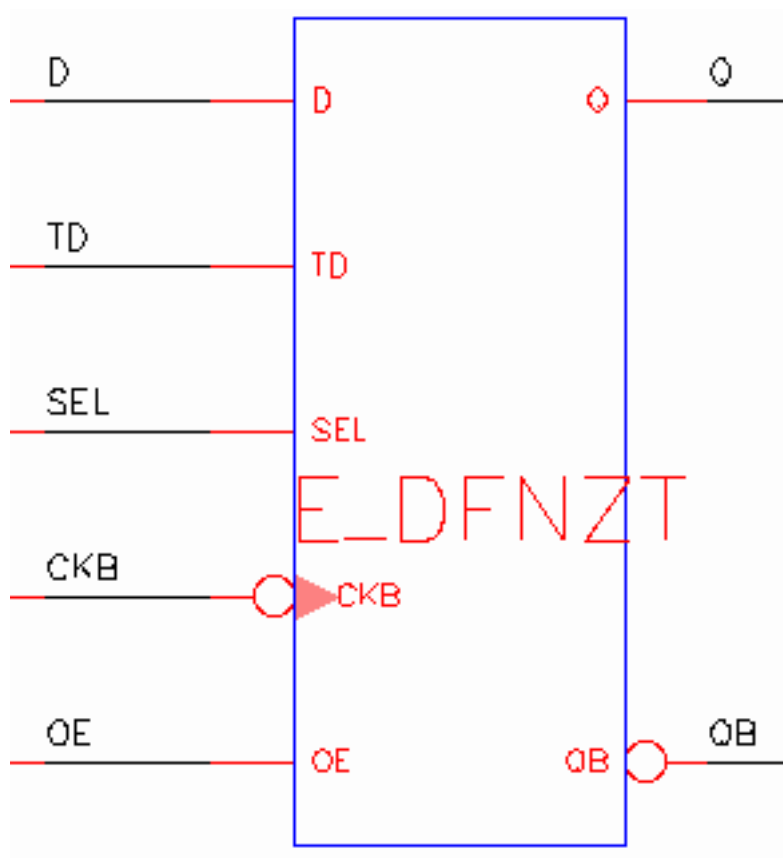
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZRSB : DFF with SCAN, negative-edge-trigger, active-LOW set & reset



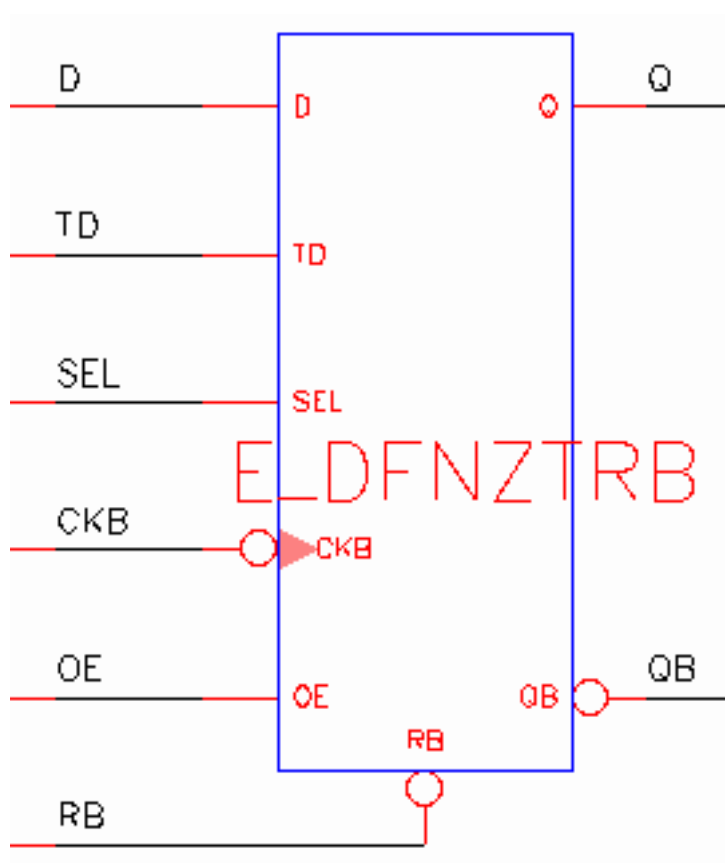
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset
SB	INPUT	async set
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZT : DFF with SCAN, negative-edge-trigger, and Output enable



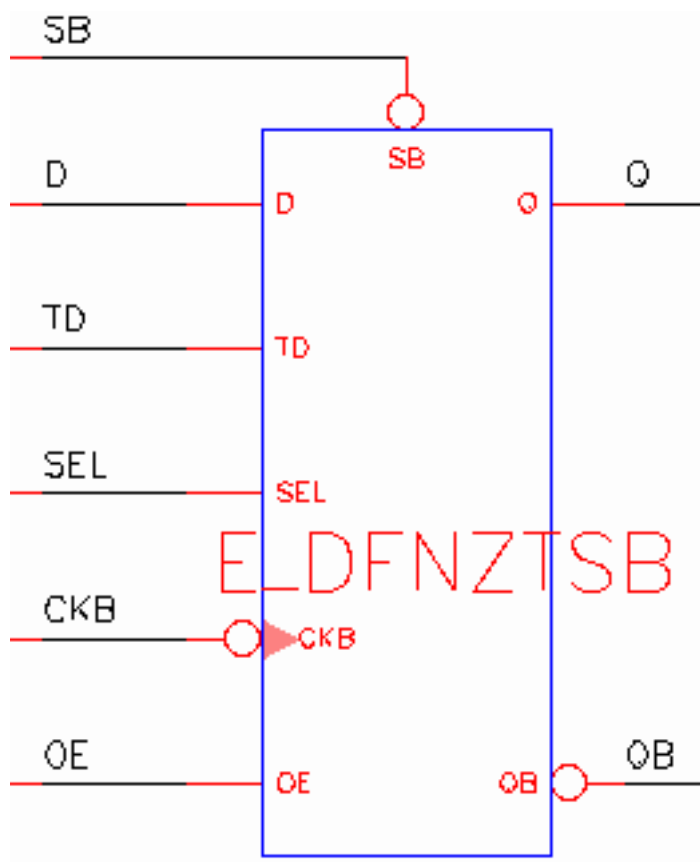
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
OE	INPUT	output enable, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZTRB: DFF with SCAN, negative-edge-trigger, active-LOW reset, and Output enable



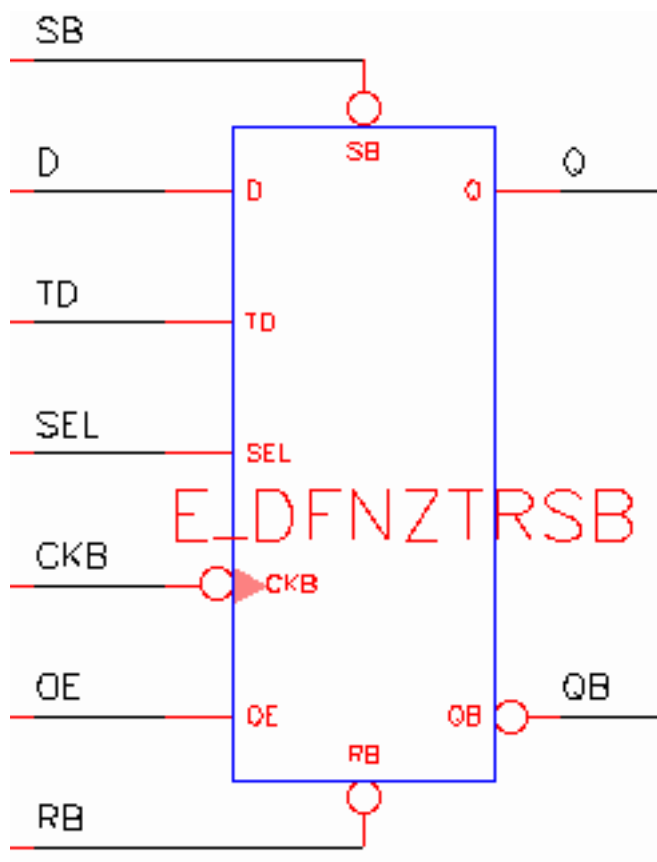
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZTSB : DFF with SCAN, negapositive-edge-trigger, active-LOW set, and Output enable

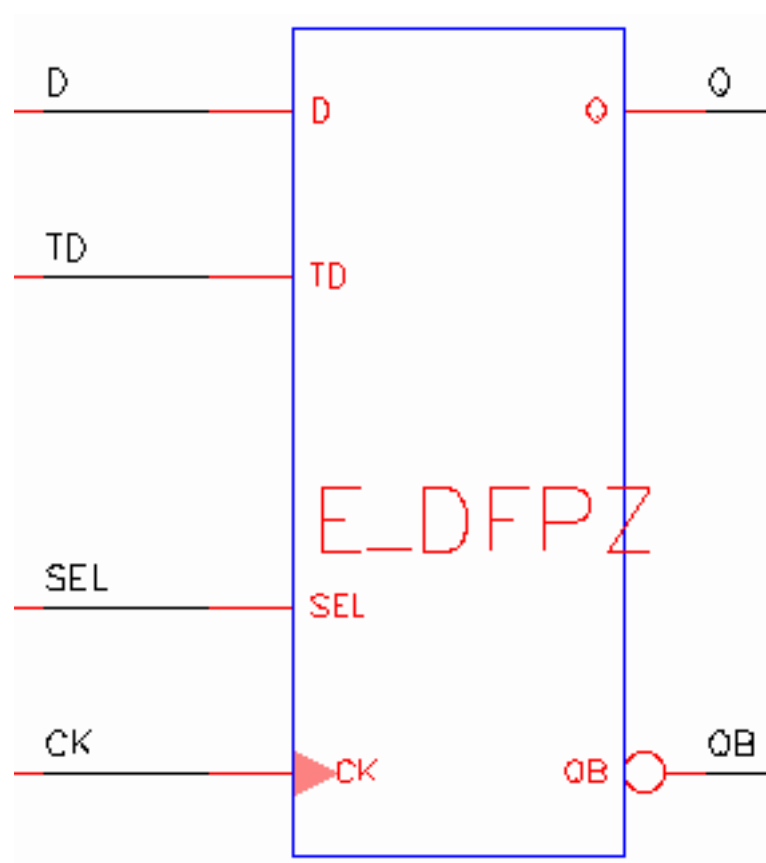


Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
OE	INPUT	output enable, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFNZTRSB : DFF with SCAN, negative-edge-trigger, active-LOW set & reset, and Output enable

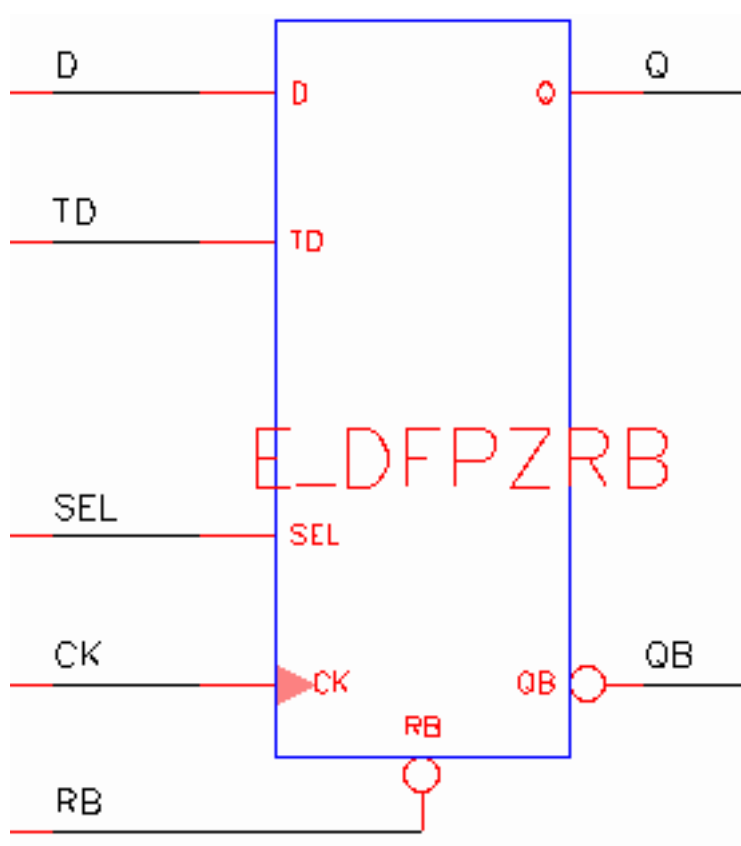


Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZ : DFF with SCAN, positive-edge-trigger

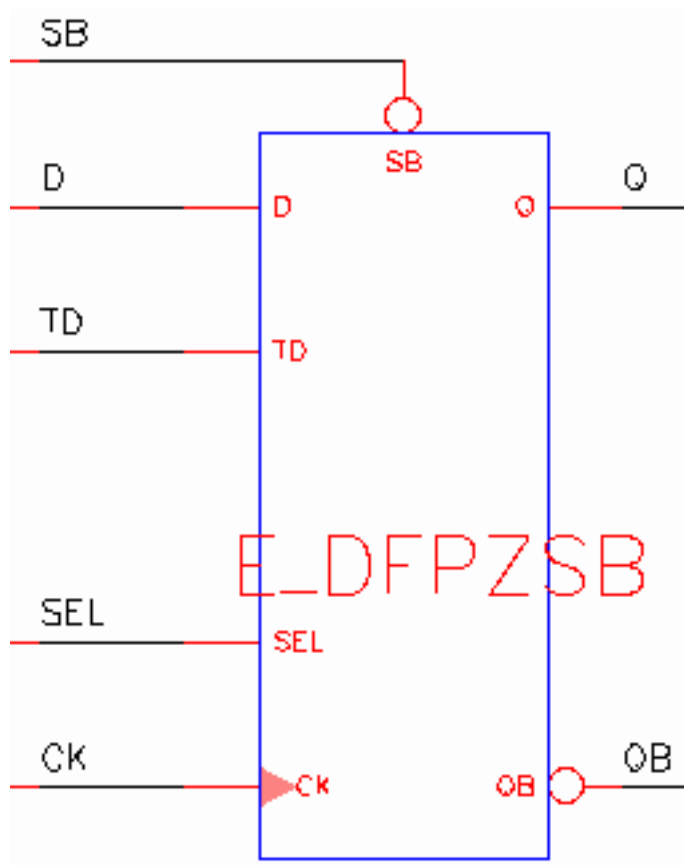
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZRB : DFF with SCAN, positive-edge-trigger, active-LOW reset



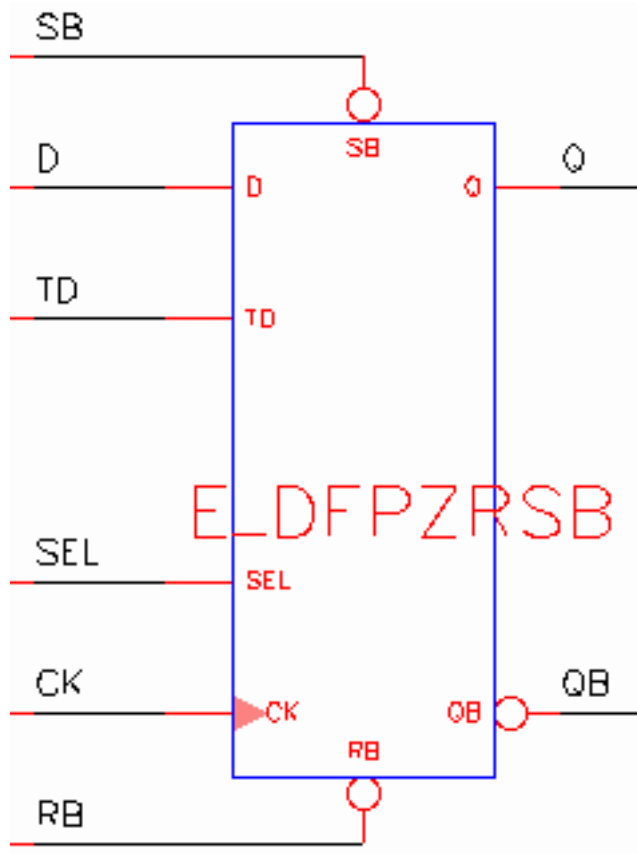
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZSB : DFF with SCAN, positive-edge-trigger, active-LOW set

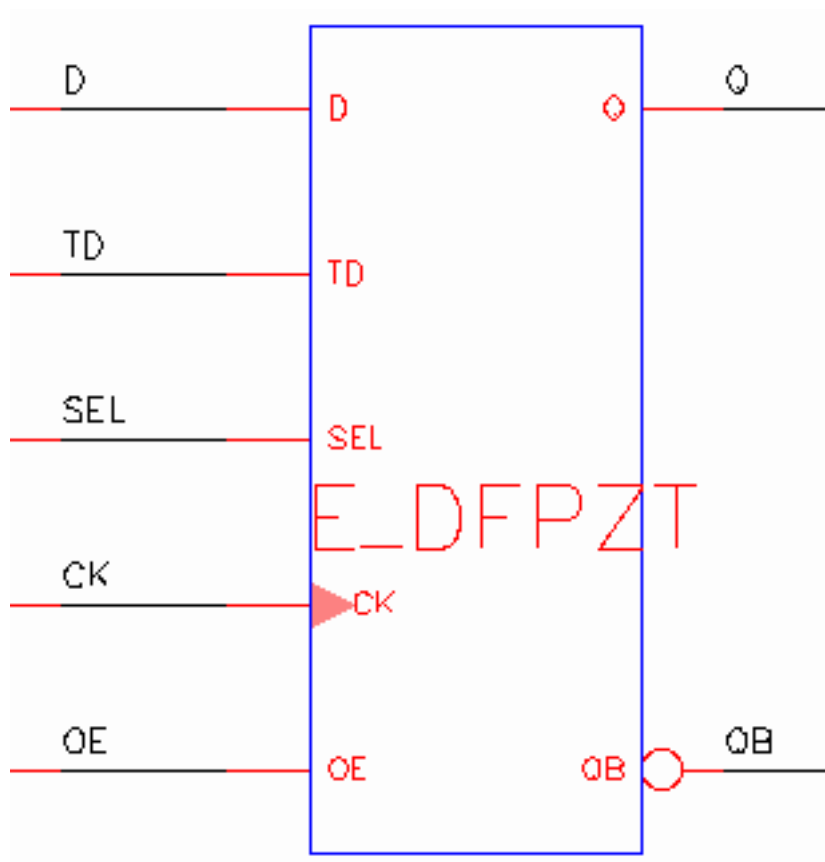


Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZRSB : DFF with SCAN, positive-edge-trigger, active-LOW set & reset

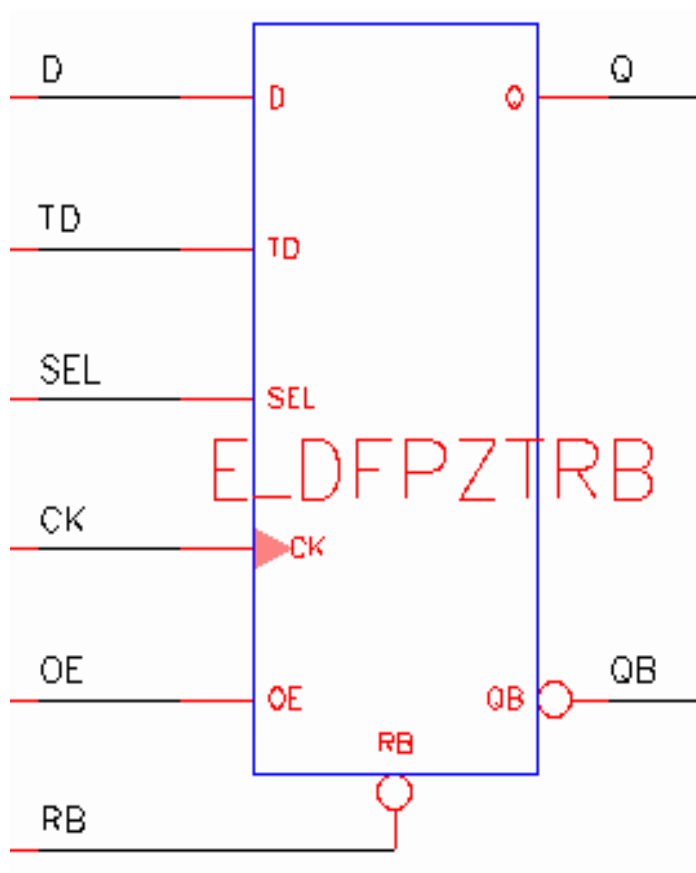


Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset
SB	INPUT	async set
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZT : DFF with SCAN, positive-edge-trigger, and Output enable

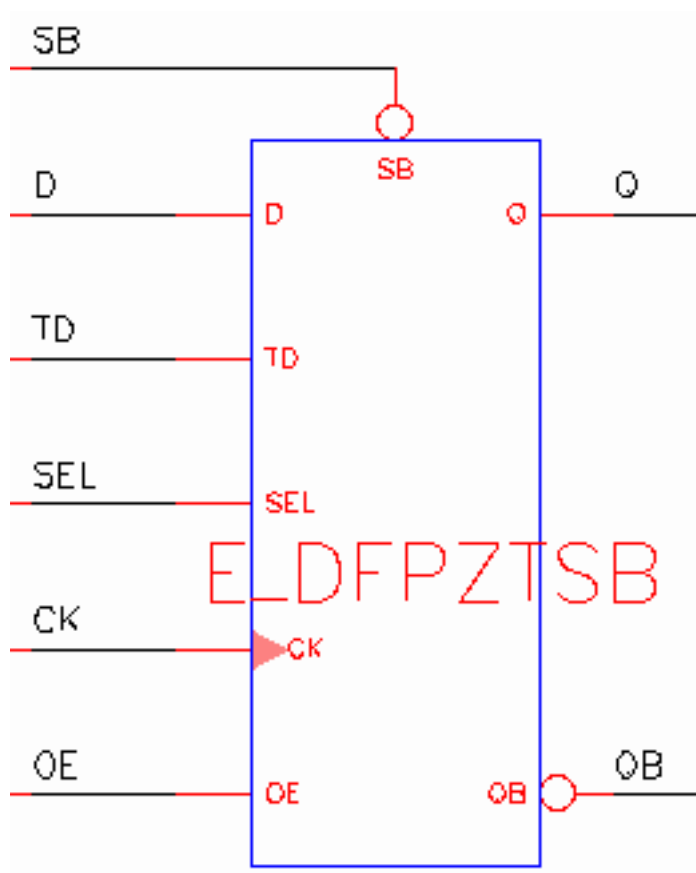
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
OE	INPUT	output enable, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZTSB : DFF with SCAN, positive-edge-trigger, active-LOW set, and Output enable



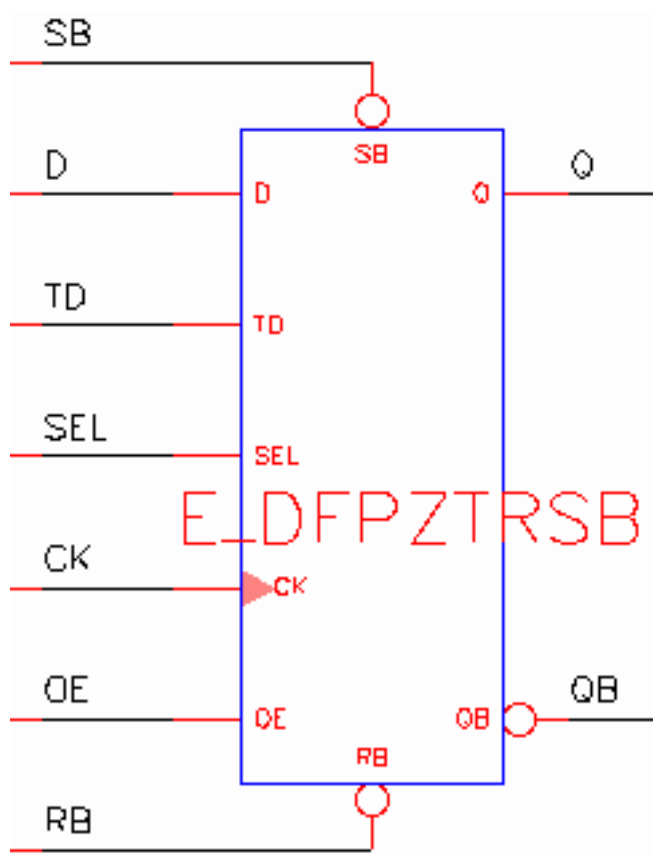
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
OE	INPUT	output enable, active-HIGH
RB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZTSB : DFF with SCAN, positive-edge-trigger, active-LOW set, and Output enable



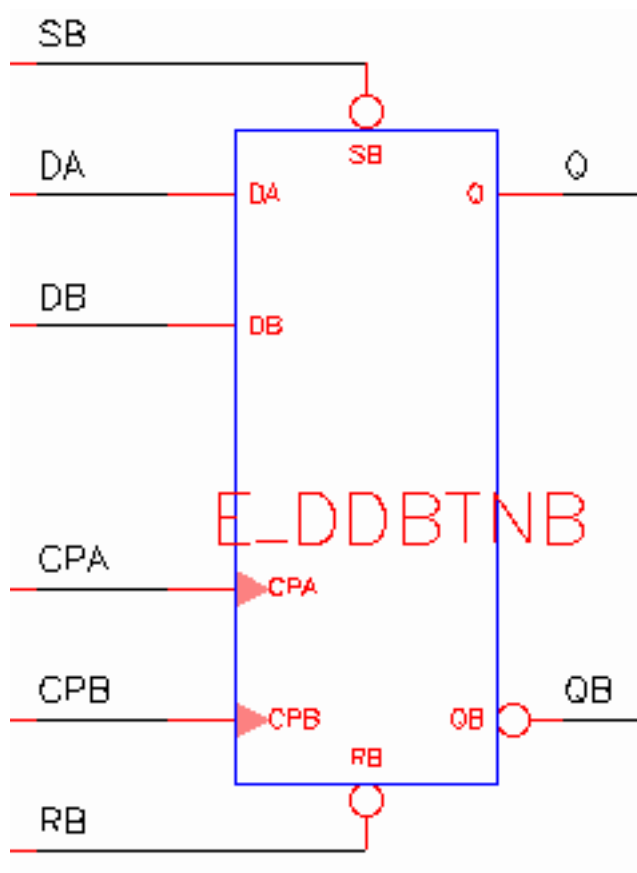
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
OE	INPUT	output enable, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFPZTRSB : DFF with SCAN, positive-edge-trigger, active-LOW set & reset, and Output enable



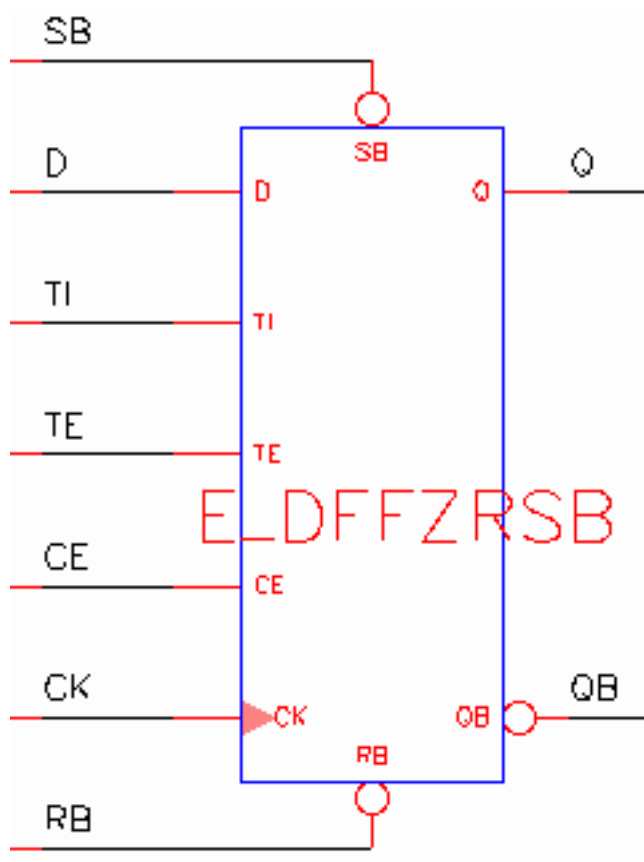
Pin	Direction	Description
D	INPUT	data
TD	INPUT	scan data in
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DDBTNB : DFF with double clock, active-LOW set & reset



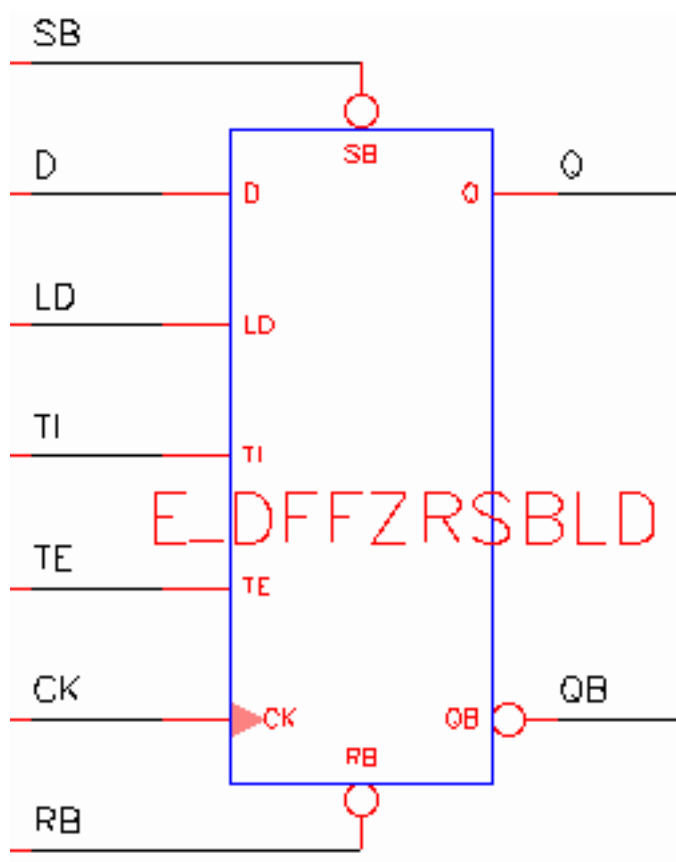
Pin	Direction	Description
DA	INPUT	data A
DB	INPUT	data B
CPA	INPUT	clock A, active-HIGH
CPB	INPUT	clock B, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFFZRSB : DFF active-HIGH scan, positive-edge-trigger, active-LOW set & reset, active-HIGH output enable



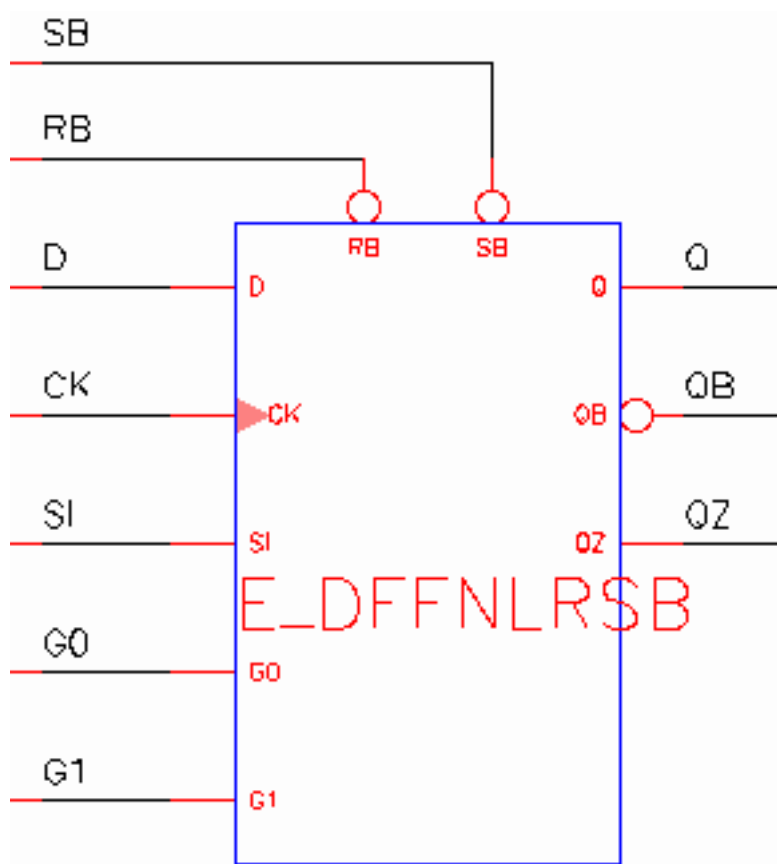
Pin	Direction	Description
D	INPUT	data
TI	INPUT	scan data
TE	INPUT	scan enable
CE	INPUT	output enable, active-HIGH
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFFZRSBLD : DFF active HIGH scan, positive-edge-trigger, active-HIGH D load, active-low reset

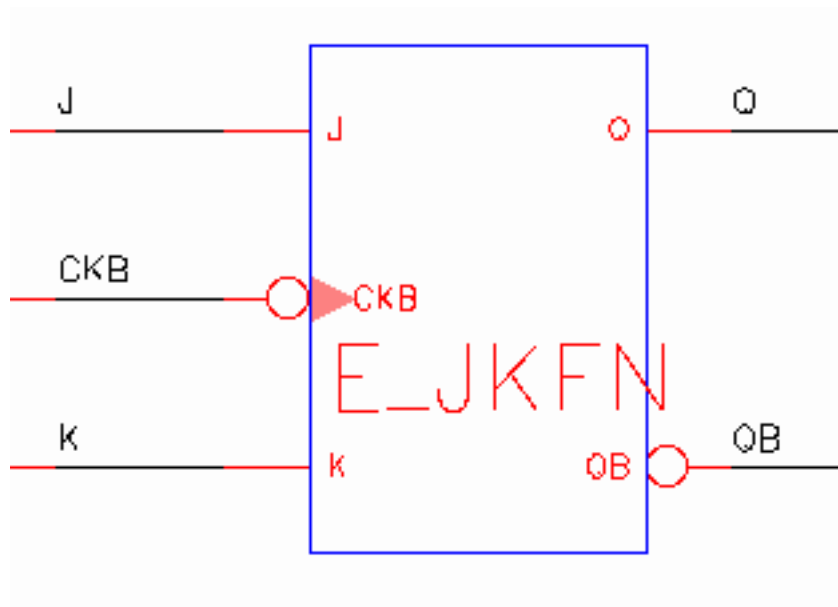


Pin	Direction	Description
D	INPUT	data
LD	INPUT	load, active-HIGH
TI	INPUT	scan data
TE	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_DFFNLRSB : D Flip-Flop, positive-edge-triggered, active-LOW set & reset , with scan

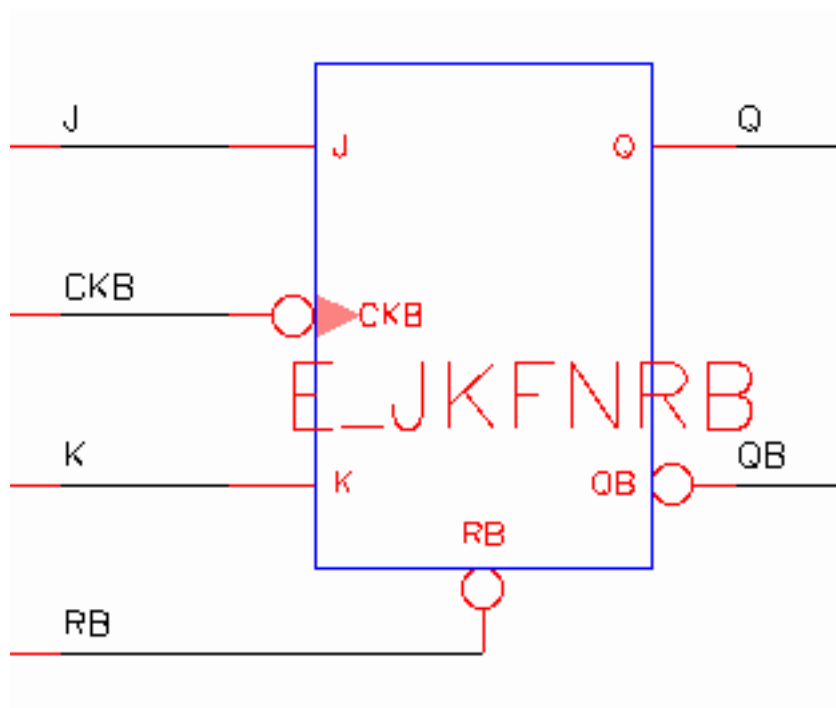


Pin	Direction	Description
D	INPUT	data
CK	INPUT	clock, positive-edge-trigger
SI	INPUT	scan data in
G0	INPUT	enable 0
G1	INPUT	enable 1
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFN : JK Flip-Flop, negative-edge-trigger

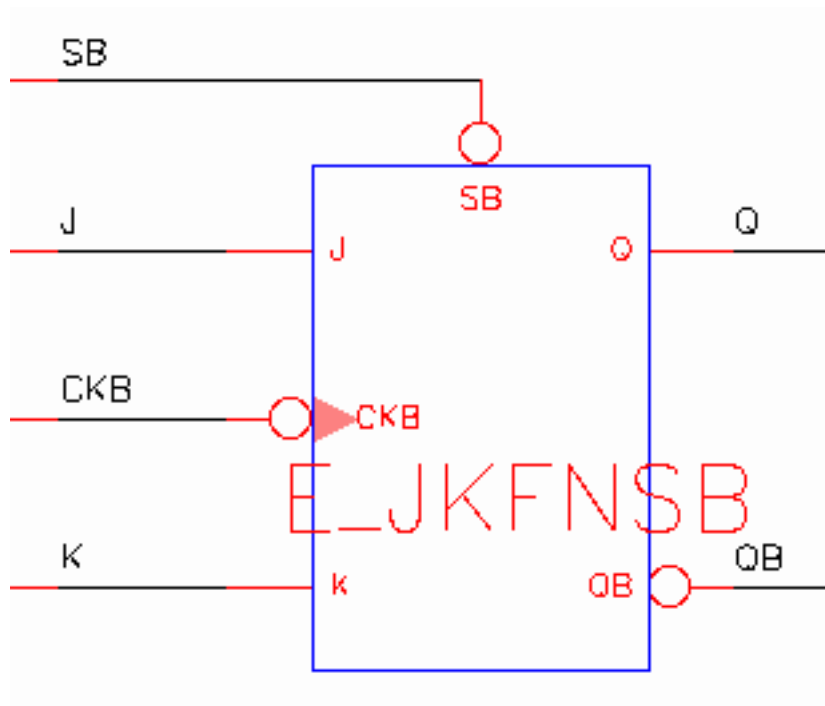
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNRB : JK Flip-Flop, negative-edge-trigger, active-LOW reset



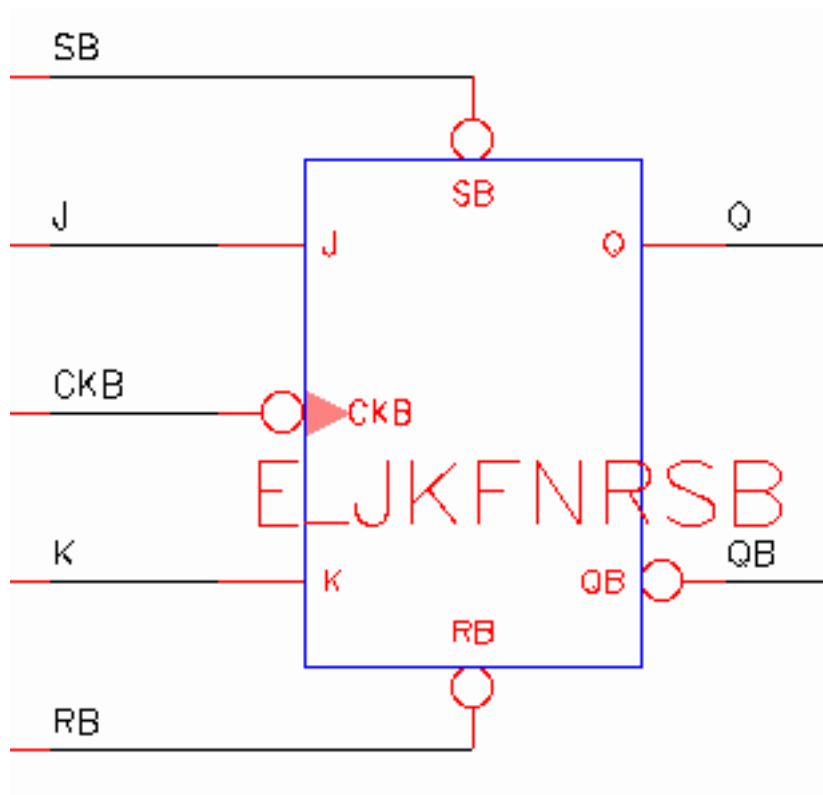
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNSB : JK Flip-Flop, negative-edge-trigger, active-LOW set



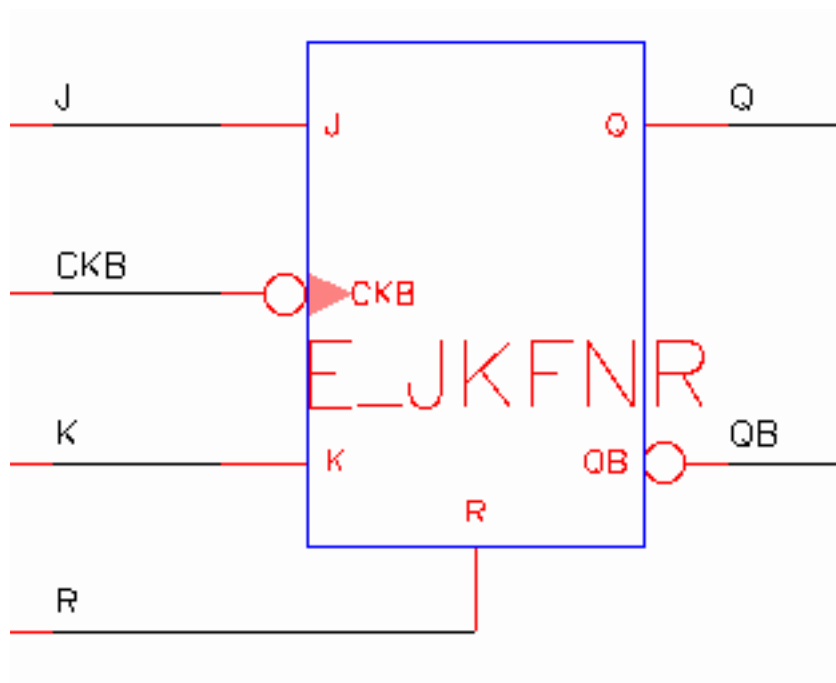
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNRSB : JK Flip-Flop, negative-edge-trigger, active-LOW set & reset



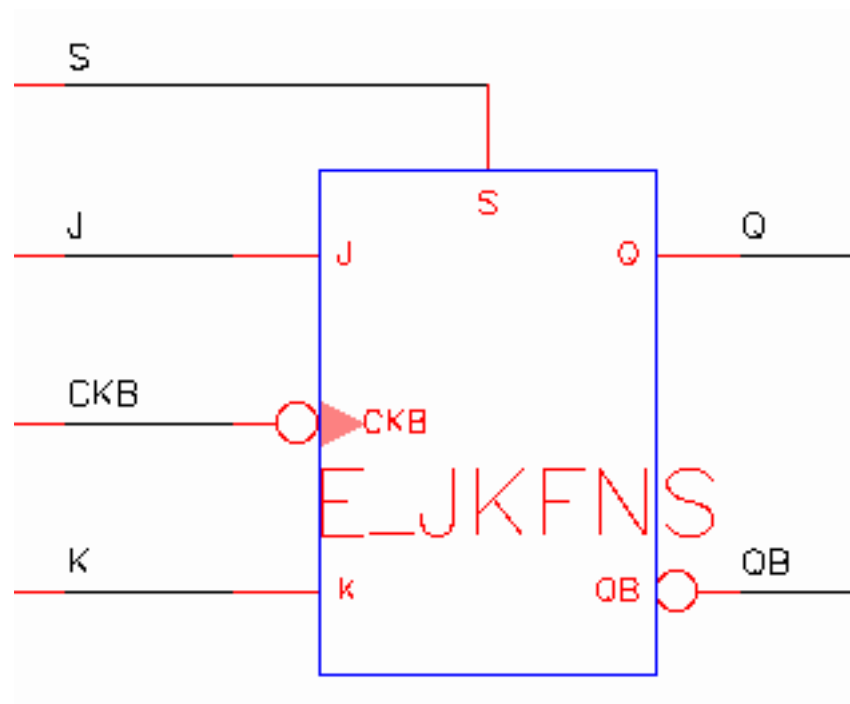
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNR : JK Flip-Flop, negative-edge-trigger, active-HIGH reset



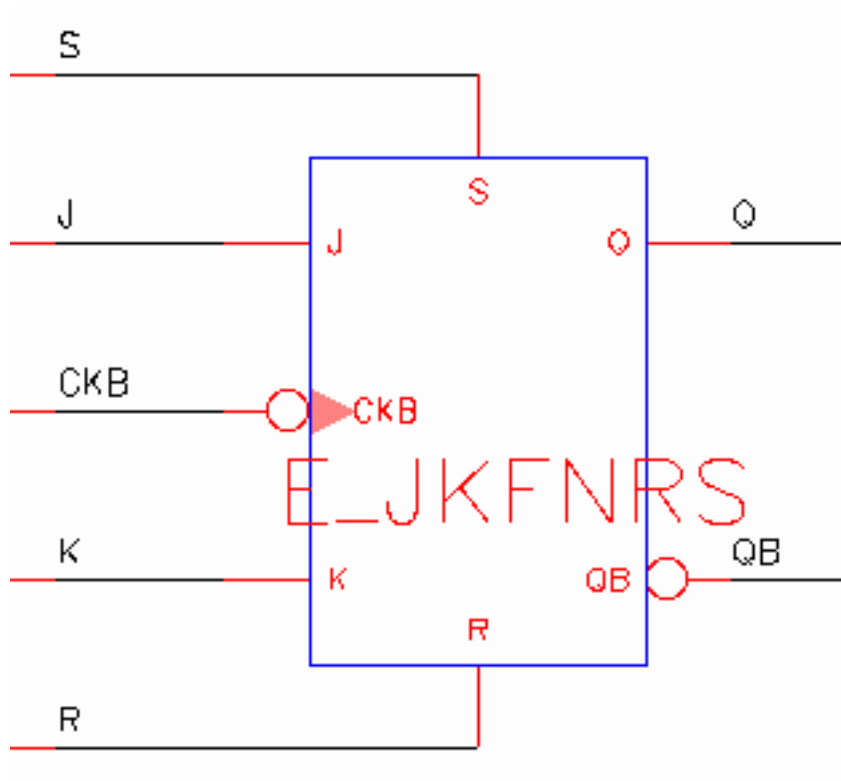
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNS : JK Flip-Flop, negative-edge-trigger, active-HIGH set

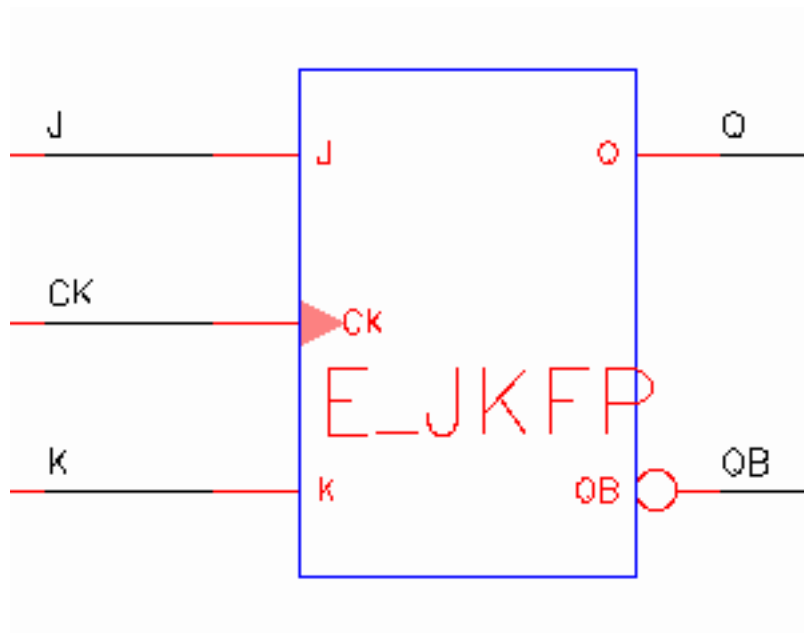


Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNRS : JK Flip-Flop, negative-edge-trigger, active-HIGH set & reset

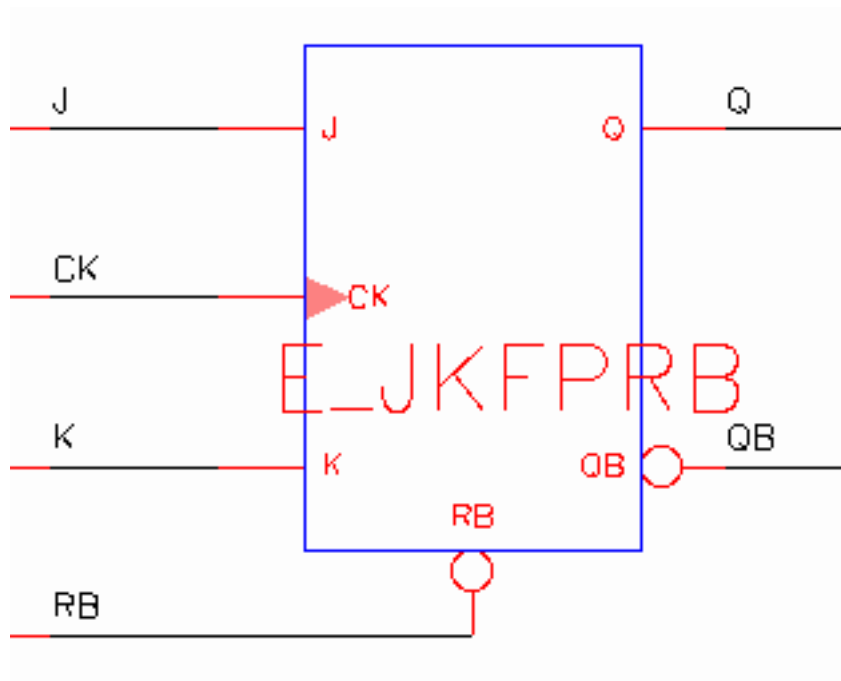


Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFP : JK Flip-Flop, positive-edge-trigger

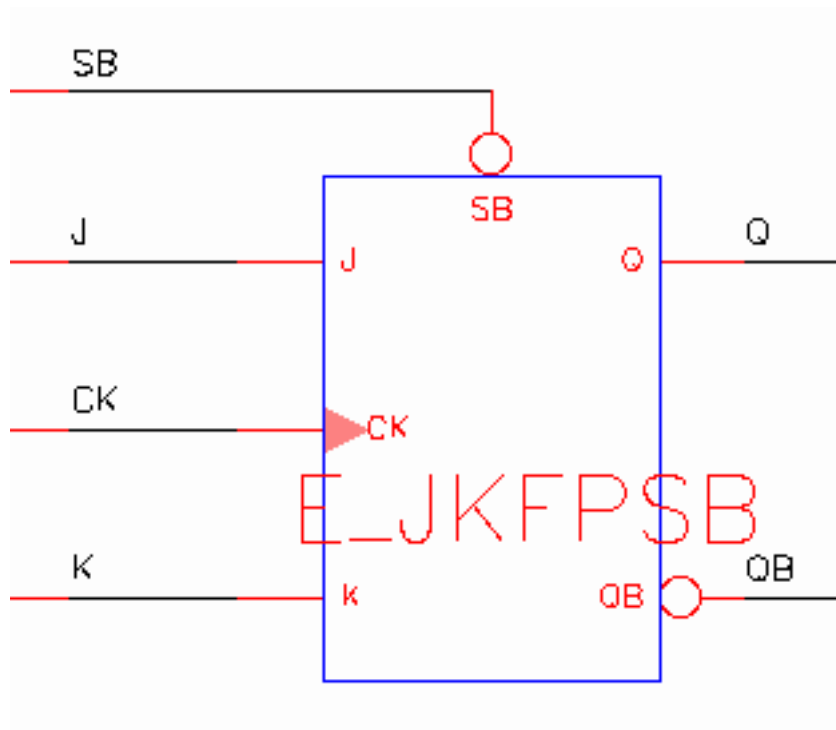
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPRB : JK Flip-Flop, positive-edge-trigger, active-LOW reset



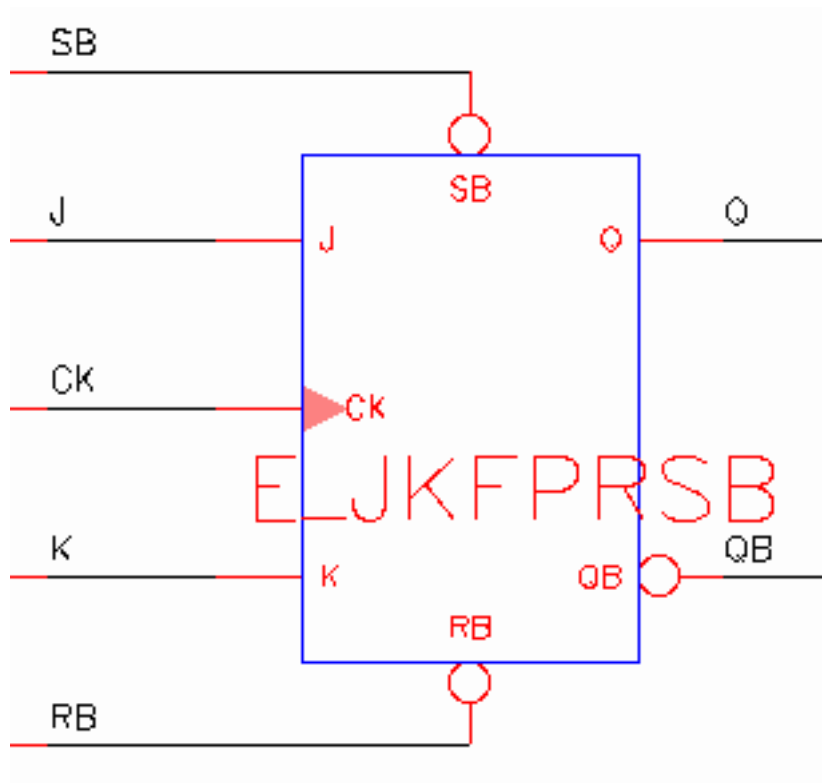
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSB : JK Flip-Flop, positive-edge-trigger, active-LOW set



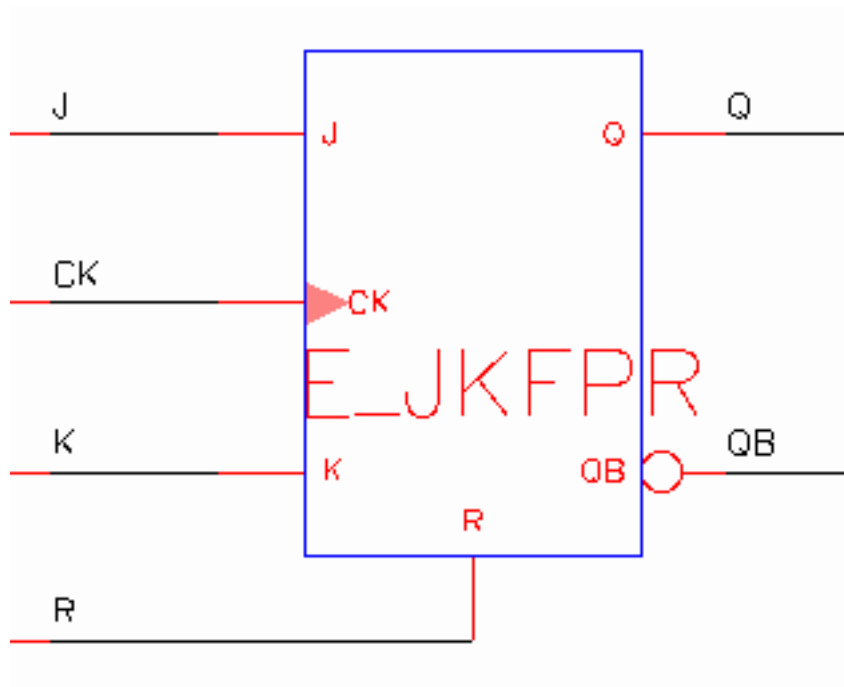
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPRSB : JK Flip-Flop, positive-edge-trigger, active-LOW set & reset



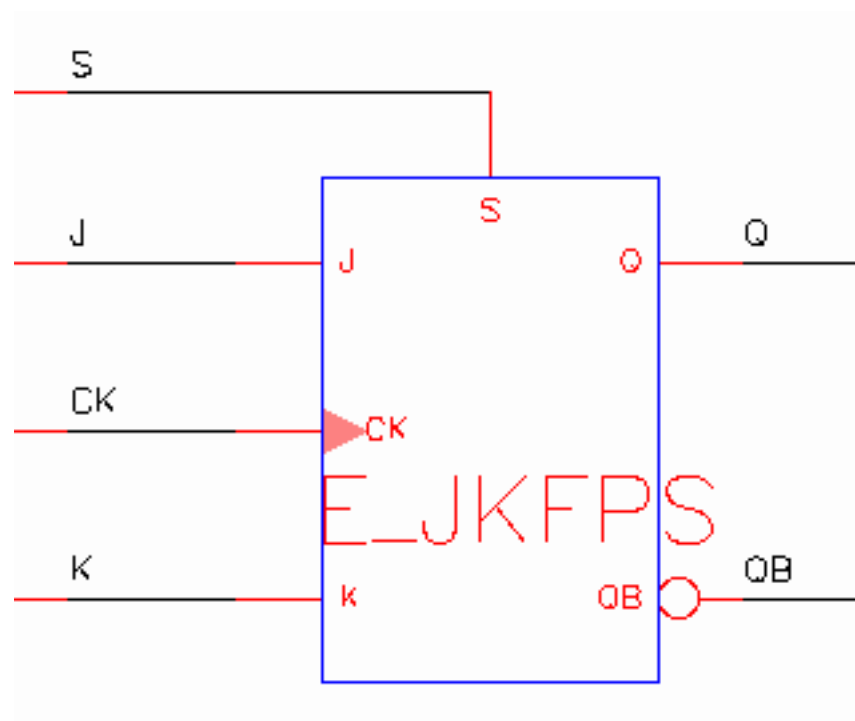
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positivie-edge-trigger
K	INPUT	data
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPR : JK Flip-Flop, positive-edge-trigger, active-HIGH reset



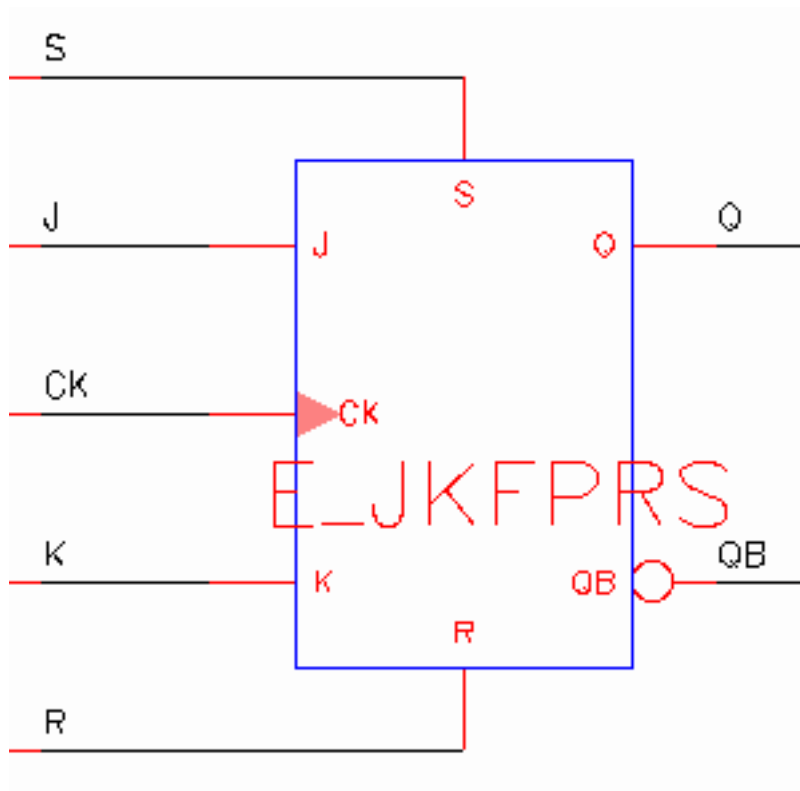
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPS : JK Flip-Flop, positive-edge-trigger, active-HIGH set



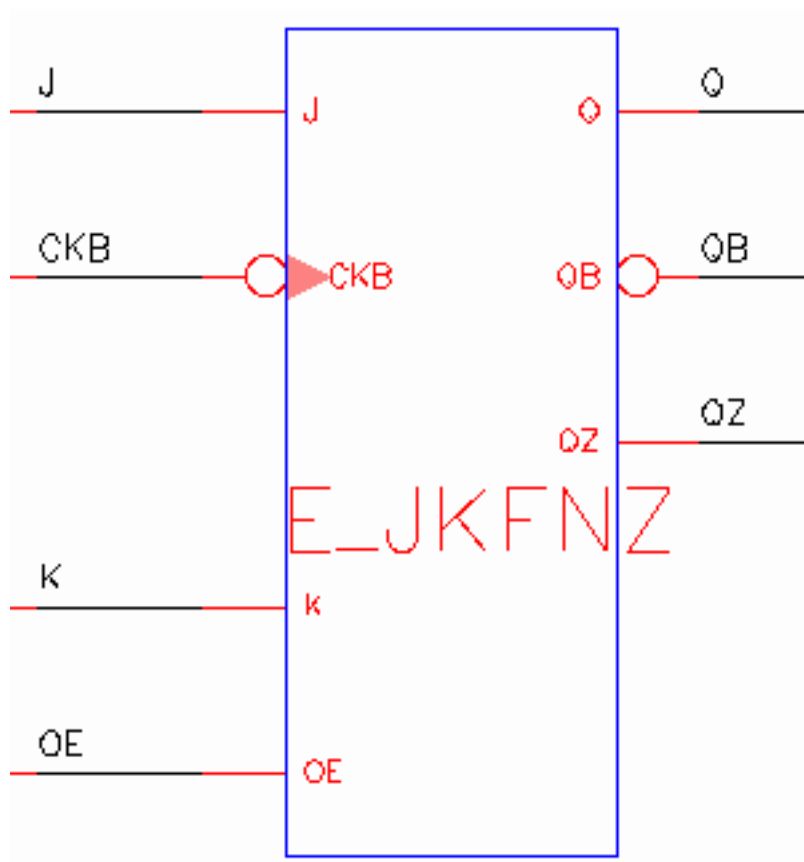
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPRS : JK Flip-Flop, positive-edge-trigger, active-HIGH set & reset



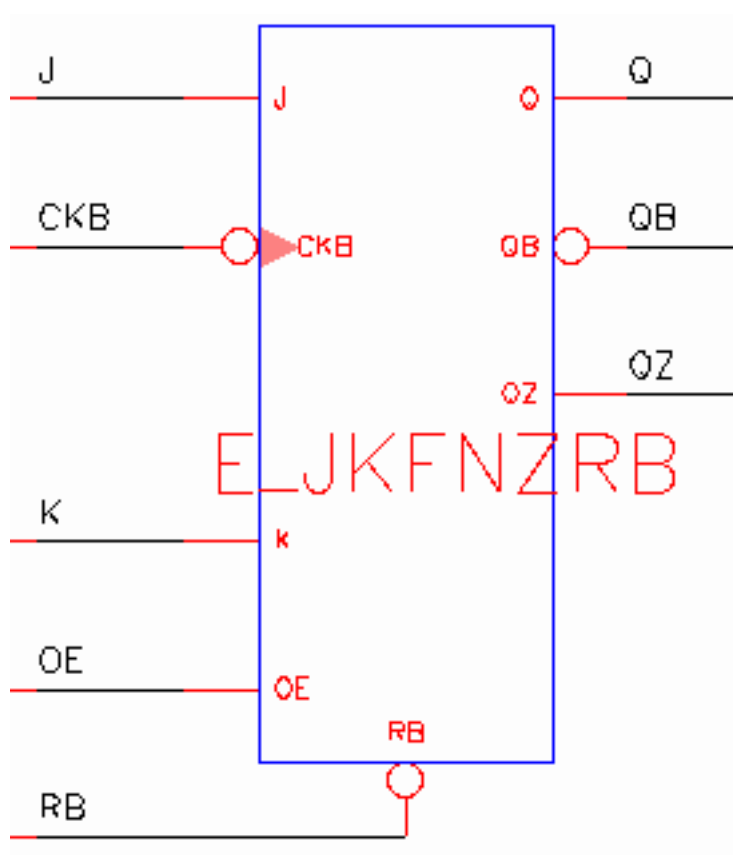
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positivie-edge-trigger
K	INPUT	data
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNZ : JK Flip-Flop, negative-edge-trigger, active-HIGH output enable



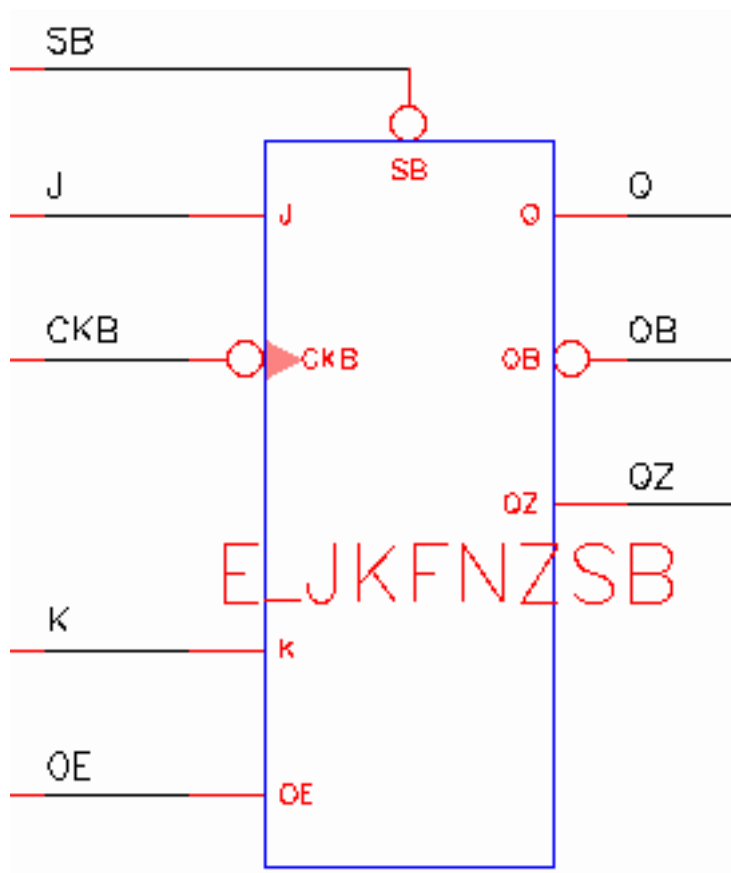
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNZRB : JK Flip-Flop, negative-edge-trigger, active-LOW reset, active-HIGH output enable



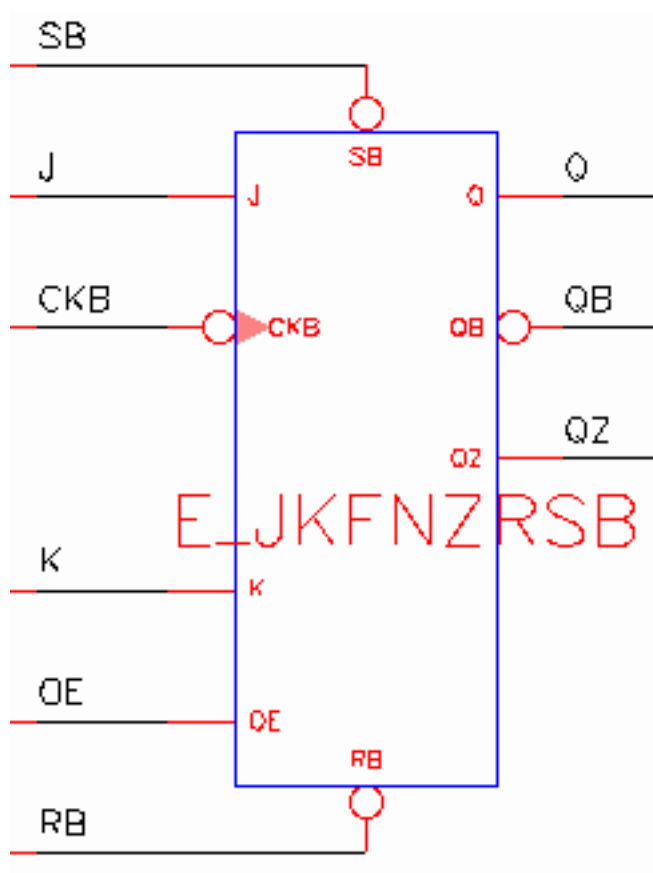
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNZSB : JK Flip-Flop, negative-edge-trigger, active-LOW set, active-HIGH output enable



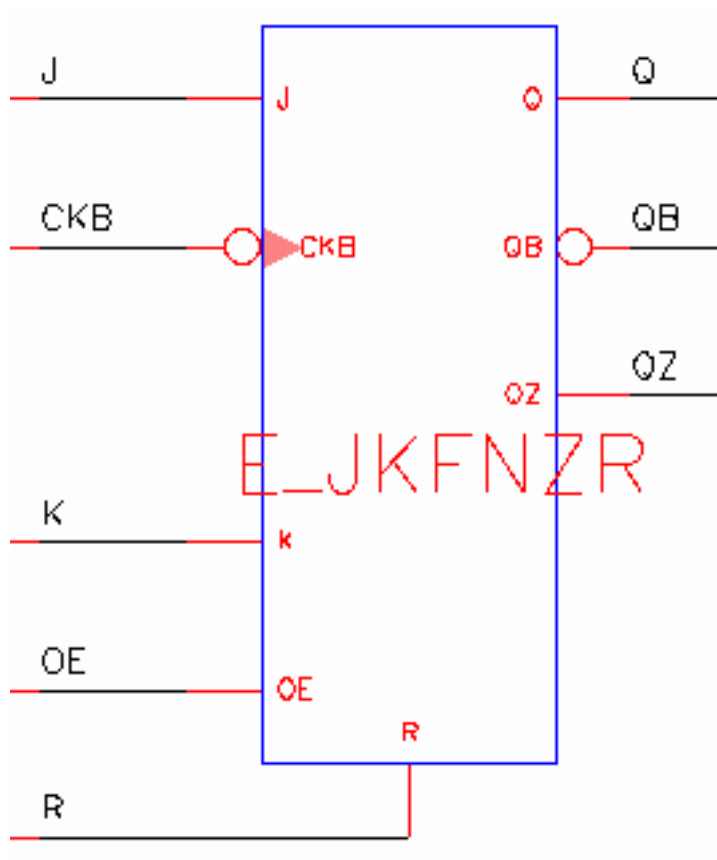
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNZRSB : JK Flip-Flop, negative-edge-trigger, active-LOW set & reset, active-HIGH output enable



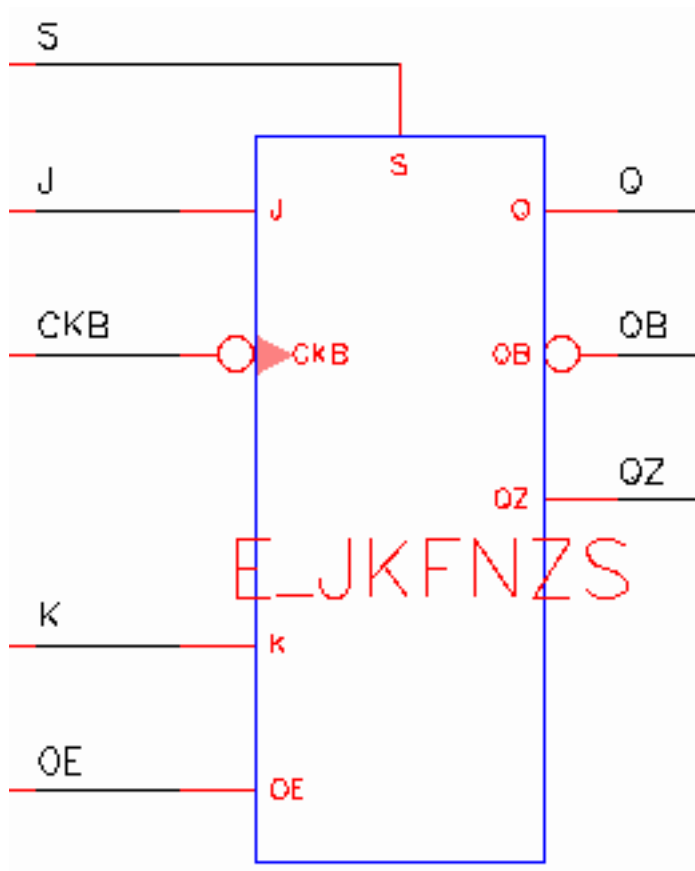
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNZR : JK Flip-Flop, negative-edge-trigger, active-HIGH reset, active-HIGH output enable

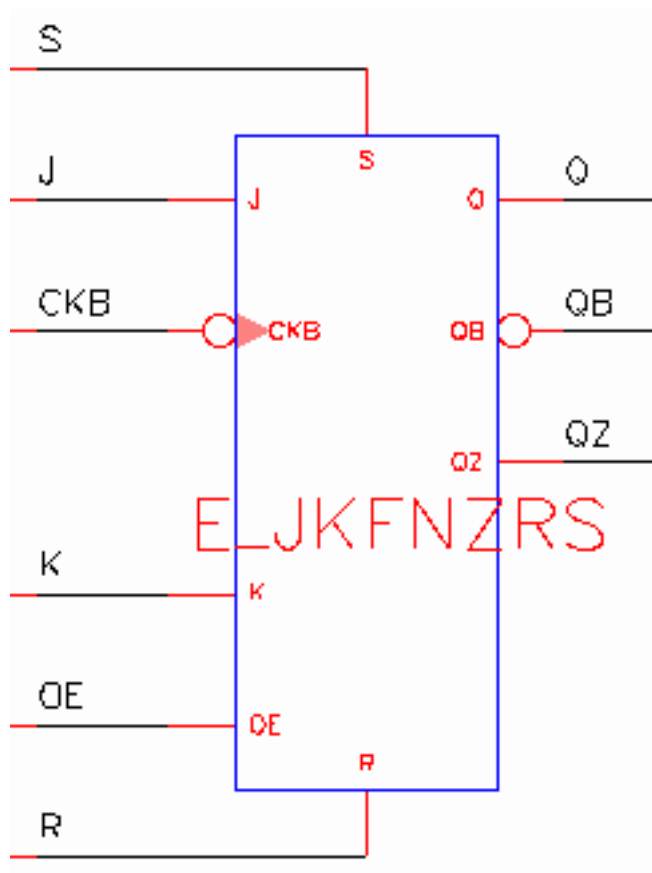


Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNZS : JK Flip-Flop, negative-edge-trigger, active-HIGH set, active-HIGH output enable

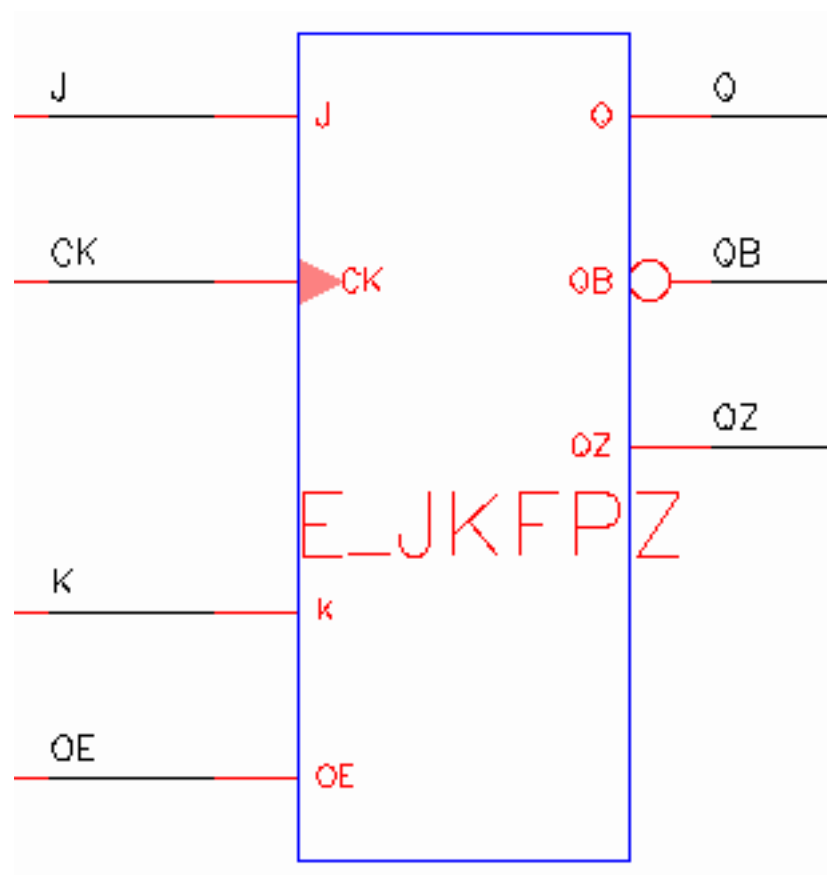


Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
S	INPUT	async set, active HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNZRS : JK Flip-Flop, negative-edge-trigger, active-HIGH set & reset, active-HIGH output enable

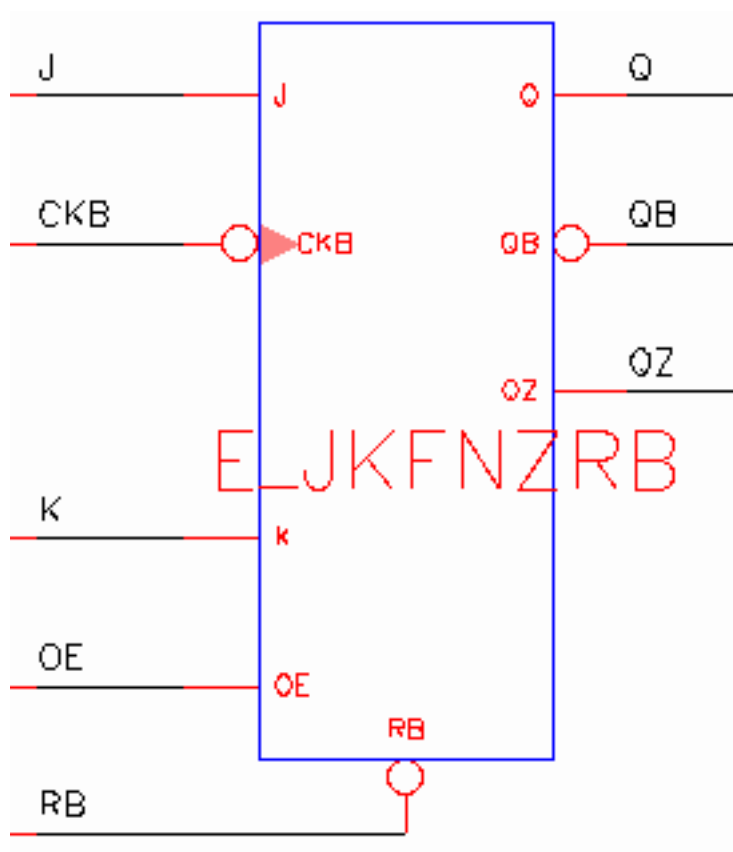
Pin	Direction	Description
J	INPUT	data
CKB	INPUT	clock, negative-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active HIGH
S	INPUT	async set, active HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFPZ : JK Flip-Flop, positive-edge-trigger, active-HIGH output enable



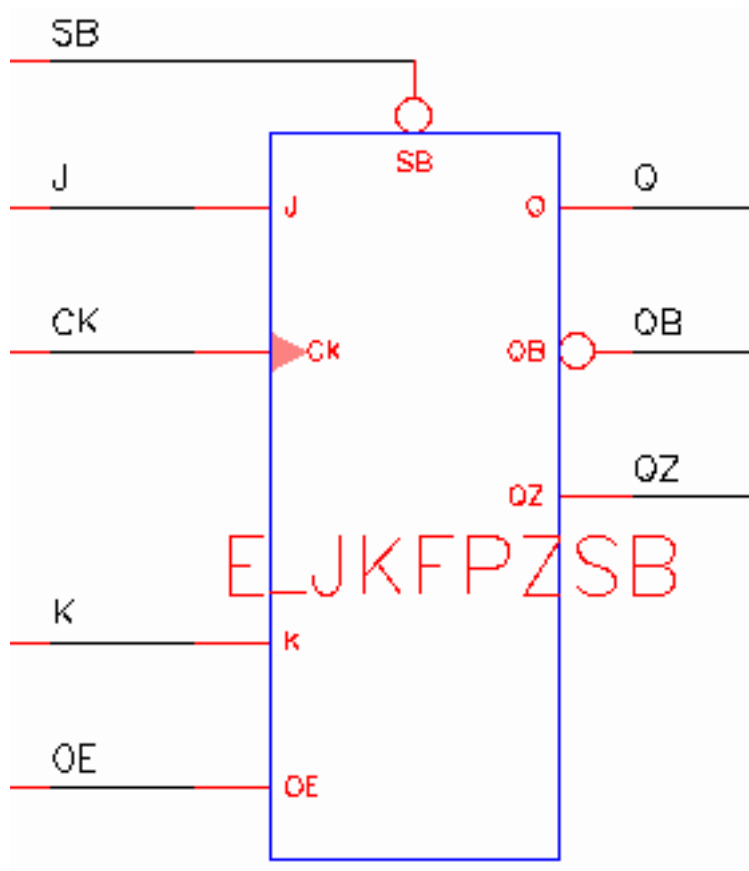
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFPZRB : JK Flip-Flop, positive-edge-trigger, active-LOW reset, active-HIGH output enable

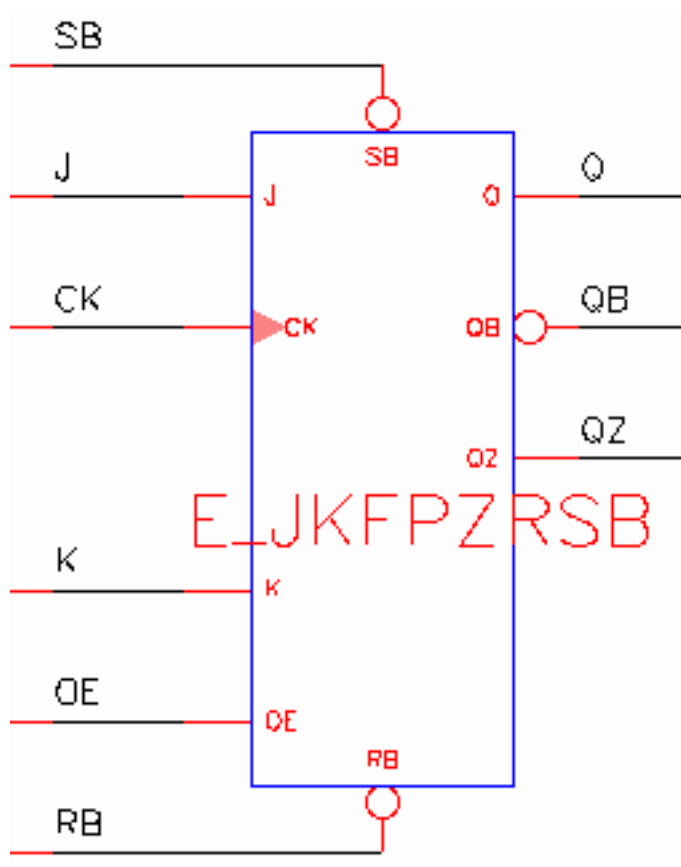


Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFPZSB : JK Flip-Flop, positive-edge-trigger, active-LOW set, active-HIGH output enable

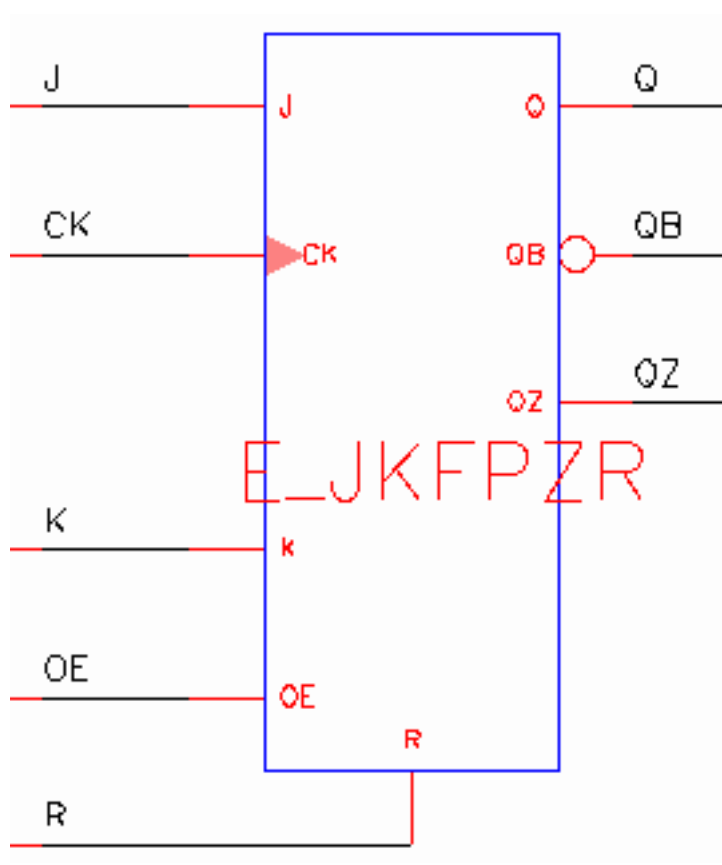


Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFPZRSB : JK Flip-Flop, positive-edge-trigger, active-LOW set & reset, active-HIGH output enable

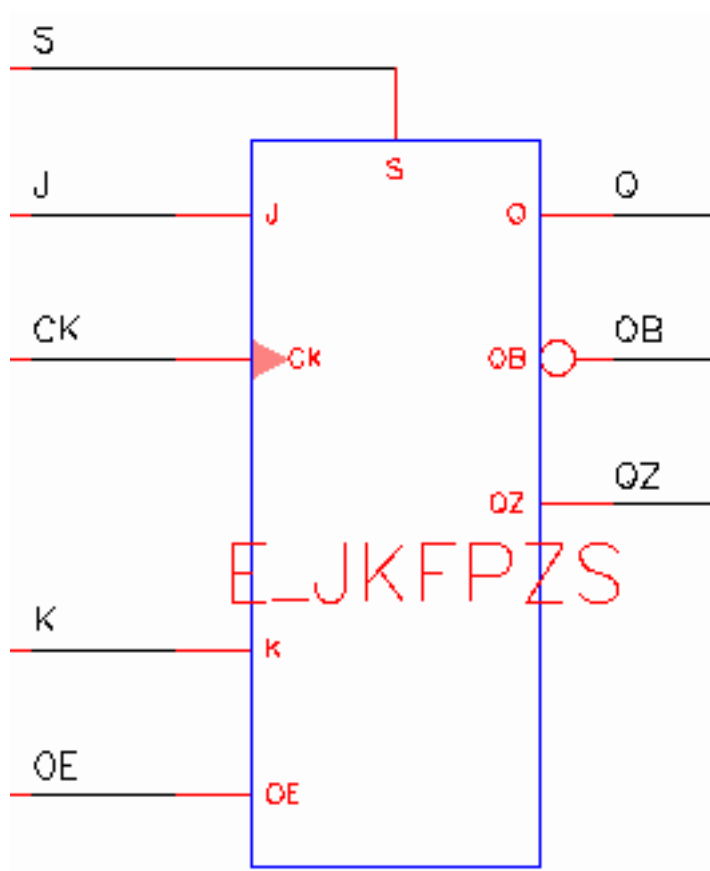
Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFPZR : JK Flip-Flop, positive-edge-trigger, active-HIGH reset, active-HIGH output enable

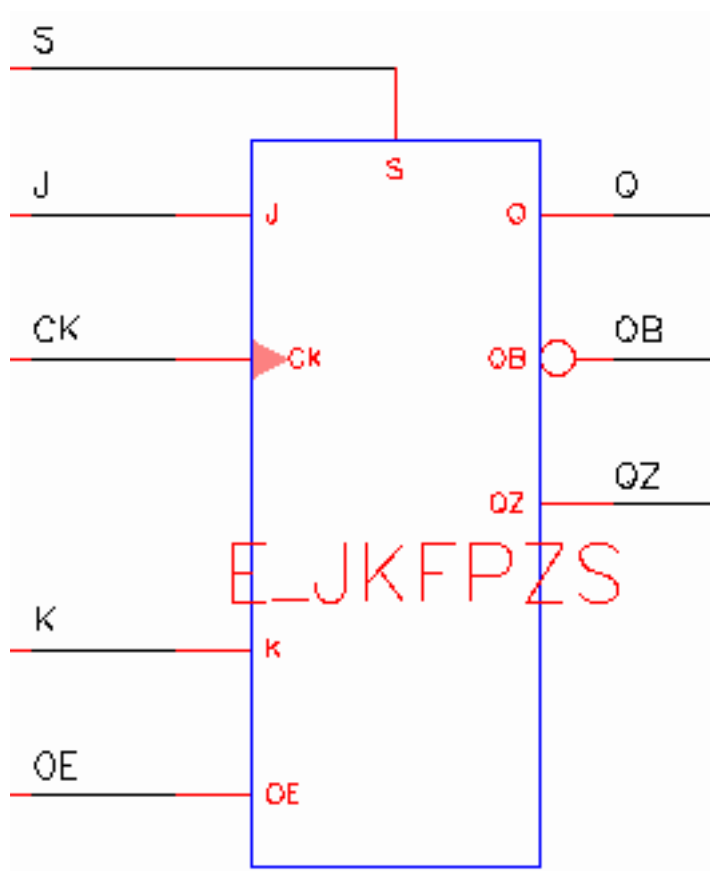


Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFPZS : JK Flip-Flop, positive-edge-trigger, active-HIGH set, active-HIGH output enable

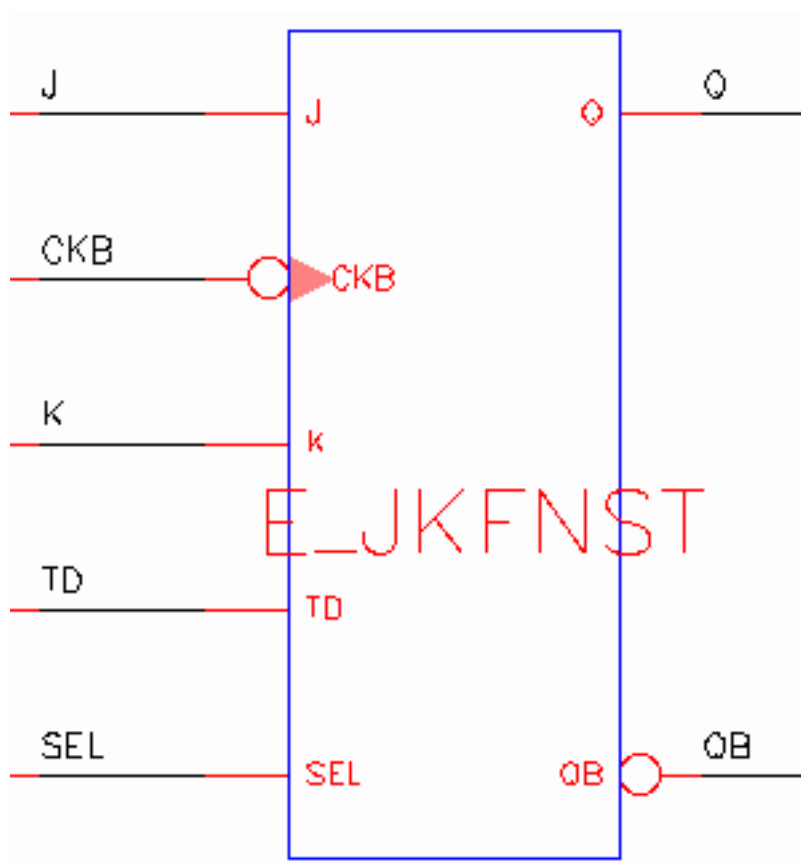


Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

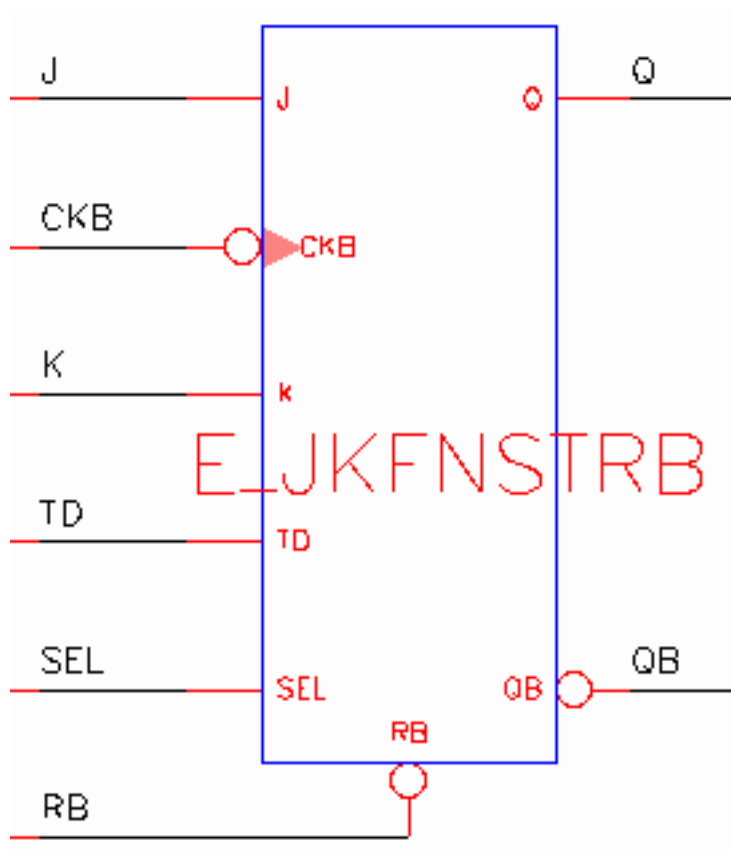
E_JKFPZRS : JK Flip-Flop, positive-edge-trigger, active-HIGH set & reset, active-HIGH output enable

Pin	Direction	Description
J	INPUT	data
CK	INPUT	clock, positive-edge-trigger
K	INPUT	data
OE	INPUT	output enable, active-HIGH
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted
QZ	OUTPUT	data, tri-state

E_JKFNST : JK Flip-Flop, active-HIGH scan, negative-edge-triggered

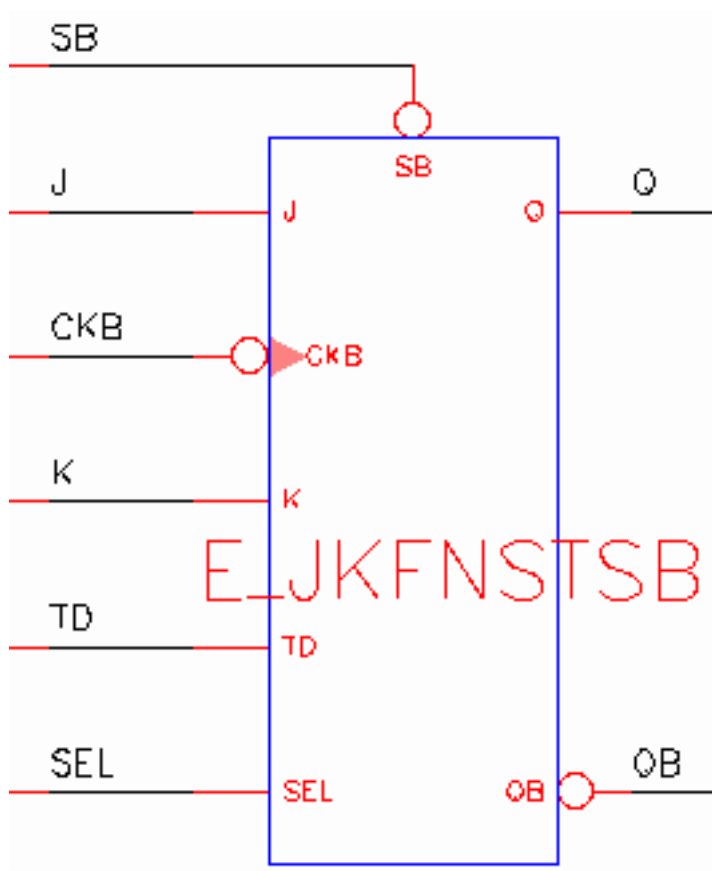


Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNSTRB : JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-LOW reset

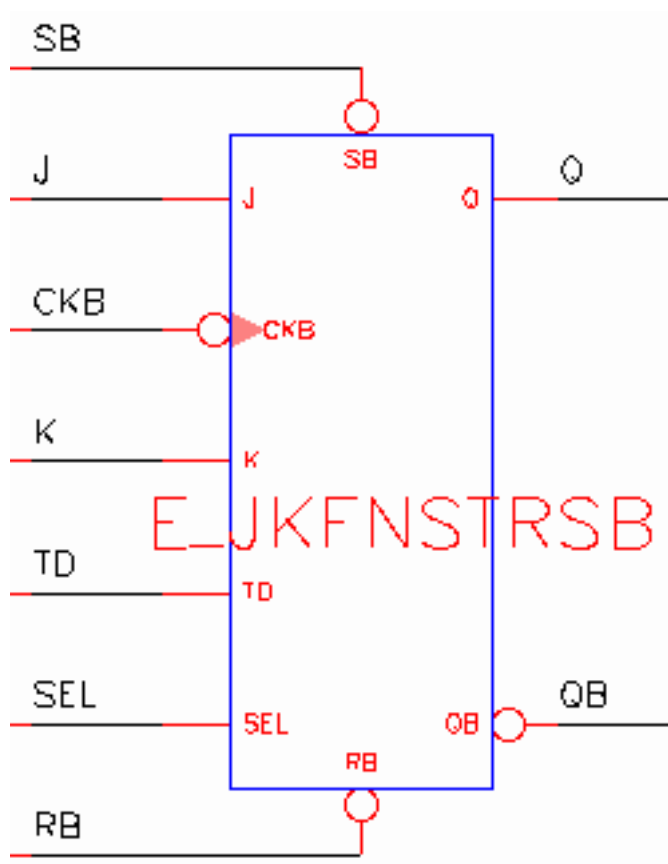
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNSTSB : JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-LOW set



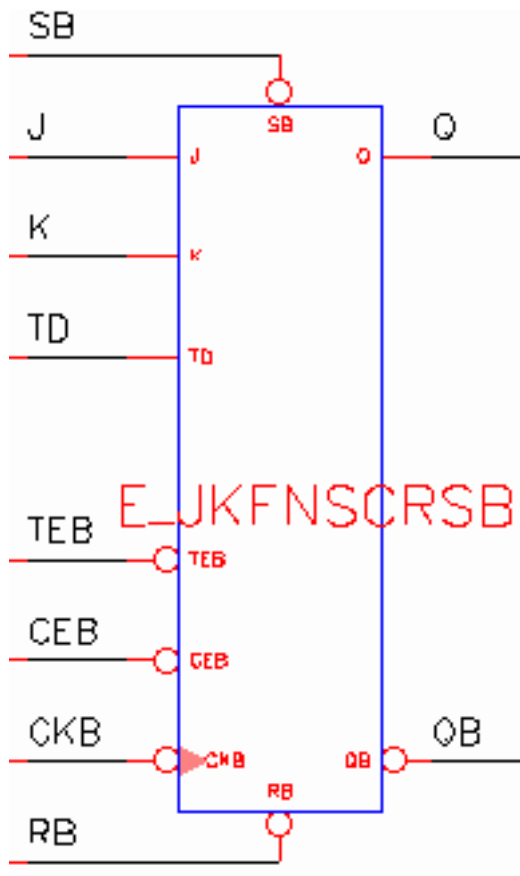
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNSTRSB : JK Flip-Flop, active-HIGH scan, negative-edge-triggered, active-LOW set & reset



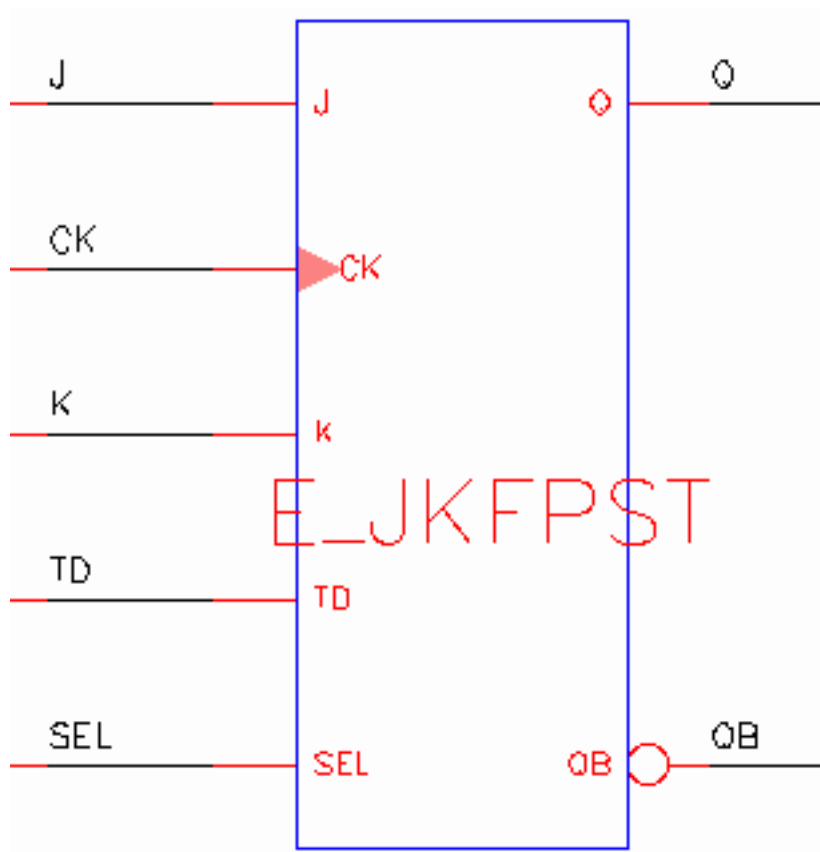
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFNSCRSB : JK Flip-Flop, active-LOW scan, negative-edge-triggered, active-LOW set & reset, active-LOW clock enable



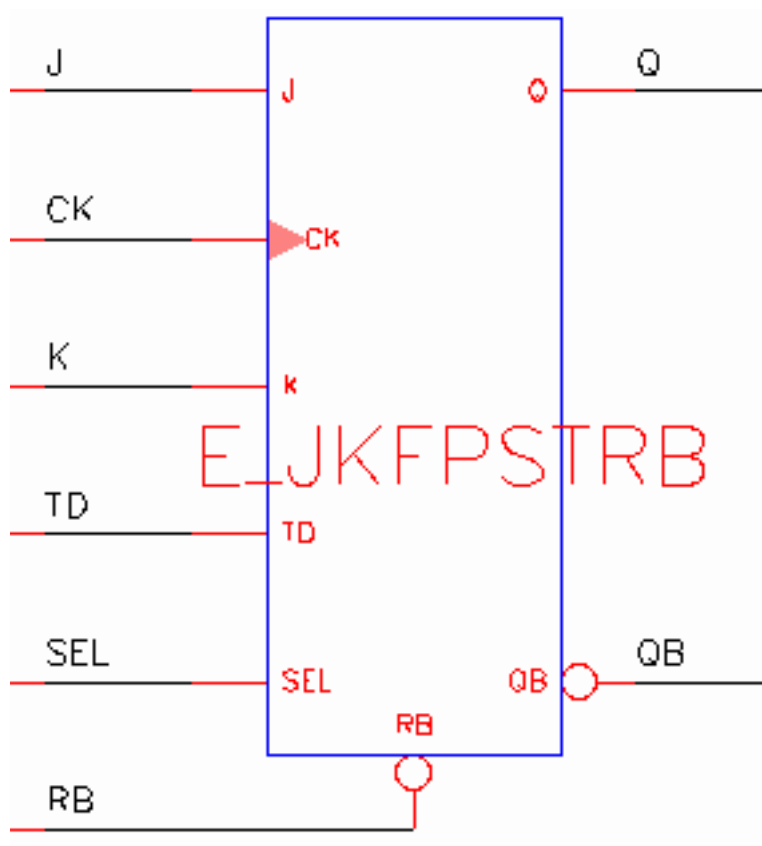
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
TEB	INPUT	scan enable, active-LOW
CEB	INPUT	output enable, active-LOW
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPST : JK Flip-Flop, active-HIGH scan, positive-edge-triggered



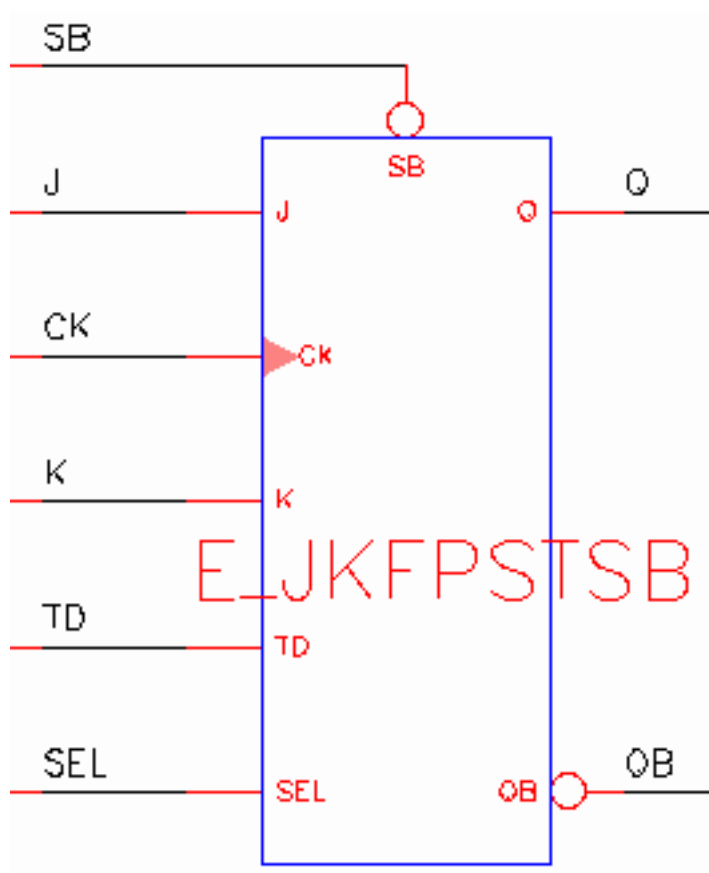
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSTRB : JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-LOW reset



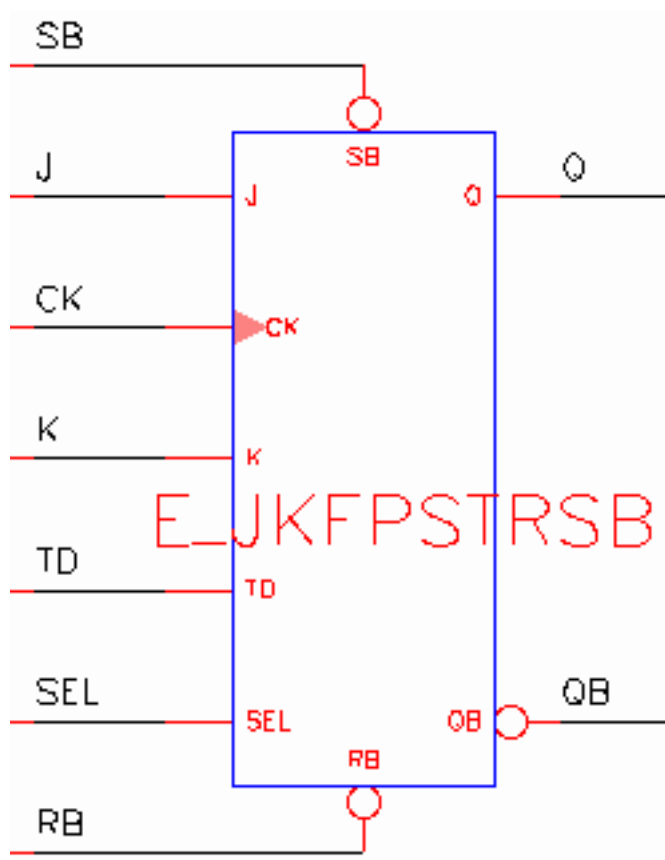
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSTSB : JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-LOW set



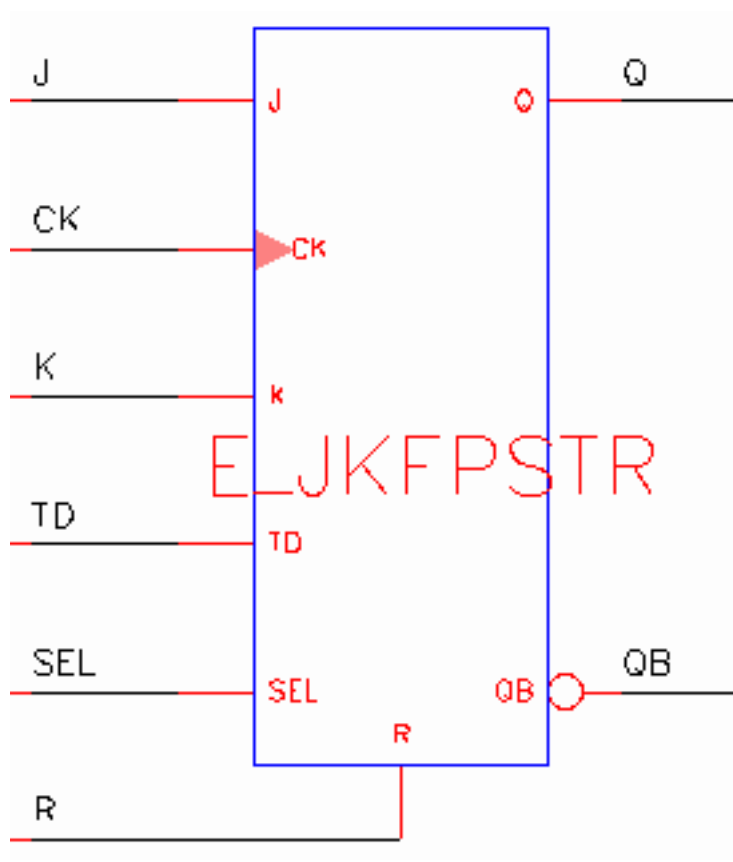
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSTRSB : JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-LOW set & reset

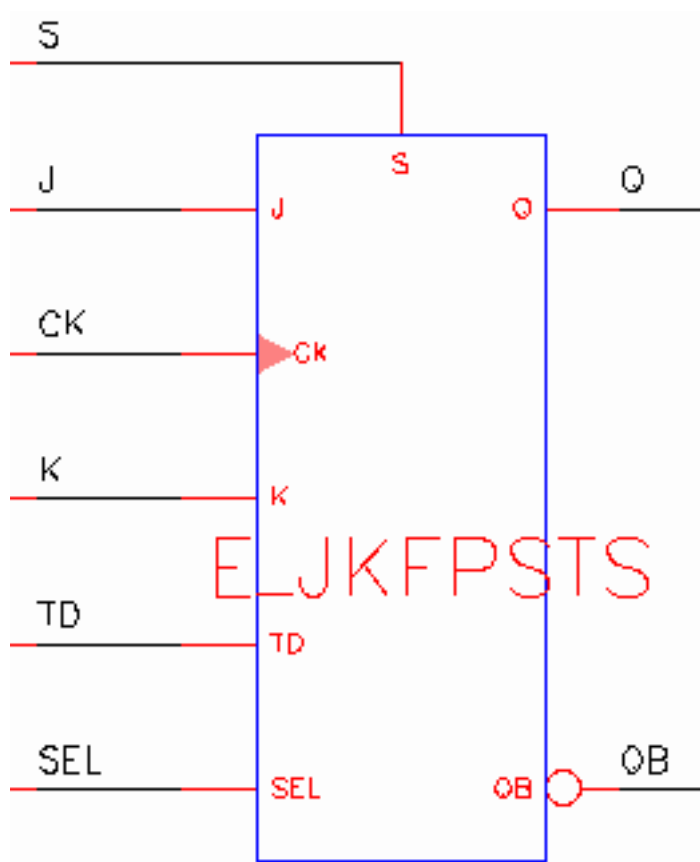


Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSTR : JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-HIGH reset

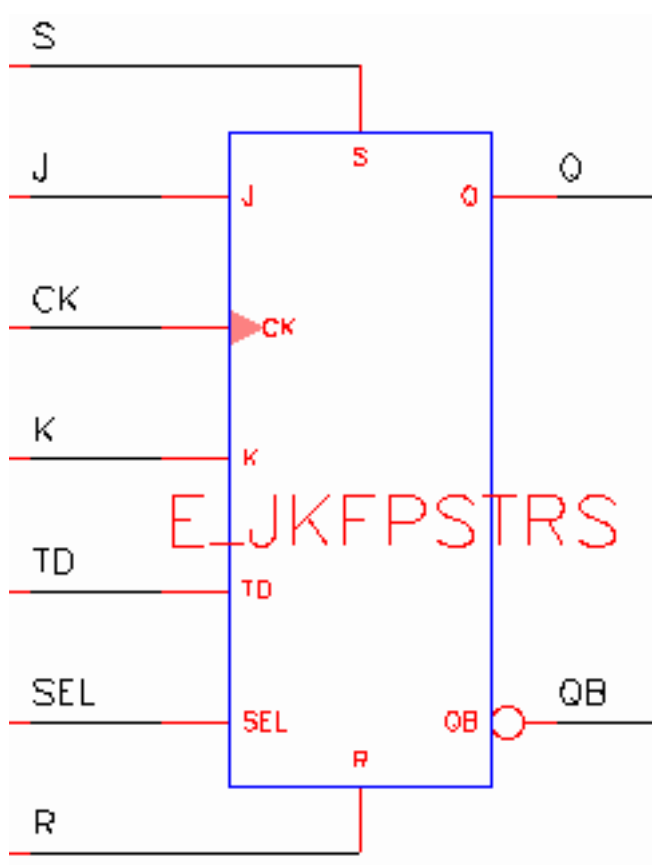


Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSTS : JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-HIGH set

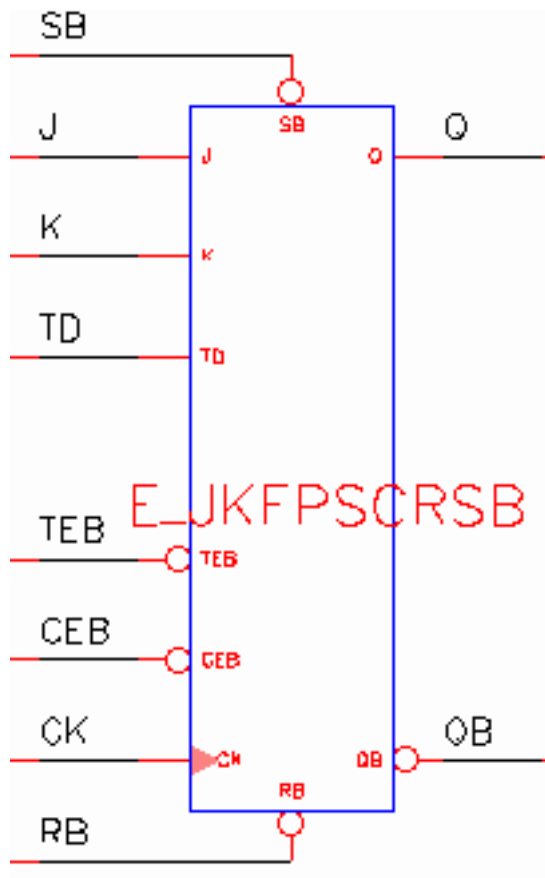
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSTRS : JK Flip-Flop, active-HIGH scan, positive-edge-triggered, active-HIGH set & reset



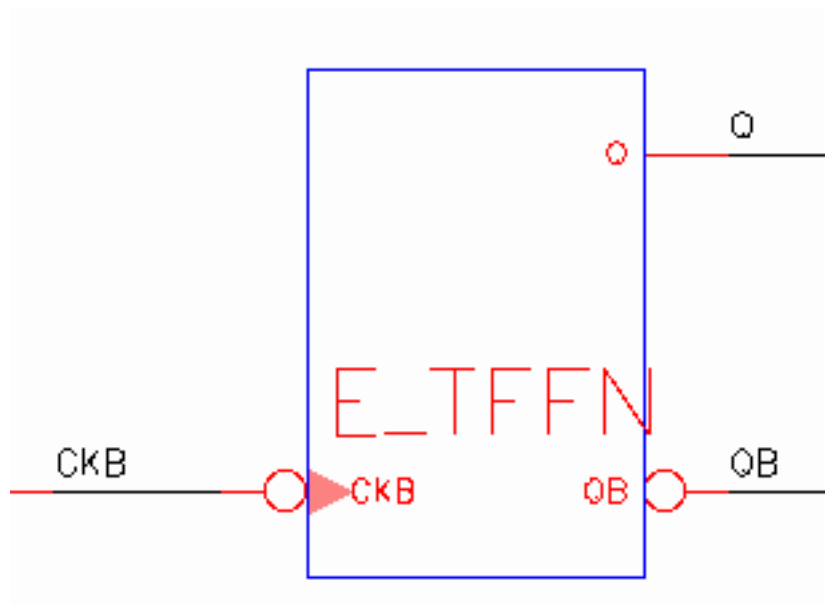
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
SEL	INPUT	scan enable
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_JKFPSCRSB : JK Flip-Flop, active-LOW scan, positive-edge-triggered, active-LOW set & reset, active-LOW clock enable



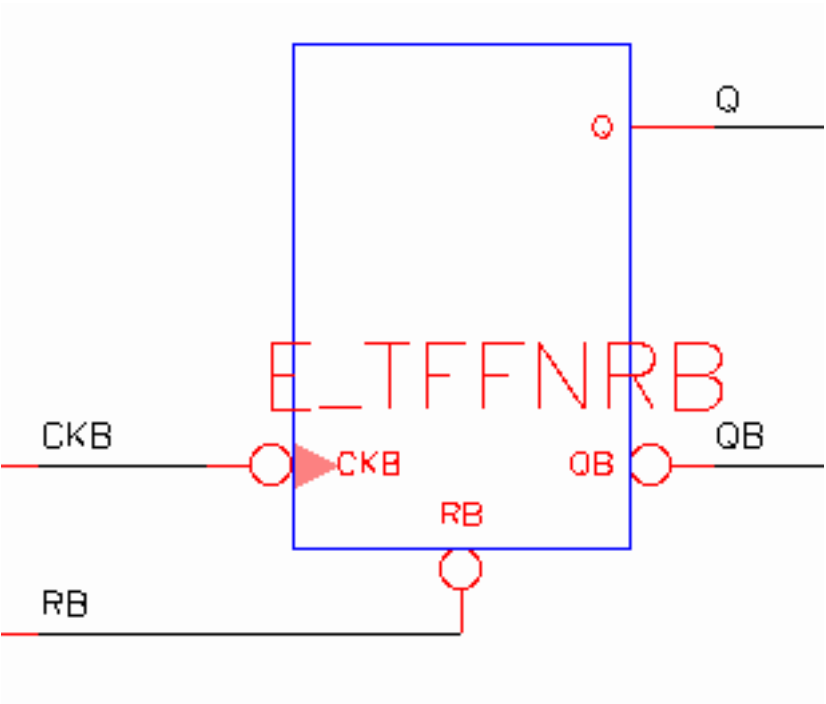
Pin	Direction	Description
J	INPUT	data
K	INPUT	data
TD	INPUT	scan data
TEB	INPUT	scan enable, active-LOW
CEB	INPUT	output enable, active-LOW
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFN : T Flip-Flop, negative-edge-trigger



Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

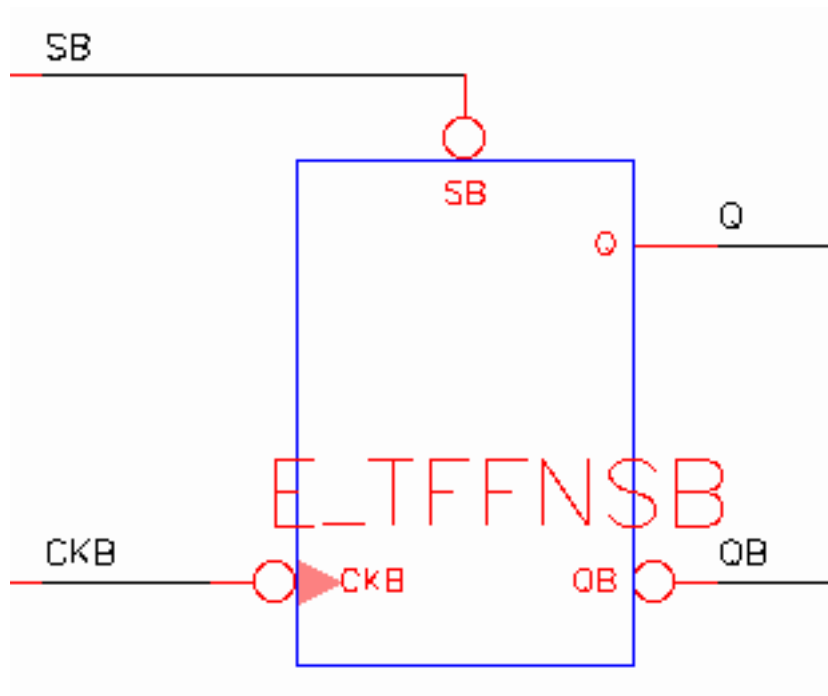
E_TFFNRB : T Flip-Flop, negative-edge-trigger, active-LOW reset



fc f

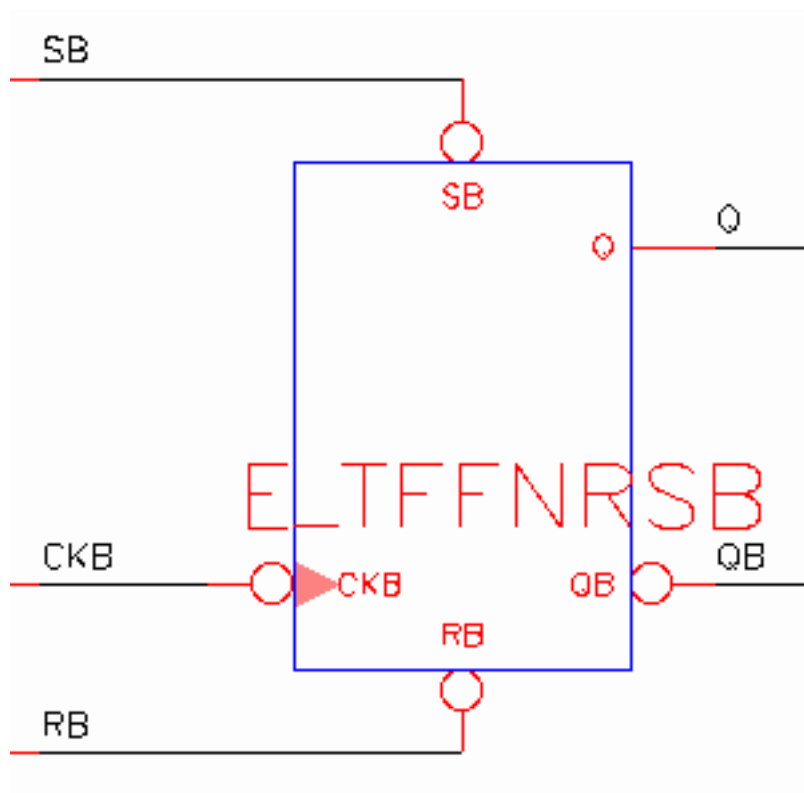
Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFNSB : T Flip-Flop, negative-edge-trigger, active-LOW set



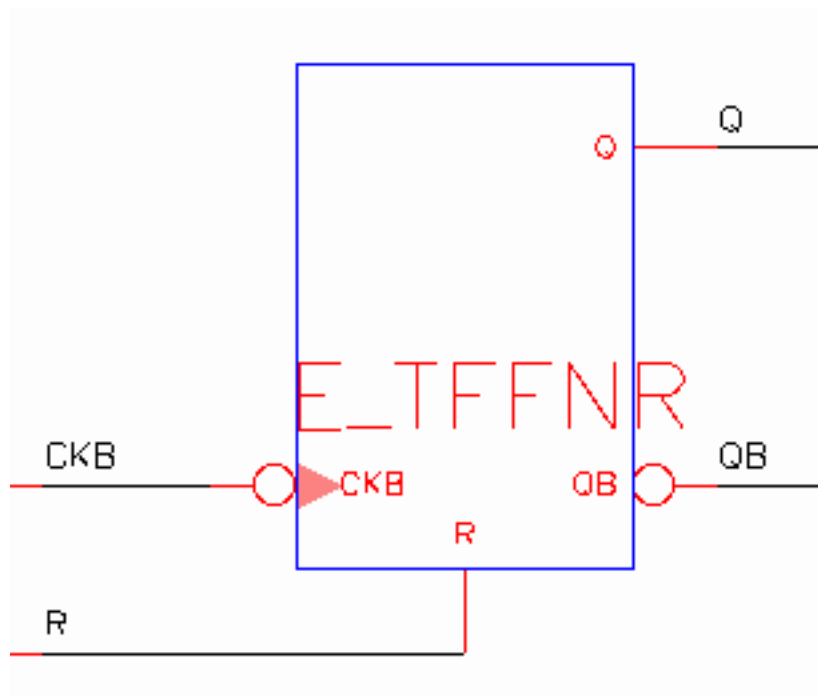
Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFNRSB : T Flip-Flop, negative-edge-trigger, active-LOW set & reset



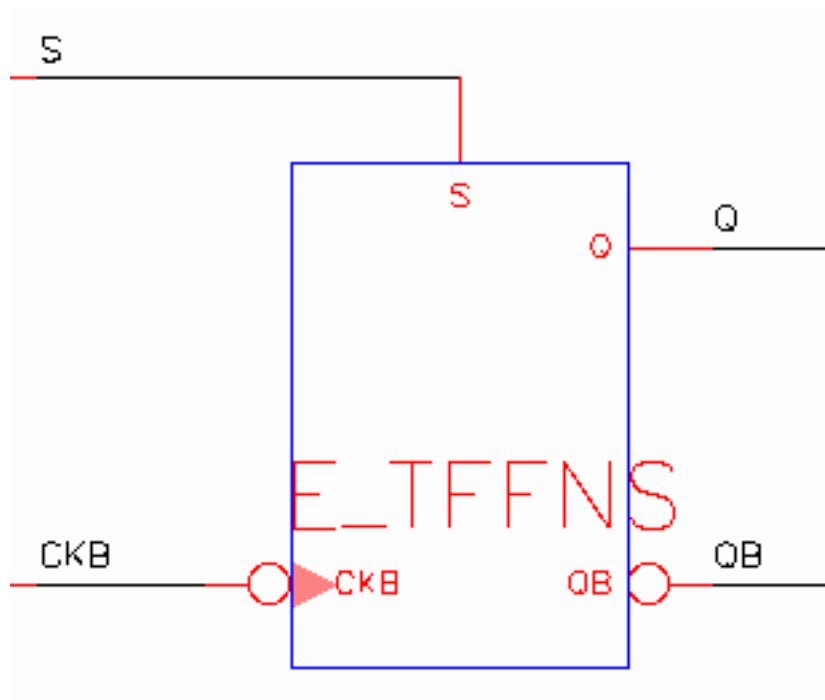
Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFNR : T Flip-Flop, negative-edge-trigger, active-HIGH reset



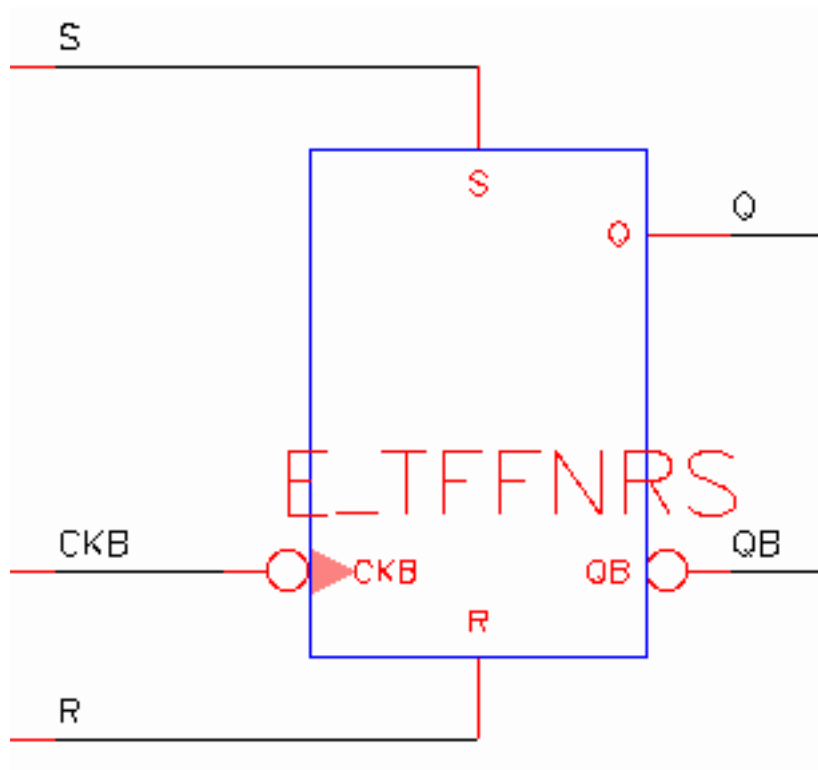
Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFNS : T Flip-Flop, negative-edge-trigger, active-HIGH set

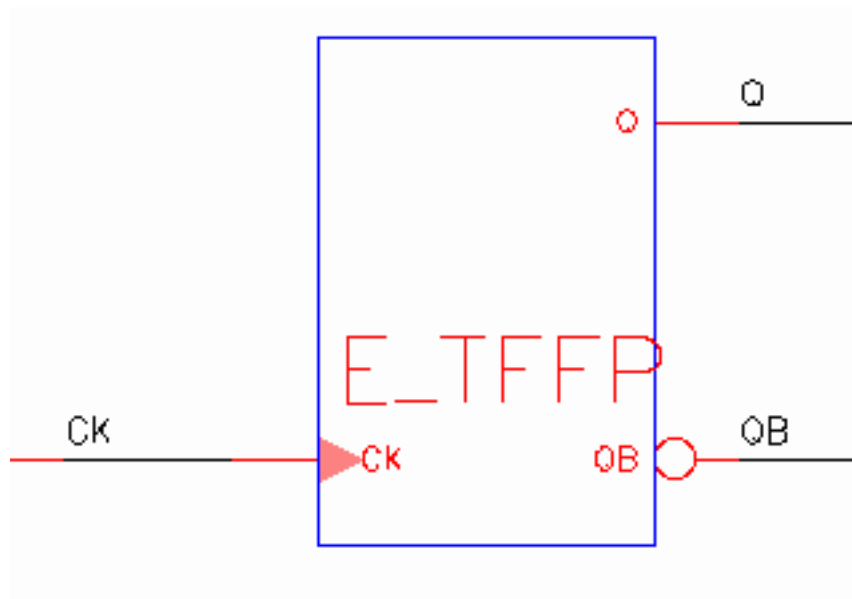


Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFNRS : T Flip-Flop, negative-edge-trigger, active-HIGH set & reset

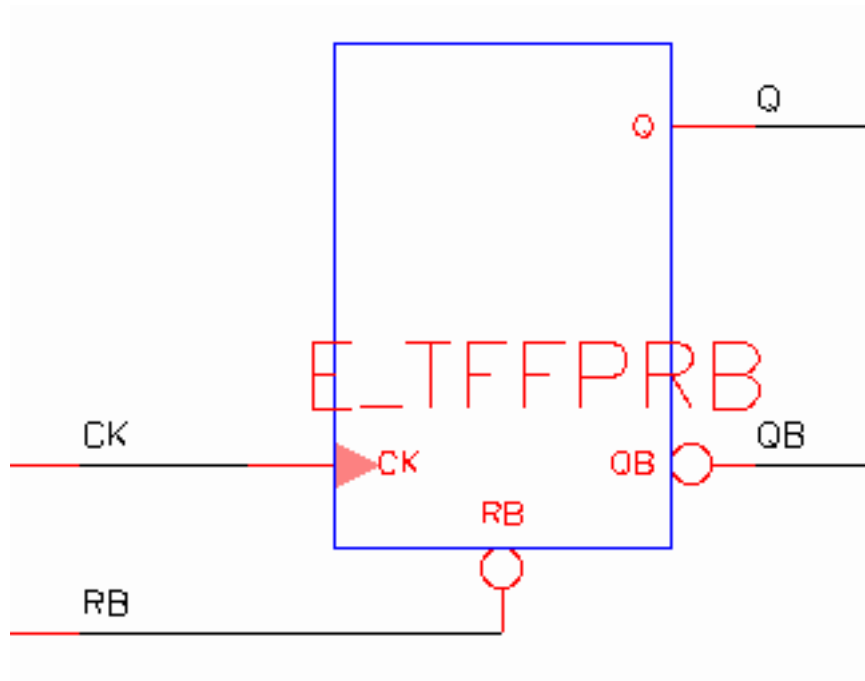


Pin	Direction	Description
CKB	INPUT	clock, negative-edge-trigger
R	INPUT	async reset, active-HIGH
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFP : T Flip-Flop, positive-edge-trigger

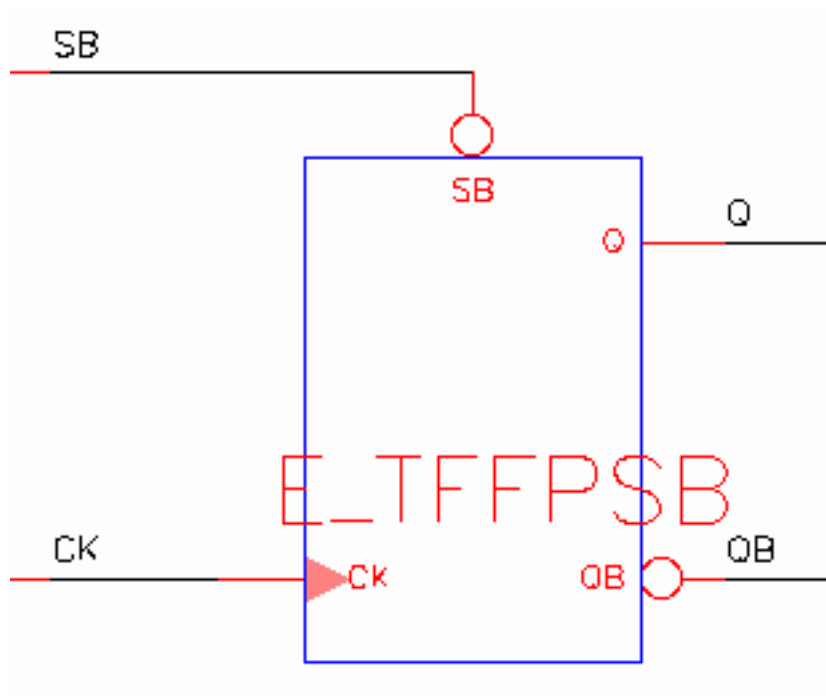
Pin	Direction	Description
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFPRB : T Flip-Flop, positive-edge-trigger, active-LOW reset



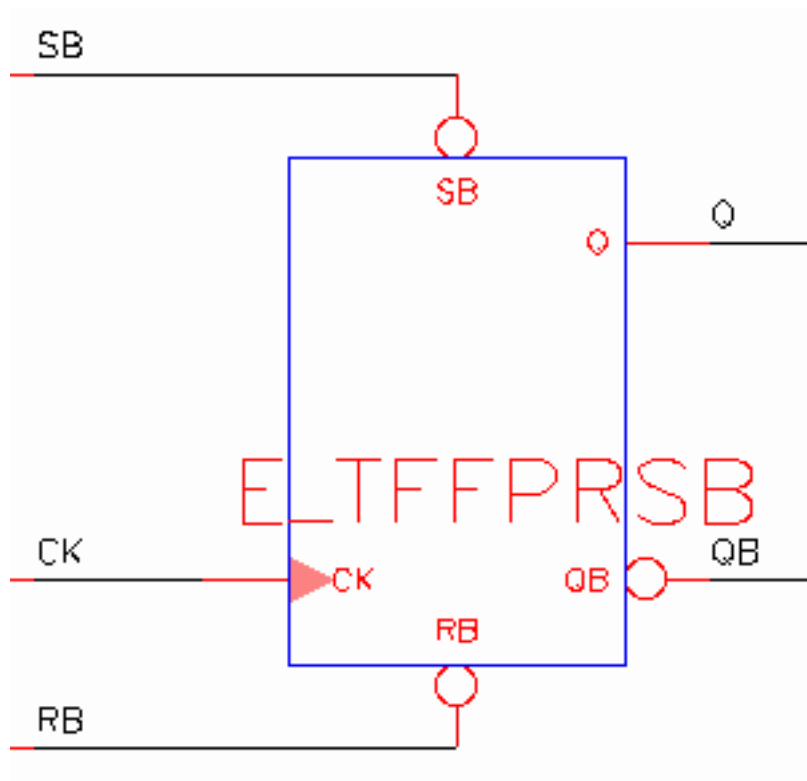
Pin	Direction	Description
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFPSB : T Flip-Flop, positive-edge-trigger, active-LOW set



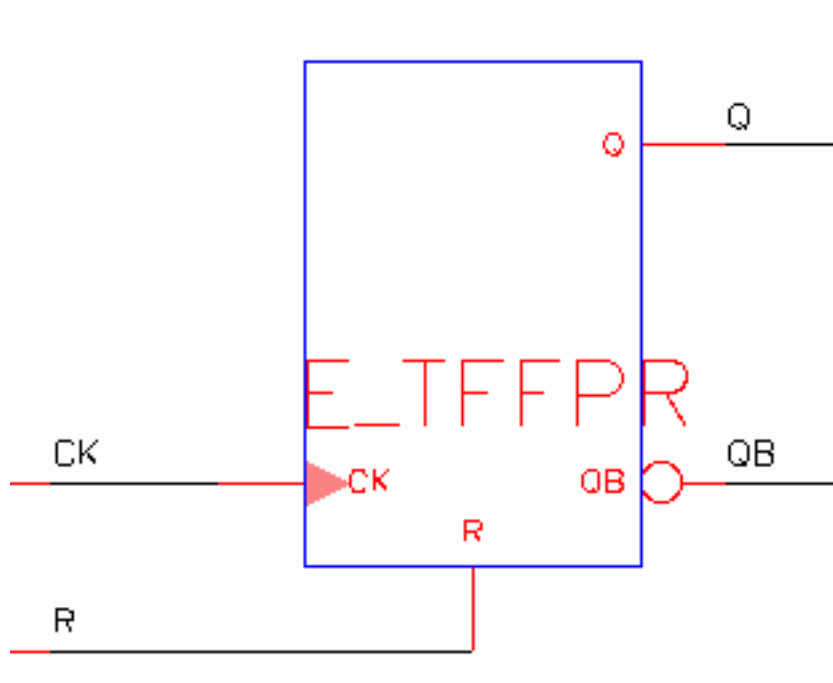
Pin	Direction	Description
CK	INPUT	clock, positive-edge-trigger
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFPRSB : T Flip-Flop, positive-edge-trigger, active-LOW set & reset



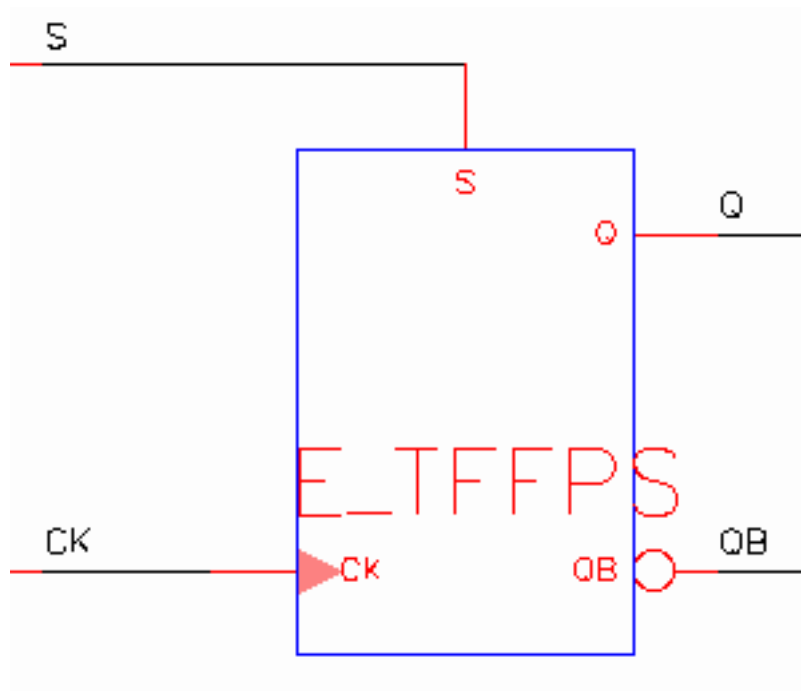
Pin	Direction	Description
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	async set, active-LOW
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_TFFPR : T Flip-Flop, positive-edge-trigger, active-HIGH reset

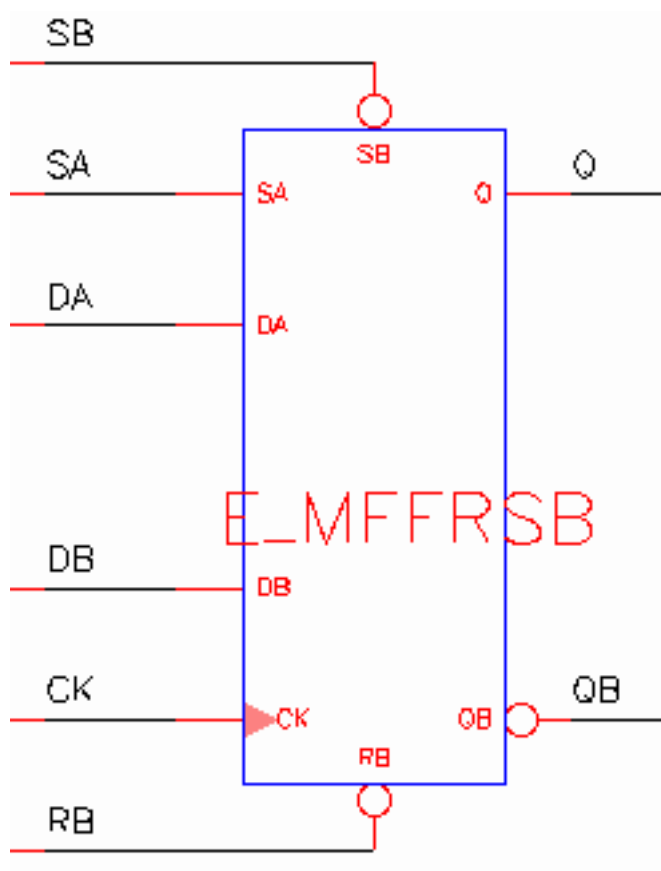


Pin	Direction	Description
CK	INPUT	clock, positive-edge-trigger
R	INPUT	async reset, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

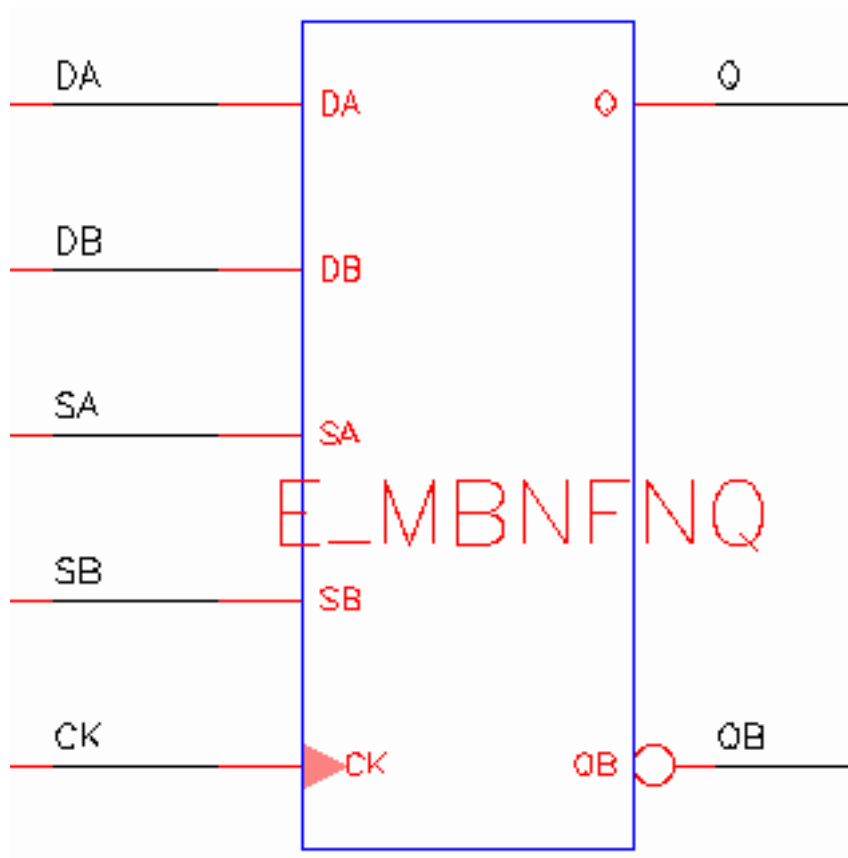
E_TFFPS : T Flip-Flop, positive-edge-trigger, active-HIGH set



Pin	Direction	Description
CK	INPUT	clock, positive-edge-trigger
S	INPUT	async set, active-HIGH
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_MFFRSB : MFBTNB

Pin	Direction	Description
SA	INPUT	select A
DA	INPUT	data A
DB	INPUT	data B
CK	INPUT	clock, positive-edge-trigger
RB	INPUT	async reset, active-LOW
SB	INPUT	select B
Q	OUTPUT	data
QB	OUTPUT	data, inverted

E_MBNFNQ : Multiplexd-FF

Pin	Direction	Description
DA	INPUT	data A
DB	INPUT	data B
SA	INPUT	select A
SB	INPUT	select B
CK	INPUT	clock, positive-edge-trigger
Q	OUTPUT	data
QB	OUTPUT	data, inverted

I/O Functions

[E_IPAD](#)

[E_CIPAD](#)

[E_IPADS](#)

[E_CDDR](#)

[E_BDEBTB](#)

[E_BDE](#)

[E_BDEB](#)

[E_OPAD](#)

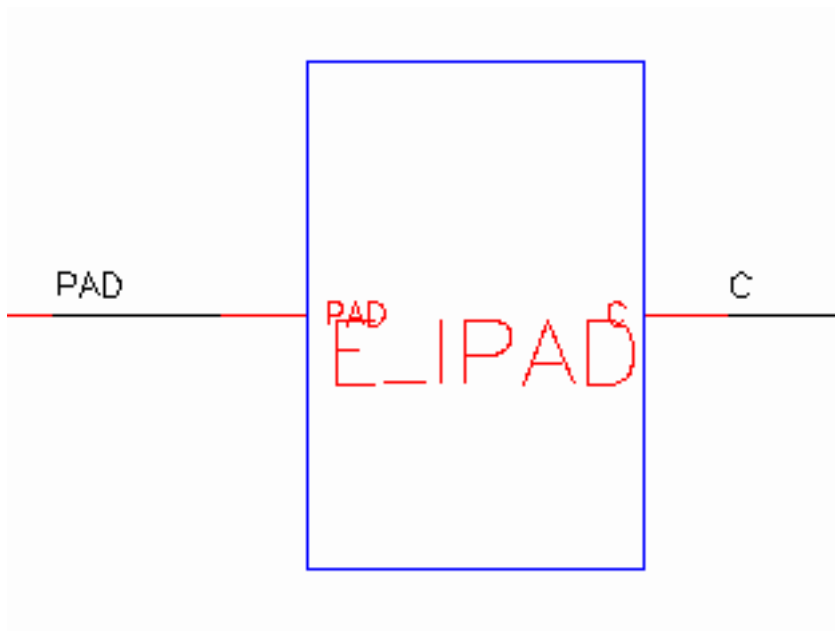
[E_OPADE](#)

[E_OPADEB](#)

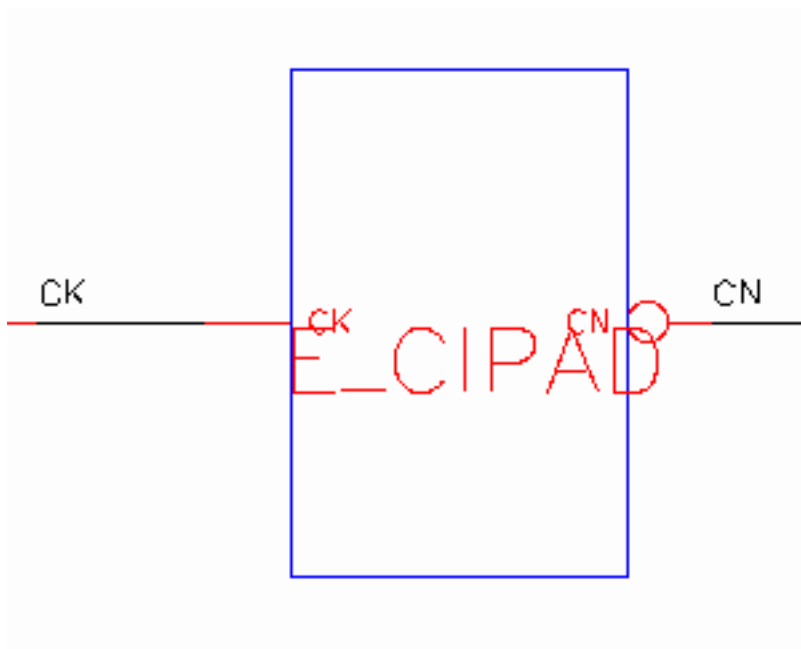
[E_VDDPAD](#)

[E_VSSPAD](#)

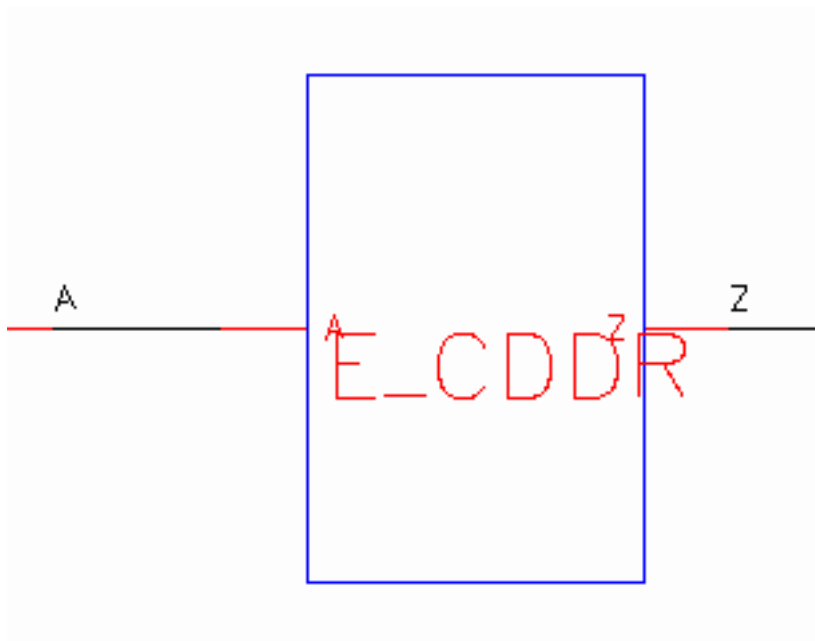
[E_OSCX](#)

E_IPAD : INPUT PAD

Pin	Direction	Description
PAD	INPUT	pad
C	OUTPUT	data

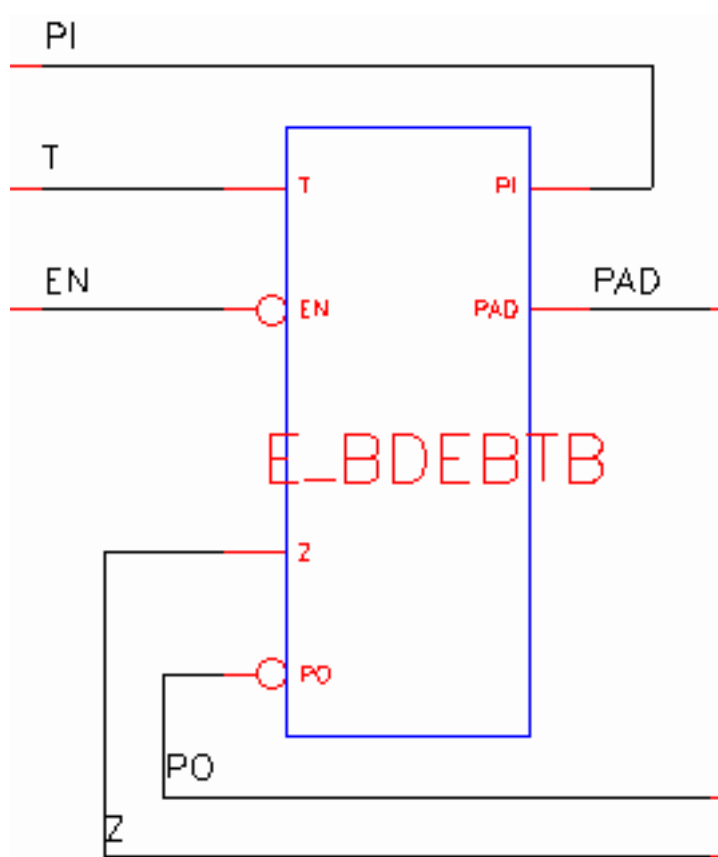
E_CIPAD : Clock Input PAD (Inverted)

Pin	Direction	Description
CK	INPUT	pad
CN	OUTPUT	data, inverted

E_CDDR : CLOCK INPUT CLOCK DRIVER

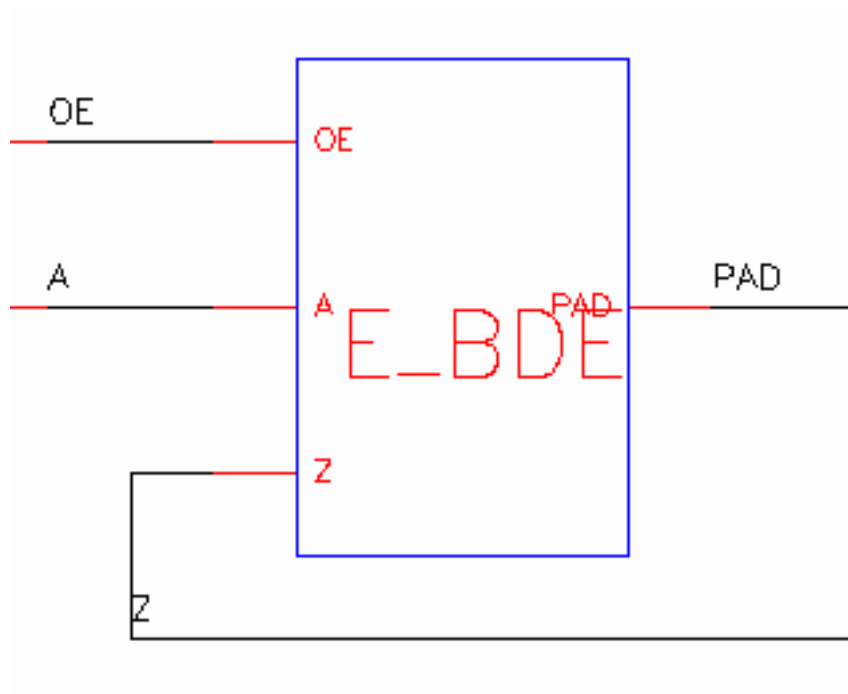
Pin	Direction	Description
A	INPUT	pad
Z	OUTPUT	data

E_BDEBTB : Bidirectional IO buffer, active-LOW test enable, parametric test gate



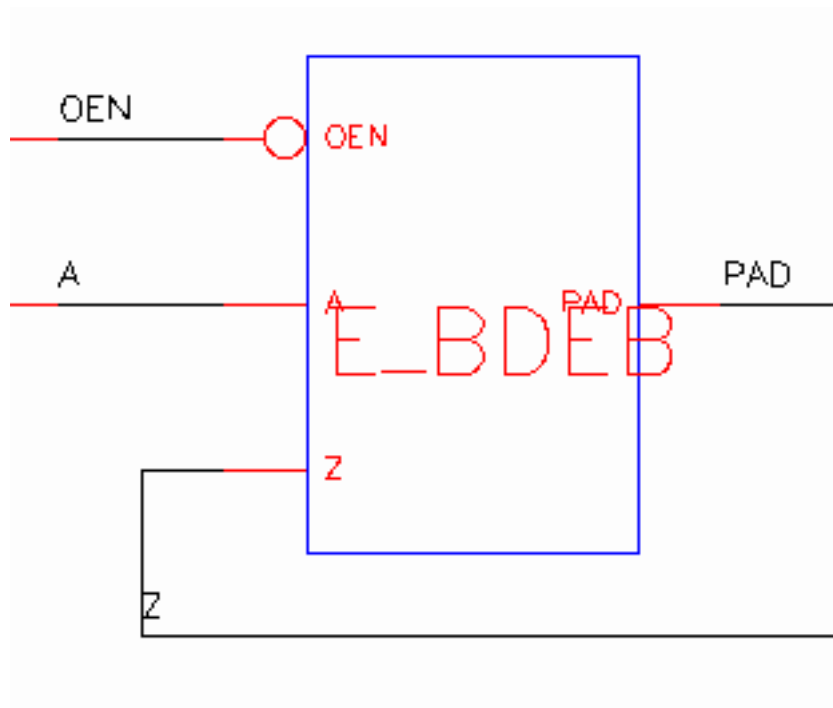
Pin	Direction	Description
T	INPUT	parametric test gate
EN	INPUT	test enable, active-LOW
A	INPUT	data
PI	INPUT	parameteric test scan in
Z	OUTPUT	data
PO	OUTPUT	parameteric test scan out
PAD	INOUT	pad

E_BDE : Bidirectional IO buffer, active-HIGH output enable

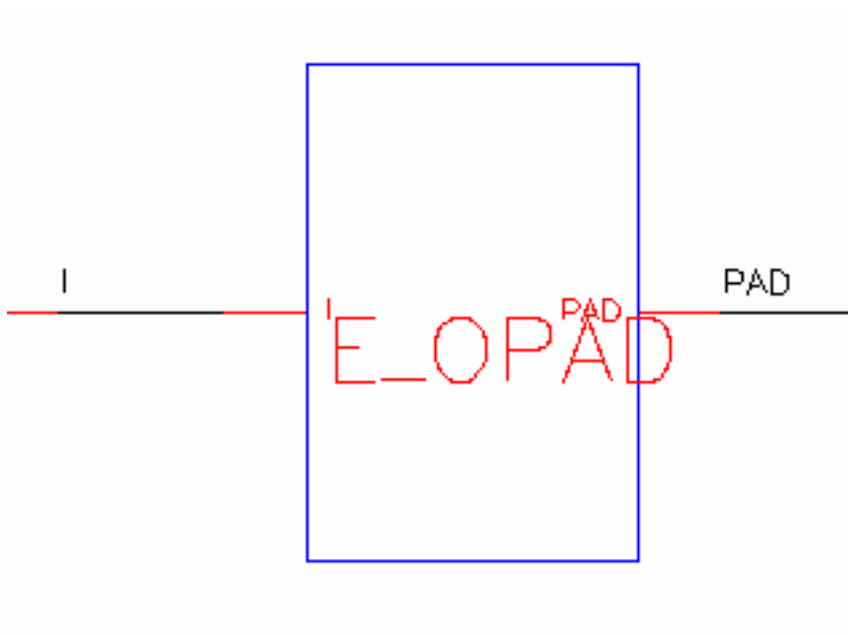


Pin	Direction	Description
OE	INPUT	output enable, active-HIGH
A	INPUT	data
Z	OUTPUT	data
PAD	INOUT	pad

E_BDEB : Bidirectional IO buffer, active-LOW output enable

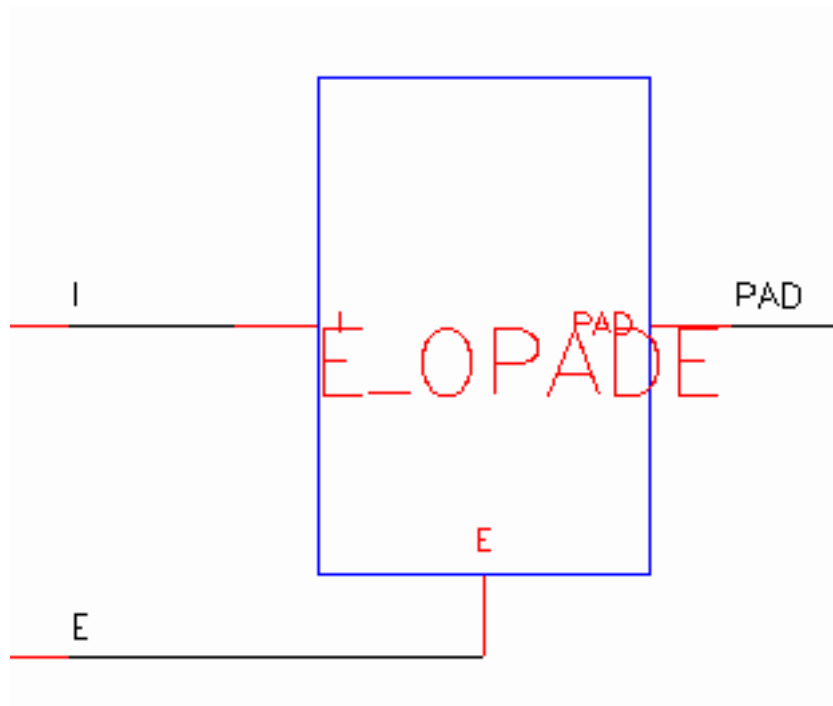


Pin	Direction	Description
OEN	INPUT	output enable, active-LOW
A	INPUT	data
Z	OUTPUT	data
PAD	INOUT	pad

E_OPAD : OUTPUT PAD

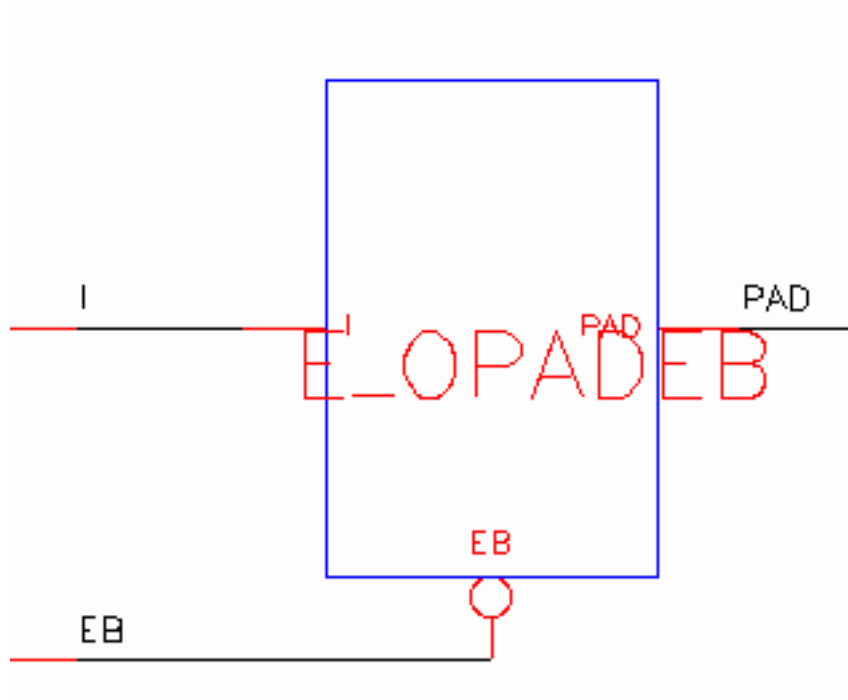
Pin	Direction	Description
I	INPUT	data
PAD	OUTPUT	pad

E_OPADE : OUTPUT PAD, active-HIGH output enable, tri-state output

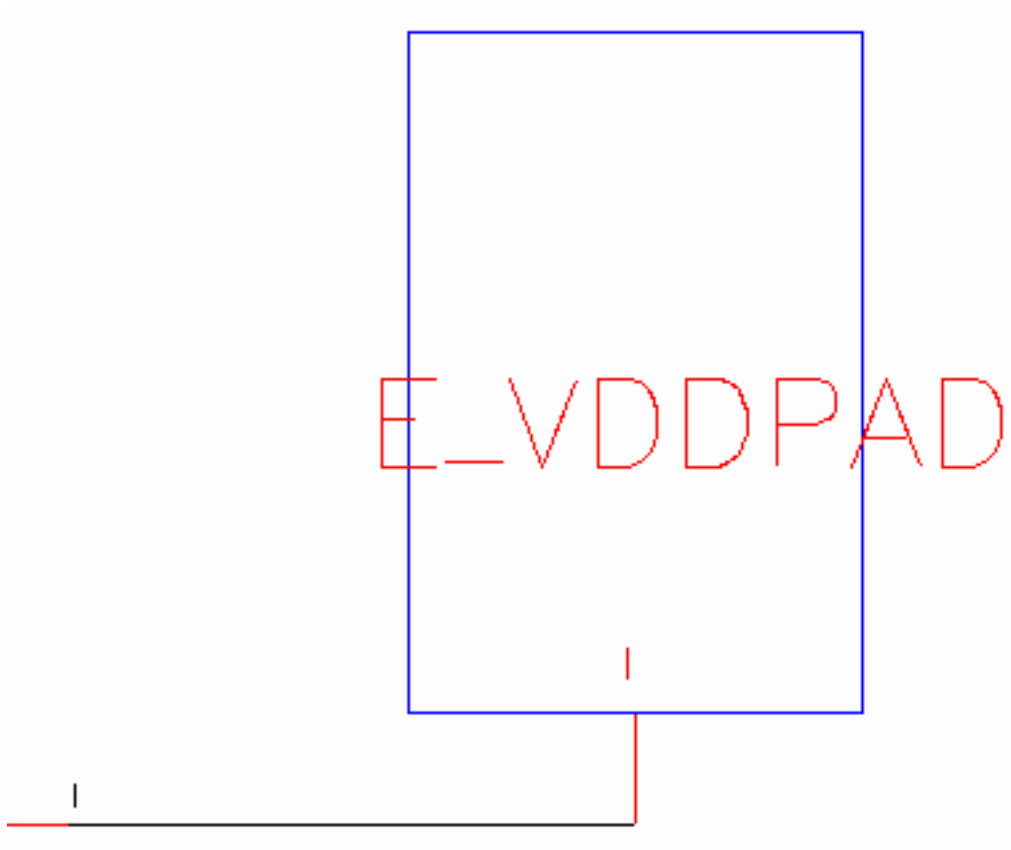


Pin	Direction	Description
I	INPUT	data
E	INPUT	output enable, active-HIGH
PAD	OUTPUT	pad

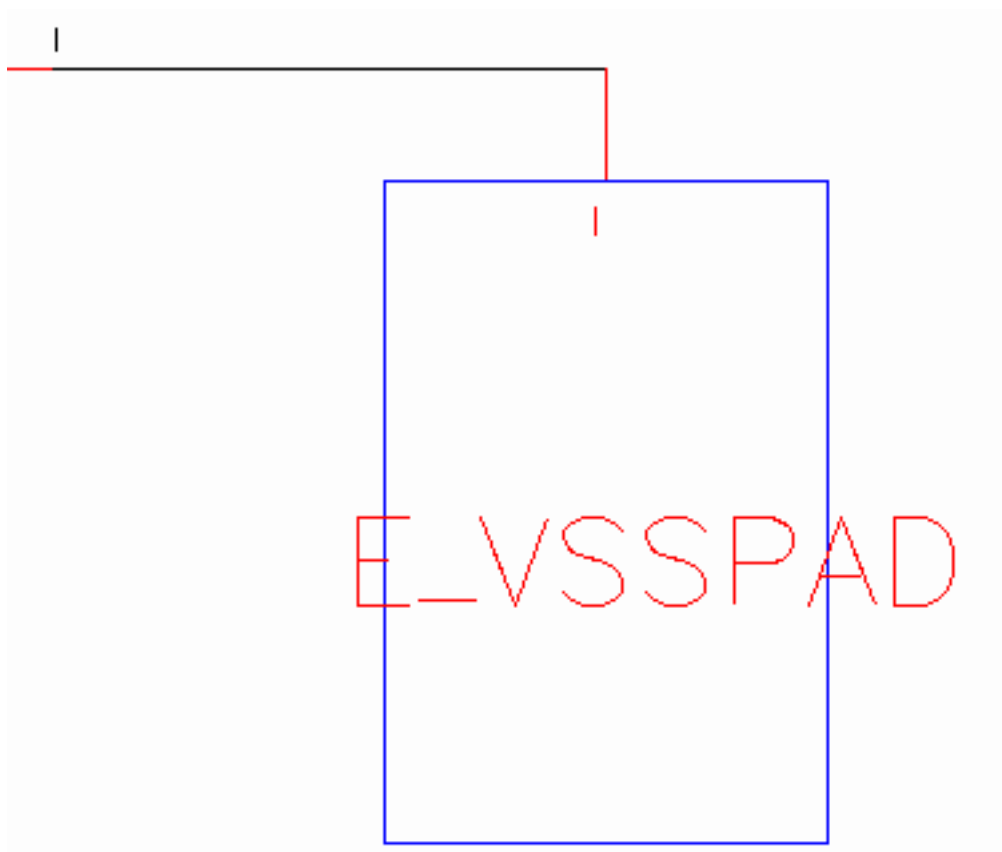
E_OPADEB : CMOS OUTPUT PAD, active-LOW ooutput anable, tri-state output



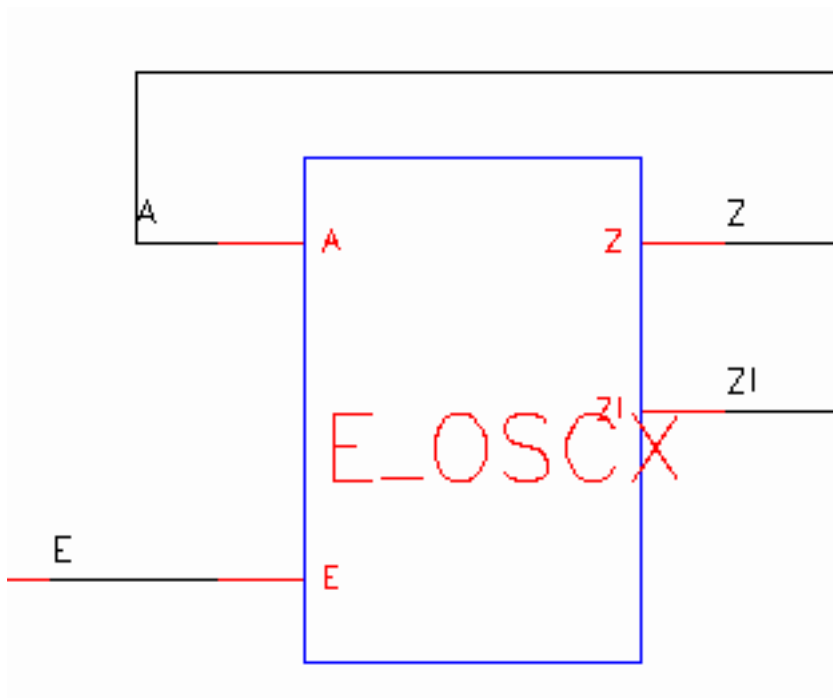
Pin	Direction	Description
I	INPUT	data
EB	INPUT	output enable, active-LOW
PAD	OUTPUT	pad

E_VDDPAD : POWER PAD

Pin	Direction	Description
I	INPUT	data

E_VSSPAD : GROUND PAD

Pin	Direction	Description
I	INPUT	data

E_OSCX : CRYSTAL OSCILLATOR PADS with Feedback Register

Pin	Direction	Description
A	INOUT	data
E	INPUT	output enable, active-HIGH
Z	OUTPUT	data
ZI	OUTPUT	data

Miscellaneous

[E_CLOAD1](#)

[E_MAXCAP](#)

[E_NETWTH](#)

[E_RP01D1](#)

[E_WEIGHT](#)

[E_VLOAD](#)

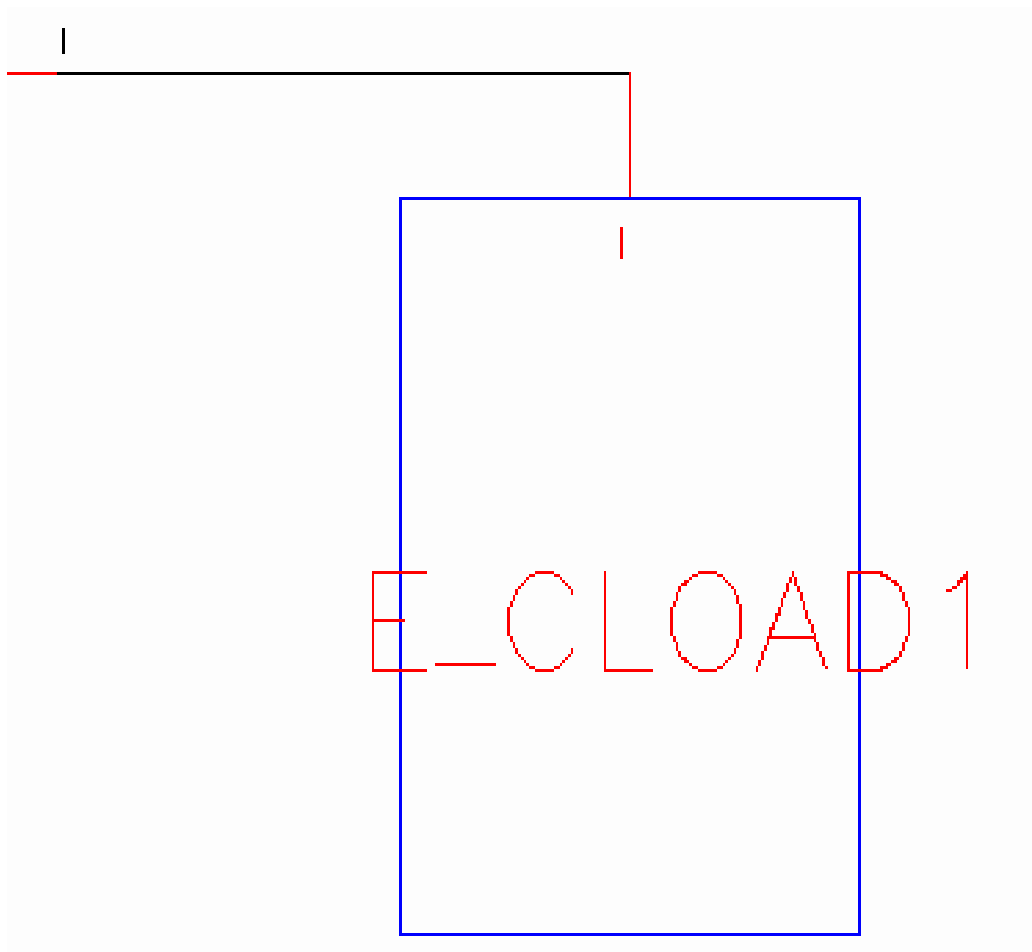
[E_PASS](#)

[E_PD](#)

[E_PU](#)

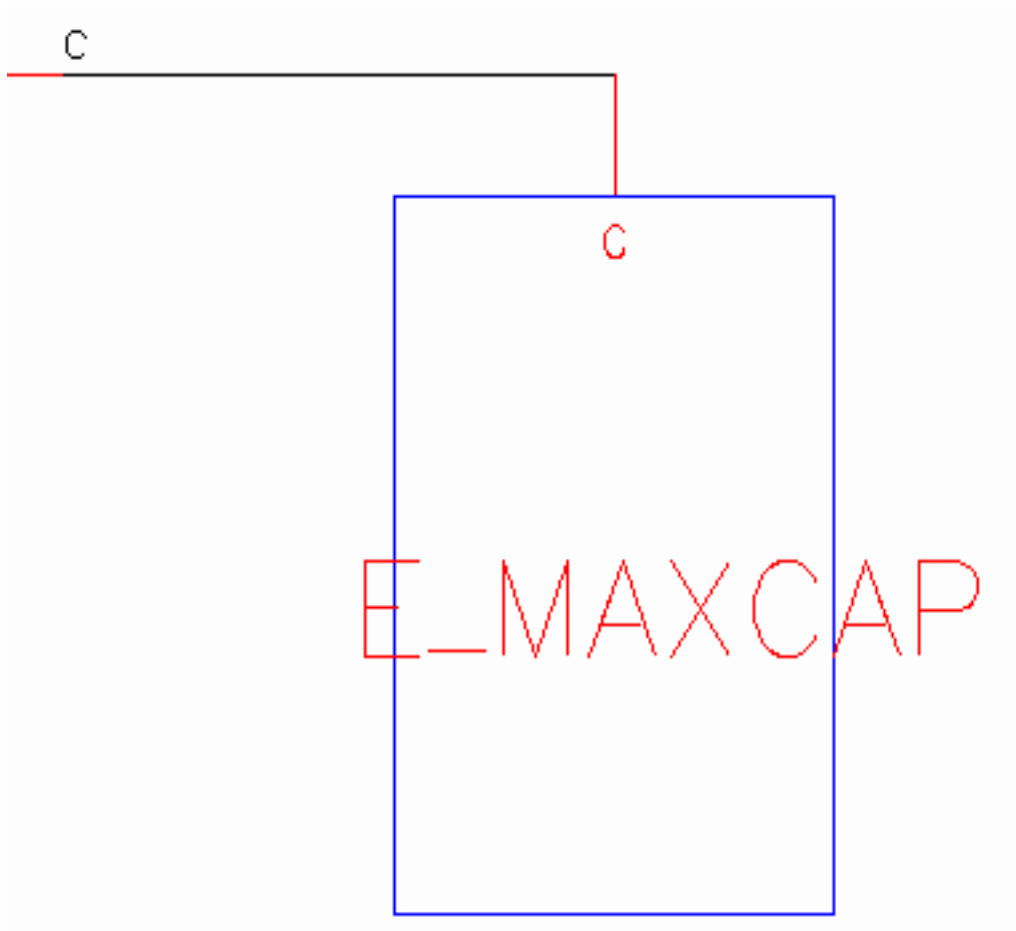
[E_TIE](#)

[E_DELAY](#)

E_CLOAD1 : clock load cell

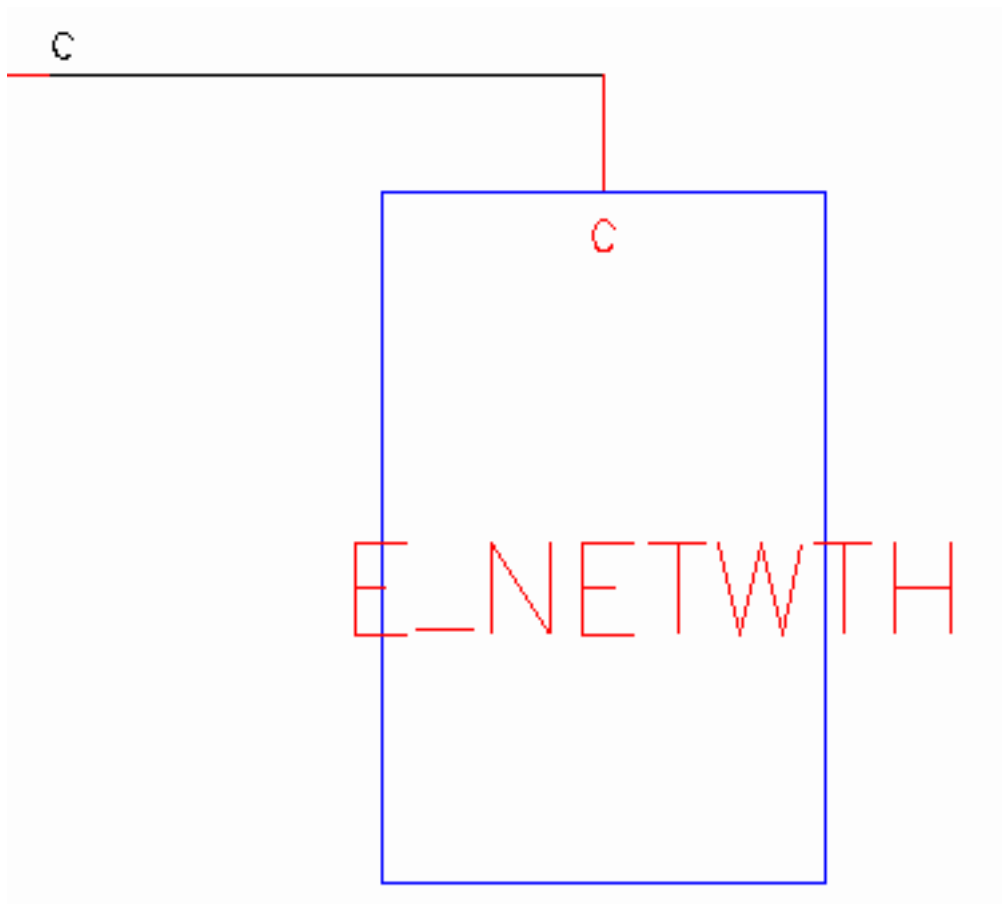
Pin	Direction	Description
I	INPUT	data

E_MAXCAP : maximum capacitance symbol

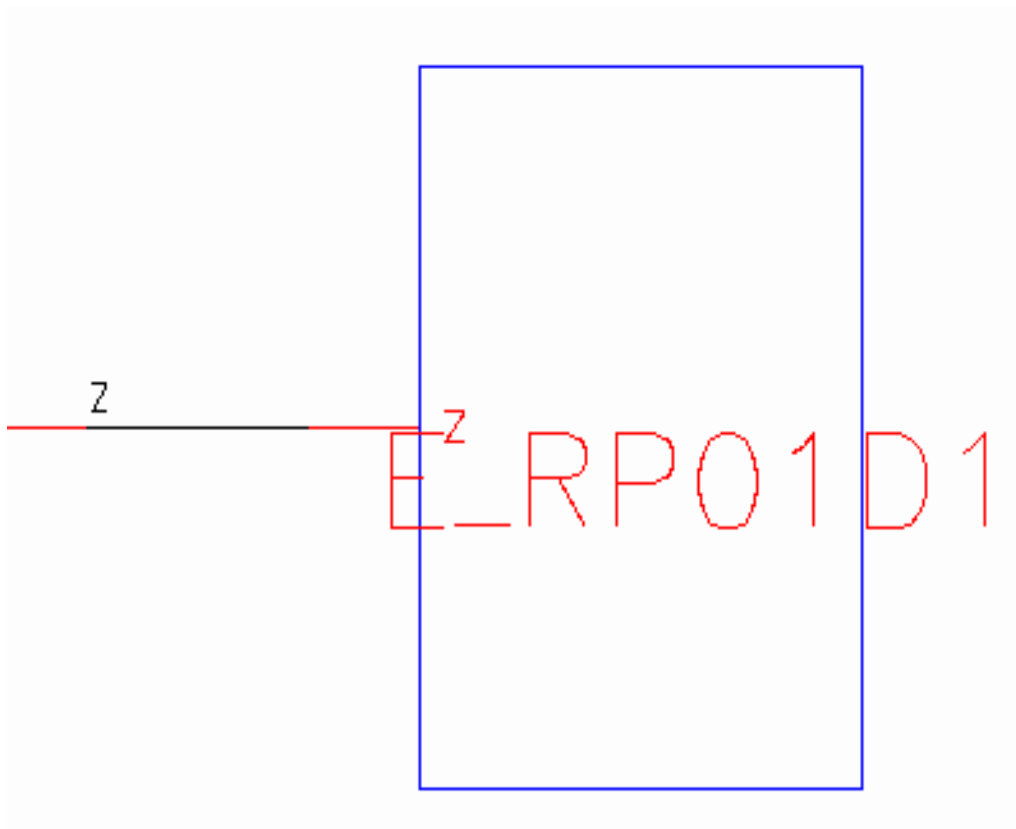


Pin	Direction	Description
C	INPUT	data

E_NETWTH : net width symbol

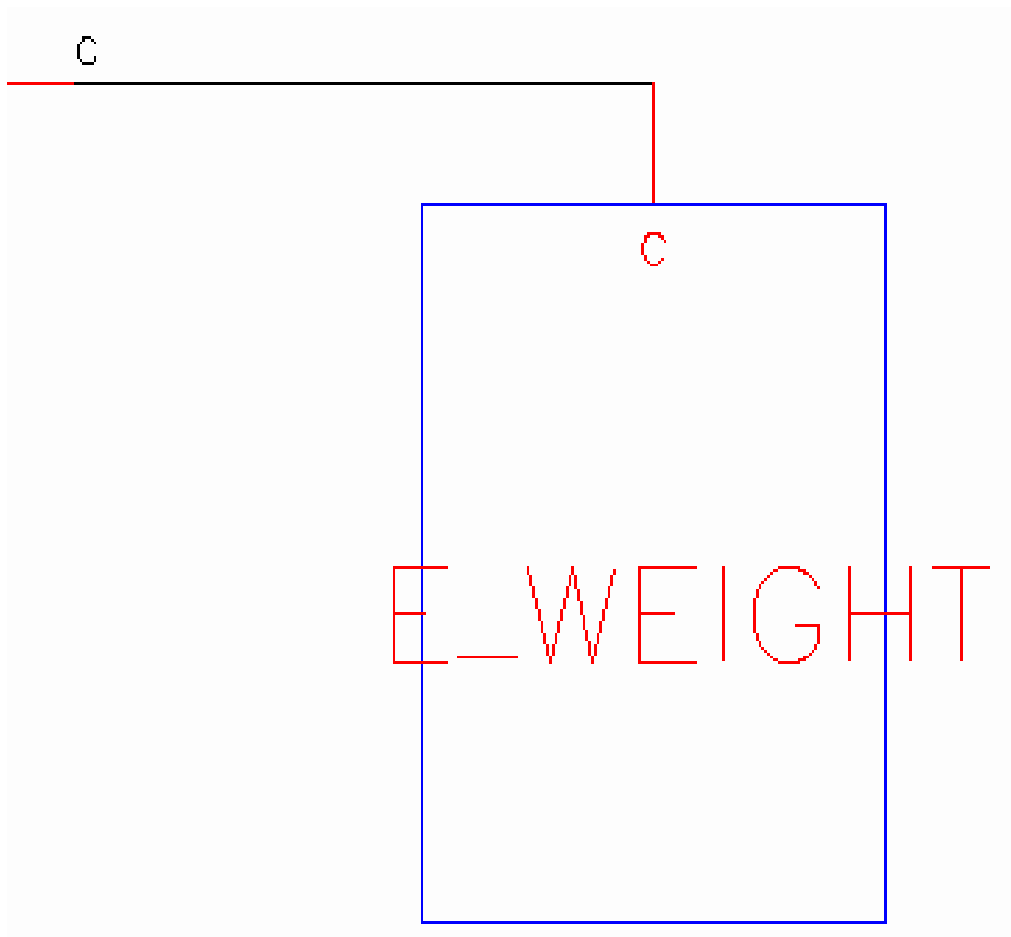


Pin	Direction	Description
C	INPUT	data

E_RP01D1 : repeater cell

Pin	Direction	Description
Z	INPUT	data

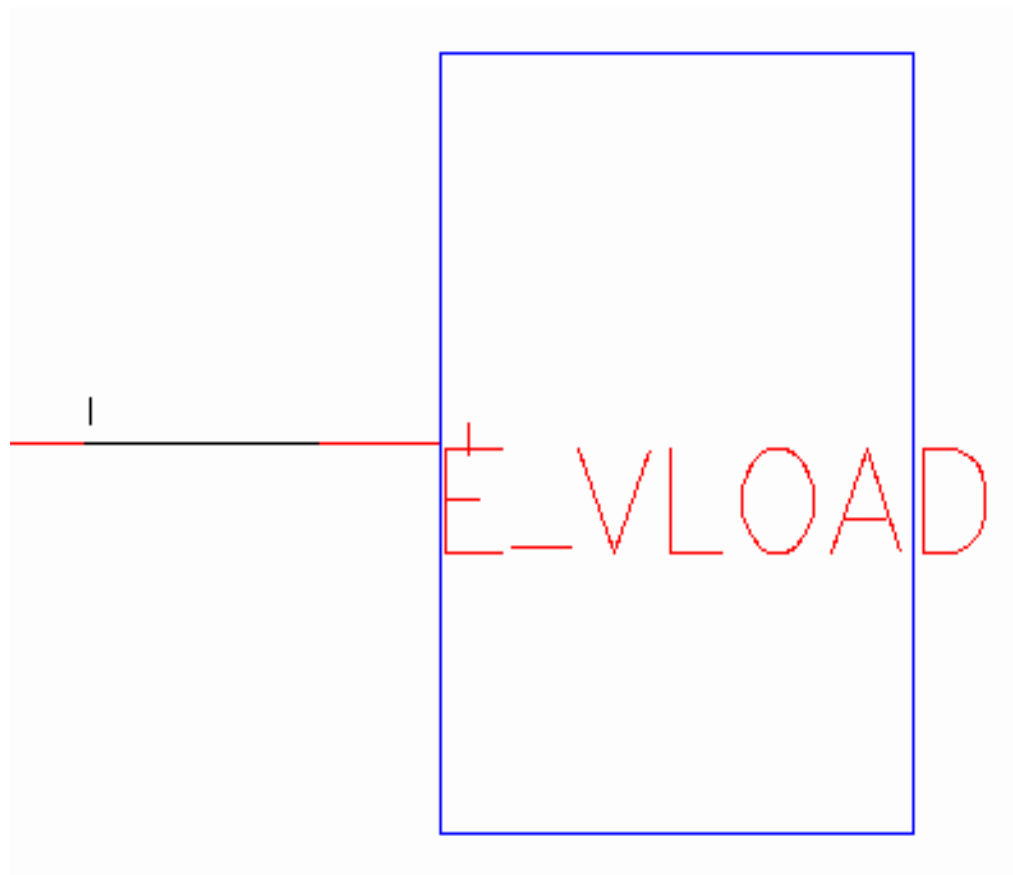
E_WEIGHT : weight icon



Pin	Direction	Description
C	INPUT	data

E_VLOAD

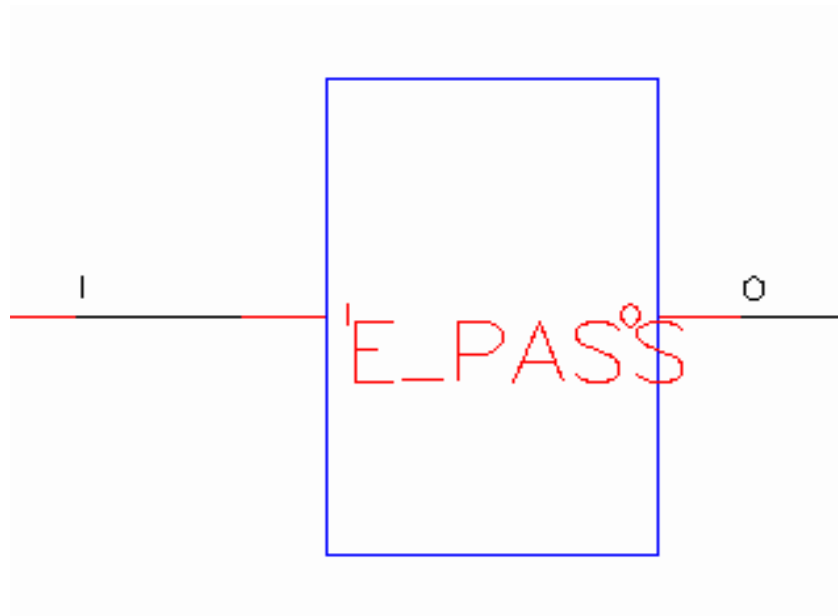
E_VLOAD : voltage load



Pin	Direction	Description
I	INPUT	data

E_PASS

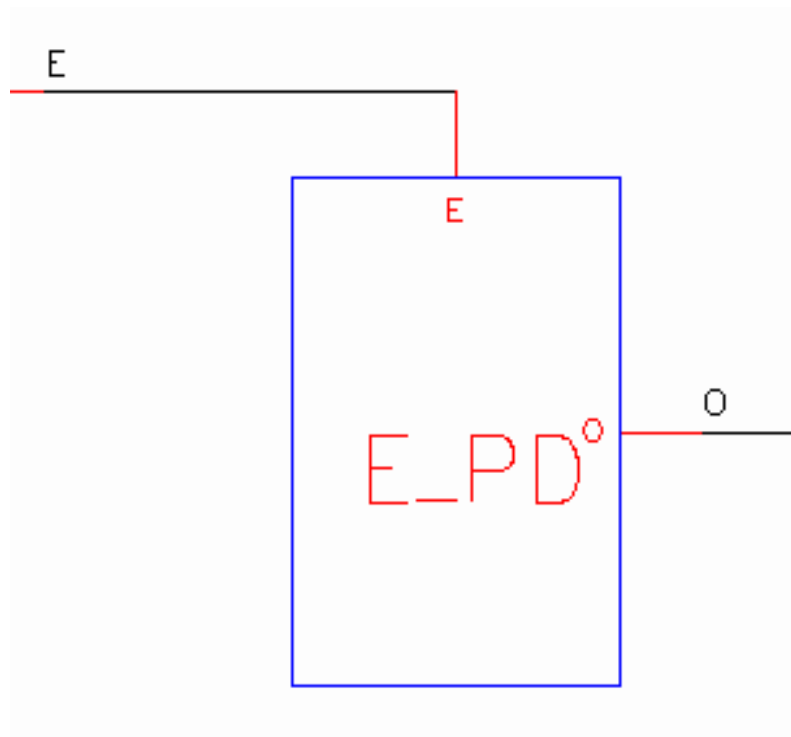
E_PASS : pass cell



Pin	Direction	Description
I	INPUT	data
O	OUTPUT	data

E_PD

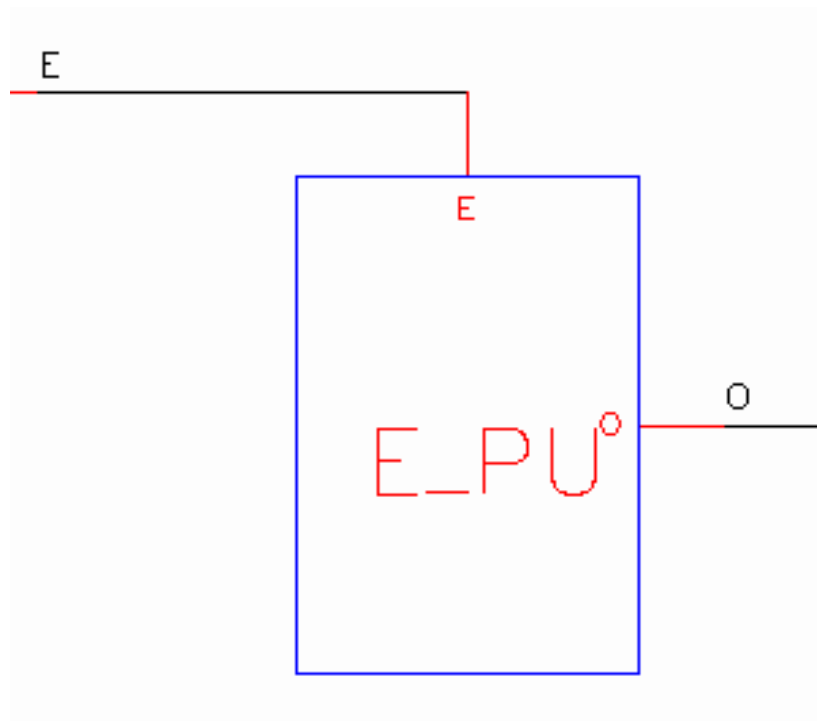
E_PD : pull/down cell



Pin	Direction	Description
E	INPUT	output enable, active-HIGH
O	OUTPUT	data

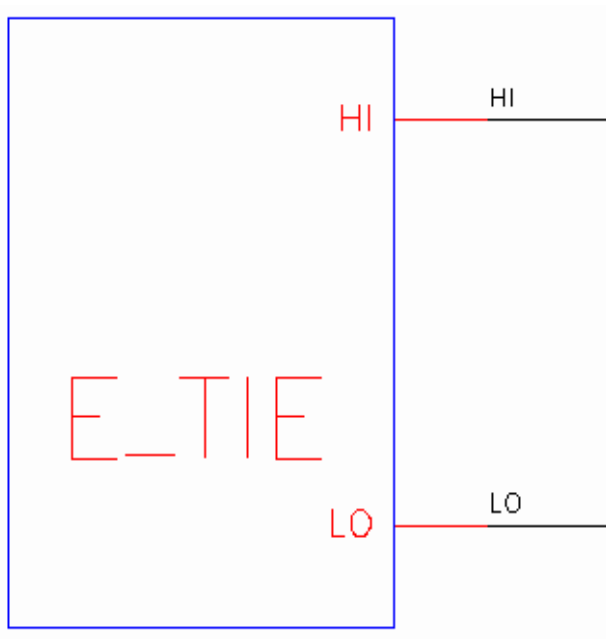
E_PU

E_PU : pull/up cell

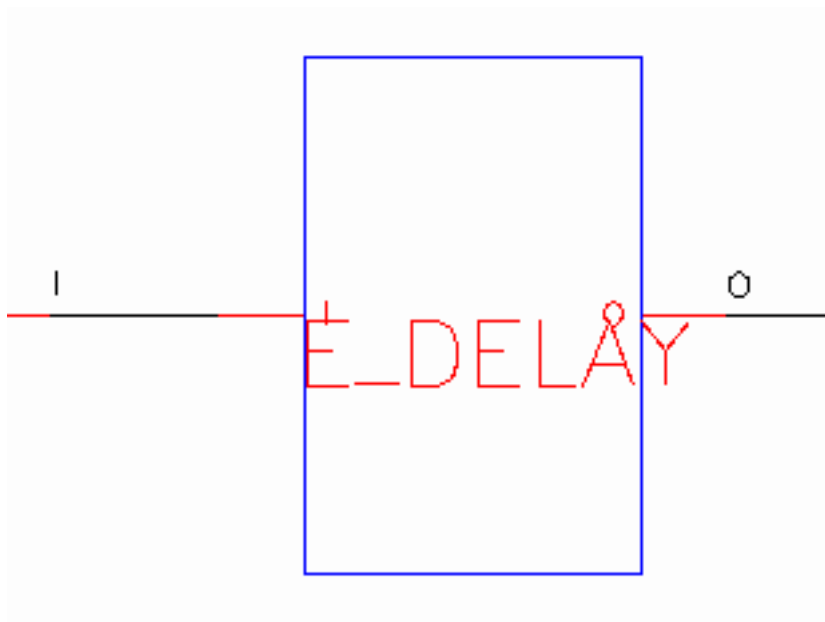


Pin	Direction	Description
E	INPUT	output enable, active-HIGH
O	OUTPUT	data

E_TIE : tie HIGH/LOW cell



Pin	Direction	Description
HI	OUTPUT	data
LO	OUTPUT	data

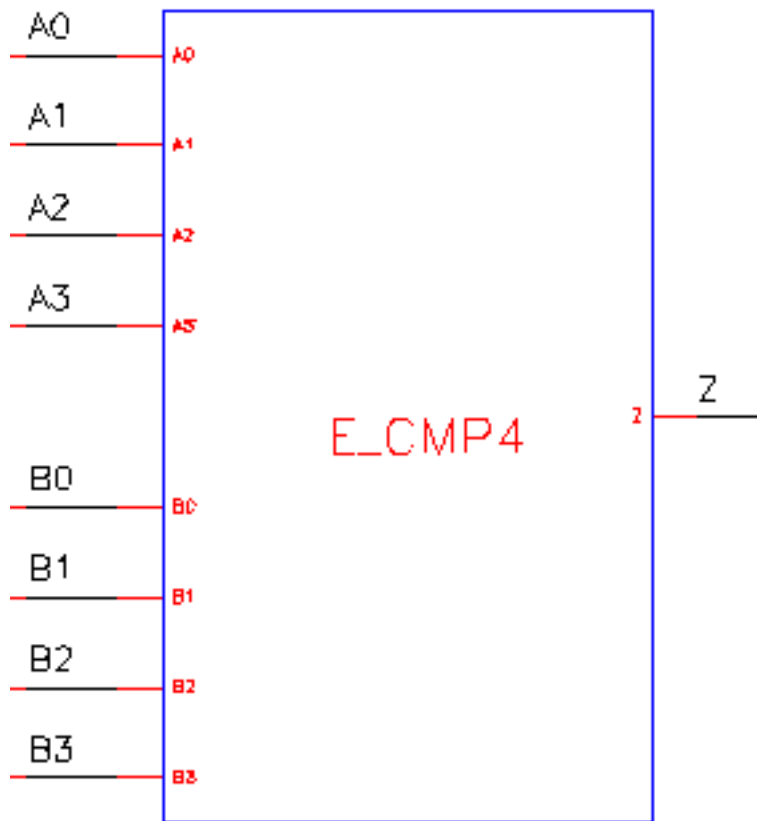
E_DELAY: DELAY

Pin	Direction	Description
I	INPUT	data
O	OUTPUT	data

Comparator Gates

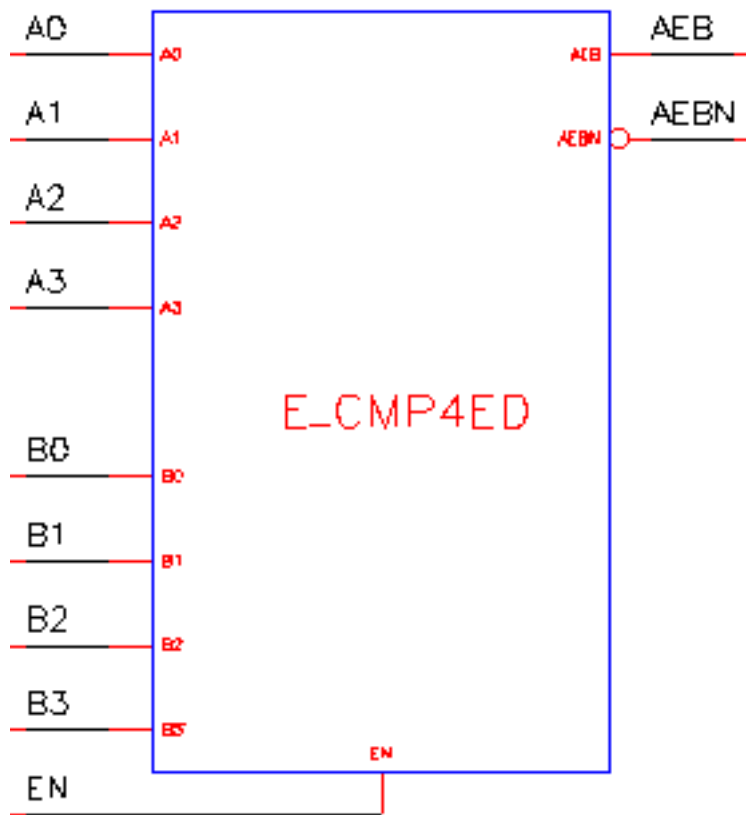
[E_CMP4](#)

[E_CMP4ED](#)

E_CMP4 : 4-bit comparator (equal only)

Pin	Direction	Description
A0	INPUT	operand, LSB
A1	INPUT	operand
A2	INPUT	operand
A3	INPUT	operand, MSB
B0	INPUT	operand, LSB
B1	INPUT	operand
B2	INPUT	operand
B3	INPUT	operand, MSB
Z	OUTPUT	data

E_CMP4ED : 4-bit comparator (equal & below), active-HIGH enable, complementary output



Pin	Direction	Description
A0	INPUT	operand, LSB
A1	INPUT	operand
A2	INPUT	operand
A3	INPUT	operand, MSB
B0	INPUT	operand, LSB
B1	INPUT	operand
B2	INPUT	operand
B3	INPUT	operand, MSB
EN	INPUT	data
AEB	OUTPUT	data

AEBN	OUTPUT	data, inverted
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